Aerospace Technology

Study notes for the first midterm

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and Associated Technologies · UPV

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Summary of the course slides, written for the exam, with extra explanations for readers new to the field. It covers the first-midterm syllabus: the slides, up to propulsive systems, plus a final unit with the additional course documents and the review questions. Red boxes mark what you must know and blue boxes are worked examples, many taken from past exams.

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UNIT 1

Introduction to aerospace technology

1 Course overview

Aerospace Technology (code 11888, UPV, academic year 2026–2027) is a **general overview** course: it does not go deep into a single topic, but gives a view of all the systems that make up an aircraft or a spacecraft, with particular emphasis on propulsion. It bridges the basic subjects (physics, mathematics, thermodynamics) and the applied ones (aircraft design, propulsion, structures). It is worth 4.5 ECTS credits (2.5 theory and 2.0 practical), and a maximum of 20 % absences is allowed, which weighs especially on the practical sessions.

1.1 Content and assessment

The slides divide the syllabus into five blocks, and these notes follow that order. The available slides cover only the first two blocks, and these notes cover only those:

Course block	Content	Unit of these notes
Introduction	industry, vehicles, infrastructure	1
Fixed-wing aircraft	environment, design, structures, flight, propulsion	2 to 6
Space vehicles	launchers, space environment, satellites	not covered (no slides)
Missiles	classification, guidance	not covered (no slides)
Miscellaneous aircraft	rotorcraft, special aircraft	not covered (no slides)

Assessment	Weight and date	Details
Multiple-choice test 1	20 %, 05/11/2026, 12:30	up to section 2.5
Multiple-choice test 2	20 %, 15/01/2027, 8:00	from section 2.6 onwards
Open-answer test	35 %, 15/01/2027, 9:00	all sections
Oral defense	20 % (group)	30/11 to 02/12/2026; plus a debate on 18/12/2026
Observation	5 %	throughout the course
Resit	27/01/2027, 11:30	any combination of the three written tests; the resit mark replaces the original one

WHAT THE MULTIPLE-CHOICE TESTS ARE LIKE
According to past exams: multiple-choice questions with **a single correct answer**, and **each wrong answer subtracts a third** of what a correct one adds. Blank answers score nothing. It is best to answer only when you can rule out at least one option.

Note: the slides do not state how sections 2.5 and 2.6 map onto these units, and no mapping has been assumed (source: course presentation, slides 6 to 11).

2 The aerospace sector

Aerospace covers two worlds: **aeronautics** (flight within the atmosphere) and **astronautics** (flight outside it). The slides divide the activity into six groups: industry, airlines, aeronautical organizations, space agencies, research centers and the military.

2.1 The aerospace industry

It is a strategic activity with very particular characteristics:

Characteristics (learn them: they appear in tests)	
Cutting-edge technology	**Long** delivery times
Highly qualified staff	High investment required
Small scales	Government intervention
Difficult automation	International cooperation

The logic is a chain: few units are built (small scales) and each one is complex (difficult automation), so each aircraft takes a long time to deliver and requires a lot of money. That is why governments intervene and projects are shared among countries.

> **COMMON TEST MISTAKE**
> "*Short* delivery times" is **not** a characteristic of the aerospace industry (it appeared in the 2018/19 and 2019/20 exams). They are long.

A fact to remember the cost. Pound for pound, an aircraft engine is worth the same as silver, and a car the same as a hamburger (Northwest Aerospace Alliance, cited on slide 5). In the 2022/23 exam, the cost per kg of an A320 turbofan was asked; the official key as collected was *"about 100 times that of a car piston engine"* (the option "like gold" was not marked), so it is worth remembering that the cost per kilogram is an order of magnitude higher than that of a car engine, not the exact figure.

Breakdown of the industry (slide 6; Figure 1.1):

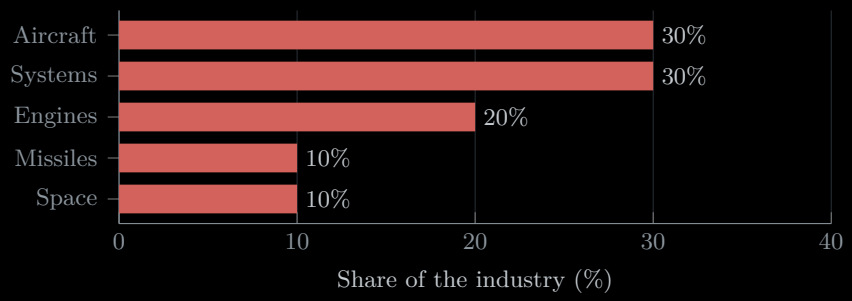


Figure 1.1. Approximate breakdown of aerospace industry activity. Source: T1, slide 6.

The teams are **highly multidisciplinary** (flight physics, propulsion, structures, avionics…). Each program requires: design and development; advanced testing and calculations (wind tunnel, flight tests, CFD, *Computational Fluid Dynamics*); production (manufacturing, assembly and certification); and maintenance.

2.2 Airlines

Airlines are the industry's **customers**. They are distinguished by their target audience: traditional (Iberia, Lufthansa), low-cost (easyJet, Ryanair, Vueling) and regional (Air Nostrum). It is a market

with a large volume of money but small profits and fierce competition, so airlines form alliances, merge, and many will disappear. For the designer this means that the customer demands **low operating cost**.

2.3 Aeronautical organizations

They were created mainly to **maintain air traffic safety**. It is worth knowing what each one does and its scope (Figure 1.2):

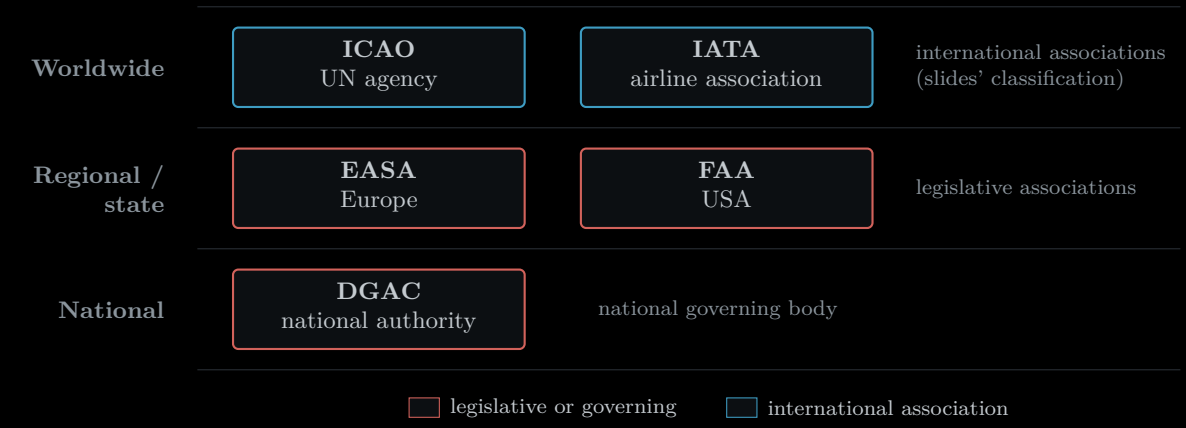


Figure 1.2. Who is who in the aeronautical organization, following the slides' classification. Source: T1, slide 9.

- **ICAO** (International Civil Aviation Organization): specialized UN agency. It is **not** a research center, nor the agency that legislates for Spain.
- **IATA** (*International Air Transport Association*): partnership of airlines for cooperation.
- **EASA** (*European Union Aviation Safety Agency*): main legislative organization in **Europe**. **FAA** (*Federal Aviation Administration*): the one in the **USA**.
- **DGAC** (General Directorate of Civil Aviation): national governing body.

> **BEWARE OF EXAM MIX-UPS**
>
> - **INTA** is a research and testing center and does **not** legislate (2023/24 question).
> - **ESA** is a **space** agency; it does not define European aviation legislation: that is EASA (2023/24 and 2024/25).
> - ICAO is an international organization **created by the UN** (2020/21). IATA does **not** legislate in Europe. DGAC is not a US body.

2.4 Space agencies and research centers

The main space agencies are **ESA** (*European Space Agency*) and **NASA** (*National Aeronautics and Space Administration*); some countries have their own. Research centers include the space agencies themselves, the French **ONERA** (*Office National d'Études et de Recherches Aérospatiales*) and **INTA** (*National Institute of Aerospace Technology*, the Spanish public institute for R&D in aeronautics, space and naval operations).

INTA tests materials, components, equipment and systems, and provides services to industry. Milestones cited in the slides: *Intasat* (the first Spanish satellite) and collaboration with NASA's Mercury and Apollo projects. Its facilities (slide 11) include a turbojet testing center, the El Arenosillo Testing Centre and space tracking stations.

2.5 Military activity

Air forces are both **customers and users** of aerospace equipment, and have maintenance and research centers. The Spanish Air Force is organized into a Force (material and human resources to prepare for and carry out operations) and a Force Support (management, administration and control of resources), according to slide 12.

3 Classification of aerospace vehicles

Three major families are used: **aircraft**, **missiles** and **space vehicles**, organized as in Figure 1.3.

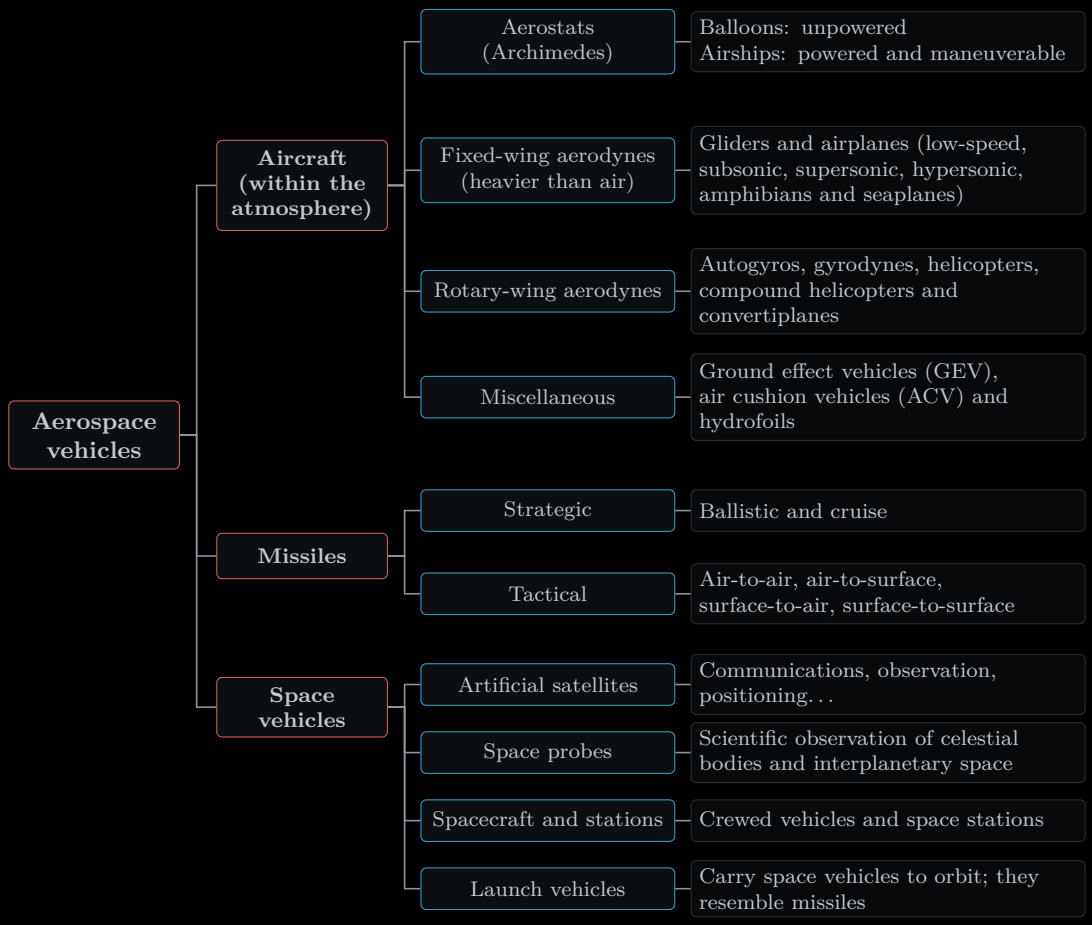


Figure 1.3. Classification tree of aerospace vehicles. Source: T1, slides 14 to 23 (slide 21 repeats "air-to-surface"; it has been completed with surface-to-air).

3.1 Aircraft: how they overcome weight

The criterion on slide 14 is *how they generate the force that overcomes gravity*. There are two fundamentally different ways. Figure 1.4 shows what each family looks like and Figure 1.5 where its supporting force comes from:

- **Aerostats**: they use **Archimedes' principle** ("any body submerged in a fluid experiences an upward vertical force equal to the weight of the fluid displaced"). They are large containers filled with a light gas, so that their average density is **lower than that of the surrounding air**. Two types: **balloons** (unpowered) and **airships** (powered and maneuverable).
- **Aerodynes** (the slide simply calls them *aircraft*): they are **heavier than air** and are supported by aerodynamic forces generated by fixed or rotating wings.

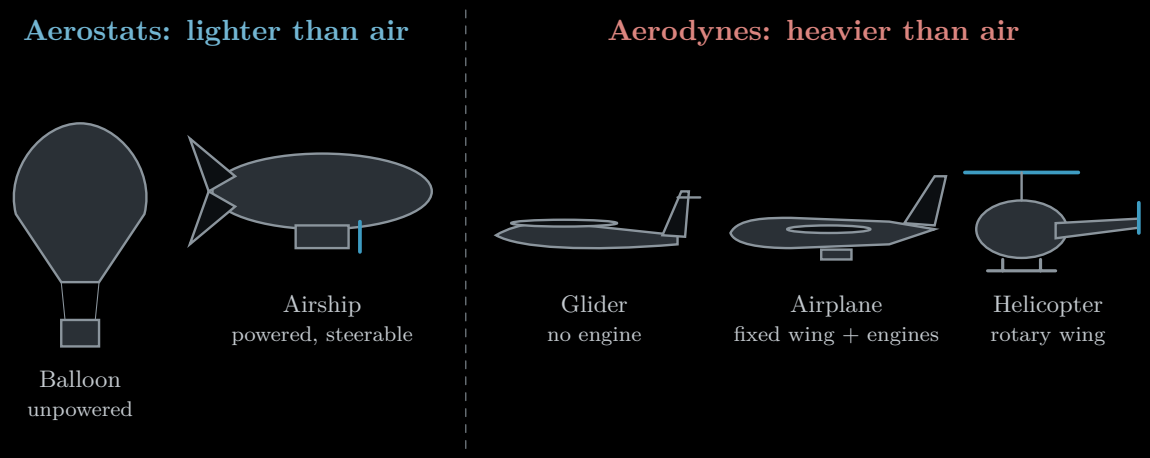


Figure 1.4. Aircraft by how they overcome weight: aerostats (lighter than air) and aerodynes (heavier than air). Side views, schematic silhouettes, not to scale.

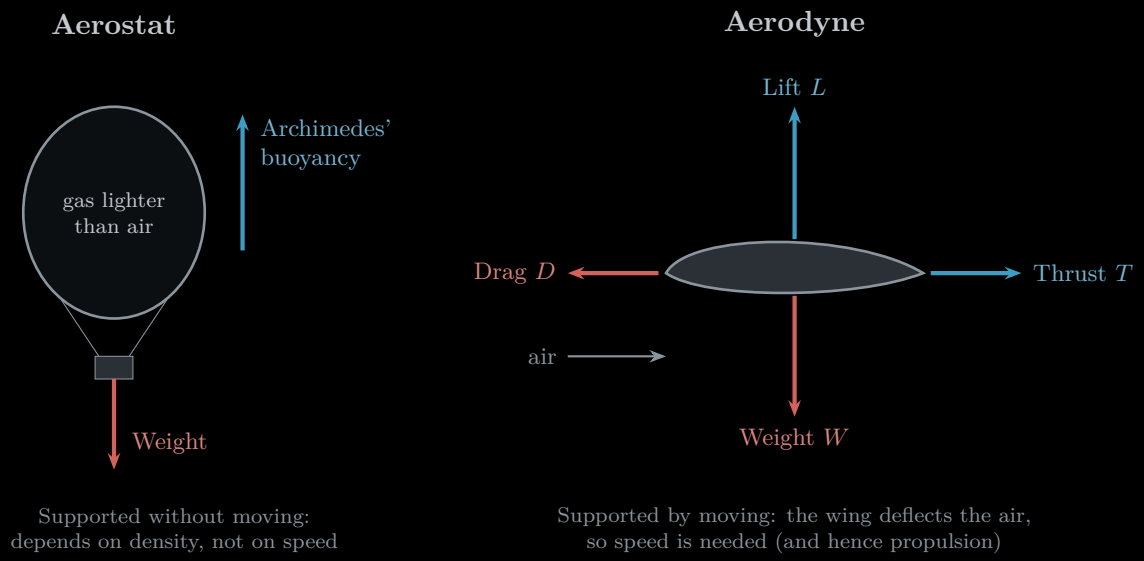


Figure 1.5. Aerostat versus aerodyne: where the force that supports the aircraft comes from.

- **Miscellaneous**: Ground Effect Vehicles (GEV), which fly very close to the surface on a layer of air; Air Cushion Vehicles (ACV, or *hovercraft*), which create a volume of pressurized air under the vehicle; and hydrofoils, wing sections that work submerged in water instead of air (Figure 1.6).

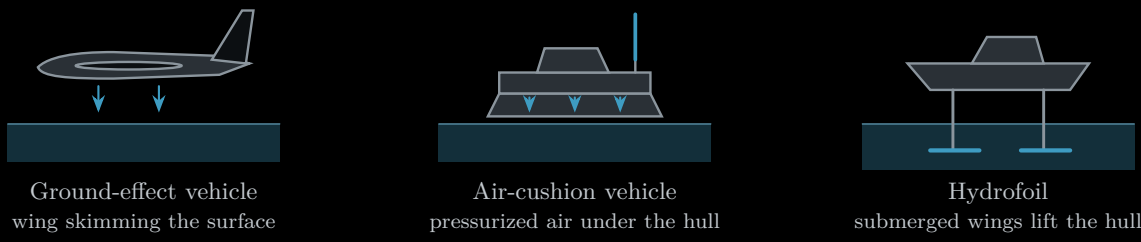


Figure 1.6. Miscellaneous vehicles: ground-effect vehicle (cyan arrows = air trapped between wing and surface), air-cushion vehicle and hydrofoil. Schematic.

Analogy. An aerostat floats like a cork in water and does not need to move. An aerodyne is supported like a planing boat: it only does so while moving forward, because the wing needs air in relative motion.

Aerodynes are subdivided into:

- **Fixed wing**: gliders (unpowered) and airplanes. They are also classified by speed regime (low speed, subsonic, supersonic, hypersonic), and there are amphibians and seaplanes.
- **Rotary wing**: autogyros, gyrodynes, helicopters, compound helicopters and convertiplanes.

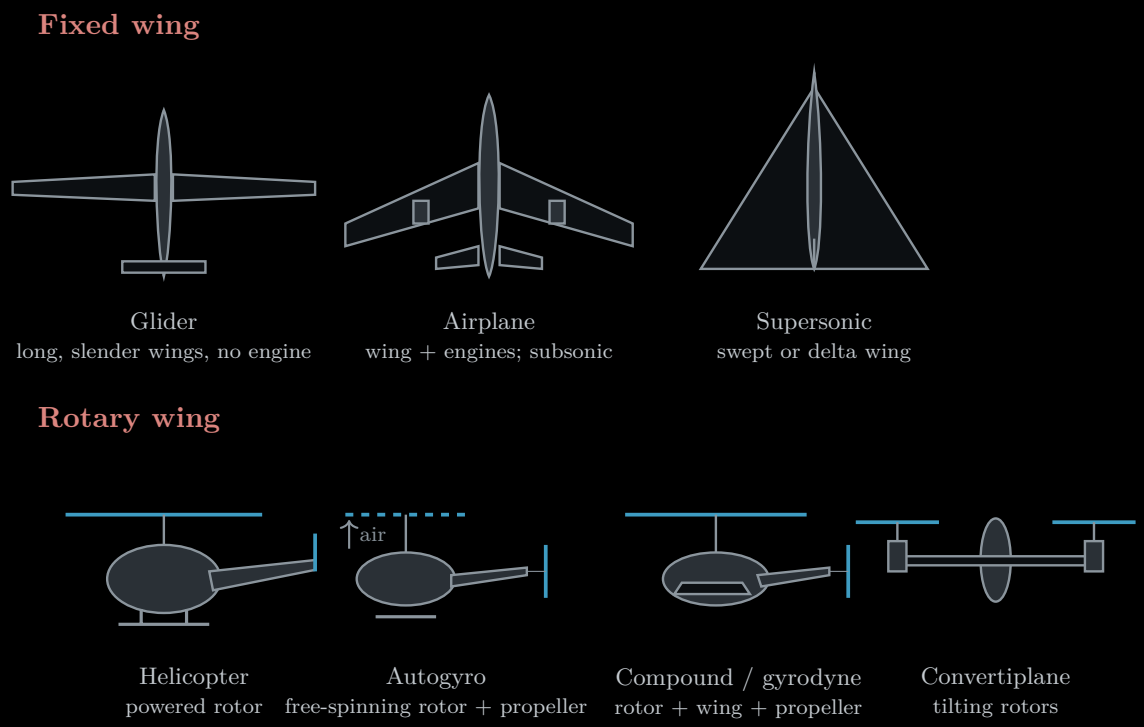


Figure 1.7. Aerodyne families: fixed wing (top views) and rotary wing (side views, blue = rotor). Schematic silhouettes, not to scale.

RECURRING EXAM QUESTION: AEROSTATS VERSUS AERODYNES
Aerostats do **not** need airflow over airfoils (they are not supported by a wing), do **not** deflect flow toward the ground, and the A320 or the B737 are **aerodynes**. It is not true that aerodynes must be lighter than air (it is the opposite: they are heavier) nor that aerostats with a less dense gas need shorter runways (they need no runway to take off). The popular explanation "the air travels a longer path over the upper surface" is not valid either (rejected in 2022/23). In 2024/25 it was marked that aerostats *cannot fly at high speeds because of their aerodynamic drag* (large volumes).

3.2 Missiles

A missile is an **unmanned, powered vehicle that guides a military payload to a target**. It is different from a guided bomb (which has no propulsion of its own) and from a rocket (which is not guided). The most common classification is by military function:

- **Strategic**: ballistic (long range, leave the troposphere) and cruise.
- **Tactical**: air-to-air, air-to-surface, surface-to-air and surface-to-surface.

In a 2020/21 test the false option was *"cruise missiles commonly reach supersonic speeds in cruise"*: they usually fly subsonic, like a small unmanned airplane.

3.3 Space vehicles

They are designed for **extra-atmospheric** flight: artificial satellites (which orbit the Earth: communications, observation, positioning…), space probes (scientific observation of celestial bodies and interplanetary space), crewed spacecraft and space stations, and **launch vehicles** (which carry the others to orbit; they resemble missiles). Figure 1.8 shows the five types.

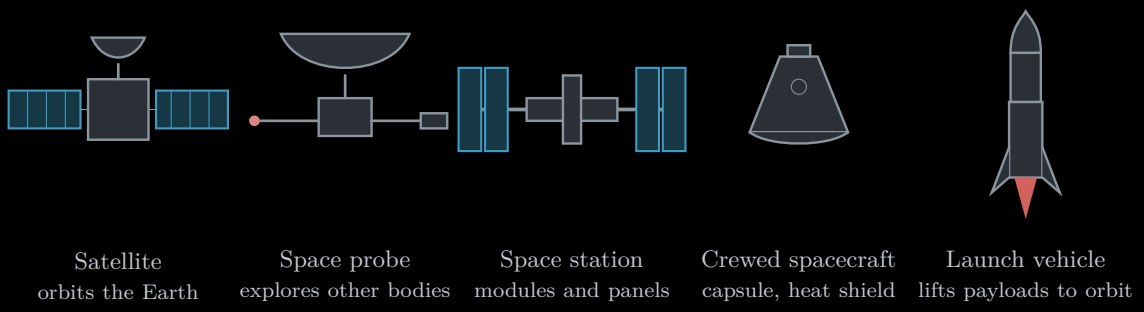


Figure 1.8. Types of space vehicles. Schematic silhouettes, not to scale.

4 Aerospace infrastructure

4.1 For aircraft

Aerodromes and airports. *Aerodrome*: a defined area used wholly or partly for the departure, arrival and surface movement of aircraft. *Airport*: an aerodrome with permanent facilities and services open to the public. It is divided into the **airside** (runways, aircraft parking), the **landside** (passenger terminal), the industrial zone and the flight control and navigation system (Figure 1.9).



Figure 1.9. Schematic airport layout in plan view: the airside (where aircraft move) and the landside (where passengers arrive).

Air Traffic Control (ATC). Three levels of control that hand the aircraft over from one to the next (Figure 1.10):

- **TWR** (control tower): takeoffs, landings and movements at the airport.
- **Terminal Control Centers**: approach/departure area around the airport.
- **En-route Air Traffic Control Centers**: the aircraft in cruise.

Navigation aids. They are the "lighthouses and roads" of the sky. The slides list them; in Unit 2 (Section 5, low-visibility flight) it is explained how each one works.

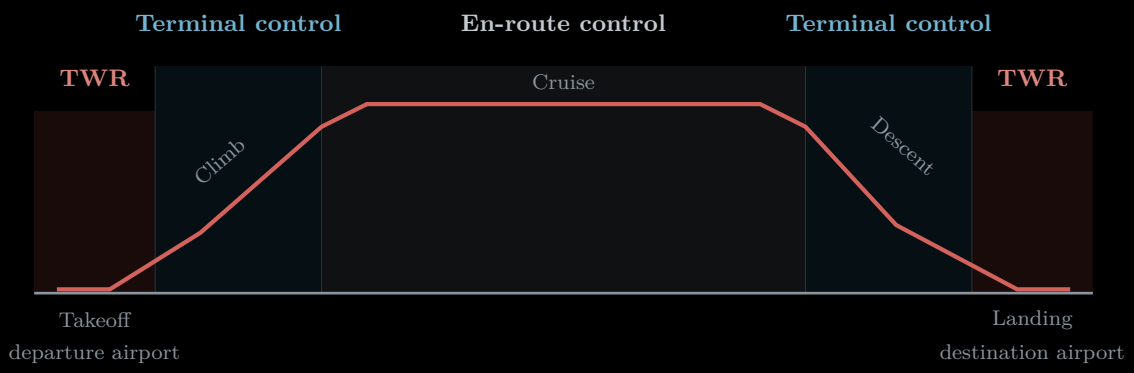


Figure 1.10. Who controls the aircraft in each phase of flight. Source: T1, slide 27 (the TWR tower is the airport's own control; the phase diagram is explanatory).

- Ground stations: **VOR** (*VHF Omni-directional Range*), **DME** (*Distance Measuring Equipment*), **NDB** (*Non-Directional Beacon*), **ILS** (*Instrument Landing System*), **PAPI** (*Precision Approach Path Indicator*).
- Satellites: the GPS constellation (and GNSS in general).

Figure 1.11 sketches the idea behind each ground aid.

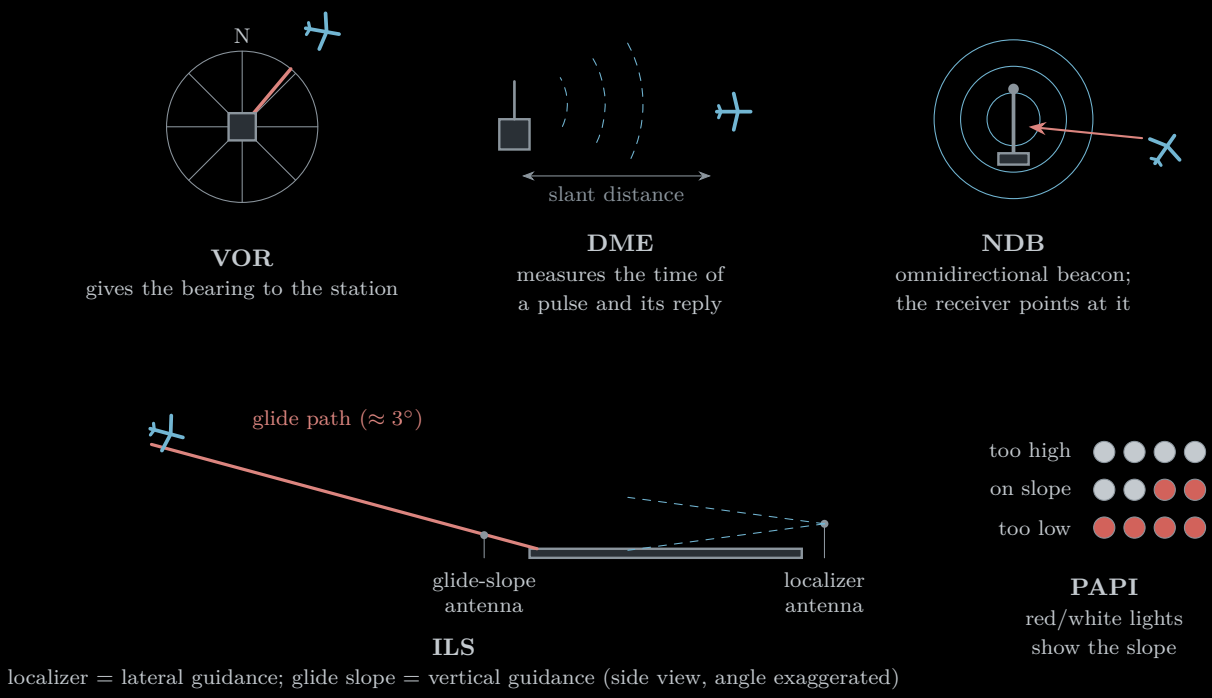


Figure 1.11. Ground navigation aids, schematic (not to scale). Detail of each in Unit 2.

4.2 For space vehicles

- **Launch sites**: they must be close to the **equator** (the Earth's rotation contributes speed: about 0.46 km/s at the equator, supplementary data) and to the **coast**, in an **uninhabited** region (if the rocket fails, it should fall into the sea). Nearby they need an industrial zone, final assembly facilities, power supply, a port and airport, and the launch pad.
- **Control centers**: they monitor the trajectory, check the operation of instruments and systems, and send commands (cameras, orbit corrections). Since a satellite moves relative to the center, **tracking stations** spread around the world are needed (example on the slide: those of the ISS).

FROM THE 2020/21 TEST
"Launch sites: (a) close to inhabited areas; (b) close to the equator to take advantage of the Earth's motion; (d) require industrial zones in the surroundings": the collected key marked "all the others are correct", but (a) contradicts slide 29 (uninhabited region). Remember: equator, coast, uninhabited region and industrial zone.

5 Future challenges

Once durability and safety are "guaranteed", the slides cite these challenges: in **propulsion**, more efficient concepts (outer rotor, boundary layer ingestion, distributed electric propulsion), alternative fuels and more electric engines; in **materials and structures**, the adoption of composite materials, new materials (metallic foams, nanomaterials) and non-destructive testing; in **logistics and manufacturing**, optimizing the supply chain and automating more; in **operations**, more efficient air traffic management and navigation; and **data management** (*big data*).

6 Summary and what usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM
This part contributes **about two stand-alone test questions** per first midterm (11 of the 115 questions in six first midterms, 2018/19 to 2024/25 except 2021/22), almost always concept review:

- Characteristics of the industry ("short delivery times" is the false one) and organizations (EASA, ICAO, IATA, INTA, ESA), always as matching an organization to its function.
- Aerostats (Archimedes) versus aerodynes (aerodynamic lift); classification of missiles and launch sites.
- Navigation aids: Unit 2. There are no numerical problems in this unit.

UNIT 2

The atmospheric flight environment

1 Why the atmosphere matters

An aircraft is an object that "swims" in the air. Everything it can do (be supported, move forward, climb) depends on **what the air is like** around it: how much mass there is in each cubic meter, what temperature it is at, whether there is wind, clouds or ice. This unit describes that environment and builds a simple **model** so that calculations can be made.

1.1 Forces in cruise

In straight and level flight at constant speed (*cruise*) there are four forces: **lift** L (upward, the force that supports the aircraft), **weight** W (downward), **drag** D (which slows it down) and **thrust** T (pushes it forward, provided by the engine). In cruise they balance: $L = W$ and $T = D$ (Figure 2.1).

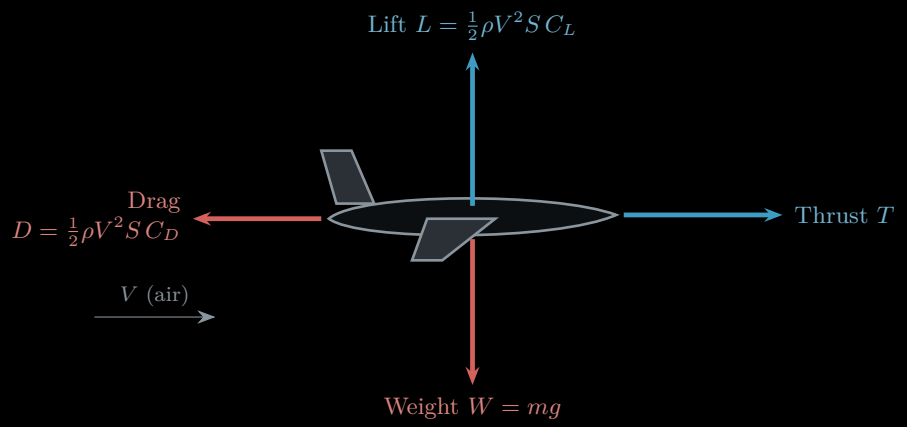


Figure 2.1. Forces in cruise flight (simplified version). Source: T2, slide 4.

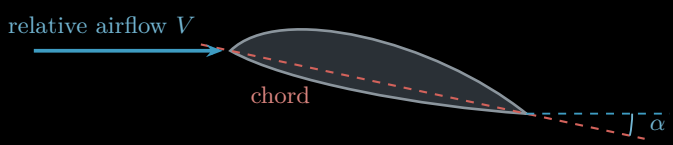


Figure 2.2. Angle of attack α: the angle between the wing chord and the relative airflow V. Schematic.

$$L = \tfrac{1}{2}\,\rho\,V^2\,S\,C_L \qquad D = \tfrac{1}{2}\,\rho\,V^2\,S\,C_D \qquad W = m\,g$$

Glossary of symbols: ρ is the air density (kg/m^3); V is the aircraft's speed relative to the air (m/s); S is the wing area (m^2); C_L and C_D are the lift and drag coefficients (dimensionless numbers that depend on the shape of the wing and the angle of attack α, defined in Figure 2.2); m is the mass (kg); g is gravity (9.81 m/s^2).

The key point: L and D are proportional to the density ρ. Engine thrust and power have a more complex dependence, but, broadly speaking, thrust also increases with density (the higher you fly, the less air enters the engine). In addition to the above, density influences:

- the **buoyancy** of aerostats: it is a direct function of the air density;
- **cabin pressurization**: a breathable pressure must be maintained inside while outside there is almost none;
- the **weather**: headwind and tailwind, icing and visibility;
- other effects specific to speed or altitude: **kinetic heating** and **radiation**.

The effects of altitude are important, day-to-day variations are significant and geographic position matters quite a lot. This is why an **atmosphere model** is needed that gives, at least, the properties of the air as a function of flight altitude.

> **WHAT HAPPENS TO THE ANGLE OF ATTACK WHEN CLIMBING (2023/24 EXAM)**
> An aircraft flies at 10 km at 850 km/h and climbs to 12 km **without changing its speed**. What does its angle of attack α (the angle between the wing and the incoming air) do?
> **Setup.** In level flight, $L = W$, that is, $\frac{1}{2}\rho V^2 S C_L = W$. With V, S and W constant, the required coefficient is $C_L = \dfrac{2W}{\rho V^2 S} \propto \dfrac{1}{\rho}$.
> **Step 1.** The higher you climb, the less dense the air is: $\rho(10\text{ km}) = 0.413$ and $\rho(12\text{ km}) = 0.311$ kg/m^3. The ratio is $0.413/0.311 \approx 1.33$: a C_L 33 % higher is needed.
> **Step 2.** In the linear region of the wing's curve, C_L grows with α, so α must **increase during the 2 km**. Answer: **it increases over the two km** (option b).
> **Common mistake.** The temperature is constant above 11 km, but that does not mean the density stops decreasing: the pressure keeps falling and therefore so does $\rho = p/(RT)$.

2 The International Standard Atmosphere (ISA)

2.1 Layers of the atmosphere

The atmosphere is divided into layers according to how the temperature varies with altitude (Figure 2.3):

Layer	Altitude	T gradient	What happens
Troposphere	0 to 11 km	−6.5 K/km	weather occurs here; small aircraft fly here, and airliners in its upper part
Stratosphere (1)	11 to 20 km	0 (isothermal, 216.65 K)	airliner cruise
Stratosphere (2)	20 to 32 km	+1 K/km	
Stratosphere (3)	32 to 47 km	+2.8 K/km	the temperature rises to ≈ 270.65 K at the stratopause
Mesosphere, thermosphere	above	—	outside the scope of the course model (slide 8)

The boundary between the troposphere and the stratosphere is the **tropopause** (11 km); the one that closes the stratosphere is the **stratopause** (47 km in ISA).

Why the temperature drops in the troposphere. The ground, heated by the Sun, warms the air from below; the farther from the ground, the colder. In the stratosphere the ozone layer absorbs ultraviolet radiation, which is why the temperature stops dropping and then rises (supplementary explanation, not in the slides).

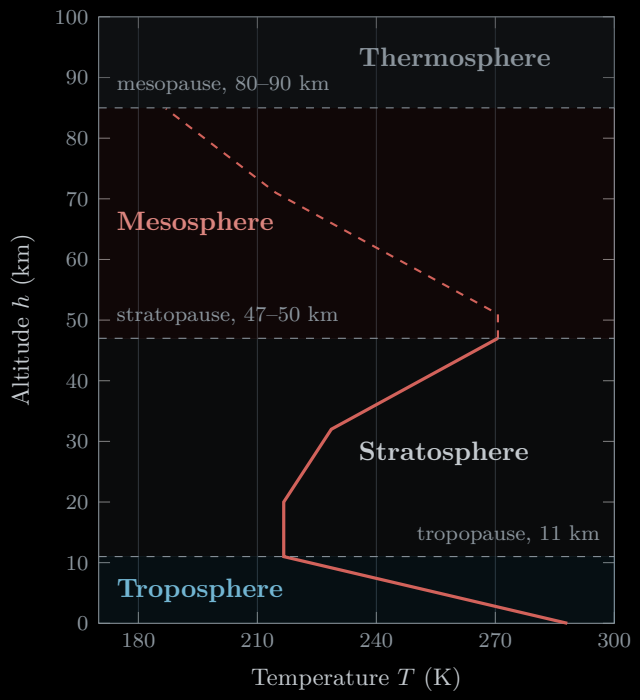


Figure 2.3. Layers of the atmosphere and ISA temperature profile (solid line, up to 47 km; the dashed lines above are the standard continuation beyond what the slides require, 47 to 85 km). Source: T2, slides 8 and 9; the upper limits (stratopause, mesopause) are approximate, as in the slide figure.

2.2 The ISA model (ISO 2533:1975)

The *International Standard Atmosphere* is a **fictitious, average** atmosphere that lets all engineers use the same numbers (design, testing, altimeters). Its assumptions:

- It is defined as a function of **altitude**, measured from sea level. **Careful**: this is not the same as **height**, which is the distance to the ground (an aircraft 3 000 m above an airport located at an altitude of 600 m is flying at an altitude of 3 600 m; Figure 2.4).
- Air behaves as an **ideal gas**: $p = \rho R T$ (with $R \approx 287$ J/(kg K), the gas constant of air; supplementary data).
- There is **hydrostatic equilibrium** (fluid at rest): the pressure at a point supports the weight of the air above it, $\dfrac{\mathrm{d}p}{\mathrm{d}h} = -\rho g$.
- The temperature follows the piecewise law in the table above.
- Sea-level values: $T_0 = 288.15$ K (15 °C), $p_0 = 101\,325$ Pa, $\rho_0 = 1.225$ kg/m^3.
- The altitude is **geopotential**, not geometric: $h = \dfrac{1}{g_0}\displaystyle\int_0^h g\,\mathrm{d}z$, with $g_0 = 9.80665$ m/s^2. This avoids having to account for g changing slightly with height (at aircraft flight altitudes the difference is negligible; for problems, $g = g_0$ is taken as constant).

Where the formulas come from. Combining the two equations: from $\dfrac{\mathrm{d}p}{\mathrm{d}h} = -\rho g$ and $\rho = \dfrac{p}{RT}$ we get $\dfrac{\mathrm{d}p}{p} = -\dfrac{g}{RT}\,\mathrm{d}h$. There are two cases:

- **Troposphere** ($T = T_0 - \lambda h$, with $\lambda = 6.5 \cdot 10^{-3}$ K/m): integrating gives a power law.
- **Isothermal layer** (T constant): integrating gives an exponential.

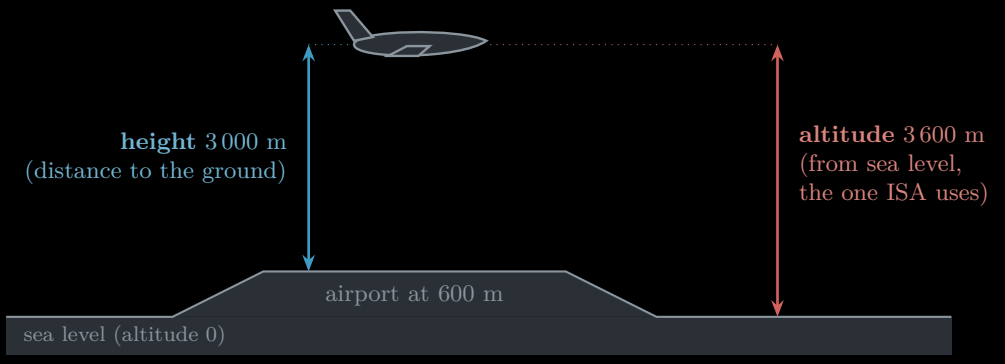


Figure 2.4. Altitude (from sea level) versus height (above the ground) for an aircraft over an airport at 600 m.

Troposphere ($0 \le h \le 11$ km), with $A = \dfrac{g_0}{6.5 \cdot 10^{-3}\, R} \approx 5.256$:

$$T(h) = 288.15 - 6.5 \cdot 10^{-3}\, h$$

$$p(h) = 101\,325 \left(1 - \frac{6.5 \cdot 10^{-3}\, h}{288.15}\right)^{A}$$

$$\rho(h) = 1.225 \left(1 - \frac{6.5 \cdot 10^{-3}\, h}{288.15}\right)^{A-1}$$

(with h in meters; p in Pa; ρ in kg/m^3). Note that the exponent of ρ is $A - 1 \approx 4.256$, one less than that of p, because $\rho = p/(RT)$ and T also decreases. **These are the formulas the slides ask for**; for the isothermal stratosphere, $p = p_{11} \exp\left[-\dfrac{g_0(h - 11000)}{R\,T}\right]$ with $T = 216.65$ K and $p_{11} = 22\,632$ Pa (supplement), and $\rho = p/(RT)$ in any case. Figure 2.5 plots these profiles.

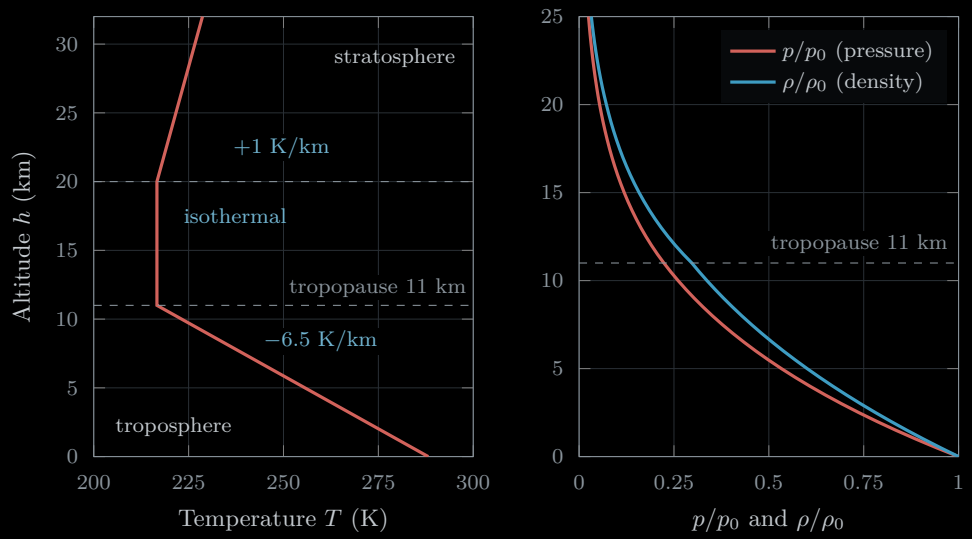


Figure 2.5. ISA profiles. Left: temperature (red). Right: relative pressure and density, which fall *nonlinearly*. Source: T2 model, slides 9 and 10.

h (km)	T (K)	p (kPa)	ρ (kg/m^3)	ρ/ρ_0
0	288.15	101.33	1.225	1.000
1	281.65	89.87	1.112	0.907
2	275.15	79.50	1.007	0.822
3	268.65	70.11	0.909	0.742
4	262.15	61.64	0.819	0.669
5	255.65	54.02	0.736	0.601
6	249.15	47.18	0.660	0.539
8	236.15	35.60	0.525	0.429
10	223.15	26.44	0.413	0.337
11	216.65	22.63	0.364	0.297
12	216.65	19.33	0.311	0.254
15	216.65	12.04	0.194	0.158
20	216.65	5.47	0.088	0.072

Table computed with the model formulas ($R = 287.053$ J/(kg K)); it is not in the slides and serves to check calculations. Rule of thumb: at 5.5 km (about 18 000 ft) half the pressure remains, and at 11 km a little more than one fifth.

COMMON MISTAKES WITH ISA

- The **temperature** drops **6.5 K per kilometer** (not per meter) in the troposphere. The pressure does **not** drop linearly ("10 kPa/km" is false; it was tested in the 2024/25 exam). It falls almost exponentially.
- Above the tropopause the **pressure keeps dropping**; only the temperature is constant up to 20 km. The pressure at 15 km is **lower** than at the tropopause (2023/24 exam).
- The temperature does **not** always decrease with altitude: in the stratosphere it rises (2024/25 exam).
- In the stratosphere "the temperature decreases 6.5 K/km" is false: it is isothermal in its first part (2022/23).

ISA PROPERTIES AT 8 KM

Compute T, p and ρ at $h = 8000$ m.

Step 1 (temperature): $T = 288.15 - 6.5 \cdot 10^{-3} \cdot 8000 = 236.15$ K ($-37\,^\circ$C).

Step 2 (ratio): $\dfrac{T}{T_0} = 1 - \dfrac{6.5 \cdot 10^{-3} \cdot 8000}{288.15} = 0.8195$.

Step 3 (pressure): $p = 101\,325 \cdot 0.8195^{5.256} = 101\,325 \cdot 0.3514 \approx 35\,600$ Pa $= 35.6$ kPa.

Step 4 (density): $\rho = 1.225 \cdot 0.8195^{4.256} = 0.525$ kg/m^3. Check: $\rho = \dfrac{p}{RT} = \dfrac{35\,600}{287 \cdot 236.15} = 0.525$. It is 43 % of the sea-level value.

COMPARING TEMPERATURES WITH ISA±Δ (2023/24 EXAM)

"The temperature at 3 km on a cold day (ISA−15) is lower than at 5 km on a standard day". True?

$T(3\text{ km}, \text{ISA} - 15) = 288.15 - 19.5 - 15 = 253.65$ K; $T(5\text{ km}, \text{ISA}) = 288.15 - 32.5 = 255.65$ K. Since $253.65 < 255.65$: **true**.

"The temperature at 10 km in ISA−10 is higher than at 13 km in ISA": $223.15 - 10 = 213.15$ K versus 216.65 K: **false**. Rule of thumb: ISA±Δ adds or subtracts Δ to the whole temperature curve.

2.3 Useful supplements (added explanation, not in the slides)

Two altitudes appear all the time in performance questions and are defined from the ISA:

- **Pressure altitude**: the ISA altitude whose pressure equals the measured pressure. It is what a barometric altimeter reads when set to the standard setting 1013.25 hPa.
- **Density altitude**: the ISA altitude whose density equals the actual air density. It is what the aircraft "feels": engine power, propeller thrust and lift all depend on ρ. On a hot or high-elevation airfield the density altitude is much higher than the geometric elevation, which is why performance degrades (next section).

3 Takeoff and landing at high altitude and in hot weather

3.1 High altitude

The density at a high-altitude airport can be estimated with ISA. Example from slide 12: Daocheng Yading airport (China), at 4411 m above sea level, has $\rho_{ISA} = 0.64\,\rho_{SL}$ (SL: sea level).

Consequence: with lower density, **more speed** is needed to support the same weight, so **long runways and higher takeoff and landing speeds** are required. Quantitatively: the stall speed (the minimum speed at which the wing can still support the weight) is

$$V_s = \sqrt{\frac{2W}{\rho\, S\, C_{L,max}}} \;\propto\; \frac{1}{\sqrt{\rho}},$$

so that $\dfrac{V_s}{V_{s,0}} = \sqrt{\dfrac{\rho_0}{\rho}}$. This result has been asked in exams (2019/20: "the stall speed **increases** with altitude"). Figure 2.6 shows how much it grows with altitude.

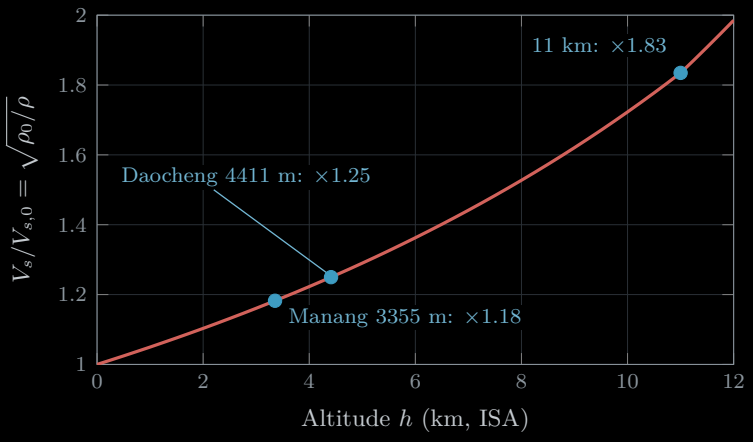


Figure 2.6. How much the stall speed rises with altitude (ISA). Daocheng: $\sqrt{1/0.64} = 1.25$. Own calculation from the ISA model.

MANANG (2020/21 EXAM)
An aircraft has a stall speed of 160 km/h at sea level. What is it at Manang airport (3355 m)?

Step 1: density at 3355 m. $T/T_0 = 1 - \dfrac{6.5 \cdot 10^{-3} \cdot 3355}{288.15} = 0.9243$; $\rho/\rho_0 = 0.9243^{4.256} = 0.715$.
Step 2: $V_s = 160\sqrt{1/0.715} = 160 \cdot 1.182 = 189$ km/h. **Answer: 189 km/h** (option b).

3.2 Hot and cold days

A nonstandard day is described as ISA+Δ or ISA−Δ (examples: ISA+15, ISA−10). This is done as follows:

- The **sea-level temperature is modified** (303.15 K for ISA+15; 278.15 K for ISA−10) and the whole T curve is shifted by the same Δ.
- The **pressure is computed as usual** (the ISA value for that altitude).
- The **density is computed with the ideal gas law**: $\rho = \dfrac{p_{ISA}(h)}{R\,(T_{ISA}(h) + \Delta)}$.

Hot day ⇒ less dense air ⇒ harder to take off and land. A cold day is the opposite (denser air), as Figure 2.7 shows.

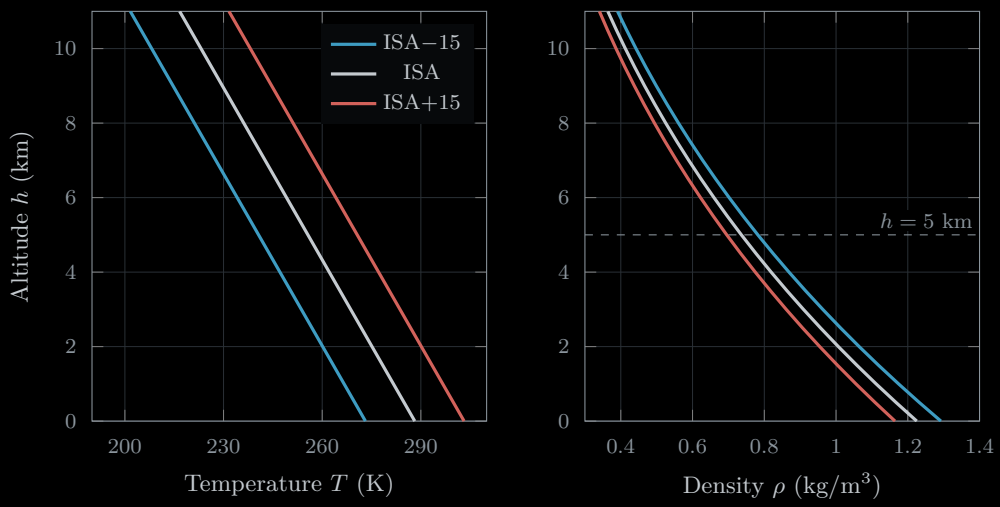


Figure 2.7. Effect of ISA±15 on temperature (left) and density (right) in the troposphere: ISA−15 is the cold day (blue) and ISA+15 the hot day (red). At the same altitude the hot day is less dense.

DENSITY ON A HOT DAY

Sea-level density in ISA+15: $\rho = \dfrac{101\,325}{287.05 \cdot 303.15} = 1.164$ kg/m^3, 5 % less than the standard value (1.225). In ISA−10: $\dfrac{101\,325}{287.05 \cdot 278.15} = 1.269$, 3.6 % more.

At 5 km with ISA+20: $\rho = \dfrac{54\,020}{287.05 \cdot (255.65 + 20)} = 0.683$ kg/m^3, versus 0.736 in ISA.

4 Clouds, winds and icing

4.1 Clouds

They form with **high humidity and low temperature**. Effects: poor visibility (instrument flight rules are needed, Section 5); possible storms, with the risk of lightning strikes (aircraft are protected against lightning, but it causes damage) and gusts; and possible icing. The risk is **lower above the tropopause**: almost all weather phenomena occur in the lower part of the troposphere (2024/25 exam) and in the stratosphere they are not significant (2022/23). That is why airliners fly high, above about 9 km: humidity and turbulence are avoided (2022/23 exam).

4.2 Winds

- They are **critical in takeoff and landing** (headwind, tailwind or crosswind). It is usual to take off and land with a **headwind**: the incoming air already gives relative speed to the wing without the aircraft moving relative to the ground, so less runway is needed.
- **Jet stream**: strong, narrow winds at high altitude. It is exploited with a **tailwind**, because it increases the ground speed; it is not "always avoided" nor does it "increase lift" (2018/19 exam).
- **Gusts**: turbulence, abrupt changes in lift and drag and possible loss of control. They are also lower above the tropopause.

Figure 2.8 illustrates the headwind on takeoff and the jet stream in cruise.

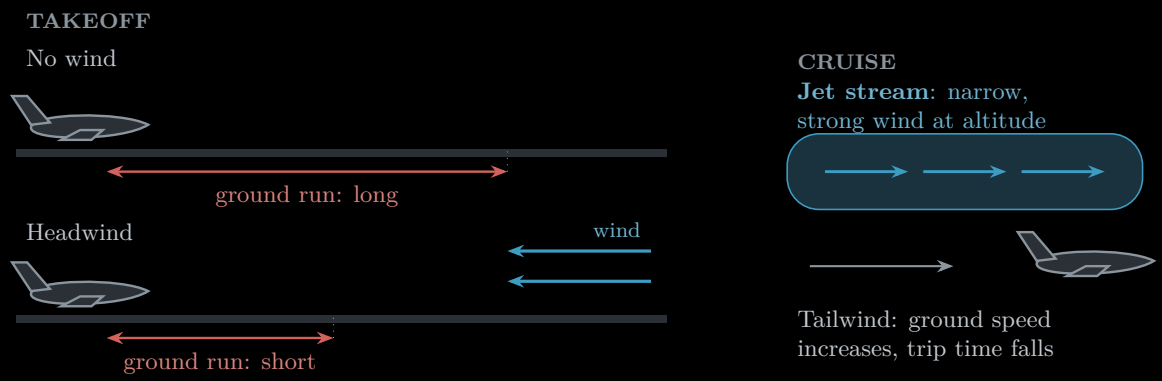


Figure 2.8. Left: with a headwind the wing reaches flying speed after a shorter ground run. Right: a jet stream used as a tailwind raises the ground speed. Schematic.

4.3 Icing

It occurs at **very low temperatures and high humidity**. When ice forms on the surface, it **changes the shape of the aircraft: lift decreases, drag and weight increase**. It can damage engines and instruments (example of the ATR-72 flight in 1994: the iced horizontal stabilizer stalled).

Prevention and protection: ice detectors and deicing systems on the ground and in flight (pneumatic boots, fluids, engine bleed air, electro-thermal, electromechanical, passive systems…). They are used on the wing, the engines and several sensors (2019/20 exam), not only on the ground. Figure 2.9 shows the effect of ice on the profile and how a pneumatic boot (an inflatable rubber strip on the leading edge) breaks it off.

> **ABOUT ICE IN THE TESTS**
>
> - Ice formation is **not** more typical at 10 km than at 6 km "because of humidity": almost all the water vapor is in the lower part of the troposphere. This was the false option in 2024/25.
> - Ice on the leading edge is a possible cause of **stall**: it deforms the airfoil and lowers $C_{L,max}$ (2023/24: the option "all the others are valid").

5 Low-visibility flight

When the pilot cannot see, flight is conducted under **Instrument Flight Rules** (IFR) instead of **Visual Flight Rules** (VFR). For each aid, the slides (17 to 19) ask: "what is its operating principle

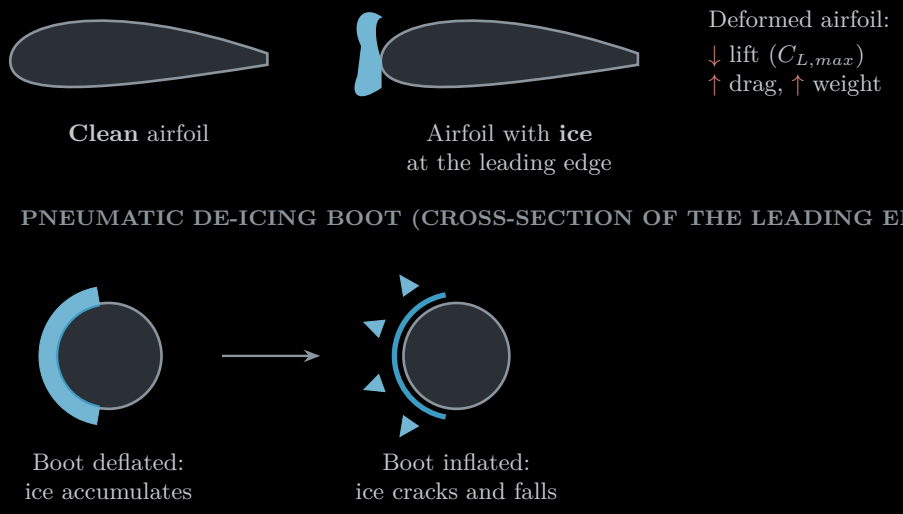


Figure 2.9. Ice on the leading edge deforms the airfoil; the pneumatic boot inflates and cracks the ice, which the airflow then carries away. Schematic.

and how do they differ?". Each one is summarized here (the detailed explanation is supplementary); Figure 2.10 shows where each aid operates.

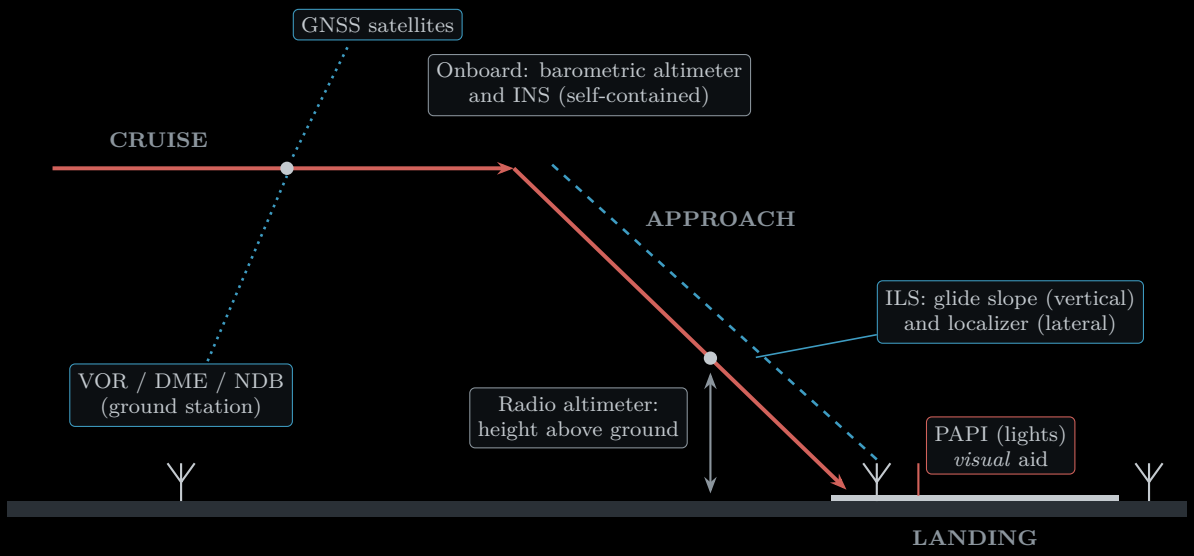


Figure 2.10. Where each aid operates. In blue, aids that work in low visibility; in red, the one that needs visibility (PAPI). Source: T2, slides 17 to 19 (own diagram).

In cruise

Aid	Principle	Remarks
Barometric altimeter	measures static pressure and converts it into altitude using ISA	cruise; it is miscalibrated if the day is not standard
NDB (*Non-Directional Beacon*)	ground beacon that transmits in all directions; the onboard receiver indicates the direction toward it	cruise; low accuracy
VOR (*VHF Omni-directional Range*)	ground station in the VHF band that lets the aircraft know on which radial (bearing from the station) it is	cruise
DME (*Distance Measuring Equipment*)	the aircraft interrogates and the station replies; the round-trip time gives the distance	cruise; with VOR it gives the position
INS (*Inertial Navigation System*)	accelerometers and gyroscopes integrate the motion from a known position	self-contained; accumulates error over time
GNSS (*Global Navigation Satellite System*)	receives signals from several satellites and computes the position from time differences (GPS is a GNSS)	cruise; depends on satellites

Figure 2.11 shows how a VOR and a DME combine to fix the position.

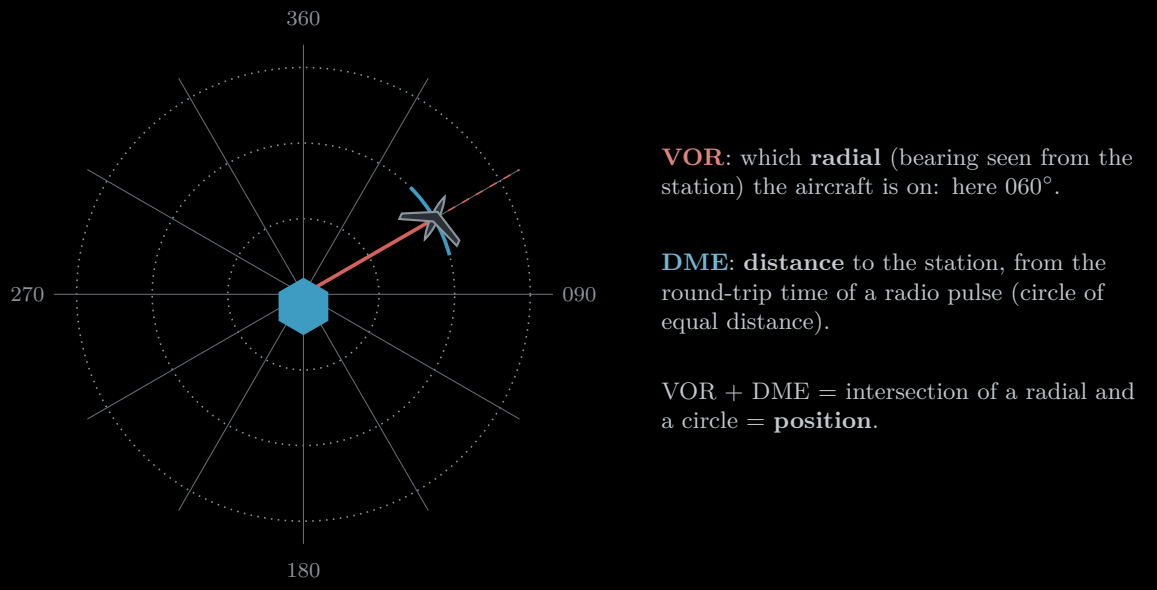


Figure 2.11. VOR and DME. Top view, north up; the aircraft is on radial 060 at the distance given by the DME circle. Schematic.

In approach and landing

Aid	Principle	Remarks
Radio altimeter	downward-pointing radar that measures the height **above the ground**; not to be confused with the barometric one	landing; short range
ILS (*Instrument Landing System*)	two radio beams: one marks the runway centerline (lateral) and the other the glide slope (vertical)	landing; works with almost zero visibility
Augmented GNSS (LAAS/GBAS)	GNSS corrected by a ground station for precision approaches	landing
PAPI (*Precision Approach Path Indicator*)	red and white lights beside the runway that indicate whether the aircraft is on the glide path	**visual aid**: landing in good visibility

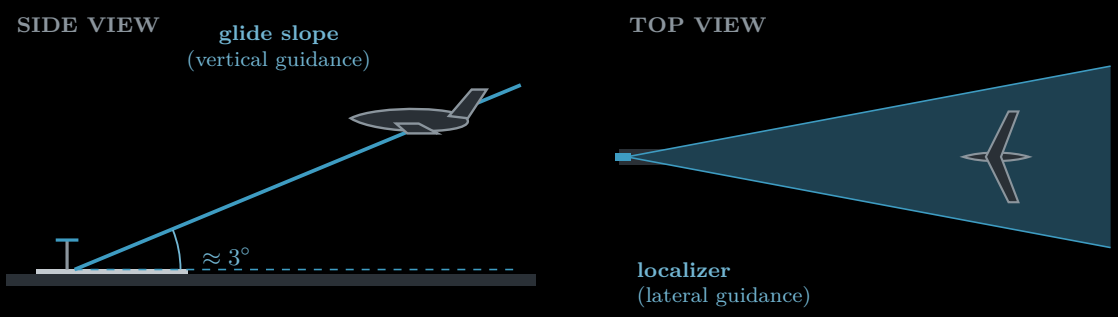


Figure 2.12. ILS: the localizer beam guides laterally along the runway centerline and the glide slope beam guides vertically (about 3°; angle exaggerated). Schematic.

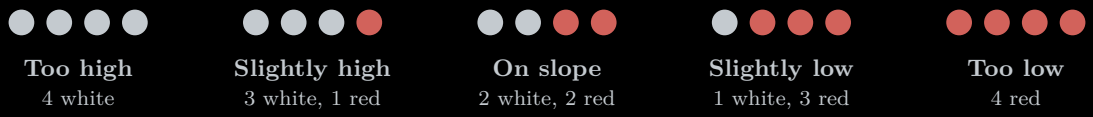


Figure 2.13. PAPI as seen by the pilot: the number of white and red lights shows whether the aircraft is above, on or below the glide path.

Figure 2.12 shows the two ILS beams and Figure 2.13 the PAPI light signals.

RECURRING QUESTION: NAVIGATION AIDS

- The **PAPI** is only a **visual** aid (VFR); the **ILS** is an instrument aid (IFR). Therefore, with visibility < 30 m the PAPI cannot be used, but the ILS, the radio altimeter or GNSS can (2023/24 exam); and in low visibility "requires good visibility" $\Rightarrow$ PAPI (2024/25).
- The **VOR** is not suitable for following the glide slope (that is done by the ILS). The barometric altimeter does not provide guidance in landing (2019/20). The ILS does not guide the takeoff.
- False in 2020/21: "the radio altimeter cannot be used in cruise because of the high speed" (its limitation is the short range above the ground, not the speed), "the barometric one cannot be used at high altitude" (it can) and "the PAPI is used in takeoff" (it is for landing).

6 Kinetic heating

When the air slows down on reaching the aircraft's surface (nose, leading edge of the wings…), its **kinetic energy is converted into internal energy** (heat). The temperature reached by air brought completely to rest is called the **total or stagnation temperature** T_0. If the deceleration is **adiabatic and isentropic** (no heat exchange and no losses), energy is conserved:

$$T_0 = T + \frac{V_\infty^2}{2\,c_p} \qquad \Longleftrightarrow \qquad T_0 = T\left(1 + \frac{\gamma-1}{2}\,M^2\right), \qquad \gamma = \frac{c_p}{c_v} \approx 1.3 \text{ to } 1.4$$

where T is the static air temperature (the ISA value at that altitude), V_∞ the flight speed, c_p the specific heat at constant pressure (air: ≈ 1005 J/(kg K)) and $M = V/a$ the **Mach number** (the ratio of the aircraft's speed to the speed of sound $a = \sqrt{\gamma R T}$). With $\gamma = 1.4$: $T_0 = T\,(1 + 0.2\,M^2)$. It is a **major problem at high speeds**: it grows with M^2 (Figure 2.14).

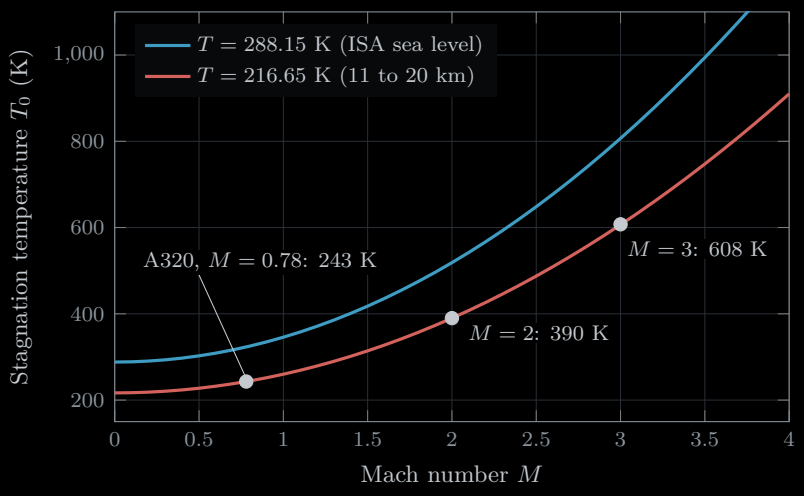


Figure 2.14. Stagnation temperature as a function of Mach number, for $\gamma = 1.4$. The two curves correspond to the ISA static temperature at sea level and at 11–20 km. The SR-71 example below (Mach 3 at 24 km, $T = 220.65$ K) gives 618 K, slightly above the lower curve.

SR-71 BLACKBIRD AT MACH 3 AND 24 KM (2020/21 EXAM)
Step 1: $T(24\text{ km}) = 216.65 + 1 \cdot (24 - 20) = 220.65$ K (the +1 K/km segment between 20 and 32 km).
Step 2: $T_0 = 220.65\,(1 + 0.2 \cdot 3^2) = 220.65 \cdot 2.8 = 618$ K. **Answer: 618 K** (option c), about 345 °C. Common mistake: forgetting the +1 K/km above 20 km and using 216.65 K, which would give 607 K.

AIRBUS A320NEO AT 11 000 M (2018/19 EXAM)
Implicit data: cruise speed $M \approx 0.78$.
$T = 216.65$ K; $T_0 = 216.65\,(1 + 0.2 \cdot 0.78^2) = 216.65 \cdot 1.122 = 243$ K (−30 °C). The collected key was "around 215 K" (the closest of the options 115, 215, 315 and 415 K). Note that 215 K is the *static* temperature; with kinetic heating about 27 K more results. At subsonic speeds the effect is small; it only matters at high speeds.

Note: in the 2023/24 exam it was stated that at Mach 2 kinetic heating "can give surface temperatures of about 600 K". With the formula: at 11–20 km, $216.65 \cdot 1.8 = 390$ K; at sea level, 519 K. Therefore 600 K does not fit Mach 2 but Mach 3 (608 K). The collected key marked it as correct; it is left here as a doubt.

7 Radiation at high altitude

Here "radiation" refers to **cosmic rays**, high-energy particles that come from space: protons (≈ 90 %), alpha particles (≈ 10 %) and heavy particles (1 to 2 %) (the figures are those on the slide and do not add up to exactly 100 %; they are orders of magnitude). Figure 2.15 sketches how the atmosphere attenuates them.

- **Why it matters**: the atmosphere acts as a shield, but the upper layers are less dense, and there is less protection near the **poles**.
- **Effects**: dangerous for electronics and for health; higher cancer risk.
- **Trend and a poor solution**: aircraft fly ever higher, which increases the dose. Shielding would be the solution, but the weight penalty is prohibitive.
- **Mitigation**: dosimeters and regular medical checkups; rerouting (for example, polar routes during a solar storm).

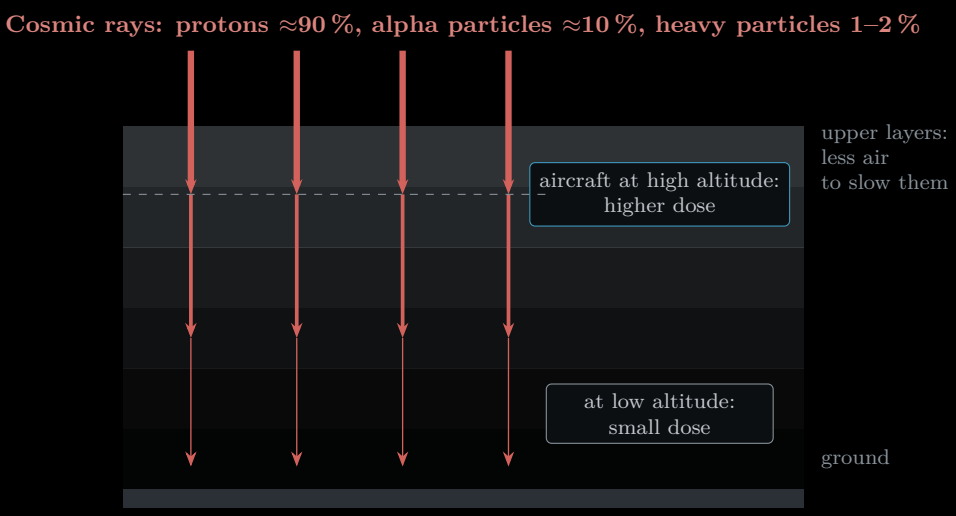


Figure 2.15. The atmosphere slows down cosmic rays; at higher altitude there is less air to stop them. Qualitative diagram.

- **Crews** are the group of workers most exposed to radiation (excluding nuclear accidents and the like).

FROM THE TEST
The absorbed radiation **increases with altitude** and is **higher near the poles** than at the equator (2019/20 exam). Excessive doses can be avoided by avoiding critical routes such as polar ones (2023/24). It is not reduced by cabin pressurization.

8 What usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM

This unit contributes **about three questions** per first midterm (17 of the 115 in six first midterms, 2018/19 to 2024/25 except 2021/22). Types observed:

- **Layers and ISA (conceptual)**: comparing temperatures or pressures at different altitudes (ISA±Δ); gradients (6.5 K/km; isothermal from 11 to 20 km; nonlinear pressure).
- **Recurring numerical problems**: total temperature (SR-71: 618 K; A320: $T_0 = T(1 + 0.2M^2)$); stall speed at another altitude ($V \propto 1/\sqrt{\rho}$; Manang: 189 km/h); angle of attack when climbing at constant speed.
- **Hot/high takeoff**: lower density, longer runways, higher speeds.
- **Weather**: headwind in takeoff and landing; jet stream with tailwind; ice and deicing systems; effects of lightning.
- **Navigation aids**: PAPI (visual) versus ILS; radio altimeter versus barometric.
- **Radiation**: grows with altitude and toward the poles.

The essentials to memorize:

- $T_0 = 288.15$ K, $p_0 = 101\,325$ Pa, $\rho_0 = 1.225$ kg/m^3; -6.5 K/km up to 11 km; 216.65 K from 11 to 20 km; $+1$ K/km from 20 to 32 km; $+2.8$ K/km up to 47 km.
- $p = p_0(T/T_0)^{5.256}$ and $\rho = \rho_0(T/T_0)^{4.256}$ in the troposphere.
- ISA+Δ: same pressure, shifted T, $\rho = p/RT$; heat $\Rightarrow$ less dense; $T_0 = T(1 + 0.2M^2)$.

UNIT 3

Aircraft design concepts

1 What aircraft design is

For Daniel P. Raymer (a reference author in the slides), *design* is **"creating the geometric description of a thing to be built"**. Aircraft design is carried out by highly multidisciplinary teams (aerodynamics, propulsion, structures, avionics...), but it is a discipline of its own, with its own history, methods and rules of thumb.

The classical problem is **iterative**: the weight depends on the engine and the aerodynamics, and these depend on the weight. It is a circle, so the question is *where do we start?* The answer, which appears on exams: **with the design specifications** (what the aircraft must do), not with the engine, the weights or the aerodynamics.

2 Phases of aircraft design

Starting from the mission specifications there are three phases, followed by manufacturing (Figure 3.1):

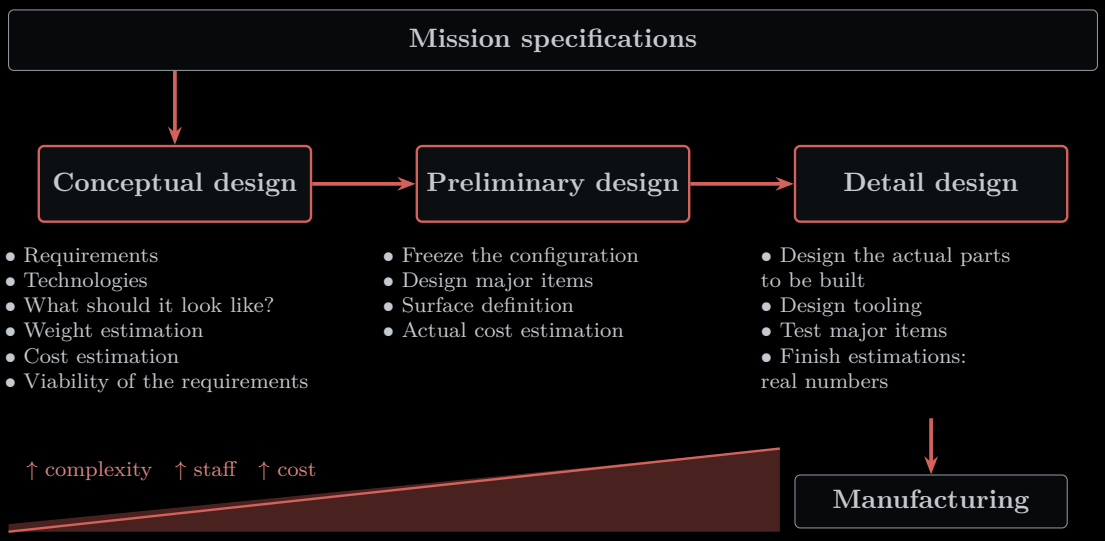


Figure 3.1. Phases of aircraft design: each phase adds complexity, staff and cost. Source: T3, slide 6.

Phase	What is done
Conceptual design	requirements, available technologies, "what it should look like", first estimate of weights and costs, and viability of the requirements. It is cheap and fast, and it is where almost everything is decided.
Preliminary design	the **configuration is frozen**; the major items are designed, the surfaces are defined and the actual cost is estimated.
Detail design	the parts to be built and the tooling are designed, the major items are tested, and the estimates are closed with real numbers.

Analogy. An analogy is building a house: first a sketch with an approximate budget (conceptual), then the plans with the layout already fixed (preliminary) and, finally, the detail of each installation (detail). Changing the idea is cheap at the sketch stage and very costly once the plans are closed.

COMMON TEST MISTAKES ABOUT THE PHASES

- The **initial weight estimate** belongs to **conceptual** design, not preliminary.
- The design is **frozen** in the **preliminary** phase (not in the conceptual one) (2022/23 exam).
- The preliminary phase **requires more staff and time** than the conceptual one (2018/19 and 2019/20): complexity, staff and cost grow with each phase.
- The main requirements are set **earlier** (specifications/conceptual). Each part is designed in the **detail** phase.
- Saying that the conceptual phase is "the most critical because it is the most expensive" is false: it is the cheapest, although its decisions condition everything.
- Available technology is critical already in the conceptual phase (it is identified in that phase), not only in the detail phase.

3 Conceptual design overview

Conceptual design is organized into four blocks that are repeated until the aircraft is viable (Figure 3.2):

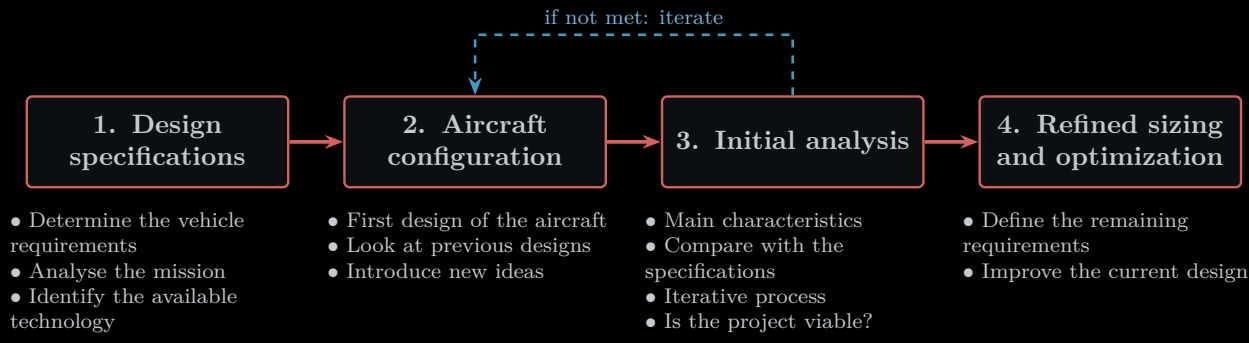


Figure 3.2. The four blocks of conceptual design. Source: T3, slide 9.

4 Design specifications

These are the **mission requirements**. Those listed in the slides (10):

- **Range** and **payload**: how much and how far.
- **Cruise speed**.
- **Runway**: length available for takeoff and landing.
- **Manoeuvrability** and stability.
- **Cost**.
- Others: number of desired operations, fashion, development time...

The mission rules: an agricultural aircraft and an airliner have such different specifications that they end up with very different shapes.

5 Aircraft configuration

This is the **first design**: where the wing, the engines, the control surfaces and the landing gear go. Previous similar aircraft are examined and new ideas are introduced. The decisions are tied to cruise speed, runway, stability/manoeuvrability, range, aerodynamic efficiency and payload.

5.1 The wing

Wing location: low, medium or high (Figure 3.3, which also shows the dihedral angle).

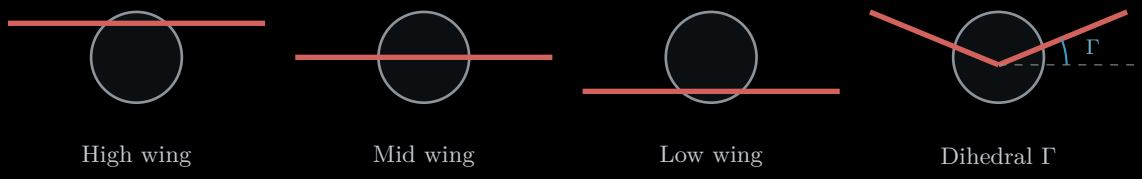


Figure 3.3. Wing position relative to the fuselage (front view) and dihedral angle. Own diagram.

Rule of thumb (supplement): cargo aircraft usually have a high wing, to keep the fuselage close to the ground, load easily and leave room for the engines; airliners usually have it low. In the 2022/23 exam the statement "a low wing is chosen for cargo aircraft" was rejected.

Wing planform: the shape of the wing seen from above (Figure 3.4).

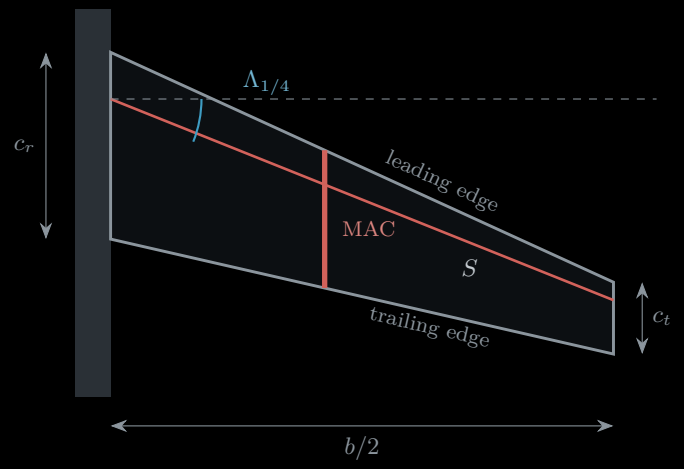


Figure 3.4. Half-wing planform with its parameters. Source: T3, slide 12 (own diagram, with MAC computed with the formula below).

Symbol	Name	Meaning
S	wing area	area of the wing seen from above
b	wingspan	from tip to tip
c_r, c_t	root chord and tip chord	
$\lambda = c_t/c_r$	taper ratio	how much the wing tapers
$AR = b^2/S$	aspect ratio	how long and narrow it is
$\Lambda_{1/4}$	sweep angle	angle of the 25 % chord line
Γ	dihedral angle	upward inclination of the wings seen from the front
$\bar{c} = S/b$	mean geometric chord	chord of a rectangular wing with the same S and b
MAC	mean aerodynamic chord	"equivalent" chord in aerodynamic effects

$$AR = \frac{b^2}{S} \quad \lambda = \frac{c_t}{c_r} \quad \bar{c} = \frac{S}{b} \quad S_{\text{trapezoidal}} = \frac{b}{2}(c_r + c_t)$$

The **MAC** (*Mean Aerodynamic Chord*), supplement: for a trapezoidal wing it is

$$\text{MAC} = \tfrac{2}{3}\, c_r\, \frac{1+\lambda+\lambda^2}{1+\lambda}.$$

PARAMETERS OF A WING PLANFORM

Trapezoidal wing with $b = 30$ m, $c_r = 6$ m and $c_t = 1.5$ m (example data).

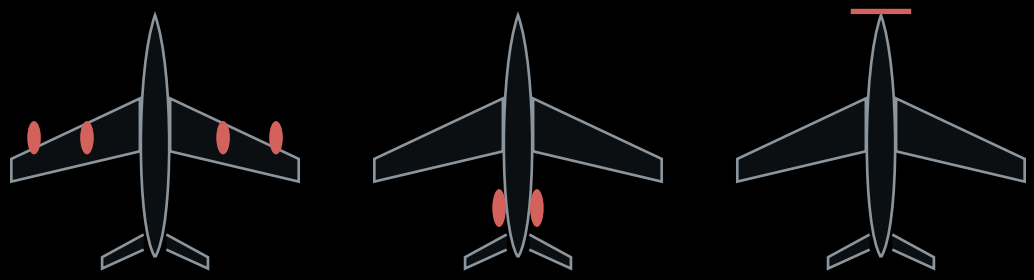
$S = \frac{b}{2}(c_r + c_t) = 15 \cdot 7.5 = 112.5$ m^2; $\lambda = 1.5/6 = 0.25$; $AR = \frac{30^2}{112.5} = 8$; $\bar{c} = \frac{112.5}{30} = 3.75$ m; $\text{MAC} = \frac{2}{3} \cdot 6 \cdot \frac{1 + 0.25 + 0.0625}{1.25} = 4.2$ m.

Check with approximate real data (A320, from memory): $b \approx 34.1$ m and $S \approx 122.6$ m^2 give $AR \approx 9.5$.

What these parameters do is studied in Unit 5, on aerodynamics (sweep, aspect ratio, taper ratio). What has been asked in tests: the typical taper ratio of an airliner is **"around 0.6"** among the options $\{0,\ 0.6,\ 1,\ >1\}$ (2023/24 exam; real values are usually somewhat lower, but 0 and 1 must be ruled out, as they correspond to the delta wing and the rectangular wing). A low taper ratio improves structural stresses and makes it easier to stow the landing gear and fuel in the wing (2022/23 exam).

5.2 The engine

The **type** (propeller, turbofan, turbojet), the **location** (under the wing, on the fuselage) and the **number** must be decided; Figure 3.5 shows three typical arrangements.



Under the wing (like the A380, 4 turbofans) | On the rear fuselage (like the Citation VII, 2 turbofans) | Propeller on the nose (like the Piper PA-18, 1 engine)

Figure 3.5. Three engine arrangements, with the examples from the slides. Source: T3, slide 13 (own silhouettes).

5.3 Control surfaces and landing gear

- **Control surfaces**: the tail ones (elevator and rudder) and the ailerons. The surface that controls roll is the **aileron** (2024/25 exam). Figure 3.6 shows where they are.
- **Landing gear**: position and type. Examples from slide 15: BAe Jetstream 31, PZL-106 Kruk (agricultural) and An-225 (very heavy cargo). **Fixed gear** is usually the best solution in **low-speed** aircraft: it weighs less and is simpler, at the cost of more drag (2022/23 exam). Figure 3.7 compares the three solutions.

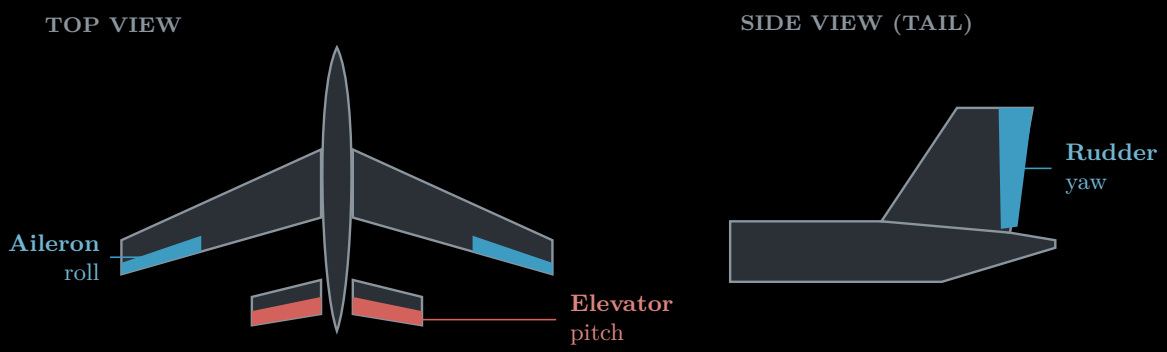
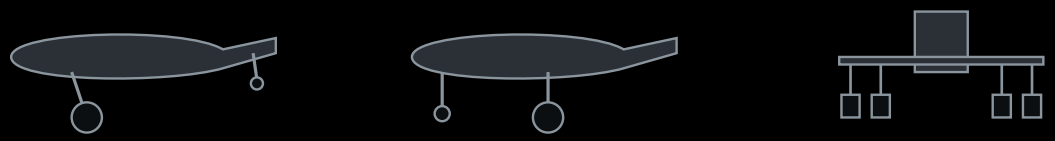


Figure 3.6. Primary control surfaces: ailerons (roll, on the wing trailing edge), elevator (pitch, on the horizontal tail) and rudder (yaw, on the fin). Schematic.



Fixed gear
light, simple, more drag
(PZL-106 Kruk)

Retractable gear
folds into the body; less drag, heavier
(BAe Jetstream 31)

Many wheels
spread a very heavy load
(An-225, 32 wheels)

Figure 3.7. Landing gear types, with the examples from the slides. Schematic.

5.4 Roskam classification

There is no single configuration: it depends on the mission. One possible classification (Jan Roskam, slide 16; "not by any means the only one"):

1. Homebuilt propeller aircraft
2. Single-engine propeller
3. Twin-engine propeller
4. Agricultural aircraft
5. Business jet
6. Regional turboprop
7. Jet transport
8. Military: trainer, transport, patrol, tanker, search and rescue
9. Military: fighters and bombers
10. Seaplanes
11. Supersonic cruise aircraft
12. Rotary-wing aircraft

6 Initial analysis

With the sketch, the main characteristics are determined and compared with the specifications. For cruise only, the loop is the one in Figure 3.8:

6.1 Weights

$$W_{TO} = W_{OE} + W_F + W_{PL}, \qquad W_{OE} = W_E + W_{crew} \quad \Rightarrow \quad W_{TO} = W_E + W_{crew} + W_F + W_{PL}$$

Figure 3.9 shows how these terms add up.

- W_{TO}: takeoff weight (its maximum is the MTOW, *Maximum Take-Off Weight*).
- W_{OE}: operating empty weight $= W_E + W_{crew}$.
- W_E: empty weight: structure, engines, systems, *seats*...
- W_{crew}: flight crew (pilots, flight attendants...).

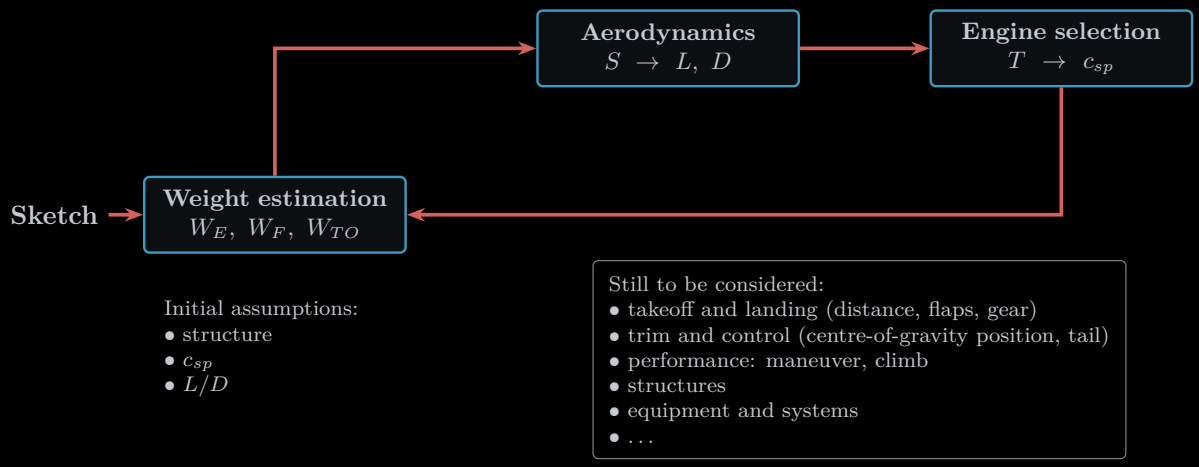


Figure 3.8. Iterative design (cruise only). Source: T3, slide 17. "The Dash-one is never built": the first design always changes.

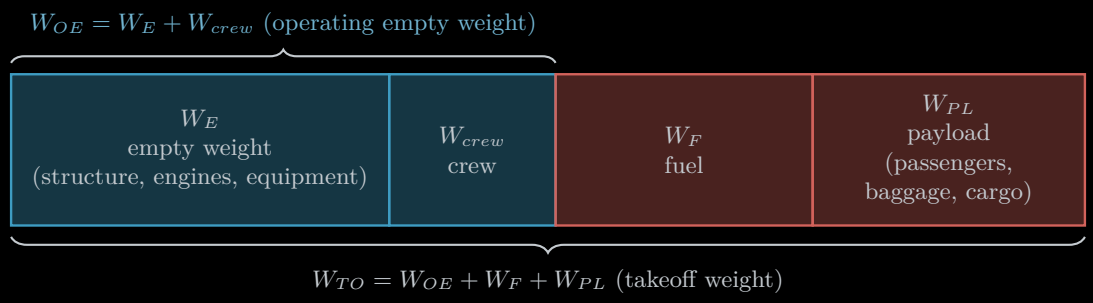


Figure 3.9. Breakdown of the takeoff weight (not to scale). Source: T3, slide 18.

- W_F: fuel. W_{PL}: payload (passengers, baggage, cargo).

Each one depends on the mission. (The slides write "weight" and the exam "mass": for problem purposes they are the same except for the factor g.)

RECURRING QUESTION: WHAT GOES INTO EACH WEIGHT?

- The **crew** is in the operating empty weight (2023/24) and **not** in the payload (2018/19). Fuel, passengers and suitcases are **not** in W_{OE}.
- **Seats** do not go in the payload: they are empty weight (2019/20).
- Empty weight fraction: the fraction W_E/W_{TO} changes with the mission and **tends opposite to the total weight** (the greater W_{TO}, the smaller the fraction; 2020/21 exam). New materials reduce it and leave more room for fuel and payload.

6.2 Fuel: mission profile and the Breguet equation

The fuel depends on the **mission profile** (Figure 3.10) and is minimized with a good specific fuel consumption and good aerodynamic efficiency.

The **Breguet equation** (under the assumptions that V, L/D and c_{sp} are constant) gives the range of a jet aircraft:

$$R \approx \frac{V}{c_{sp}} \frac{L}{D} \ln \frac{W_i}{W_f}$$

where V is the cruise speed, c_{sp} the **specific fuel consumption** (fuel burned per unit of thrust

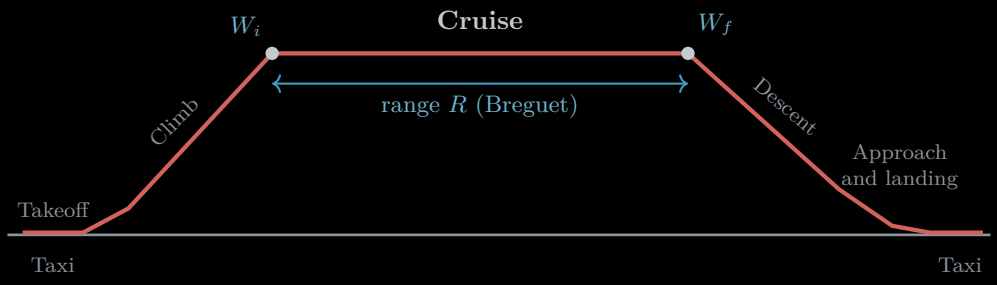


Figure 3.10. Typical mission profile. Source: T3, slide 20.

and time; here in units of 1/s, referred to weight), L/D the **aerodynamic efficiency** (lift divided by drag) and W_i, W_f the weights at the start and end of cruise (the difference is the fuel burned).

How to read it: more range with more speed, more efficiency L/D, less consumption c_{sp} and more fuel ($\ln W_i/W_f$ grows slowly, because it is a logarithm). Solving, the fuel fraction is $1 - \dfrac{W_f}{W_i} = 1 - \exp\Big(-\dfrac{R\,c_{sp}}{V\,L/D}\Big)$.

NUANCES OF THE BREGUET EQUATION (2020/21 TO 2024/25 TESTS)

- The range is proportional to the **logarithm** of the weight ratio, not to W_f/W_{TO} nor directly to the fuel.
- In a jet it is proportional to the **flight speed**; in a propeller engine it is **not** (for a propeller, the propeller efficiency and the power-specific consumption appear, and the speed cancels; supplement).
- In the absence of wind, the range of a jet is directly proportional to the average speed (2024/25). It is *inversely* proportional to the specific fuel consumption.
- The operating empty weight and the payload **do** influence the range (via W_i): the opposite statement was false (2020/21).

RANGE AND FUEL OF AN AIRLINER (ILLUSTRATIVE DATA)

$V = 230$ m/s, $L/D = 17$, $W_i/W_f = 1.25$ and $c_{sp} = 1.67 \cdot 10^{-4}$ s^{-1} (equivalent to about 0.6 lb/(lbf h), typical of a turbofan).

$R = \dfrac{230}{1.67 \cdot 10^{-4}} \cdot 17 \cdot \ln 1.25 = 1.377 \cdot 10^6 \cdot 17 \cdot 0.2231 = 5.2 \cdot 10^6$ m ≈ 5200 km. The fuel fraction is $1 - 1/1.25 = 0.20$, that is, 20 % of the weight at the start of cruise.

2023/24 EXAM: LOWER CONSUMPTION, LOWER AERODYNAMIC EFFICIENCY

Jet airliner: the specific fuel consumption drops by 3 % and the aerodynamic efficiency drops by 5 %, with the same speed and the same design mission. What happens?

Step 1. With V and R constant, $\dfrac{L/D}{c_{sp}} \ln \dfrac{W_i}{W_f}$ must remain the same. The ratio $\dfrac{L/D}{c_{sp}}$ becomes $\dfrac{0.95}{0.97} = 0.979$ of the previous one: it has worsened by 2 %.

Step 2. To compensate, $\ln \dfrac{W_i}{W_f}$ must be $1/0.979 = 1.021$ times greater: **more fuel** is needed.

Step 3 (example numbers). If it was previously 20 % ($\ln = 0.2231$), it is now $\ln = 0.2278$, that is $1 - e^{-0.2278} = 0.204$, that is 20.4 %. With the payload fixed and, for example, $W_E/W_{TO} = 0.5$, $W_{TO} = \dfrac{W_{PL} + W_{crew}}{1 - 0.5 - W_F/W_{TO}}$ goes from $\dfrac{x}{0.300}$ to $\dfrac{x}{0.296}$: it rises 1.4 %.

Answer: the **takeoff mass will have increased** (option c). The payload does not drop

because the aircraft is designed for the load the mission requires.

INITIAL WEIGHT SIZING (MADE-UP NUMERICAL EXAMPLE)
An aircraft must carry $W_{PL} = 18\,000$ kg and a crew of $W_{crew} = 500$ kg. From similar aircraft, $W_E/W_{TO} = 0.50$ is estimated and, with Breguet, $W_F/W_{TO} = 0.20$. What is W_{TO}?
$W_{TO} = W_E + W_{crew} + W_F + W_{PL} \;\Rightarrow\; W_{TO}\,(1 - 0.50 - 0.20) = 18\,500$, hence $W_{TO} = \dfrac{18\,500}{0.30} = 61\,667$ kg, with $W_E = 30\,833$ kg and $W_F = 12\,333$ kg. The calculation is circular: if W_E/W_{TO} turns out greater than assumed, W_{TO} increases and it must be repeated with the new estimate.

2019/20 EXAM: VALENCIA–PARIS AND BACK
An aircraft takes off at its MTOW on VLC–CDG, with 15 000 kg of cargo, and burns **all** its fuel. After refueling it flies the return leg with 10 000 kg.
The MTOW is a **maximum**. For the return it will take on the same fuel (same route), with $W_{TO} = W_{OE} + W_F + 10\,000 < W_{OE} + W_F + 15\,000 =$ MTOW: it **takes off at a weight lower than its MTOW** (option a). Neither the MTOW nor W_{OE} changes.

7 Refined sizing and optimization

Once the revised sketch makes sense, the remaining elements are characterized: aerodynamics, weights, propulsion, stability and control, structures and cost, ... Cost and technological availability are critical **before** moving on to preliminary design. The process is repeated for **several configurations**, and only when candidates have been designed is the one with the best properties chosen. Many designs fulfill the same mission (slide 22 compares an Avro with a B-52).

8 What usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM
There are **about three test questions** per first midterm (17 of the 115 in six first midterms), almost always conceptual; the numerical ones are short:

- What to start with (specifications) and what each phase does (conceptual: weights and viability; preliminary: freeze; detail: parts).
- Weight composition: what W_{OE} includes (crew yes; fuel, passengers and suitcases no) and the payload (without crew or seats).
- Breguet: proportionality (V, L/D, $1/c_{sp}$, logarithm of the weight ratio); the problem of changing c_{sp} and L/D; round-trip route with MTOW.
- Configuration: fixed gear in slow aircraft; high/low wing; mission-dependent wing planform; typical taper ratio; aileron for roll.

The essentials: $W_{TO} = W_E + W_{crew} + W_F + W_{PL}$; $W_{OE} = W_E + W_{crew}$; $R \approx \frac{V}{c_{sp}} \frac{L}{D} \ln \frac{W_i}{W_f}$; $AR = b^2/S$; $\lambda = c_t/c_r$; $\bar{c} = S/b$.

UNIT 4

Aircraft structures

1 What a structure does and the vocabulary you need

The structure of an aircraft is its **skeleton and skin**: it holds everything else (engines, cargo, passengers, landing gear) and carries forces from one place to another without breaking or deforming excessively. Its design is always a compromise: it must be **very strong and very stiff yet as light as possible**, because every kilogram of structure is one kilogram less of payload or fuel (which is why the operating empty weight, OEW, was so important in Unit 3).

MINIMUM GLOSSARY FOR THIS UNIT

- **Stress** (σ or τ): force per unit area inside the material, $\sigma = F/A$ (Pa or N/mm^2). It measures *how hard the material is working*.
- **Strain** (ε): relative elongation, $\varepsilon = \Delta L/L$ (dimensionless). It measures *how much the material has stretched*.
- **Strength**: maximum stress the material withstands before deforming permanently or breaking. **Stiffness**: how hard it is to deform. **They are not the same**: a steel cable is very strong but not at all stiff in bending; a thick plastic ruler is stiff but not very strong.
- **Beam**: long, slender member. **Cantilever**: beam fixed at one end and free at the other, like a diving board. Structurally, a wing is a cantilever fixed to the fuselage.
- **Longitudinal** = along the long direction of the member; **transverse** = perpendicular to it. **Spar** = main longitudinal beam of the wing.

2 Stresses, stiffness and buckling

2.1 Types of stress

Any structural member undergoes one of these stresses or a combination of them (Figure 4.1).

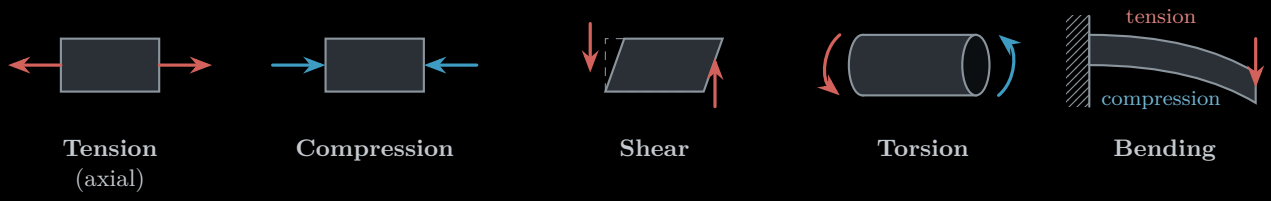


Figure 4.1. Basic stresses: tension, compression, shear, torsion and bending (tension on one side and compression on the other). Source: Unit 4, slide 4.

- **Tension** (also called axial stress): stretches. **Compression**: squashes, and in slender members it causes buckling.

- **Shear**: two parallel, opposite forces try to slide one section relative to the adjacent one, like the blades of a pair of scissors.
- **Torsion**: twisting, like wringing out a towel.
- **Bending**: flexing. In a bent beam, one face stretches (tension) and the opposite face shortens (compression); in between there is a neutral fibre with no stress. That is why the *useful* material in bending is the material far from the centre (see the I shape below).

2.2 Hooke's law and the concept of stiffness

A linear spring obeys $F = kx$ (applied force F, spring stiffness k, extension x). A beam under axial load behaves in the same way in the elastic region:

$$\sigma = E\,\varepsilon \qquad \Longrightarrow \qquad \frac{F}{A} = E\,\frac{\Delta L}{L} \qquad \Longrightarrow \qquad k_{axial} = \frac{EA}{L}$$

with E the Young (elastic) modulus (how much the material "resists stretching", in Pa) and A the cross-sectional area. For shear the same holds with the **shear modulus** G and the shear strain γ_{xy}:

$$\tau_{xy} = G\,\gamma_{xy} \qquad\qquad k_{shear} = \frac{GA}{L}$$

For bending, in a cantilever with a force F at the free end:

$$y_{max} = \frac{FL^3}{3EI_x} \qquad \Longrightarrow \qquad k_{bending} = \frac{3EI_x}{L^3}$$

In the cantilever the maximum moment is at the fixed end, $M_{z,max} = -FL$, and I_x is the **moment of inertia** (second moment of area) of the section about its axis, which measures how the area is distributed relative to the neutral fibre.

Stress	Stiffness	What increases it
Axial	EA/L	$E\uparrow$, $A\uparrow$, $L\downarrow$
Shear	GA/L	$G\uparrow$, $A\uparrow$, $L\downarrow$
Bending (cantilever)	$3EI/L^3$	$E\uparrow$, $I\uparrow$, $L\downarrow\downarrow$ (cubic)

KEY IDEA
The higher the stiffness, the less the structure deforms under a given load. High stiffness is sought to limit deformations, but *elastic* deformations are a "good" mechanism (the wing flexes and recovers its shape, absorbing gusts). What must be avoided are **permanent deformations** and breakage. Stiffness depends on E (or G), on the geometry (A, I) and on the length L, which enters cubed in bending.

The stress–strain curve

As Figure 4.2 shows, as long as the stress stays below P (the **elastic limit**) the material returns to its shape when the load is removed, like a spring. If P is exceeded, on unloading the curve descends *parallel to the elastic region* and a permanent deformation S remains. The *yield stress* is the stress beyond which this happens appreciably, and the maximum stress reached is the ultimate strength.

Common mistake: confusing the *elastic limit* (permanent damage begins) with the *ultimate strength* (fracture). In design, members are sized to stay always below the elastic limit.

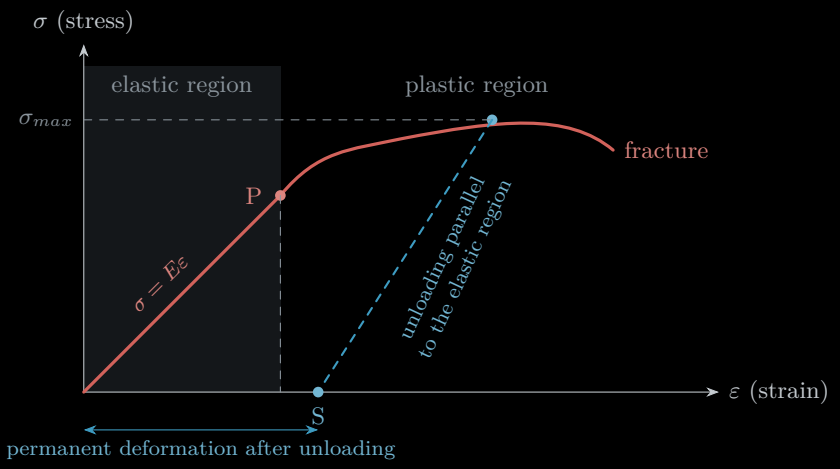


Figure 4.2. Material behaviour: linear and reversible up to the elastic limit (P); beyond P, unloading leaves a permanent deformation (S). Source: Unit 4, slide 6.

Moment of inertia: why beams are I-shaped

Supplement (not developed in the slides). For a rectangle of base b and height h (the dimension in the bending direction), $I = bh^3/12$. The factor h^3 appears: **doubling the height multiplies the bending stiffness by 8** while only doubling the material, whereas doubling the width only doubles it. That is why the I-beam (Figure 4.3; flanges far from the axis, thin web) or the tube is so efficient: it puts the material where it works hardest and wastes no weight at the centre.

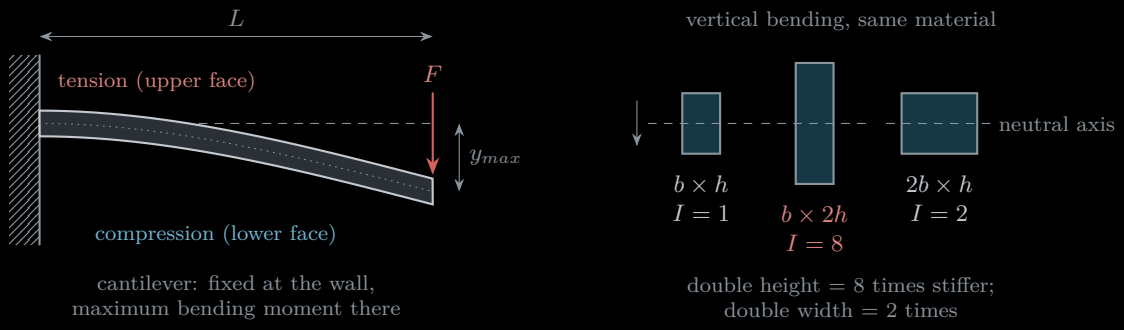


Figure 4.3. Left: a cantilever beam loaded at the free end (tension on one face, compression on the other, maximum moment at the wall). Right: the same material and the same bending direction; $I = bh^3/12$ grows with the cube of the height (own supplement).

2024/25 EXAM, TEST 1, QUESTION 7: INCREASING BENDING STIFFNESS
For a member subjected to a bending moment, stiffness is proportional to EI/L^3. Effect of each option:

Option	Factor on stiffness	Comment
Doubling the length L	$1/2^3 = 0.125$	becomes 8 times worse
Cross-sectional area ×3	depends on how	A does not appear in $3EI/L^3$; it only matters if I grows
Material with a 20 % higher E	1.2	small improvement
Moment of inertia ×2	2	**the best**

Answer: increase I by a factor of 2 (option d). The area option is misleading: more area does not imply a larger moment of inertia.

2.3 Buckling

A slender column under **compression** can suddenly give way by bowing sideways (*buckling*), without the material having reached its strength. This is what happens with a plastic ruler squeezed between the hands: it bends long before it is crushed. The same happens with the skin and stringers under compression or shear. It is an *instability* (a sudden change of shape), not a failure of the material. Figure 4.4 shows how the support conditions change the effective length.

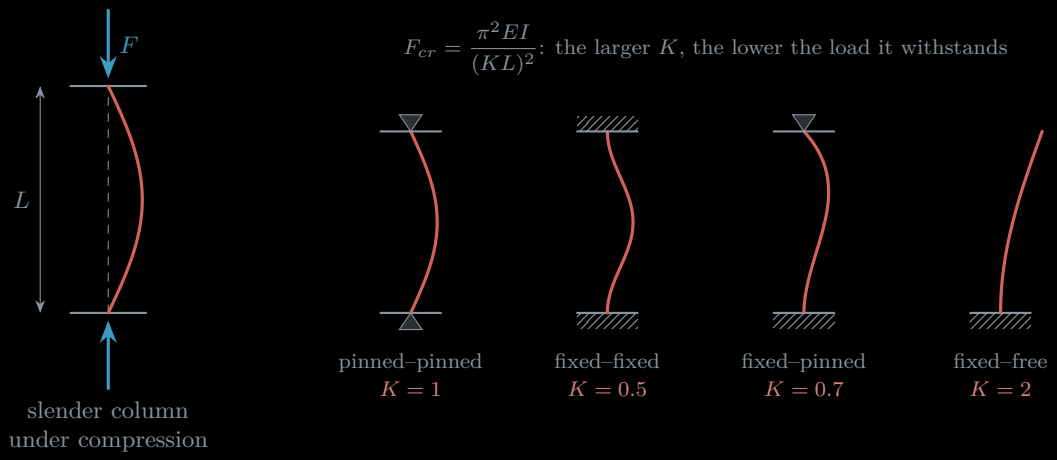


Figure 4.4. Buckling of a column and effect of the support conditions on the effective length KL (the values of K are the classical ones from Euler theory; the slide only says that K depends on the supports). Source: Unit 4, slide 9.

$$F_{cr} = \frac{\pi^2 \, E \, I}{(K \, L)^2}$$

E Young modulus, I moment of inertia, L length, K effective length factor (depends on the supports). Consequences:

- The critical buckling load **may be lower than the load that would reach the elastic limit**: members must be sized taking buckling into account.
- The parameters are the same ones that give stiffness (E, I, L), and similar relations hold for plates (the skin). That is why stringers and frames are added: they **shorten the free length** of the panel and prevent it from buckling.
- **Length is the most powerful parameter**, because it enters squared: half the length between supports = four times the buckling load.

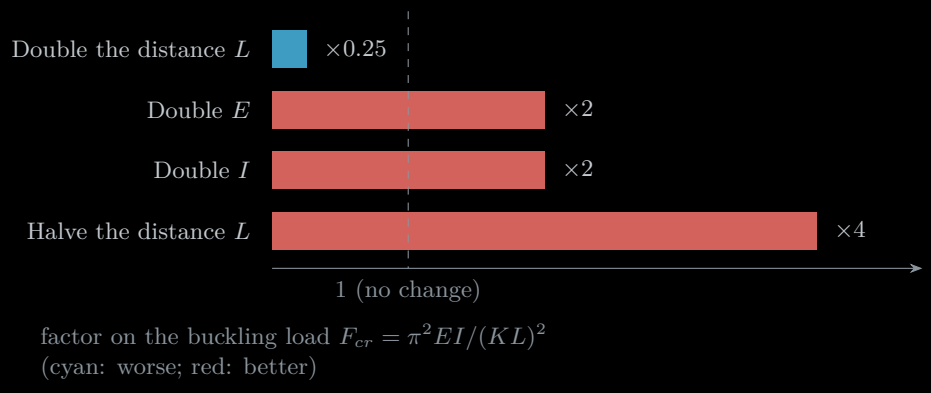


Figure 4.5. Effect of each modification on the buckling load (2023/24 exam, question 10).

2023/24 EXAM, TEST 1, QUESTION 10: HOW TO INCREASE THE BUCKLING LOAD OF A STRINGER?
With total cost and mass held constant, the options are compared (Figure 4.5; all relative to the initial situation, $F_0 = \pi^2 EI/(KL)^2$):

- Doubling the distance between supports: $L \to 2L$, $F \to F_0/4$.
- **Halving the distance between supports: $L \to L/2$, $F \to 4F_0$.**
- Doubling the Young modulus: $F \to 2F_0$. Doubling the moment of inertia: $F \to 2F_0$.

Answer: halve the distance between supports (option b).

3 Fuselage structure

The fuselage houses the cargo, passengers and crew and joins the wings and tail. It bears bending (due to its weight and to the forces from the wings and tail), torsion, shear and, if pressurized, the internal pressure. It has gone through several historical architectures.

3.1 Truss structures

A **truss structure** is a set of bars in which each bar undergoes only axial stress (tension or compression), **never bending**. It usually has at least 4 **main longerons** running along the whole fuselage, joined by bars that form the lattice. **The skin** (usually fabric, sometimes wood or metal) **does not contribute**: it only provides the aerodynamic shape. It is light but not very aerodynamic and does not make use of the skin. The two classic layouts, Pratt and Warren, are compared in Figure 4.6.

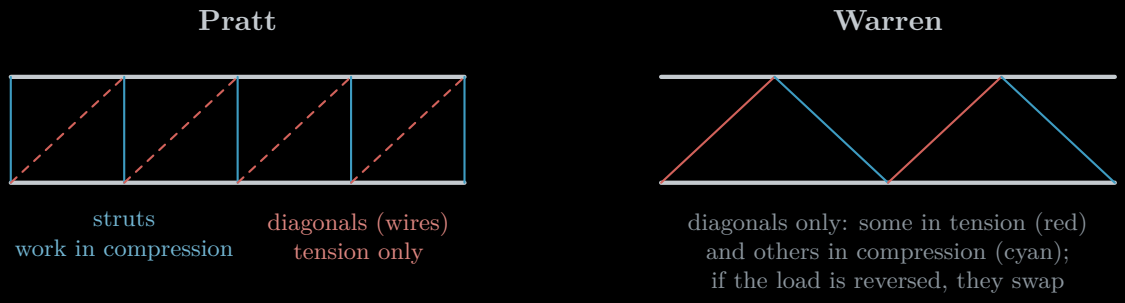
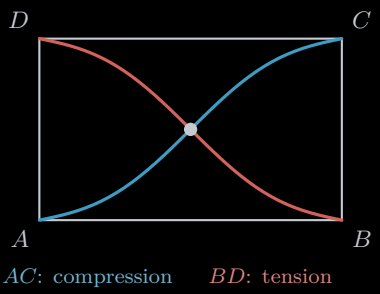


Figure 4.6. Fuselage truss structures. Pratt: vertical struts (in compression) and diagonals that are wires (working only in tension). Warren: diagonals only, working in tension or compression depending on the load. Source: Unit 4, slides 11 to 13.

- **Pratt**: the longerons are joined by vertical **struts**, which work mainly in compression, and **diagonals** that work only in tension and are usually wires. Examples: **Blériot XI**, **Wright Flyer** (first controlled aircraft of the 20th century). Biplanes used this *wire-braced* structure in their wings too.
- **Warren**: the longerons are joined only by **diagonals** that work in both tension *and* compression: under a given load some members are in tension and others in compression, and if the load is reversed they swap roles. Example: Piper PA-18.

3.2 Geodesic structure

It is made of **geodesic** arcs (minimum-length lines over a curved surface). In each section one member works in compression and another in tension, crossing each other (Figure 4.7).



1. AC is compressed and its curvature increases.
2. Tension appears in BD, which tries to straighten AC.
3. They are joined at the centre: the forces oppose and balance each other (equilibrium, also in torsion).

Figure 4.7. Principle of the geodesic structure: the compression of AC and the tension of BD balance each other because they are joined at the centre. Source: Unit 4, slides 14 and 15.

- **Torsional loads are cancelled**.
- It achieves **maximum strength with minimum weight**, and **it still performs well even if part of the structure is lost** (as stated in a test from previous years: the geodesic structure is the one that best withstands the loss of part of it).
- **Problem**: it is very slow to build; it was restricted to World War II (Vickers Wellington, Vickers Warwick).

3.3 Monocoque and reinforced monocoque

Monocoque ("a single shell"): **the skin bears the stresses** and is the main structural element. It is a thin layer, ideally without longitudinal or transverse load-bearing members. Advantages: stability and aerodynamic efficiency. It was originally built in plywood (Deperdussin) and is now made of light alloys.

Reinforced monocoque: the skin is stiffened transversally by vertical rings, the **frames**. Since the skin withstands tension but *not compression* (it buckles), angular profiles are added to the frames. Today it is only used where there are no openings (windows) or where they are few and small, so that the stress distribution is uniform: it is light but **difficult to build and repair** and limits the design. The evolution from the monocoque to the semi-monocoque is summarised in Figure 4.8.

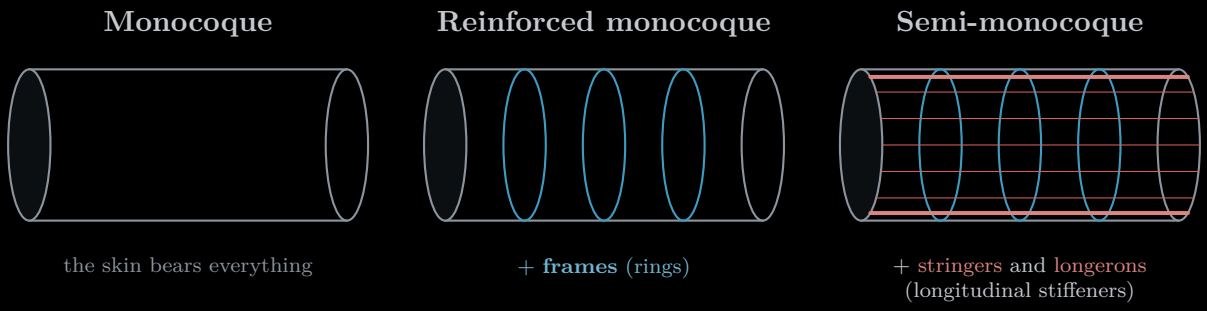


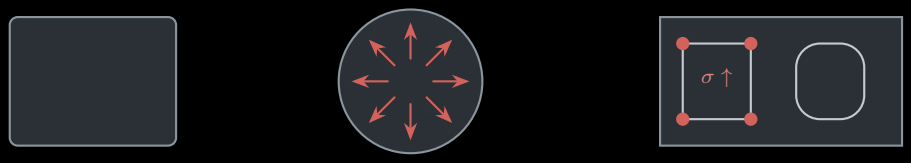
Figure 4.8. Evolution of the skin: monocoque, reinforced monocoque (frames) and semi-monocoque (frames plus longitudinal stiffeners). Source: Unit 4, slides 16 to 18.

3.4 Semi-monocoque

This is the current architecture: a **thinner skin** (with a structural role) with **frames** and also **stringers** and, sometimes, **longerons** as longitudinal stiffeners. It greatly increases buckling resistance at low weight.

- **Non-pressurized**: it can have a **square** shape (makes better use of the space; example: CASA C-212).

- **Pressurized**: the **internal overpressure** is added as an extra load; the fuselage inflates like a balloon and therefore has a **rounded** shape, as Figure 4.9 illustrates (a cylindrical section distributes the pressure uniformly: in a square box, the pressure would try to round it off and concentrate stresses at the corners). Concentrated shear stresses appear on the frames: **floating frames** are used so that the shell can expand uniformly. Any discontinuity causes problems (stress concentrators, sealing issues, corrosion) and **the windows are still the weakest point**.



Non-pressurized
square shape acceptable

Pressurized
round section
(inflates like a balloon)

Windows
sharp corner = stress concentrator;
rounded: stress spread out

Figure 4.9. Why the pressurized fuselage is round and the windows have rounded corners. Source: Unit 4, slides 19 and 20.

> **CASE STUDY: THE 1953–54 COMETS**
> In 1953 and 1954 three de Havilland Comets broke up in flight soon after taking off, killing everyone on board. Construction techniques were improved, the windows were changed from **square to rounded**, and other improvements made later versions safe. Sharp corners concentrate stress (fatigue cracks) and rounded ones do not. (Slide 21; the slide only mentions "square windows"; fatigue as the mechanism is added information.)

Parts of a semi-monocoque fuselage

Figure 4.10 shows the main elements in a side section; the table below gives the function of each one.

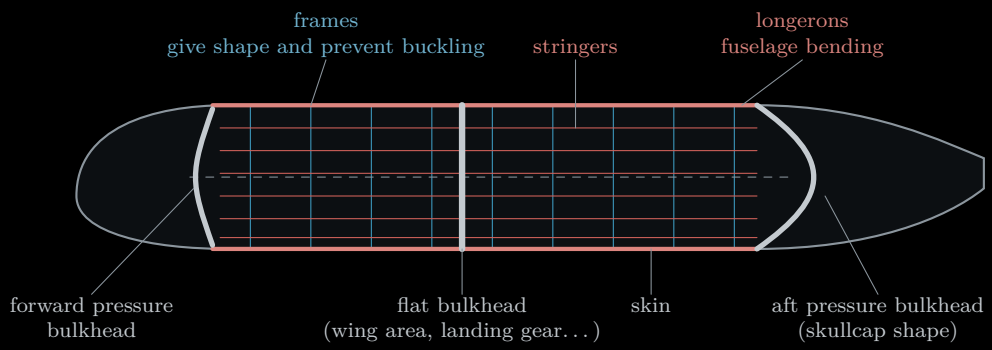


Figure 4.10. Parts of a pressurized semi-monocoque fuselage, in side section (schematic; in reality there are many more frames). Source: Unit 4, slides 22 to 25.

Element	Function
Skin	Bears most of the tension, compression and torsion, and **maintains the cabin pressure**.
Frames (or *former rings*)	Transverse members that give the skin its **circular shape**. They support the skin and stringers, **raising their resistance to buckling**.
Stringers	Small longitudinal beams. They support the skin and raise its buckling resistance; they bear tension and compression, including the **fuselage bending**.
Longerons	Larger longitudinal beams that bear bending and distribute axial loads. **They do not always exist.**
Pressure bulkheads	Close the cylinder at the front and rear to maintain pressurization. The rear one has a **skullcap** shape.
Flat bulkheads	Structural planes in stress-concentration zones (engine pylons, landing gear, wings, tail surfaces).

All of them are joined with **rivets or bolts**. Other elements shown in the illustrations of slide 23: seat track, floor beam, window frames and columns.

SUMMARY OF FUSELAGE TYPES

Type	What bears the loads	Notes
Pratt	Bars (struts in compression, wire diagonals in tension); skin does not contribute	Wright Flyer, Blériot XI
Warren	Diagonal bars, in tension or compression depending on the load	Piper PA-18
Geodesic	Crossed arcs (compression and tension)	Cancels torsion; slow to build (WWII)
Monocoque	The skin	Aerodynamic; thin skin without stiffeners
Reinforced monocoque	Skin + frames	No windows or very few
Semi-monocoque	Skin + frames + stringers (+ longerons)	Current; pressurized or not

4 Wing structure

The wing must **generate lift and bear loads**: it behaves as a **beam** (a cantilever fixed to the fuselage) that must resist the stresses. The loads come from:

- **Aerodynamic loads** (lift and drag).
- **Engine thrust** (if the engines are mounted on the wing).
- **Landing gear reaction**.
- **Deflection of the high-lift devices** (flaps, slats).

In addition there are the self-weight and the fuel inside. In flight, lift bends the wing upwards: the upper surface is compressed and the lower one is in tension (the upper flange of the spar is in compression and the lower one in tension). Figure 4.11 shows the main parts of the wing in plan view.

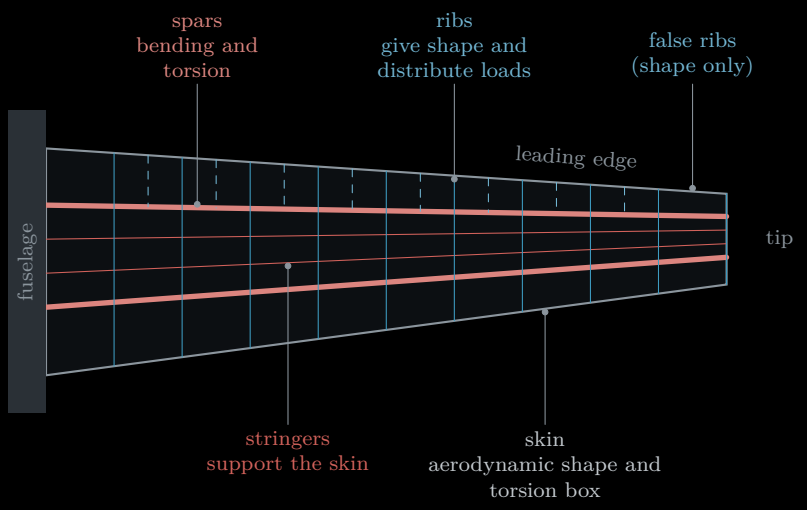


Figure 4.11. Parts of a semi-monocoque wing seen in plan view (half-wing): spars, ribs, stringers, false ribs and skin. Source: Unit 4, slides 28 to 31.

4.1 Parts of a semi-monocoque wing

Element	Function
Skin	Gives the **aerodynamic shape** and maintains it along the wingspan; it **can contribute to the structural strength**.
Spars	Beams running along the wing: **the main support of the structure**. They bear **bending and torsion**.
Ribs	Transverse members. They give the **airfoil shape** and transmit the loads from the skin to the spars.
Stringers	Longitudinal members that transmit the load from the skin to the ribs and **stiffen the skin against buckling**.
Fittings	Metal joining parts: wing–fuselage, control surfaces–wing... They bear stresses, vibrations and deflections.

The closed region between the spars and the skin forms the **torsion box**, which resists torsion and also serves as a fuel tank (the ribs create baffles in the tanks). Figure 4.12 shows it in cross-section.

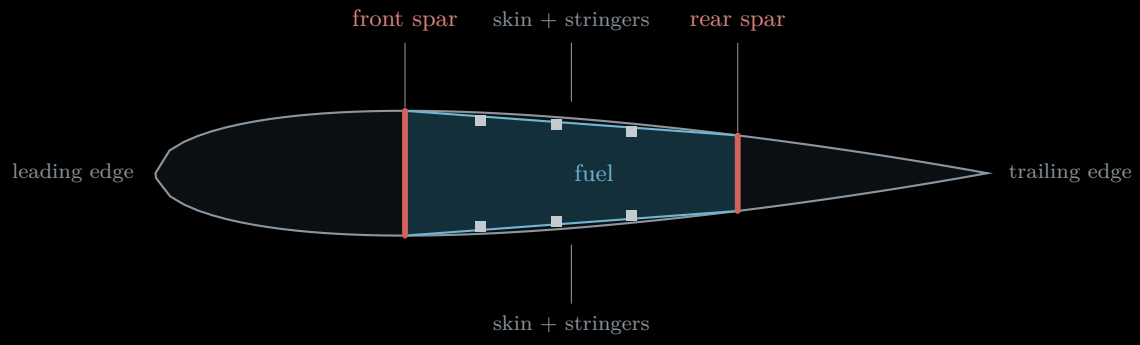


Figure 4.12. Cross-section of a wing: the torsion box (between the two spars, closed by the upper and lower skin with stringers).

Spars

A spar has two main parts: the **web**, which bears the **shear stresses**, and the **flanges**, upper (compression in flight) and lower (tension), which bear the tension and compression due to bending.

It is like an I-beam: the flanges do the bending work and the web keeps them apart and absorbs the shear. Figure 4.13 shows typical spar sections and the two spar types according to the web.

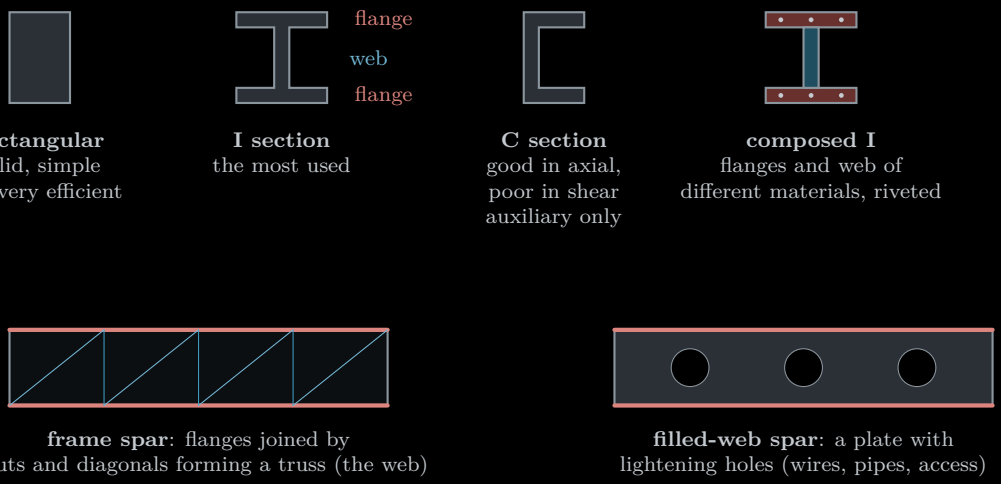


Figure 4.13. Typical spar sections and spar types according to the web. Source: Unit 4, slides 32 to 34.

- **Frame spars**: the flanges are joined by vertical and/or diagonal members (riveted, bolted or welded) that form a truss which is the web.
- **Filled-web spars**: the flanges are joined by a plate. If the web is very tall, it carries vertical reinforcements. **Holes** are made in it to reduce weight, to pass wires and pipes through, and to make access easier for maintenance.

The forces vary along the span, so the spar section can be **variable** (smaller towards the tip) to save weight. Its shape depends on the airfoil, its thickness, the required strength and the material:

- **Rectangular**: solid, economical and simple, not very efficient.
- **I section**: upper and lower flanges joined by the web.
- **C section**: good for axial stresses, but **unstable under shear loads**; it is only used as an **auxiliary spar**.
- **Composed I section**: the flanges are made of one material and the web of another, joined with rivets.

Ribs

- They keep the **airfoil shape** (thickness, spacing between spars).
- They **transmit the aerodynamic forces** to the spars.
- They **distribute concentrated loads** (landing gear, tanks...).
- They **close the wing cells** (torsion box) and create baffles in the fuel tanks.

Type (by function)	Purpose
Compression ribs	Link the spars and distribute the stresses to them; placed where there are **local stresses** (engine mounts, landing gear).
Master ribs	Keep the distance between spars and give stiffness to the skin and stringers, **preventing their buckling**.
Common ribs	Less robust: they keep the airfoil shape and transmit the internal forces to the spars.
False ribs	Only used to **keep the shape of the skin**; located between the spar and the leading or trailing edge.

Like spars, ribs can be of the **truss** or **filled-web** type, and have a **web** (stiffness against vertical and diagonal deformations; if metallic, it is manufactured by pressing) and **nerves**. In swept wings they can be placed **parallel to the fuselage axis** or **perpendicular to the spars** (Figure 4.14).

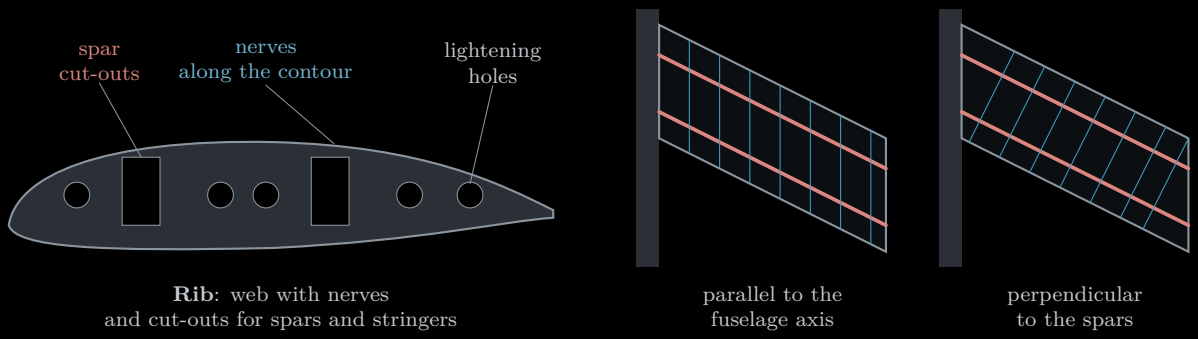


Figure 4.14. Anatomy of a rib and arrangement in swept wings. Source: Unit 4, slides 35 to 37.

5 Materials

5.1 Requirements for aircraft structural materials

- **Specific strength** (strength/weight) **and stiffness** *at the same time* (the combination, not just one).
- **Predictable** behaviour.
- **Good ageing** (against the environment: corrosion…).
- **Fireproof**.
- Easy to obtain, readily available and at a **reasonable cost**.
- **Fatigue resistance** (thousands of load cycles) and **thermal resistance**.

Specific strength $= \sigma_{\text{strength}}/\rho$: how much stress the material withstands per unit of density. The same applies to specific stiffness E/ρ. This is what matters because the goal is to withstand what is needed with the least weight.

5.2 Materials used

- **Origins of aviation**: **wood** (problems with humidity), **fabric** and **steel wire** (very dense).
- **Light aluminium alloys**: high specific strength with low density; they are the most widely used.
- **Titanium alloys**: where **thermal loads** are expected.
- **Stainless steel**: extremely strong, but with a high density; only a few components (the **landing gear**).
- **Composite materials**: fabric sheets (carbon fibre) bound by a **resin matrix** (generally epoxy), sometimes embedded in a honeycomb core (sandwich panel). **Extremely light**, but with problems of **impact resistance and lightning resistance**.

Figure 4.15 shows how a sandwich panel is built.

Treatments and coatings required: **aluminium** against **corrosion**; **composites** against **humidity**.

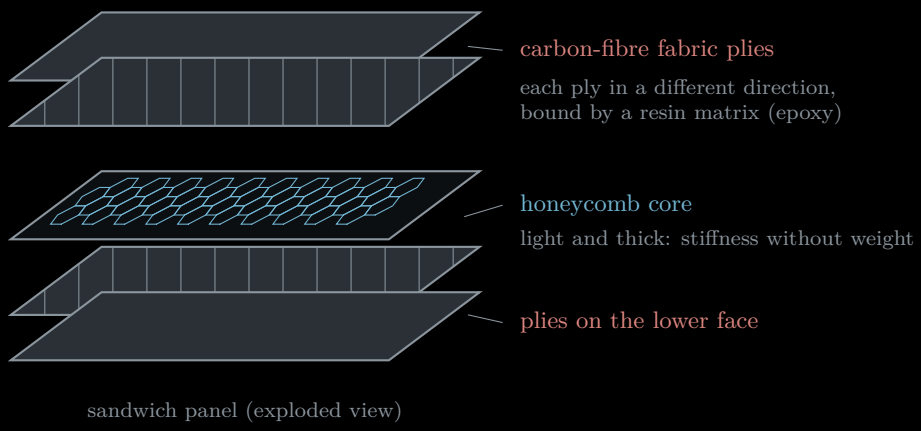


Figure 4.15. Sandwich panel of composite material: fabric plies bound by resin on both faces of a honeycomb core.

Material	Advantage	Drawback / use
Wood, fabric, wire	Available; light (wood)	Humidity (wood); wire very dense. Early aviation.
Aluminium alloy	High specific strength, low density	Corrosion: needs treatment. Most aircraft.
Titanium alloy	Withstands thermal loads	Thermal loads (expensive: added information, from exams of other years).
Stainless steel	Very strong	Dense: landing gear only.
Composites (CFRP)	Very light	Impact, lightning, humidity.

2023/24 EXAM, TEST 1, QUESTION 9: WING SKIN THAT ALWAYS WORKS IN TENSION

If the region always works in tension it cannot buckle, so buckling does not matter. What must not be exceeded is the elastic limit (so that no permanent deformation remains). With the goal of "high strength at low weight", the **specific yield strength** σ_y/ρ is maximized. (This is the answer given in the students' compilation; the official key is not in the exam PDF.)

2024/25 EXAM, TEST 1, QUESTION 8: WHICH ONE IS NOT CORRECT

"Aluminium alloys are used in most aircraft because they are the materials with the highest strength" is the false statement: they are used for their high **specific** strength with low density, not because they have the highest absolute strength (steel is stronger, but much heavier). Steel in the landing gear is correct (high strength per unit volume), titanium for thermal loads is also correct, and composites do need a coating against humidity.

WHAT USUALLY APPEARS ON THE EXAM (UNIT 4)

All past exams are **multiple-choice tests** (4 options, a wrong answer deducts 1/3). From this unit there are **three to four questions** per first midterm (21 of the 115 in six first midterms), almost always conceptual:

- **Fuselage**: which type *cancels torsion* (geodesic) and which *withstands the loss of part of the structure best* (geodesic); which type was used by the first controlled aircraft (Pratt, Wright Flyer); what bears the pressurization in a semi-monocoque (the skin, and also the pressure bulkheads; *not* the flat bulkheads nor the stringers as such); what the frames bear (they give shape and prevent buckling, *not* the bending moment); **the spacing between windows is chosen according to the spacing between frames**.

- **Wing**: the spar **web** bears **shear**, the **flanges** bear tension and compression due to bending; the spars bear the wing bending (a 2022/23 test, however, gives as correct that *the skin* bears the bending moment from lift: in a semi-monocoque the skin and stringers work together with the spars); the **ribs** give shape and distribute loads; stringers prevent skin buckling; false ribs only maintain the shape.
- **Stiffness and buckling, with numbers**: $k = 3EI/L^3$ in bending (doubling I doubles it; doubling L divides it by 8); $F_{cr} \propto EI/(KL)^2$ (half the distance between supports $\rightarrow \times 4$, the most effective option).
- **Materials**: aluminium for *specific* strength; titanium for thermal loads; steel only in the landing gear; composites with a coating against humidity (and problems with impact and lightning); surface treatments against corrosion and humidity.

Essentials: $\sigma = E\varepsilon$; $k_{axial} = EA/L$; $k_{bending} = 3EI/L^3$; $F_{cr} = \pi^2 EI/(KL)^2$; skin + frames + stringers = semi-monocoque; web = shear, flanges = bending; specific strength = strength/density.

UNIT 5

Fundamentals of atmospheric flight

1 Introduction: where the force that keeps an aircraft flying comes from

The air surrounding a vehicle exerts forces on it. There are two big families, which give rise to the most basic classification of aircraft:

- **Aerostats**: the force is **independent of the velocity** of the air relative to the body. It is Archimedes' principle (buoyancy from being lighter than the displaced air): balloons and airships.
- **Aerodynes**: the force **depends on the relative velocity** between air and body (aerodynamic force): aeroplanes, helicopters, drones. These are the ones studied in this unit. An aerodyne does not have to be lighter than air; it needs to move relative to it.

Common mistake: saying that aerostats need flow around an airfoil to stay aloft (false) or that an A320 is an aerostat (false: it is an aerodyne). Aerostats also cannot fly fast because of their large aerodynamic drag (exam question).

MINIMUM GLOSSARY FOR THIS UNIT

- **Airfoil**: the shape of the section of a wing cut by a plane parallel to the direction of motion. A wing of infinite span is a 2D airfoil.
- **Angle of attack** α: angle between the airfoil chord and the incident airstream.
- **Coefficient** (lift, drag…): the force divided by $\frac{1}{2}\rho V^2 S$, that is, the force *without units* and independent of how much air there is, how fast it moves or how large the body is. It depends mainly on the shape and on the angle of attack.
- **Mach number** $M = U_\infty / a_\infty$: flight speed divided by the speed of sound ($a = \sqrt{\gamma R T} \approx 340\,\text{m/s}$ at sea level; Unit 2).
- **Reynolds number** Re: ratio of inertial to viscous forces; it indicates the relative importance of the air viscosity at that scale.
- **Boundary layer**: very thin layer of air attached to the surface where viscosity slows the air down until it stops at the wall.
- **Stall**: the flow separates from the wing and a lot of lift is lost.

1.1 Aerodynamic forces and axes

Four forces act on an aircraft: **lift** L (perpendicular to the airstream), **drag** D (parallel to the airstream and in the same direction), **thrust** T (from the engine) and **weight** $W = mg$ (Figure 5.1). The first two are written with coefficients:

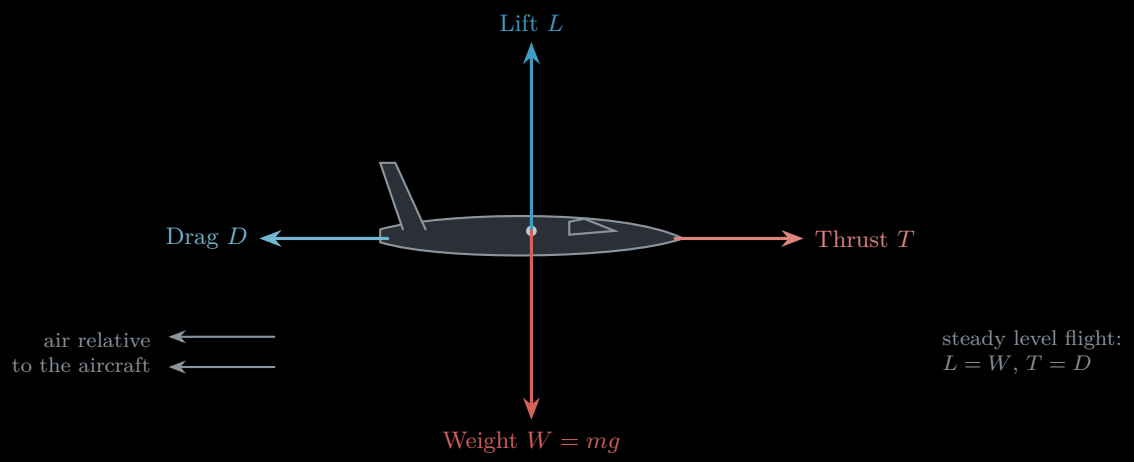


Figure 5.1. The four forces in steady level flight. Source: Unit 5, slide 5.

$$L = \tfrac{1}{2}\rho V^2 S\, C_L \qquad D = \tfrac{1}{2}\rho V^2 S\, C_D \qquad W = mg$$

ρ air density, V velocity, S reference area (the wing area), C_L and C_D coefficients (dimensionless). The quantity $\frac{1}{2}\rho V^2$ is the **dynamic pressure**, which measures how much kinetic energy per unit volume the air carries. In steady level flight $L = W$ and $T = D$.

Three axes passing through the centre of gravity are defined (Figure 5.2): the **longitudinal** or roll axis (from tail to nose), the **lateral** or pitch axis (from wingtip to wingtip, towards the right) and the **vertical** or yaw axis (perpendicular to both, pointing to the ground). The three **aerodynamic moments** are roll, pitch and yaw.

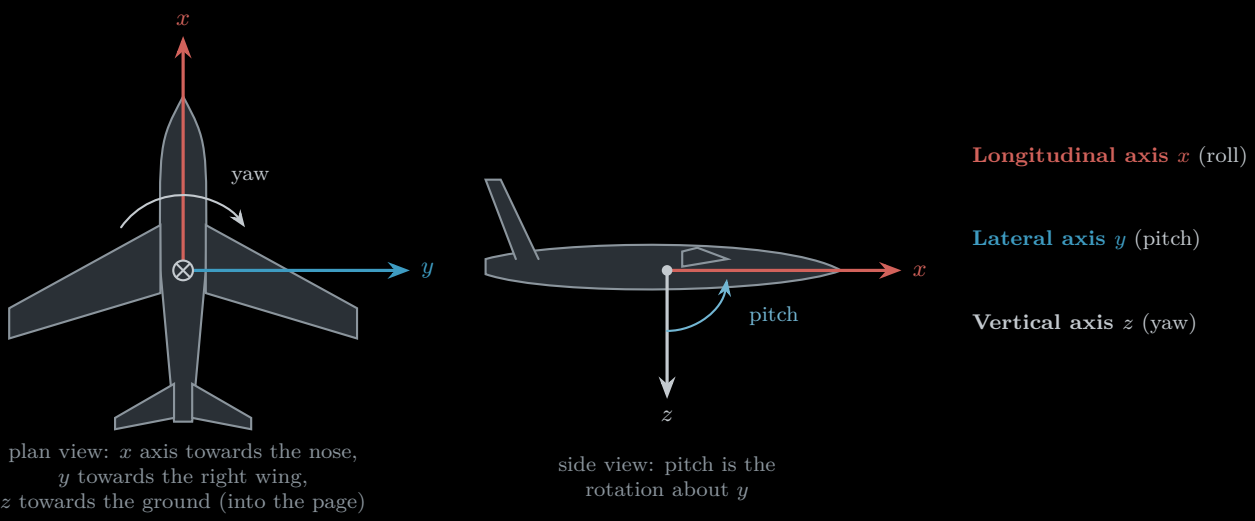


Figure 5.2. Aircraft axes and rotations. Source: Unit 5, slide 6.

1.2 Blunt and streamlined bodies

A **blunt body** (a plate perpendicular to the wind) has **a lot of drag**. A **streamlined body** (like a wing) has **little drag** and can generate **a lot of lift**: this is the one of interest for the flight of aerodynes (Figure 5.3).

1.3 Flight regimes

Behaviour changes greatly with the Mach number $M_\infty = U_\infty / a_\infty$ (Figure 5.4):

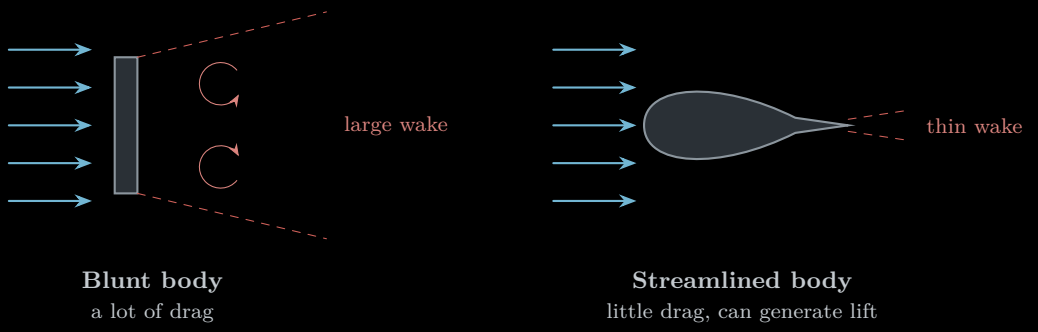


Figure 5.3. Blunt versus streamlined body: the blunt one leaves a large wake behind it.

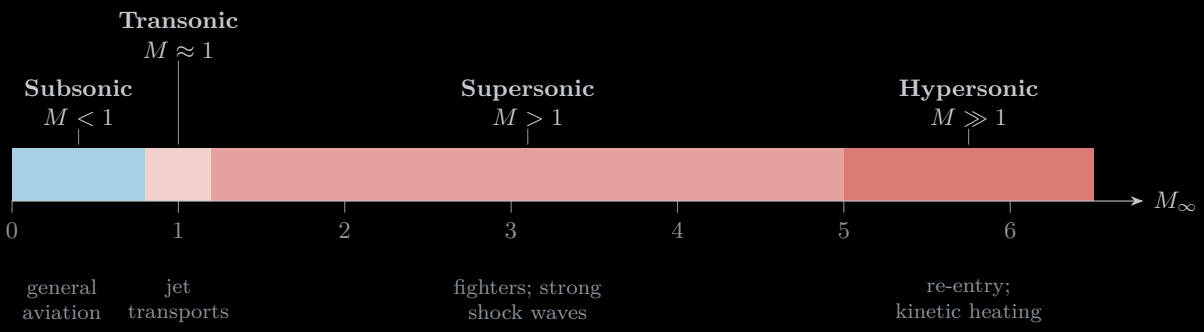


Figure 5.4. Regimes according to the Mach number (the numerical limits are indicative; the slides only give the conditions $M < 1$, $M \approx 1$, $M > 1$ and $M \gg 1$). Source: Unit 5, slide 8.

- **Subsonic** ($M < 1$): the usual case in general aviation.
- **Transonic** ($M \approx 1$): almost the case of commercial jet transports.
- **Supersonic** ($M > 1$): typical of fighters; strong shock waves may appear.
- **Hypersonic** ($M \gg 1$): typical of re-entry vehicles; there are always strong shock waves and kinetic heating (Section 6) is a problem.

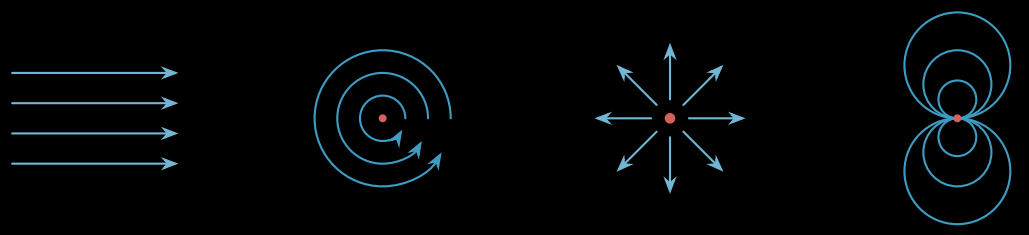
1.4 Potential flow, boundary layer and Reynolds number

Far from streamlined bodies the velocity field **derives from a potential**, just like a gravitational or electric field. The flow is assumed incompressible ($\nabla \cdot \mathbf{v} = 0$) and irrotational ($\nabla \times \mathbf{v} = 0$), so that there exists a φ with $\mathbf{v} = \nabla\varphi$ and the **Laplace equation** holds:

$$\nabla^2\varphi = \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} + \frac{\partial^2\varphi}{\partial z^2} = 0 \qquad U_x = \frac{\partial\varphi}{\partial x},\ U_y = \frac{\partial\varphi}{\partial y},\ U_z = \frac{\partial\varphi}{\partial z}$$

Since the equation is linear, solutions can be **added**: combining simple flows yields complex flows (e.g. uniform flow + doublet = flow around a cylinder). The elementary solutions (slide 10) are drawn in Figure 5.5 and summarised in the table below:

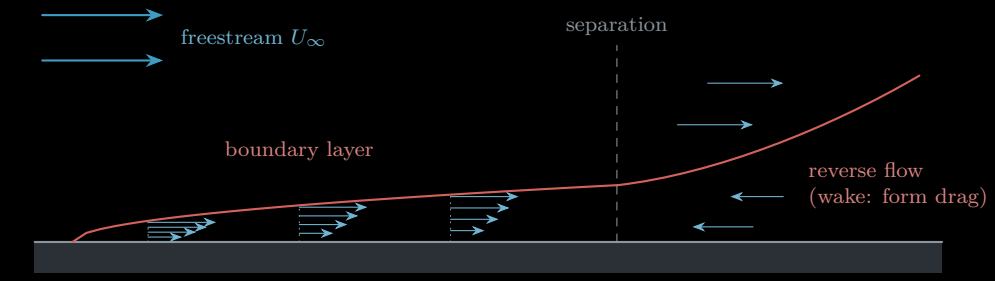
Flow	Potential	Velocity
Uniform	$\varphi = U_0 x$	$U_x = U_0$
Line vortex	$\varphi = \dfrac{\Gamma}{2\pi}\theta$	$U_\theta = \dfrac{\Gamma}{2\pi r}$
Source/sink	$\varphi = \dfrac{Q}{2\pi}\ln r$	$U_r = \dfrac{Q}{2\pi r}$
Doublet (dipole)	$\varphi = -\dfrac{\mu}{r}\cos\theta$	–



uniform $\varphi = U_0\, x$ | **line vortex** $\varphi = \dfrac{\Gamma}{2\pi}\theta$ | **source / sink** $\varphi = \dfrac{Q}{2\pi}\ln r$ | **doublet (dipole)** $\varphi = -\dfrac{\mu}{r}\cos\theta$

Figure 5.5. Elementary solutions of potential flow. The slide writes the source with a minus sign, but then $U_r = Q/(2\pi r)$ would come out negative; the usual sign is used here. Source: Unit 5, slides 9 to 11.

Potential flow **has no viscosity**: it only gives forces perpendicular to the surface, from pressure. But close to the body, viscous effects are important: this is the **boundary layer**, where the air goes from being at rest at the wall to the freestream velocity over a very short distance; it generates **tangential (shear) stresses**. Figure 5.6 shows how the velocity profile grows downstream and what happens at separation.



velocity goes from 0 at the wall to U_∞ over a small thickness: shear stress (friction drag).
With an adverse pressure gradient the layer separates.

Figure 5.6. Boundary layer: velocity profile growing downstream and separation with reverse flow. Source: Unit 5, slides 12 to 14.

$$Re = \frac{\rho_\infty U_\infty\, c}{\mu_\infty}$$

ρ_∞ density, U_∞ velocity, μ_∞ dynamic viscosity and c characteristic length (the wing chord). If Re is **small**, viscous effects matter and the boundary layer is **thick**; if Re is **large**, the layer is **thin**. The aerodynamic behaviour depends on Re.

- The boundary layer can be **laminar** (ordered layers) or **turbulent** (chaotic mixing). A laminar layer *transitions* to turbulent with roughness and with high Re.
- Boundary layers can **separate** from the body; this happens more easily if they are laminar. The higher Re is, the later the flow separates.

WHAT IS THE VALUE OF Re ON A TYPICAL WING
Chord $c = 1.5\,\mathrm{m}$, $V = 50\,\mathrm{m/s}$, sea level ($\rho = 1.225\,\mathrm{kg/m^3}$, $\mu = 1.789 \cdot 10^{-5}\,\mathrm{Pa\,s}$):

$$Re = \frac{1.225 \cdot 50 \cdot 1.5}{1.789 \cdot 10^{-5}} \approx 5.1 \cdot 10^6$$

This is a "high" value: viscous effects are limited to a very thin layer and the flow outside it is almost potential. In a wind-tunnel lab session, with chords of a few centimetres and low speeds, Re is several orders of magnitude smaller, so the boundary layer matters more.

2 Two-dimensional aerodynamics (airfoils)

2.1 Airfoil nomenclature

The wing section is called an **airfoil**; it is essentially a wing of infinite span, and much of the wing behaviour is extrapolated from that of the airfoil. Airfoils usually have low thickness and low camber. Their nomenclature is shown in Figure 5.7.

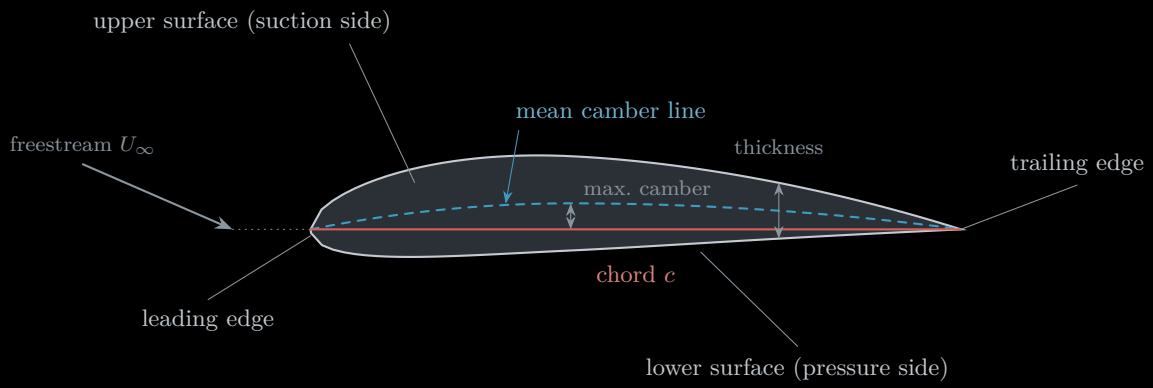


Figure 5.7. Nomenclature of a cambered airfoil (here with exaggerated camber and thickness). Source: Unit 5, slide 17.

- **Leading edge**: front part, also defined by its radius. **Trailing edge**: rear part, normally sharp.
- **Chord**: line joining both edges; its length is also called the chord c.
- **Upper surface** (suction side): top face. **Lower surface** (pressure side): bottom face.
- **Mean camber line**: line equidistant from the upper and lower surfaces. **Maximum camber**: measured relative to the chord (0 = symmetric airfoil); also its position.
- **Thickness**: distance between the upper and lower surfaces (perpendicular to the camber line in the American convention, perpendicular to the chord in the British one). **Maximum thickness** and its position, relative to the chord.

2.2 Airfoil aerodynamic coefficients

The forces and moments on an airfoil change greatly with the conditions (density, velocity…), but once non-dimensionalized they depend almost **only on the airfoil shape and its orientation relative to the airstream**. Lowercase letters are used for the airfoil (per unit span: l, d in N/m, m in N·m/m):

$$c_l = \frac{l}{\frac{1}{2}\rho_\infty U_\infty^2\, c} \qquad c_d = \frac{d}{\frac{1}{2}\rho_\infty U_\infty^2\, c} \qquad c_m = \frac{m}{\frac{1}{2}\rho_\infty U_\infty^2\, c^2}$$

This makes it possible to have airfoil **databases** (airfoils have been tested in wind tunnels for a century).

2.3 NACA airfoils

The NACA (National Advisory Committee for Aeronautics, founded in 1915) tested a great variety of airfoils in wind tunnels and created families of great historical interest (some are still in use). The most important ones here are the **4-digit and 5-digit** series; Figure 5.8 shows some examples.



NACA 0006 symmetric, thin | NACA 0015 symmetric, thick | NACA 2412 camber 2 % | NACA 4412 camber 4 %

Figure 5.8. Some 4-digit NACA airfoils: the first digit is the maximum camber as a % of the chord, the second the position of that camber in tenths of the chord, and the last two the maximum thickness as a %. Source: Unit 5, slides 20 and 21 (own drawings).

Series	Meaning of the digits
4-digit (e.g. 2412)	1st: **maximum camber** as a % of the chord. 2nd: **position** of the maximum camber in tenths of the chord. 3rd and 4th: **maximum thickness** as a % of the chord.
5-digit (e.g. 24012, 24112)	1st $\times 0.15$ = **design lift coefficient**. 2nd $\times 5$ = position of the maximum camber as a % of the chord. 3rd: 0 = normal camber; 1 = reflex camber (with a minimum near the trailing edge). 4th and 5th: **maximum thickness** as a % of the chord.

Examples: NACA 2412 → camber 2 % at 40 % of the chord, thickness 12 %. NACA 0012 → symmetric (camber 0), thickness 12 %. NACA 24012 → design c_l of $2 \cdot 0.15 = 0.3$, maximum camber at 20 % of the chord ($4 \cdot 5$), normal camber, thickness 12 %.

> **2018/19 AND 2021/22 EXAMS: "A NACA 21518 AIRFOIL"**
> The options were: (a) $c_{l,max} = 0.3$; (b) maximum thickness of 15 %; (c) maximum camber at 0.15 of the chord; (d) none is correct. With the 5-digit rule: 1st digit 2 → *design* $c_l = 0.3$ (not the maximum: it is the c_l for which the camber is optimized); 2nd digit 1 → maximum camber at $1 \cdot 5 = 5$ % of the chord (not 15 %); last two digits, 18 → thickness of 18 % (not 15 %). Answer: (d). (Note: the middle digit, 5, is neither 0 nor 1 as the slides' rule requires; it is a deliberately tricky question.)

2.4 Lift generation

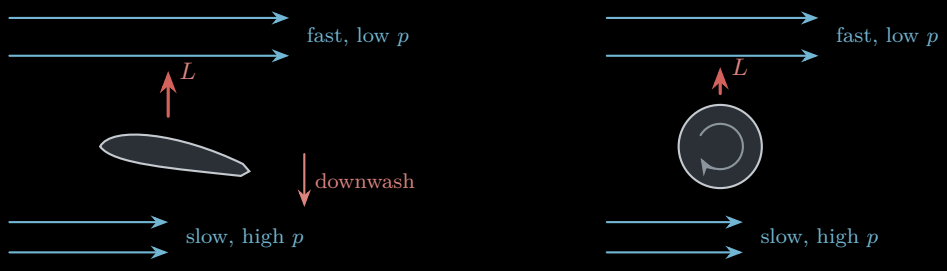
An airstream reaching the airfoil exerts an upward force on it; the airfoil exerts a force **downwards on the air** (action and reaction: the wing deflects the air downwards, *downwash*). A **circulation** Γ appears in the fluid around the airfoil (this is what the potential-flow panel method computes). It is observed (and can be proven from first principles) that the air **accelerates over the upper surface, where its pressure drops**, and decelerates under the lower surface, where it rises.

Beware of causality: the velocity changes because the pressure changes, but the pressure distribution appears because there is relative motion between the air and the airfoil. For this to happen, the airfoil must have **camber** or **a certain angle of attack** α relative to the airstream. *An everyday example is the Magnus effect*: a spinning cylinder or ball drags the air along and creates circulation; that is why balls with spin curve (Figure 5.9).

Subsonic: lift

In purely potential flow it can be shown that the only force is lift (the suction force at the leading edge cancels the drag): this is **d'Alembert's paradox**, because real airfoils do have drag (due to viscosity). **Most of the lift is generated by the upper surface, near the leading edge**: the suction peak, as the pressure distribution in Figure 5.10 shows.

The c_l–α relation is **linear** (found experimentally and provable):



Airfoil
circulation Γ appears and
the air is deflected downwards

Magnus effect
a spinning cylinder drags the air
along: circulation, and so lift

Figure 5.9. Circulation around an airfoil and around a spinning cylinder: the air goes faster (lower pressure) on one side and slower (higher pressure) on the other.

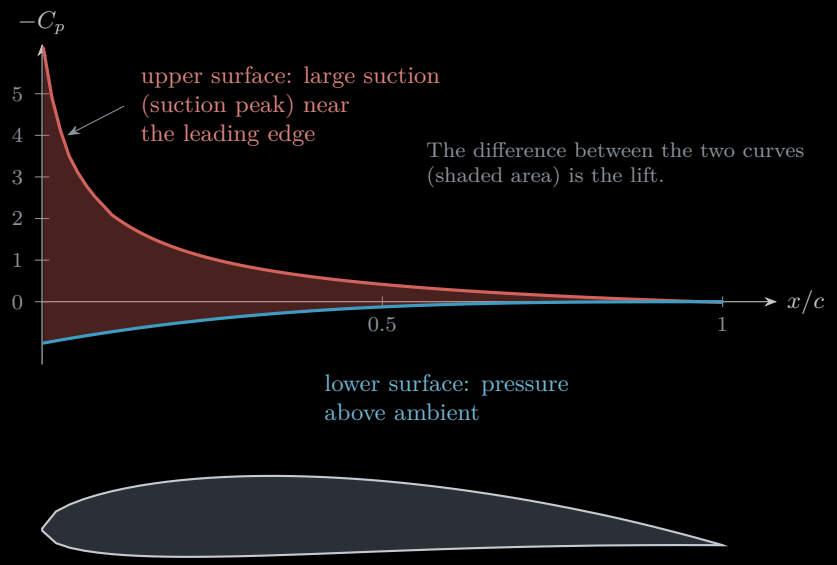


Figure 5.10. Schematic pressure distribution ($-C_p$): large suction on the upper surface near the leading edge; the lower surface contributes less. It is an illustrative curve, not a measured one. Source: Unit 5, slide 26.

$$c_l = c_{l,0} + c_{l,\alpha}\,\alpha \qquad c_{l,\alpha} \approx 2\pi \text{ (with } \alpha \text{ in radians)} \approx 0.11 \text{ per degree}$$

$c_{l,0}$ is the coefficient at zero angle of attack. It is **0 for symmetric airfoils and > 0 for airfoils with positive camber**. For a cambered airfoil, there is a negative angle of attack (the *zero-lift angle*) where $c_l = 0$ (Figure 5.11).

Stall. Downstream of the suction peak the pressure rises again: an **adverse pressure gradient** appears on the upper surface (the air moves towards higher pressure and slows down) and the boundary layer may **separate**; where there is separation, the upper surface stops "working". If the separation is massive, there is a **stall**; the **critical angle of attack** is the stall angle and gives $c_{l,max}$.

- It tends to be **higher for thick airfoils**; airfoils with **large camber** usually have a higher $c_{l,max}$.
- The **higher** Re **is, the later the flow separates** (higher $c_{l,max}$).

Among the NACA airfoils of one exam (0006, 0010, 2410, 2414) the one with the highest $c_{l,max}$ is the **2414**: it has camber *and* it is the thickest of the cambered ones.

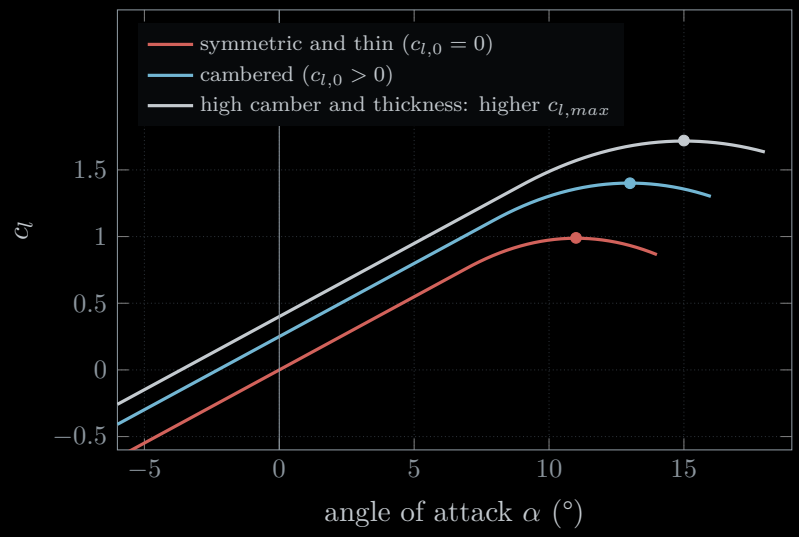


Figure 5.11. Lift curve of three airfoil types: same slope for all, different $c_{l,0}$ and different $c_{l,max}$; the dots mark the stall. Qualitative sketch inspired by slide 28.

Subsonic: drag

- **Skin friction drag**: due to viscosity (shear stresses in the boundary layer).
- **Form drag** (pressure drag): in the separated-flow region a wake forms; the larger it is, the more drag. At low angles of attack it is smaller than the friction drag.
- Together they make up the **parasitic drag**: $c_{d,par}$ = friction + form. It decreases as Re rises and is larger with a turbulent boundary layer than with a laminar one. For constant Re it is **very constant up to large angles of attack** (Figure 5.12).

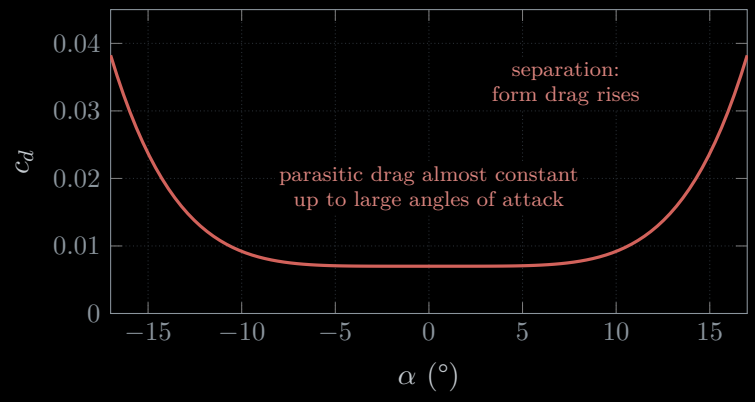


Figure 5.12. Airfoil drag coefficient versus α: almost flat and then it shoots up when the flow separates. Qualitative sketch (slide 31).

Subsonic: pitching moment

The aerodynamic forces are applied at a point, the **centre of pressure** (CP). There is another point, the **aerodynamic centre** (AC), about which the moment coefficient **does not change when α varies**. Once the AC and the moment about it are known, it is trivial to find the moment about any other point. The AC is **approximately at 1/4 of the chord** and **coincides with the CP in symmetric airfoils**.

- **Symmetric**: $c_{m,ac} = 0$ (far from stall).
- **Positive camber**: $c_{m,ac} < 0$, a moment that makes the nose **pitch down** (nose-down).

With $c_{m,ac} < 0$, the CP position is $x_{cp}/c \approx \frac{1}{4} - c_{m,ac}/c_l$ (supplement; it comes from taking moments). Answering the questions on slide 33 (Figure 5.13 plots $c_{m,ac}$ and x_{cp}/c against c_l):

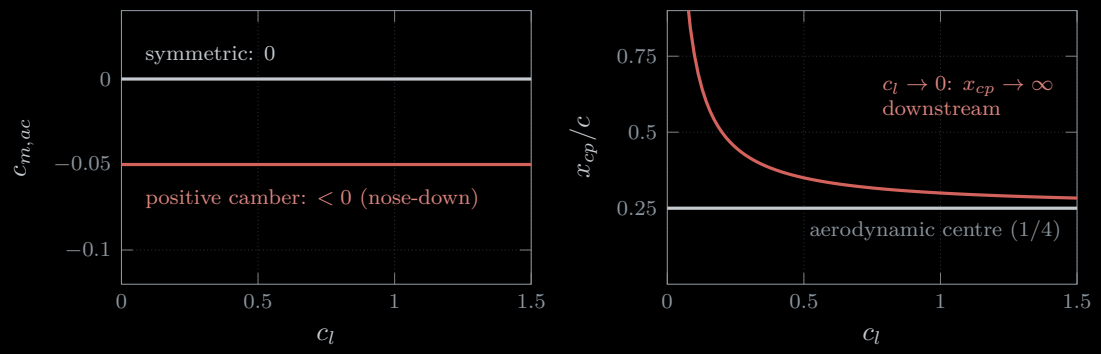


Figure 5.13. Moment about the aerodynamic centre and centre-of-pressure position (the value $c_{m,ac} = -0.05$ is illustrative). Source: Unit 5, slides 32 and 33.

- *At the zero-lift α of a cambered airfoil:* $c_l = 0$ but there is a moment, so the force is zero and the moment is not: the CP is **at infinity downstream**.
- *When α increases from the zero-lift value:* the CP **comes from infinity downstream towards the aerodynamic centre** (and approaches it as c_l grows). This is the question that appears in the tests (option "from infinity downstream towards the aerodynamic centre").
- *A 2D airfoil in level flight without drag:* moment equilibrium requires (own derivation, the slide only poses the question) the CP to coincide with the point of application of the weight.

Common mistake: getting the direction wrong. In airfoils with positive camber the CP does *not* move away from the AC towards the front as α increases; it approaches it *from behind*.

2.5 Airfoils in the transonic regime

The flow accelerates over the upper surface, so sonic conditions are reached at some point even when the flight is at $M < 1$. Two important Mach numbers:

- **Critical Mach number** M_{crit}: the flight Mach number at which $M = 1$ is reached *locally* at some point of the airfoil.
- **Drag divergence Mach number** M_{DD}: above the critical Mach number, shock waves appear on the upper surface; when they interact with the boundary layer they may separate it and the drag **grows a lot** from M_{DD}. This is the origin of the so-called "sound barrier". Once the transonic regime is passed, the drag falls again (Figure 5.14 shows the drag rise).

Always $M_{crit} < M_{DD}$. Aircraft usually fly between the two (there are shock waves but weak ones, still without massive separation); it is not true that they cannot exceed M_{crit}. The presence of shocks and separation also strongly affects the pitching moment: **controlling transonic flight is difficult**. Both problems are mitigated and delayed to higher Mach numbers with **supercritical airfoils** (Figure 5.15). **Their main goal is to increase** M_{DD} (not M_{crit}). In addition, a *thinner* airfoil delays compressibility effects (that is why a NACA 2412 is preferable to a 2416 in transonic flight).

2.6 Airfoils in supersonic and hypersonic flow

- **Supersonic, lift**: the pressure force is roughly normal to the *mean line* (in subsonic flow it was almost normal to the airstream). In a symmetric airfoil, the upper and lower surfaces contribute equally. $c_{l,max}$ decreases.

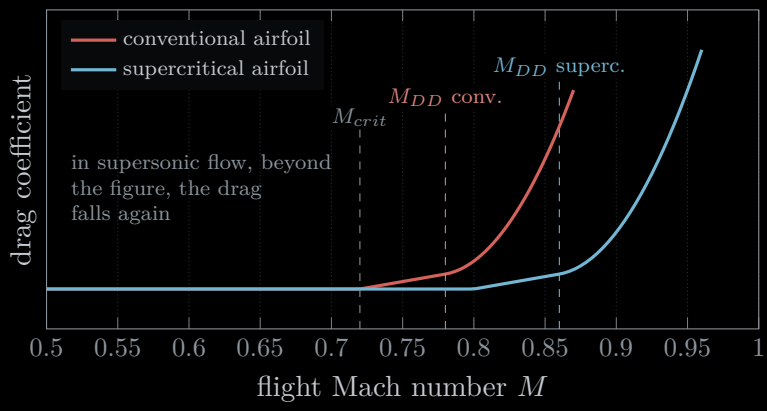


Figure 5.14. Drag rise in the transonic regime and effect of the supercritical airfoil. Qualitative sketch (slides 35 and 36).

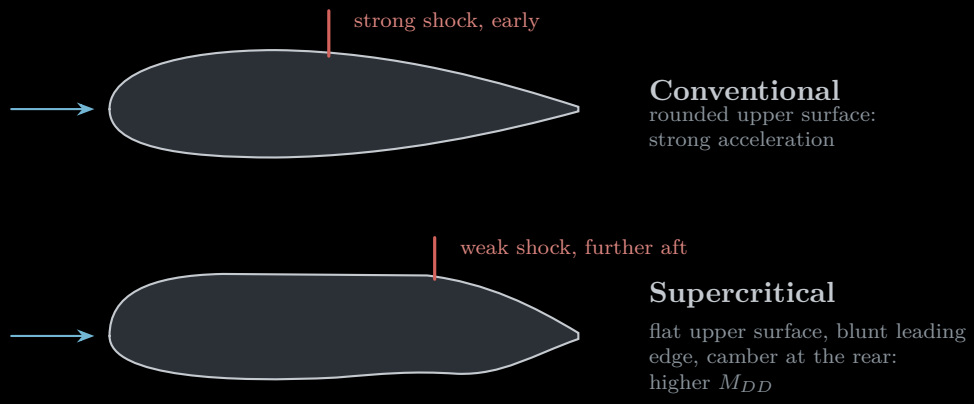
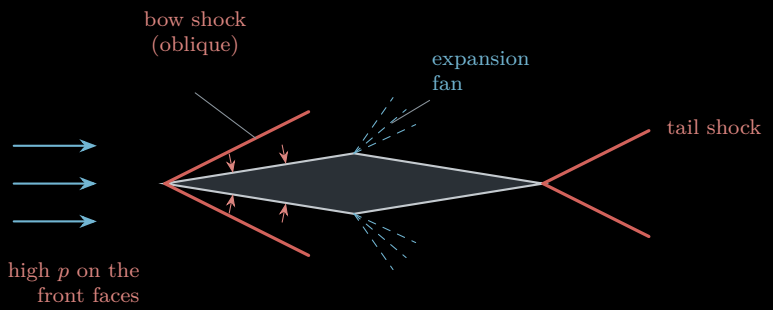


Figure 5.15. Conventional versus supercritical airfoil (schematic, own drawing): the flatter upper surface makes the shock weaker and moves it aft.

- **Supersonic, drag**: there is no d'Alembert paradox here: drag appears even in potential flow, the **wave drag** (form drag and drag due to lift; it is a potential drag). Even at zero lift there is an increase due to the shock waves. The more lift, the more drag. **A thinner airfoil reduces the form wave drag.** The origin of the wave drag is sketched in Figure 5.16.
- **Hypersonic**: the upper surface stops generating lift ("surface in shade"); c_l stops being linear with α and is very low; the drag is very high.



supersonic flow over a diamond airfoil: the pressure on the front faces is higher than on the rear faces, so there is *wave drag* even without viscosity (no d'Alembert paradox)

Figure 5.16. Supersonic flow over a diamond airfoil (schematic, own drawing).

2.7 Aerodynamic efficiency

$$AE = \frac{c_l}{c_d} \quad \text{(airfoil)} \qquad\qquad E = \frac{L}{D} = \frac{C_L}{C_D} \quad \text{(aircraft)}$$

It is the lift-to-drag ratio. If it is high, **less thrust** is needed to fly (because $T = D = W/E$ in level flight). Figure 5.17 shows how it varies with α for an airfoil.

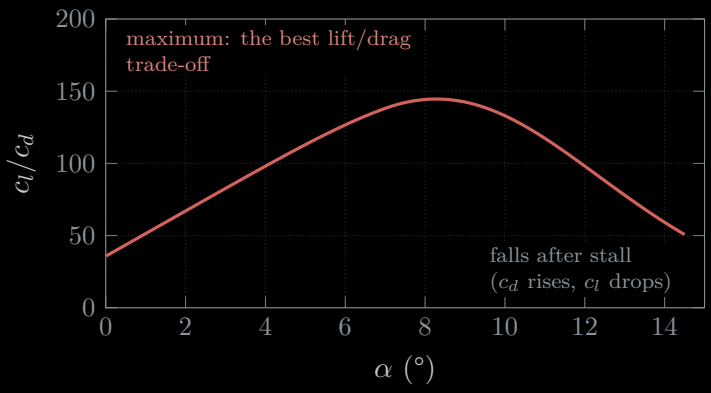


Figure 5.17. Shape of c_l/c_d versus α for a subsonic airfoil: maximum at a moderate angle, falling at stall. Qualitative sketch (slide 40).

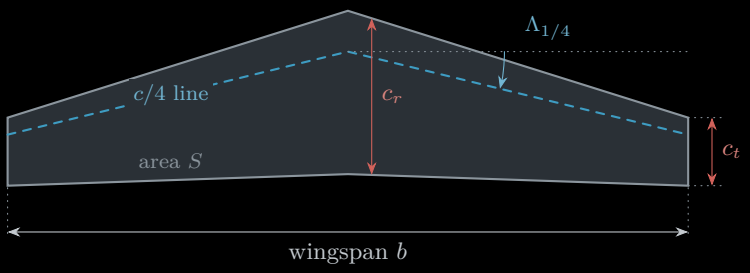
Regime	Airfoil (2D)	Aircraft/wing (3D)
Subsonic	very high, 50–100 or more	10–20 (gliders: up to 70)
Supersonic	≈ 10	≈ 10
Hypersonic	0.5–4	0.5–3

Exam value: the cruise aerodynamic efficiency of a subsonic jet airliner is **between 10 and 20** (option "20" in the test as a typical value). It is always lower in 3D than in 2D because induced drag appears.

3 Three-dimensional aerodynamics (the complete wing)

3.1 Wing geometric parameters

Only more or less conventional wings are studied. A wing is described by its position (low, mid or high) and by its **planform**, whose parameters are defined in Figure 5.18 and in the table below.



$$AR = \frac{b^2}{S} \quad \text{(aspect ratio)}$$

$$\lambda = \frac{c_t}{c_r} \quad \text{(taper ratio)}$$

$$\bar{c} = \frac{S}{b} \quad \text{(mean geometric chord)}$$

Figure 5.18. Planform parameters. Source: Unit 5, slide 43.

Parameter	Definition
Wing area S	Planform area of the wing (reference area for C_L and C_D).
Wingspan b	Distance from tip to tip.
Root chord c_r, tip chord c_t	Chords at the junction with the fuselage and at the tip.
Taper ratio λ	$\lambda = c_t/c_r$.
Aspect ratio AR	$AR = b^2/S$. Long, narrow wing = high aspect ratio.
Sweep angle $\Lambda_{1/4}$	Angle of the quarter-chord line relative to the perpendicular to the aircraft axis.
Dihedral	Inclination of the half-wings relative to the horizontal, seen from the front.
Mean geometric chord $\bar{c}$	$\bar{c} = S/b$: planform area divided by the span.
Mean aerodynamic chord (MAC)	Equivalent chord of the whole wing in aerodynamic terms (the slides refer to a separate document for computing it). **It is not the same as the geometric mean.**

Wing twist: variation of the incidence angle of the local chord along the span (**geometric twist**) or of the airfoil shape (**aerodynamic twist**) to modify the lift distribution. The usual goal is to **increase the margin to the critical angle of attack at the wingtip** so that the stall does not start there. Some wings are not trapezoidal; for example the **elliptic wing** (like the Spitfire's).

QUICK CALCULATION OF PLANFORM PARAMETERS

Trapezoidal wing of root chord $c_r = 2\,\text{m}$, tip chord $c_t = 1\,\text{m}$ and span $b = 10\,\text{m}$:

$$S = b\,\frac{c_r + c_t}{2} = 10 \cdot 1.5 = 15\,\text{m}^2 \qquad \bar{c} = \frac{S}{b} = 1.5\,\text{m}$$

$$AR = \frac{b^2}{S} = \frac{100}{15} = 6.67 \qquad \lambda = \frac{c_t}{c_r} = \frac{1}{2} = 0.5$$

(Supplement: in a trapezoid the mean aerodynamic chord is $\text{MAC} = \frac{2}{3} c_r \frac{1+\lambda+\lambda^2}{1+\lambda} = 1.56\,\text{m}$, slightly larger than $\bar{c}$.)

3.2 Subsonic: wingtip vorticity and 3D lift

The pressure difference between the lower surface (high) and the upper surface (low) favours **air recirculating around the wingtip**, from the lower to the upper surface. **Vorticity** is generated (wingtip vortices) which induces a change in the velocity field (*downwash*) in their vicinity, as Figure 5.19 shows.

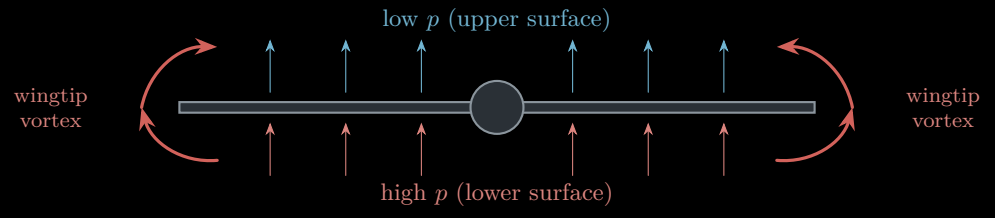


Figure 5.19. Origin of the wingtip vortices. Source: Unit 5, slide 46.

For the complete wing:

$$C_L = \frac{L}{\frac{1}{2}\rho_\infty U_\infty^2 S} \qquad C_L = C_{L,0} + C_{L,\alpha}\,\alpha \qquad C_{L,\alpha} = \frac{c_{l,\alpha}}{1 + \frac{c_{l,\alpha}}{\pi\,AR}} \text{ (long wings)}$$

Everything is the same as in 2D (linear relation with α), but:

- The **slope is lower** than that of the airfoil, and the lower AR is, the lower the slope (Figure 5.20).
- The **stall usually occurs at lower angles of attack** than for the airfoil.

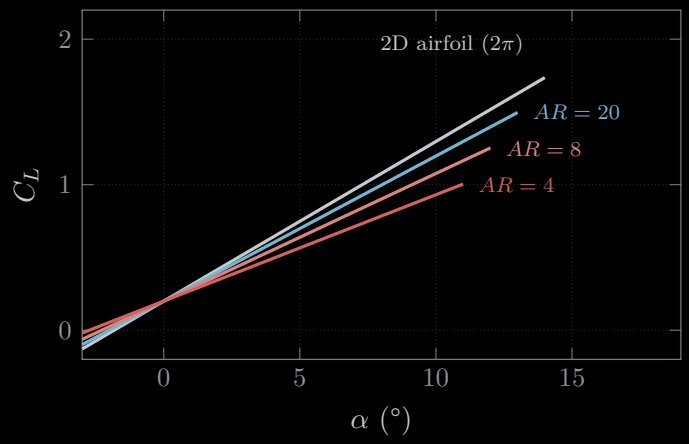


Figure 5.20. Slope of the wing C_L for several aspect ratios, with $c_{l,\alpha} = 2\pi$ (sketch; same $C_{L,0}$ for comparison). With $AR = 8$: $C_{L,\alpha} = 2\pi/(1 + 2/8) = 5.03$ rad^{-1} $= 0.088$ °$^{-1}$.

3.3 Subsonic: 3D drag

$$C_D = \frac{D}{\frac{1}{2}\rho_\infty U_\infty^2 S} \qquad C_D = C_{D,0} + C_{D,i}$$

There is **parasitic drag** $C_{D,0}$ (similar to the 2D one: friction + form) and **induced drag** $C_{D,i}$ is added, due to the vortices thrown into the wake. There is also **interference drag** between elements (wing–fuselage, etc.), which is normally added to the parasitic drag.

Parasitic drag estimation: the flat-plate analogy

The $C_{D,0}$ of the complete aircraft is estimated with the **flat-plate method**:

1. The friction coefficient C_f of a flat plate with the **same wetted area** S_{wet} (area in contact with the air) is estimated.
2. It is corrected for **3D effects** with form factors FF (from 1 to 1.35).
3. It is corrected for the **interference** between elements with interference factors FI.
4. It is "translated" into a drag coefficient by changing the reference area ($S_{wet} \to S$).

1) Flat-plate friction coefficient.

$$\text{Laminar (Blasius): } C_f = \frac{1.328}{\sqrt{Re}}$$

$$\text{Turbulent: } C_f = \frac{0.445}{(\log_{10} Re)^{2.58}\,(1 + 0.144M^2)^{0.65}}$$

(The slide writes 0.445 in the numerator; in the classical correlation from the design literature 0.455 usually appears. The slide's value is used; the difference is 2 %.) C_f **decreases as** Re **increases**

(Figure 5.21). In a turbulent layer (only there) there is a point beyond which it stops decreasing: if $Re > Re_{cutoff}$ the coefficient stays constant, and this minimum **depends on the surface finish**:

$$\text{Subsonic: } Re_{cutoff} = 38.25 \left(\frac{c}{k}\right)^{1.053}$$

$$\text{Transonic: } Re_{cutoff} = 44.62 \left(\frac{c}{k}\right)^{1.053} M^{1.16}$$

with k the equivalent surface roughness. Slide 51 gives (in feet) $k \approx 3.3 \cdot 10^{-5}$ for camouflage paint on aluminium, $2.1 \cdot 10^{-5}$ smooth paint, $1.3 \cdot 10^{-5}$ production sheet metal, $0.5 \cdot 10^{-5}$ polished sheet metal and $0.17 \cdot 10^{-5}$ smooth moulded composite; and low-Mach C_f ranges of 0.008–0.010 (propeller, fixed landing gear), 0.0045–0.007 (propeller, retractable landing gear), 0.0035–0.0045 (jet, nacelle-mounted engines) and 0.0030–0.0035 (jet, internal engines) (read from the scanned table on the slide; check them if precision is needed).

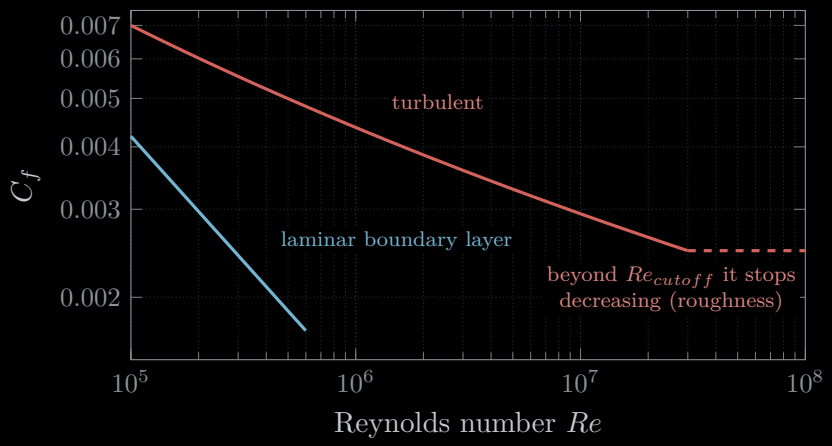


Figure 5.21. C_f of a flat plate versus Re: the laminar value is lower than the turbulent one at equal Re; in the turbulent case a plateau due to roughness is reached (the value of Re_{cutoff} here is illustrative). Source: Unit 5, slides 51 and 52.

2) Form factors.

$$FF_{fus} = \left(1 + \frac{60}{f^3} + \frac{f}{400}\right), \quad f = \frac{L}{D_{eq,max}} = \frac{L}{\sqrt{\frac{4}{\pi} A_{max}}}$$

$$FF_{wing,stab.} = \left[1 + \frac{0.6}{(x/c)_m}\frac{t}{c} + 100\left(\frac{t}{c}\right)^4\right]\left[1.34\, M^{0.18} (\cos \Lambda_m)^{0.28}\right]$$

$$FF_{engine,ext.} = 1 + \frac{0.35}{f}$$

(fuselage and cabin; wing, horizontal and vertical stabilizer, where m indicates the position of maximum thickness; engines and external elements.) **There is no need to memorize them**: what matters is to understand that they increase the flat-plate friction the thicker the element is.

3 and 4) Sum over all elements.

$$C_{D,0} = \frac{\sum_i S_{wet,i}\, C_{f,i}\, FF_i\, FI_i}{S}$$

Other additional terms may be added (protuberances, roughness, etc.).

Slide 56 also has a bar chart with the **typical cruise drag breakdown of a transport jet**: flat-plate friction (of the order of 45 %, values estimated from the chart) and induced drag (around 30 %)

dominate; the rest (profile, roughness, excrescences, interference, 3D effects and compressibility) are small contributions of a few per cent each.

ILLUSTRATIVE EXAMPLE (MADE-UP DATA FOR PRACTICE): $C_{D,0}$ OF WING + FUSELAGE

Light aircraft at $V = 60\,\text{m/s}$ at sea level ($M = 0.176$) with $S = 16\,\text{m}^2$. Wing and fuselage only, with $FI = 1$.

- **Wing**: $c = 1.6\,\text{m}$, $S_{wet} \approx 2.04S = 32.6\,\text{m}^2$, $t/c = 0.12$, $(x/c)_m = 0.3$, $\Lambda_m = 0$. $Re = \dfrac{1.225 \cdot 60 \cdot 1.6}{1.789 \cdot 10^{-5}} = 6.57 \cdot 10^6$; $C_f = \dfrac{0.445}{(6.82)^{2.58}(1 + 0.144 \cdot 0.176^2)^{0.65}} = 3.14 \cdot 10^{-3}$. $FF = [1 + 0.6/0.3 \cdot 0.12 + 100 \cdot 0.12^4] \cdot [1.34 \cdot 0.176^{0.18}] = 1.261 \cdot 0.980 = 1.236$.
- **Fuselage**: $L = 8\,\text{m}$, $A_{max} = 1.1\,\text{m}^2 \Rightarrow D_{eq} = \sqrt{4A_{max}/\pi} = 1.18\,\text{m}$, $f = 6.76$, $S_{wet} = 20\,\text{m}^2$. $Re = 3.29 \cdot 10^7$, $C_f = 2.44 \cdot 10^{-3}$, $FF = 1 + 60/6.76^3 + 6.76/400 = 1.211$.

$$C_{D,0} = \frac{32.6 \cdot 3.14 \cdot 10^{-3} \cdot 1.236 + 20 \cdot 2.44 \cdot 10^{-3} \cdot 1.211}{16} = 0.0079 + 0.0037 \approx 0.0116$$

Tail surfaces, landing gear, engine, interference… would still be missing: the real value would be considerably higher (up to ∼0.02–0.03 for a light aircraft). Friction decreases slightly with Re and the wing dominates because of its large wetted area.

Induced drag

It has been observed (and can be proven) that:

$$C_{D,i} = \frac{C_L^2}{\pi\, AR\, e} = \frac{C_L^2}{\pi\, AR}(1 + \delta)$$

where e is the **Oswald factor**: $e = 1$ for the elliptic wing (the optimal, reference case) and $e < 1$ otherwise. Consequences:

- **How can it be reduced?** By increasing the aspect ratio AR, by approaching an elliptic distribution (larger e), and with wingtip devices (they increase e).
- **It varies with C_L^2**: in cruise at higher speed less C_L is needed (because $L = W$ fixes $C_L \propto 1/V^2$), so $C_{D,i} \propto 1/V^4$, and the force $D_i \propto 1/V^2$ (the red curve in Figure 5.22). That is why it decreases as speed increases.
- The **parasitic** $C_{D,0}$ is more or less **constant with speed** (as a coefficient); the force $D_0 \propto V^2$.
- **The sum** (total drag) has a minimum: the **minimum-drag speed** V_{md}. At lower speeds the induced drag dominates; at higher speeds, the parasitic drag.

Common mistake (exam): wingtip vorticity does *not* improve efficiency; and with $e = 1$ the induced drag does *not* disappear (it is just the minimum possible for that AR and C_L).

The parabolic drag polar

The relation between C_L and C_D is called the **drag polar**. With the above, it is a parabola:

$$C_D = C_{D,0} + k\, C_L^2, \qquad k = \frac{1}{\pi\, AR\, e}$$

$$\text{Maximum efficiency when } C_{D,0} = kC_L^2: \qquad C_L^* = \sqrt{\frac{C_{D,0}}{k}}, \qquad \left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{k\, C_{D,0}}}$$

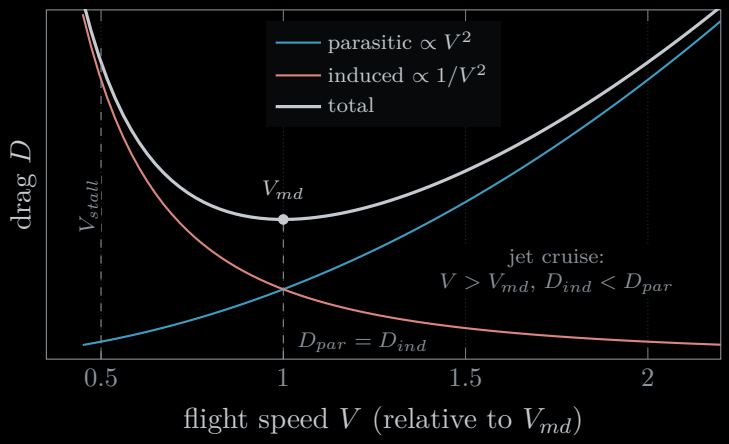


Figure 5.22. Parasitic ($\propto V^2$) and induced ($\propto 1/V^2$) drag versus speed: the sum has a minimum at V_{md} (minimum-drag speed), where both are equal. Source: Unit 5, slide 56.

(The last two, which give the tangency point of a line through the origin, are a supplement not developed in the slides, but they justify the exam question on the minimum-drag speed.) Figure 5.23 shows the polar of the example below.

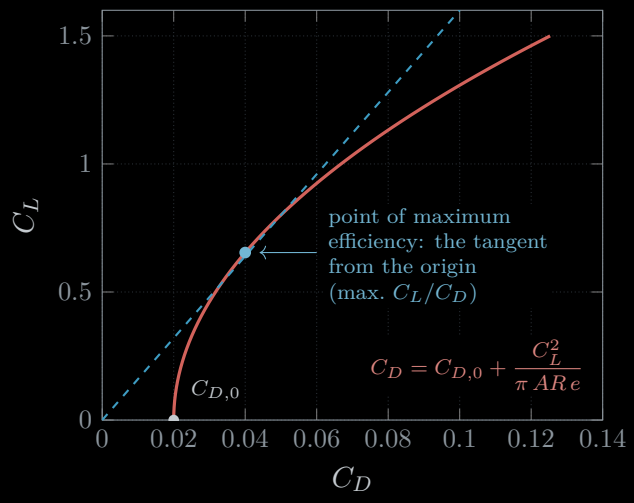


Figure 5.23. Parabolic C_L–C_D polar of the example ($C_{D,0} = 0.02$, $AR = 8$, $e = 0.85$); the tangent from the origin marks the maximum efficiency. Source: Unit 5, slide 57 (own sketch).

MAXIMUM EFFICIENCY AND MINIMUM-DRAG SPEED

Wing with $C_{D,0} = 0.02$, $AR = 8$, $e = 0.85$, $S = 16\,\mathrm{m}^2$, $m = 1000\,\mathrm{kg}$ ($W = 9810\,\mathrm{N}$):

$$k = \frac{1}{\pi \cdot 8 \cdot 0.85} = 0.0468 \qquad C_L^* = \sqrt{\frac{0.02}{0.0468}} = 0.654 \qquad \left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{0.0468 \cdot 0.02}} = 16.3$$

The corresponding speed comes from $L = W$:

$$V_{md} = \sqrt{\frac{2W}{\rho S C_L^*}} = \sqrt{\frac{2 \cdot 9810}{1.225 \cdot 16 \cdot 0.654}} = 39.1\,\mathrm{m/s} \approx 141\,\mathrm{km/h} \text{ (sea level)}$$

At 5 000 m ($\rho = 0.736$): $V_{md} = 50.5\,\mathrm{m/s} \approx 182\,\mathrm{km/h}$. The **maximum efficiency does not change with altitude**, but the speed at which it is achieved **does increase** (approximately $\propto 1/\sqrt{\rho}$), and also with $\sqrt{W/S}$. That is why in cruise at higher altitude one flies faster for the same C_L (a statement that appears in a test).

Stall on the wing

The 3D stall depends on the planform (Figure 5.24).

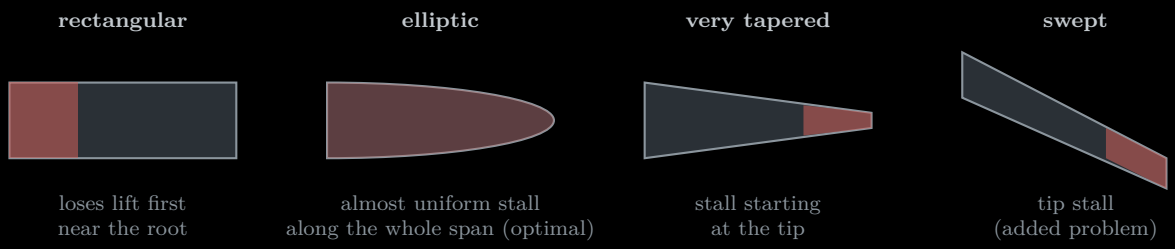


Figure 5.24. Where the stall starts depending on the planform (shaded region). Qualitative sketch based on slides 58 and 67; for the rectangular wing, the root is what the tests give.

- **Rectangular wing**: the stall starts near the **root**.
- **Elliptic wing**: uniform lift distribution; separation appears almost simultaneously along the whole span.
- **Very low taper ratio (delta type)** and swept wings: the stall starts at the **tip**, which is worse (roll control is lost).

Twist is used to avoid tip stall: with **geometric twist** the zero-lift angle of each airfoil is modified (the tip at a lower incidence, *washout*, right panel of Figure 5.26); with **aerodynamic twist** the airfoil shape (e.g. its camber) is changed along the span. A test asks which is which: *aerodynamic = changes the airfoil; geometric = changes the angle.*

3.4 Transonic: sweep delays M_{DD}

With a **swept wing** the airstream is decomposed (Figure 5.25) into a component **normal** to the leading edge, $V\cos\Lambda$, and a **tangential** one, $V\sin\Lambda$. Only the normal one accelerates the flow over the airfoil and causes compressibility effects: the wing "feels" a lower Mach number $M\cos\Lambda$. Therefore the velocity normal to a transonic shock wave is lower than the incident one and the M_{DD} **is delayed**.

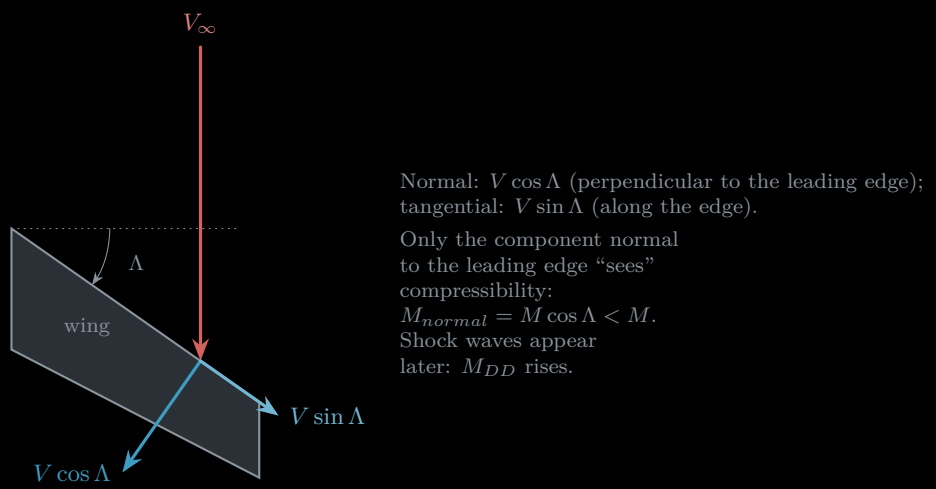


Figure 5.25. Decomposition of the velocity on a swept wing. Source: Unit 5, slides 59 and 60.

The rule $M_n = M\cos\Lambda$ is the standard model of "simple sweep theory" (slide 60 only gives sweep angles and curves, not formulas). If $M = 0.8$ and $\Lambda = 30°$: $M_n = 0.8\cos 30° = 0.69$. Slide 60 shows that as the sweep increases (0°, 10.5°, 40°, 49.25°) the transonic drag peak decreases and shifts to

a higher Mach number. Drawbacks: more structural weight, risk of tip stall, it may destabilize the aircraft and it complicates the landing gear (slide 68).

3.5 Supersonic and 3D efficiency

- **The wingtip recirculation disappears: there is no induced drag**.
- As in 2D, there is **wave drag** (form drag and drag due to lift).
- If the **sweep** is high enough: leading-edge suction appears, and the drag decreases (the photo on the slide is the Concorde).
- 3D efficiency: **lower than that of the airfoil**: subsonic 10–20 (gliders up to 70), supersonic ≈ 10, hypersonic 0.5–3.

3.6 How the wing design parameters influence it

There is no universally optimal wing: **the optimum depends on the mission**.

Parameter	High / large value	Low / small value
Area S	More space for fuel; worse behaviour in **gusts**	Low weight and structural loads; **higher stall speed** (and longer runway)
Aspect ratio AR	**Reduces induced drag**; increases parasitic drag	More fuel in the wing; easier to store the landing gear; better against **flutter** and more manoeuvrability; lower wave drag
Thickness t/c	More stiffness, more fuel volume, **higher** $c_{l,max}$ at low speed	Lower compressibility drag
Taper ratio λ	High ($\approx$ rectangular): easier to manufacture and maintain. Moderate ($\approx$ elliptic): less drag	Low ($\approx$ delta): stall from the tip, **lower structural weight**, easier to store landing gear and fuel, harder to manufacture
Sweep Λ	Raises M_{DD} and lowers drag; generates leading-edge suction in supersonic flow	Structurally simpler; (with high sweep: more weight, tip stall, may destabilize, difficult landing gear)

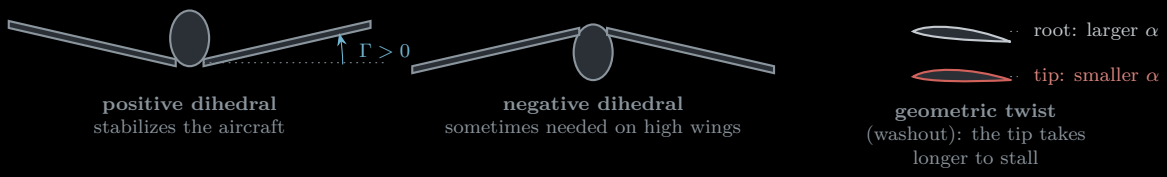


Figure 5.26. Dihedral and geometric twist. Source: Unit 5, slides 63 and 69.

Dihedral (Figure 5.26, left and centre): if **positive**, it tends to **stabilize** the aircraft (roll corrects itself); on high wings **negative dihedral** may be needed. The **typical taper ratio** of a subsonic passenger transport is **moderate** (in one test, the correct option was "around 0.6").

4 Movable surfaces, high-lift devices and other elements

4.1 High-lift devices

The minimum flight speed is linked to the stall:

$$U_{stall} = \sqrt{\frac{W}{\frac{1}{2}\rho_\infty S\, C_{L,max}}} = \sqrt{\frac{2W}{\rho_\infty S\, C_{L,max}}}$$

In practice one flies somewhat above it, as a margin for manoeuvring. To lower U_{stall} one must **raise** $C_{L,max}$; but a wing with an extreme C_L is not optimized for cruise. Solution: **temporarily** modify the wing with **high-lift devices**: **passive** (they just move surfaces) or **active** (they use the engine jet, suck the boundary layer…, with better results and higher cost). Figure 5.27 shows two of the most widely used passive devices.

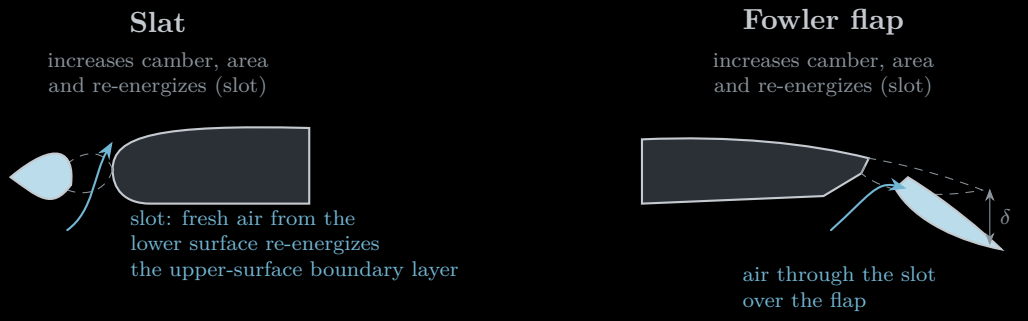


Figure 5.27. Two of the most widely used high-lift devices, schematic. Source: Unit 5, slides 72 to 75.

What each one does (table from slides 72 and 74). The columns are: "increases camber", "increases chord (area)" and "**Slot**" = re-energizes the boundary layer through the slot effect:

Position	Device	Camber	Area	Slot
Leading edge	Slot			X
	Leading-edge flap	X		
	Slat	X	X	X
	Krueger flap	X	X	
Trailing edge	Simple flap	X		
	Simple flap (slotted)	X		X
	Fowler flap (slotted; double or triple slotted)	X	X	X
	Lower-surface flap	X		

The devices not drawn above (slot, leading-edge flap, Krueger, simple, slotted and lower-surface flaps) are shown in Figure 5.28.

Their effect on the lift curve is sketched in Figure 5.29.

- They are used in **take-off and landing**; in landing they are deployed further. They are *not* used in cruise.
- They cost: **weight** and **high manufacturing and maintenance costs**; they also **generate drag** (hence larger deflections are used for landing, where the drag is welcome).
- Improvement of $C_{L,max}$: from **50 %** (simple flaps or slats) to **100 %** (double-slotted Fowler).

Frequent test mistakes: the Krueger flap *increases camber and* $C_{L,max}$; slots carry air from the lower to the upper surface (not the other way round) to delay separation; simple flaps increase camber but *not* the critical angle; the Fowler is a **trailing-edge** device and increases camber *and* area; the lower-surface flap only increases camber, not area.

4.2 Other movable surfaces

Figure 5.30 shows where the control and lifting surfaces are located; the table below gives the function of each.

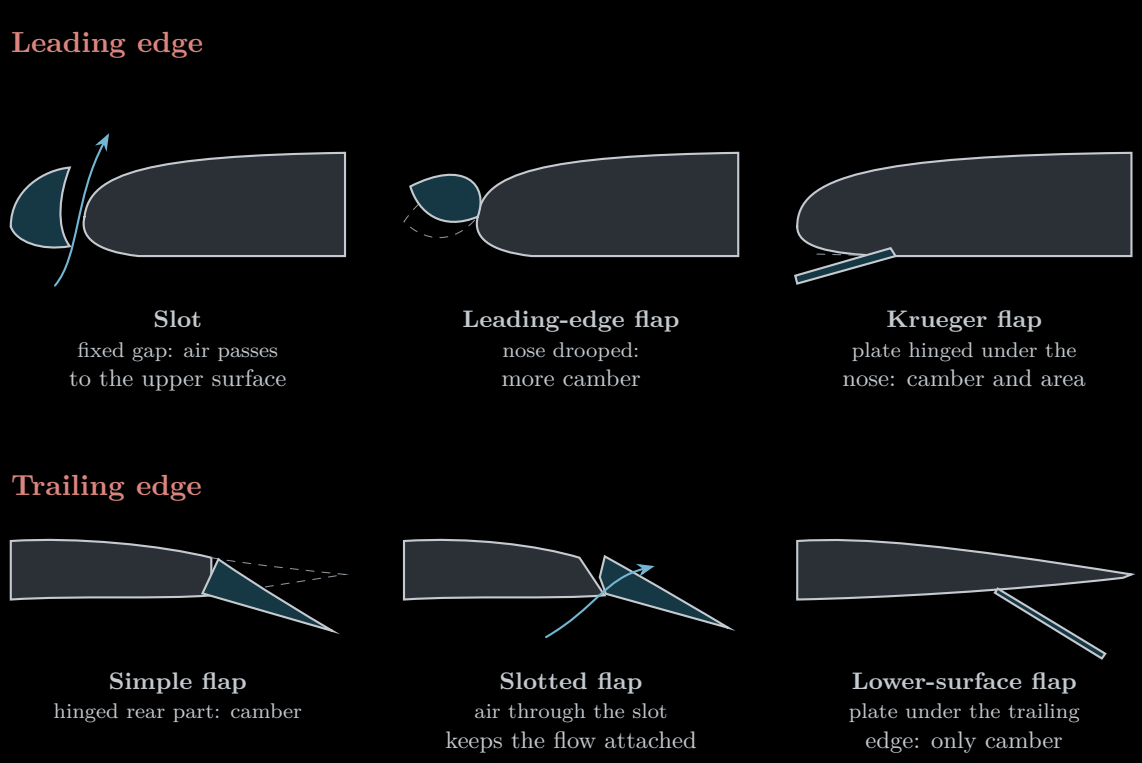


Figure 5.28. The remaining leading-edge and trailing-edge devices of the table (schematic, own drawing; dashed: stowed position).

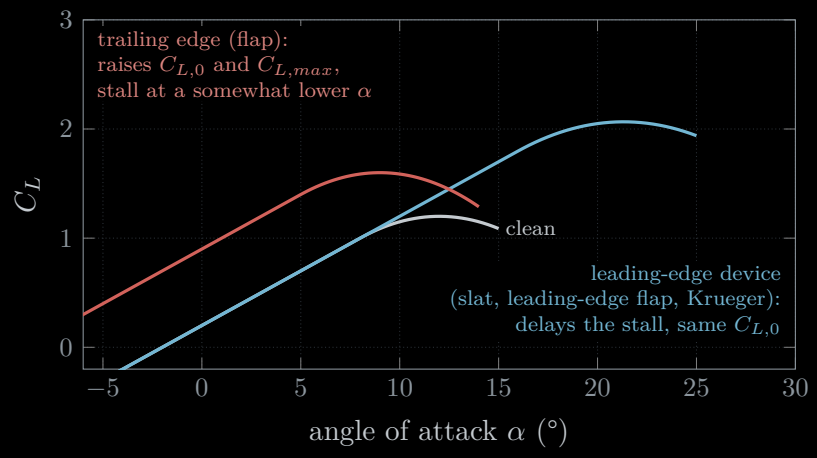


Figure 5.29. Effect on the C_L–α curve (qualitative sketch): leading-edge devices *delay the stall* (extend the α range); trailing-edge devices *raise the curve* and shift the stall to a lower α. Source: Unit 5, slides 73 and 75.

Surface	Function
Ailerons	Control the **roll** moment; close to the tip (at low speed) or close to the root (at high speed). They deflect in opposite directions.
Elevator	Controls the **pitch** moment.
Rudder	Controls the **yaw** moment.
Trim tabs	Set the neutral position of the controls (the stick).
Spoilers	Destroy lift (*dump lift*) and brake; if they act as ailerons they are called *spoilerons*.

Other elements: **wingtip devices** (winglets, Hoerner wingtips, wingtip fences…), which also **increase the Oswald factor** (they reduce induced drag), and **boundary-layer control**: strakes or LEX, dogtooth, wing fences. Wingtip devices are drawn in Figure 5.31 and boundary-layer control elements in Figure 5.32.

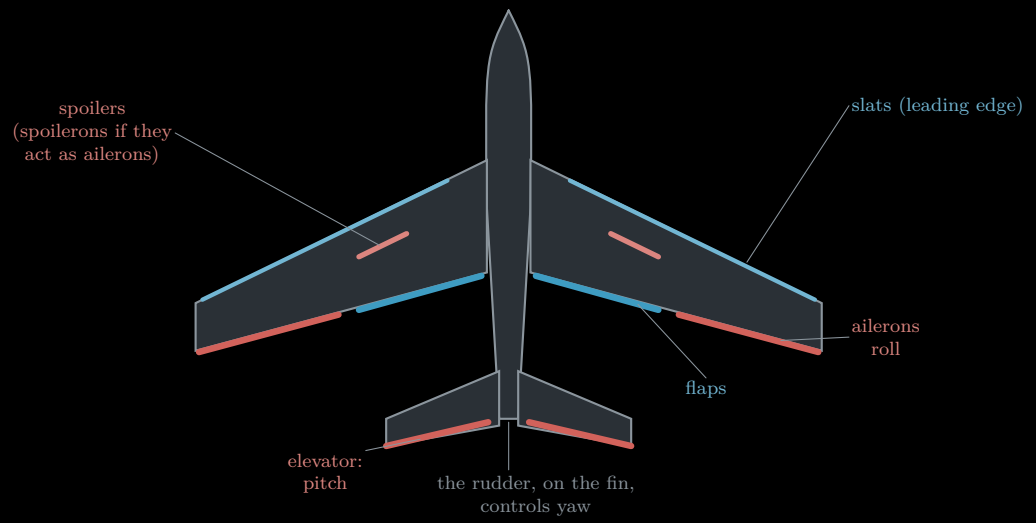


Figure 5.30. Control and lifting surfaces. Source: Unit 5, slides 77 to 79.

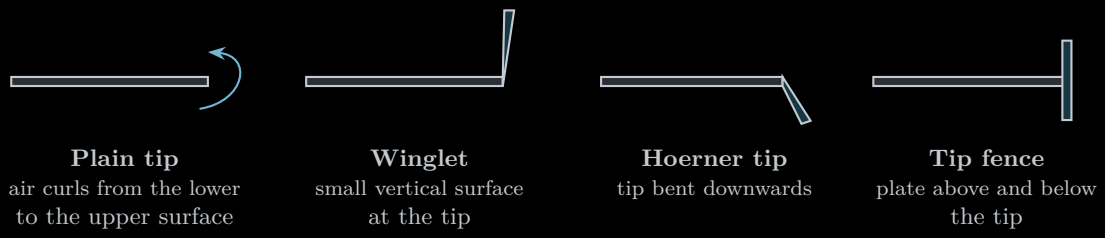


Figure 5.31. Wingtip devices, front view of the tip (schematic, own drawing).

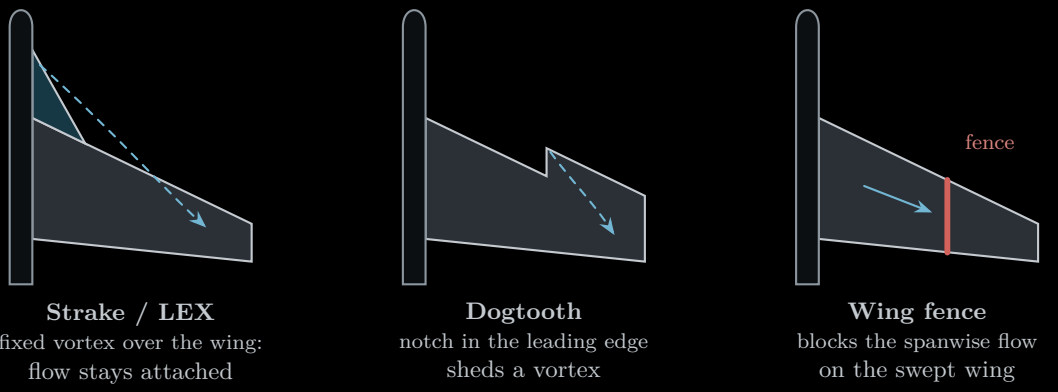


Figure 5.32. Boundary-layer control elements, plan view of half an aircraft (schematic, own drawing).

5 Solved exam-type problems

Past exams are 4-option tests; the "calculations" are quick and are solved with proportions. Here are the recurring ones.

1. STALL SPEED WHEN $C_{L,max}$ IS DOUBLED (2023/24 EXAM, QUESTION 16)
An aircraft has $V_{stall} = 280\,\text{km/h}$ in the clean configuration. What if high-lift devices that **double** $C_{L,max}$ are deployed?

$$V_{stall} = \sqrt{\frac{2W}{\rho S C_{L,max}}} \propto \frac{1}{\sqrt{C_{L,max}}} \quad \Rightarrow \quad V'_{stall} = \frac{280}{\sqrt{2}} = 198\,\text{km/h} \approx 200\,\text{km/h}$$

Frequent mistake: the option "140" comes from dividing by 2 (wrong: the speed is not proportional to $C_{L,max}$ but to its square root).

2. CLIMBING AT CONSTANT SPEED: WHAT DOES α DO? (2023/24 EXAM, QUESTION 4)
Solved in Unit 2 (Section 1.1). Summary: in nearly level flight $C_L = 2W/(\rho V^2 S)$; when climbing, ρ decreases with V, W and S constant, so C_L increases and, since $C_L = C_{L,0} + C_{L,\alpha}\alpha$ grows with α, α **increases during the 2 km** (option b).

3. BETTER FUEL CONSUMPTION BUT WORSE AERODYNAMIC EFFICIENCY (2023/24 EXAM, QUESTION 8)
Solved in Section 6.2. Summary: with R and V fixed, the Breguet equation requires $\ln(W_i/W_f) \propto c/(L/D)$, which increases by a factor $0.97/0.95 = 1.021$ (the consumption c is defined in Unit 6). Slightly more fuel is needed and, with the same payload and empty mass, the **take-off mass increases** (option c); the payload cannot increase because the design mission is fixed.

4. REQUIRED THRUST FROM THE EFFICIENCY
In level cruise $T = D = W/E$. For a 70 t aircraft with $E = 17$: $T = \dfrac{70\,000 \cdot 9.81}{17} \approx 40\,\text{kN}$ in total. If the efficiency rises to 20, $T \approx 34.3\,\text{kN}$: 15 % less thrust, and therefore less fuel. Hence the interest in increasing L/D.

5. CRUISE ABOVE THE MINIMUM-DRAG SPEED (2023/24 EXAM, QUESTION 14)
If the jet flies at $V > V_{md}$, to the right of the minimum of the drag curve (Figure 5.22): the **parasitic** drag ($\propto V^2$) **exceeds** the induced drag ($\propto 1/V^2$). Answer: **the induced drag is lower than the parasitic drag**. At V_{md} they are equal; below it the induced drag dominates.

6. WHERE IS THE CENTRE OF PRESSURE? (QUESTION REPEATED IN 2023/24 AND 2024/25)
Airfoil with positive camber and α increasing from the zero-lift value. $x_{cp}/c = \frac{1}{4} - c_{m,ac}/c_l$ with $c_{m,ac} < 0$: when $c_l \to 0^+$, $x_{cp} \to +\infty$ (downstream); as c_l grows it approaches $c/4$ from behind. Answer: **it moves from infinity downstream towards the aerodynamic centre**.

6 Summary and what usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM (UNIT 5)
This is **the unit that contributes the most questions**: **49 of the 115 questions** in six first midterms (about 8 of each 20) are about aerodynamics and wings, and the data recur almost every year. Everything is multiple-choice; the "problems" are proportionalities.

- **Airfoils**: decoding a NACA code (2412, 0012, 21518…); which has the highest $c_{l,max}$ (cambered and thicker: 2414); 0012 is used for aerobatics and stabilizers (symmetric), 2412 has $c_{l,0} > 0$; **the region that lifts most is the upper surface near the leading edge**; **CP movement** (from infinity downstream to the AC); $c_{l,max}$ of the lab's NACA 0015 ≈ 1 (near 14°); c_l of a symmetric airfoil at 4° ≈ 0.4, not 0.
- **Boundary layer and Re**: viscous effects are more noticeable at *low* Re; layers are thin at large Re; separation depends on the adverse gradient and on viscosity and can be influenced by compressibility; potential flow does not represent drag (d'Alembert).

- **Transonic**: $M_{crit} < M_{DD}$; M_{DD} is reached when the shock separates the boundary layer; supercritical airfoils raise M_{DD}; sweep and low thickness delay the effects.
- **Drag**: induced drag is *only* 3D; $C_{D,i} = C_L^2/(\pi AR\,e)$; at V_{md} parasitic = induced; $V > V_{md} \Rightarrow$ induced smaller; the minimum-drag speed grows with altitude and with $\sqrt{W/S}$.
- **Efficiency**: transport aircraft 10–20 (option "20" as the typical value, not 0.2 nor 200); gliders up to 70; hypersonic < 4.
- **High-lift devices**: table of effects; Krueger; Fowler = trailing edge + camber + area; $V_{stall} \propto 1/\sqrt{C_{L,max}}$ (280 km/h $\rightarrow$ 198); used in take-off and landing.
- **Wing parameters**: aerodynamic twist = varying airfoil; geometric = angle; high aspect ratio $\rightarrow$ less induced but more parasitic drag; low taper ratio = less structural weight and easier to store landing gear and fuel, but tip stall; typical taper ratio on a transport ~ 0.6; positive dihedral stabilizes.
- **Surfaces**: the **aileron controls roll**, the elevator pitch, the rudder yaw.

Essentials: $L = \frac{1}{2}\rho V^2 S C_L$; $c_{l,\alpha} \approx 2\pi$; $C_{L,\alpha} = c_{l,\alpha}/(1+c_{l,\alpha}/\pi AR)$; $C_D = C_{D,0}+C_L^2/(\pi AR\,e)$; $V_{stall} = \sqrt{2W/(\rho S C_{L,max})}$; $AR = b^2/S$; $Re = \rho U c/\mu$; $M = U/a$.

DOUBTS AND OPEN POINTS (UNIT 5)

- Slide 51 writes 0.445 in the turbulent C_f (the literature has 0.455) and the point-source potential with a minus sign; these have been corrected/flagged in the text.
- In one of the students' compilations (2018/19) the official key for "which aircraft has the highest cruise aerodynamic efficiency" marks the B787 over the Orlican VT-16 glider; the slides say gliders reach 70, so **that key is not trusted**.
- The values of k and of the C_f ranges on slide 51 and the percentages in the chart on slide 56 were read from scanned images: the order of magnitude is right, exact figures are not guaranteed.

UNIT 6

Propulsive systems

1 Introduction: what an aircraft engine does and how it is measured

To fly in level flight, the aerodynamic drag must be overcome (Unit 5): that is the job of **thrust**. Propulsive systems fall into two groups (Figure 6.1):

- **Air-breathing** (non-autonomous): **they use the atmosphere** (air is the working mass). These are the engines of aircraft and most drones.
- **Non-air-breathing** (autonomous): they carry **their own reaction mass** (and oxidizer), e.g. missiles and rockets.

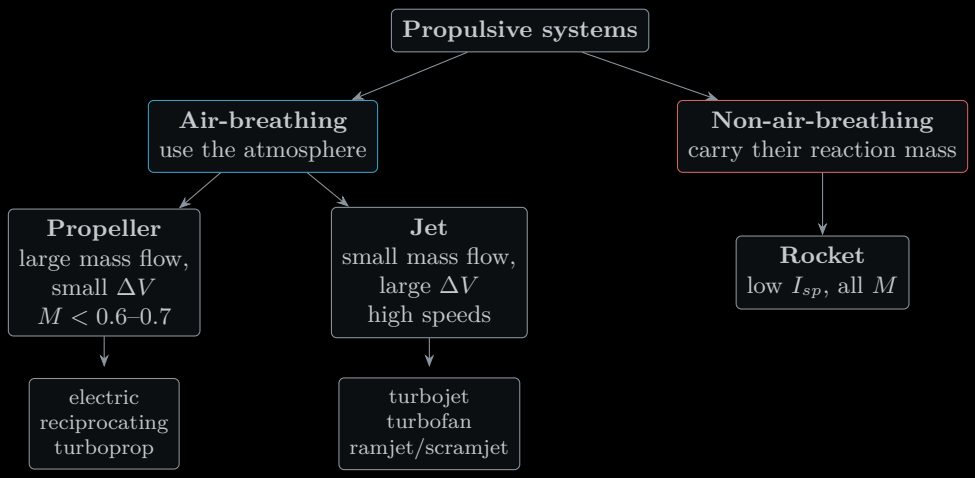


Figure 6.1. Classification of the propulsive systems studied in this unit. Source: T6, slides 3, 7 and 21.

1.1 Operating principle

The principle of air-breathing systems has two steps: (1) the system **increases the kinetic energy and momentum** of a flow of air taken from the atmosphere; (2) as a **reaction**, the atmosphere "pushes" the system (action and reaction, like a boat pushing water backwards). With the mass flow rate $\dot{m}$ (kg/s) through the engine:

$$T = \dot{m}\,V_{out} - \dot{m}\,V_{in} + p\,A_{out} - p\,A_{in} \;\approx\; \dot{m}\,(V_{out} - V_{in}) = \dot{m}\,\Delta V$$

The last two terms are the pressure thrust, which is neglected to a first approximation (it can be important in jets). Hence **the exit velocity must be greater than the inlet velocity**, $V_{out} > V_{in}$. There are two ways to obtain the same thrust (Figure 6.2):

- **a large mass flow** $\dot{m}$ with a **small** velocity increase ΔV (propeller, fan);
- **a small mass flow** with a **large** ΔV (jet).

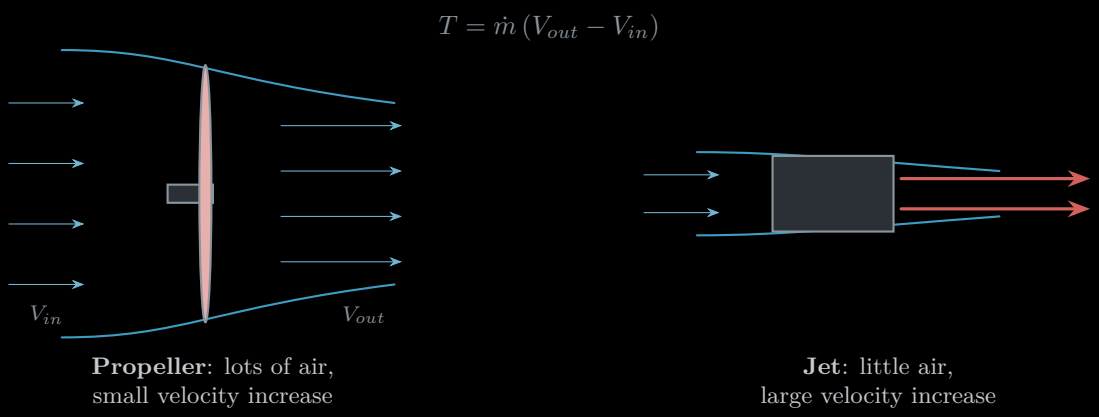


Figure 6.2. The same thrust is obtained by accelerating a lot of air a little (propeller) or a little air a lot (jet). Source: T6, slides 4 and 5.

1.2 Propulsive efficiency

Neglecting the enthalpy change, the **propulsive efficiency** η_p compares the useful power ($T \cdot V_\infty$: what actually pushes the aircraft) with the power given to the air (the increase in kinetic energy of the flow):

$$\eta_p = \frac{T \cdot V_\infty}{\frac{1}{2}\left[(\dot{m}V^2)_{out} - (\dot{m}V^2)_{in}\right]} \simeq \frac{2}{1 + \dfrac{V_{out}}{V_{in}}}$$

If $V_{out} \to V_{in}$, $\eta_p \to 1$; if V_{out} is much greater, $\eta_p \to 0$. Intuition: the kinetic energy left in the wake is energy *lost*: for the same thrust, "**a little velocity to a lot of air**" leaves less energy in the wake than "**a lot of velocity to a little air**". Figure 6.3 plots the formula: the efficiency falls steadily as V_{out}/V_{in} grows.

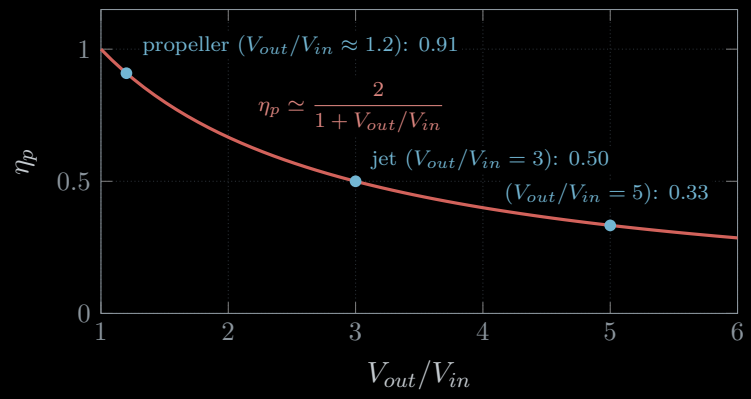


Figure 6.3. Propulsive efficiency versus V_{out}/V_{in} (from the formula on slide 6; the dots are examples). Source: T6, slides 5 and 6.

KEY IDEA

Higher propulsive efficiency = smaller velocity change with a larger mass flow: less kinetic energy is given to the atmosphere for the same thrust. Hence:

- **Propeller**: large mass flow, small ΔV, almost constant pressure between inlet and outlet. Very efficient at low speed (below Mach 0.8 and normally below 0.6), **impossible at high speed**.
- **Jet**: small mass flow, large ΔV, the pressure change can be large. Suitable for high-speed flight.

1.3 Figures of merit: specific fuel consumption and specific impulse

Two measures are used to compare engines ($\dot{m}_{fuel}$ is the fuel or propellant flow rate):

$$\text{BSFC} = \frac{\dot{m}_{fuel}}{P}\ \text{(propeller)} \qquad \text{TSFC} = \frac{\dot{m}_{fuel}}{T}\ \text{(jet)} \qquad I_{sp} = \frac{T}{\dot{m}_{fuel}\, g}$$

- **BSFC** (*Brake Specific Fuel Consumption*): fuel burned per unit of **shaft power** P (propeller); typical units g/kWh or kg/(W·s).
- **TSFC** (*Thrust Specific Fuel Consumption*): fuel per unit of **thrust** (jet); typical units kg/(N·s) or lb/(lbf·h).
- **Specific impulse** I_{sp}: thrust divided by the fuel weight flow rate. **Depending on whether g is included, it has different units**: with g it comes out in **seconds**; without g it comes out in m/s.

High specific consumptions are bad (a lot of fuel for the same power or thrust); by contrast **a high I_{sp} is good**: $I_{sp} = 1/(\text{TSFC} \cdot g)$. *Common mistake*: thinking that "more is better" for both. Link with Unit 3: the Breguet specific consumption, referred to weight, is $c_{sp} = \text{TSFC} \cdot g = 1/I_{sp}$ (in 1/s); with $\text{TSFC} = 1.70 \cdot 10^{-5}$ kg/(N s) it gives $1.67 \cdot 10^{-4}$ s^{-1}, the value in the Breguet example.

CONVERTING TSFC TO I_{sp} AND BSFC TO EFFICIENCY (ILLUSTRATIVE EXAMPLE)
A turbofan with $\text{TSFC} = 0.6\,\text{lb/(lbf} \cdot \text{h)}$. Since $1\ \text{lb/(lbf} \cdot \text{h)} = 2.83 \cdot 10^{-5}$ kg/(N · s):

$$\text{TSFC} = 1.70 \cdot 10^{-5}\ \text{kg/(N} \cdot \text{s)} \ \Rightarrow\ I_{sp} = \frac{1}{1.70 \cdot 10^{-5} \cdot 9.81} \approx 6\,000\ \text{s}$$

(of the order of the Boeing 747 turbofan marked on the chart on slide 23).
A reciprocating engine with BSFC = 270 g/kWh = $7.5 \cdot 10^{-8}$ kg/J and a fuel of 42 MJ/kg (the slide 23 value for hydrocarbons) has an efficiency

$$\eta = \frac{1}{\text{BSFC} \cdot \text{LHV}} = \frac{1}{7.5 \cdot 10^{-8} \cdot 42 \cdot 10^{6}} \approx 0.32$$

which agrees with the 0.30–0.35 in the table of Unit 1 of the UAS Propulsion notes. (Both numbers are examples; no slide gives a specific BSFC.)

2 Propeller propulsion

The **propeller** works as a **small rotating wing**: it pushes the air mass backwards, increases its velocity and obtains thrust as a consequence. The blades can be turned by an **electric** motor, a **reciprocating** engine or a **turbine**. The blade tips operate in the transonic regime, and this **sets the maximum flight speed with a propeller (up to M 0.7)**; at higher speeds complex 3D geometries are needed (the A400M).

- **Electric motor** (battery or fuel cell → electric motor → propeller): a high-efficiency conversion, but the problem is **storing the electrical energy**. The specific energy of batteries is very low: **1 kg of fuel contains the same energy as 70–80 kg of LiPo batteries**, which would give an unacceptable weight in large aircraft. It is used in **small unmanned aircraft**.
- **Reciprocating engine**: similar to automotive engines. Most are **spark-ignition** (gasoline); some are **compression-ignition** (diesel). The fuel energy is released by burning it in a **variable-geometry combustion chamber** (the cylinder with its piston) and is transmitted

to the propeller through a **connecting-rod mechanism** (Figure 6.4). Typical of ultralights and general aviation.

- **Turboprop**: **continuous-flow** combustion with a kerosene-derived fuel (typically **Jet A-1**). A turbine extracts part of the energy to drive the propeller; the same turbine also drives the **compressor**, which draws in the air and pressurizes it for efficient combustion (not all of the turbine energy reaches the propeller). It may be less efficient than a reciprocating engine, but it is **lighter**. Typical of regional aircraft (Figure 6.5).

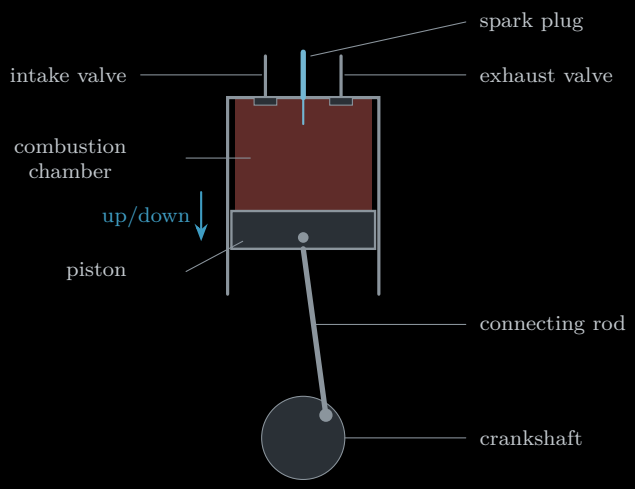


Figure 6.4. Reciprocating engine cylinder (one cylinder, four-stroke, schematic). The burning gas pushes the piston down; the connecting rod and the crank turn that linear motion into the rotation that drives the propeller.

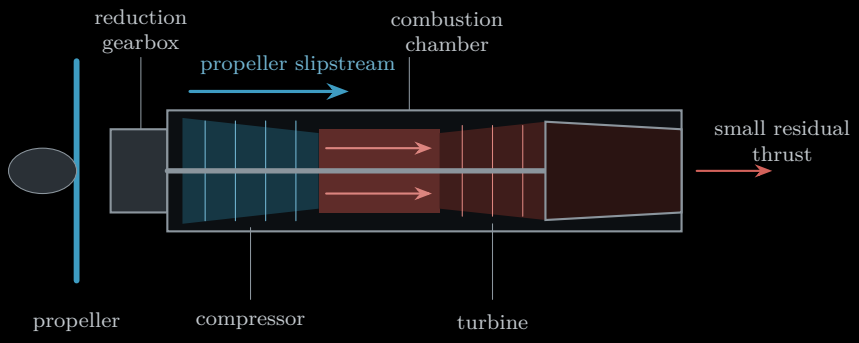


Figure 6.5. Turboprop: the turbine drives both the compressor and, through a reduction gearbox, the propeller, which produces most of the thrust. Schematic.

Common mistake: in a turboprop it is *not* the pressure difference between engine inlet and outlet that gives the thrust; the thrust comes from the propeller, which accelerates a lot of air (the exhaust jet contributes little).

2.1 Propeller coefficients (laboratory supplement)

This is not in these slides (it comes from the lab sessions and is asked in tests), but it helps in understanding the questions. For a propeller of diameter D rotating at n revolutions per second with advance (forward) speed V:

$$J = \frac{V}{nD} \qquad C_T = \frac{T}{\rho n^2 D^4} \qquad C_P = \frac{P}{\rho n^3 D^5} \qquad \eta = \frac{J\,C_T}{C_P}$$

J is the **advance ratio** (how far the aircraft advances per revolution, relative to the diameter). With a **fixed-pitch** propeller turning at constant rpm, **as flight speed increases J grows and**

C_T **always *decreases*** (each blade sees the air at a smaller angle of attack): **the thrust drops**. The efficiency, in contrast, first *rises* and then falls to zero when the thrust vanishes; it is not monotonic, so "the efficiency always increases/decreases" is false (Figure 6.6).

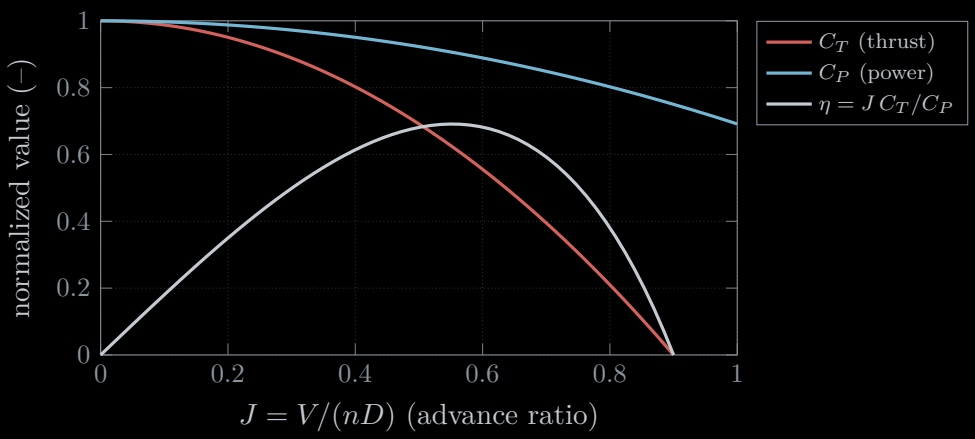


Figure 6.6. Typical curves of a fixed-pitch propeller (schematic, no real values; C_T and C_P are normalized to their value at $J = 0$).

WHICH ENGINE FOR WHICH AIRCRAFT (TEST QUESTIONS)

- Small UAS (5 kg payload, 30 km): **electric**.
- 500 km ultralight: **reciprocating engine + propeller**.
- Regional aircraft (Valencia–Mallorca): **turboprop**.
- High, fast commercial aircraft (M 0.8): **turbofan**.
- Mach 3 at 10 km altitude: the one with the highest I_{sp} is the **ramjet**.

3 Jet propulsion

When the aircraft flies so fast that the propeller suffers transonic effects, its efficiency drops rapidly. The flow can then be accelerated **internally** by means of a **thermodynamic cycle**: air is compressed, energy is added by burning fuel, and the gas is expanded. Figure 6.7 shows the elements of the simplest such engine, the turbojet, numbered as in the list below.

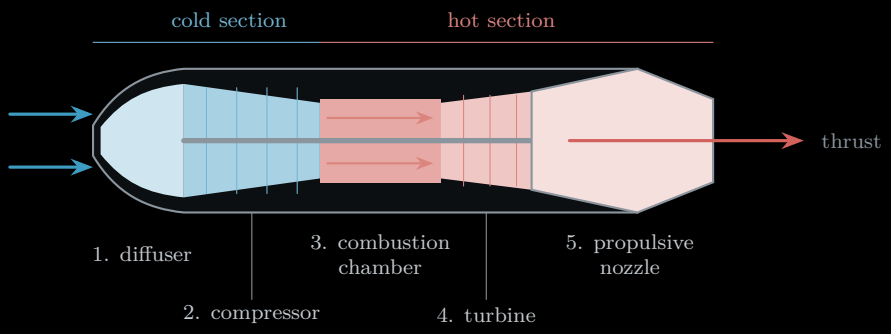


Figure 6.7. Elements of a turbojet. The air enters through the diffuser, is compressed, is heated in the combustion chamber, drives the turbine (which drives the compressor), and the remaining energy is converted into velocity in the nozzle. The blue (cold) part is upstream of the combustion chamber and the red (hot) part is downstream. Source: T6, slide 15.

1. **Diffuser**: decelerates the incoming air and raises the pressure.

2. **Compressor**: raises the pressure much further.
3. **Combustion chamber**: the fuel is burned and the temperature rises.
4. **Turbine**: extracts just enough energy to drive the compressor.
5. **Propulsive nozzle**: accelerates the flow (extracting the energy released in combustion) to high velocities.

A remark from slide 16: the propulsive nozzle accelerates the flow, but *the thrust is produced by "pushing" on the compressor blades* (that is, the force is transmitted to the engine and from there to the aircraft). And in a turbofan, **the element that increases the kinetic energy and momentum of the air is the compressor rotor** (not the stator, nor the turbine) (test question).

3.1 Nozzle

The propulsive nozzle can have two basic shapes, depending on the flight regime (Figure 6.8).

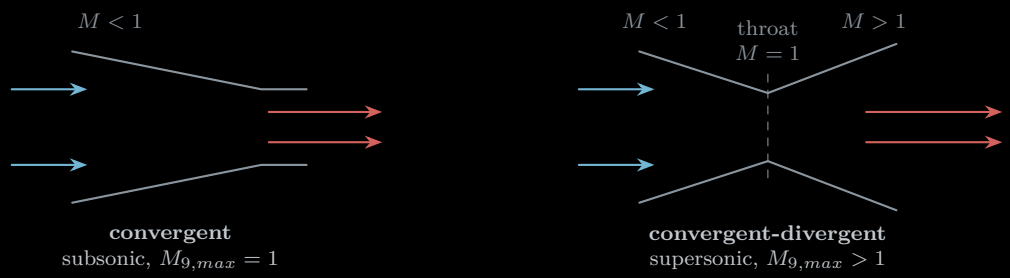


Figure 6.8. Convergent and convergent-divergent nozzles. Source: T6, slide 16.

- **Convergent**: **subsonic** case. The flow cannot exceed $M = 1$ at the exit.
- **Convergent-divergent**: **supersonic** case; the flow reaches $M = 1$ at the throat and accelerates to $M > 1$ in the divergent part.

(The subscript 9 denotes the nozzle exit: V_9 and M_9 are the exit velocity and Mach number. The detailed calculation of V_9 and of the adapted nozzle belongs to other subjects.)

3.2 Turbofan, turbojet and afterburner

The **turbofan** is something in between a turboprop and a turbojet: the air first goes through the first row of compressor blades, the **fan** (somewhat like a propeller), and **one part** goes through the engine **core** while **the rest bypasses** the core (Figure 6.9). High efficiencies are achieved with a high **bypass ratio** (more flow around the core, less through it): this is again the "a lot of air, small ΔV" principle.

When flying above the speed of sound, **the fan becomes useless** (low efficiency in supersonic flow, little thrust for its size). Only the core is then kept: the **turbojet**, or in some cases a low-bypass turbofan. The turbojet is **less efficient**, but it was developed before the turboprop and the turbofan.

To obtain extra thrust for a limited time (military applications only) an **afterburner** (*reheat*) is used: it burns more fuel after the turbine to raise the thrust, but it **considerably reduces the efficiency while in use**.

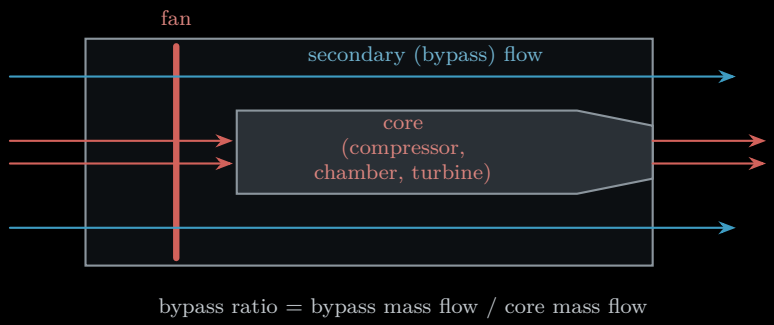


Figure 6.9. Turbofan: two streams, the core stream (red) and the bypass stream (blue). Source: T6, slide 17.

3.3 Ramjet and scramjet

At supersonic speed the intake air can be compressed **without moving parts**, using the shock waves in the diffuser (Figure 6.10). Such an engine consists only of a **diffuser, a combustion chamber and a nozzle**. It is very simple and its efficiency is low, but it produces thrust at higher speeds than the turbojet.

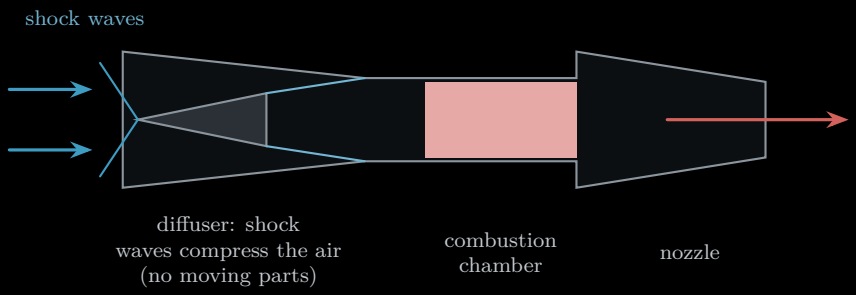


Figure 6.10. Ramjet: the diffuser compresses with shock waves; no compressor or turbine is needed. Source: T6, slides 19 and 20.

- **Ramjet**: combustion takes place at **subsonic** speed.
- **Scramjet** (supersonic-combustion ramjet): combustion takes place in **supersonic** flow.

A ramjet **has no turbine** (it does not need a starter motor to spin up its turbines; this error appears in tests) but **it cannot operate at rest**: it must be launched at supersonic speed by another system; this is why a missile such as the **Meteor** needs a **rocket** (*booster*) to reach the speed at which the ramjet ignites.

3.4 Jet engine details (added explanation, not in the slides)

- **Combustion chamber air split** (Figure 6.11): only a fraction of the air (the *primary* air) is burned with the fuel; *secondary* air enters through holes in the liner to complete combustion and cool the walls, and *dilution* (tertiary) air lowers the gas temperature to what the turbine can withstand.
- **Diffuser at high Mach**: a supersonic intake slows the flow through several oblique shocks followed by a final normal shock, which loses less total pressure than a single normal shock.
- **Variable stator vanes**: in the first compressor stages the stators can be rotated to keep the blades at a good angle of attack at different engine speeds and avoid compressor stall.
- **Blades and high-pressure turbine**: the gas is hotter than the melting point of the blade metal, so the blades are made of nickel **superalloys** (often single-crystal), with thermal coatings and internal **cooling** by air bled from the compressor.

- **Bleed air**: air tapped from the compressor is used for cabin pressurization and air conditioning, wing and engine anti-icing, and engine starting.

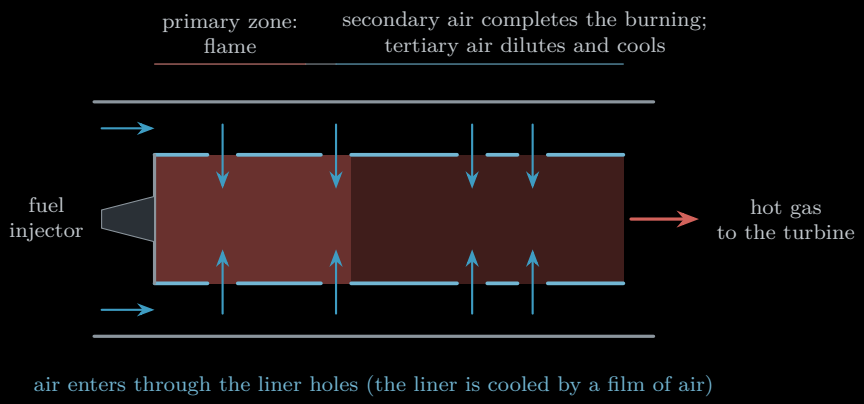


Figure 6.11. Combustion chamber air split: primary, secondary and dilution air (schematic, no real values).

3.5 Rocket

A **rocket** is a **non-air-breathing** system: it uses the atmosphere neither to react with the fuel nor as reaction mass. It has a **low specific impulse** because it carries **all of the reaction mass and the oxidizer**, not just the fuel. It is suitable **for the whole Mach range**, from almost 0 to high values.

Propellants and the rocket equation (added explanation, not in the slides)

A rocket carries fuel and oxidizer (Figure 6.12). Two families:

- **Liquid bipropellants**: LOX/LH_2 gives the highest I_{sp} (about 450 s in vacuum); LOX/RP-1 (kerosene) about 300–350 s and is denser; *hypergolic* pairs ignite on contact (no igniter, storable), at the price of toxicity. They can be throttled and restarted.
- **Solids** (APCP, *ammonium perchlorate composite propellant*): ammonium perchlorate oxidizer, HTPB rubber binder and aluminium powder. Simple and storable, but once lit they **cannot be throttled or stopped**.

The velocity gain follows the **Tsiolkovsky equation**:

$$\Delta v = I_{sp}\, g_0 \ln \frac{m_0}{m_f}$$

with m_0 the initial mass and m_f the final mass (after burning the propellant). Example: to reach orbit one needs $\Delta v \approx 9.4$ km/s (including losses); with $I_{sp} = 350$ s, $m_0/m_f = \exp(9400/(350 \cdot 9.81)) = 15.5$, so about 94 % of the launch mass is propellant. The structure and payload must fit in the remaining 6 %, which is why rockets are **staged**: discarding empty tanks and engines improves m_f for the next stage. Figure 6.13 shows how the propellant fraction saturates as Δv grows.

4 Summary: which system for which speed

Figure 6.14 places each engine type on a specific impulse versus Mach chart (the higher the I_{sp}, the less fuel for the same thrust); the table below gives the same information in words.

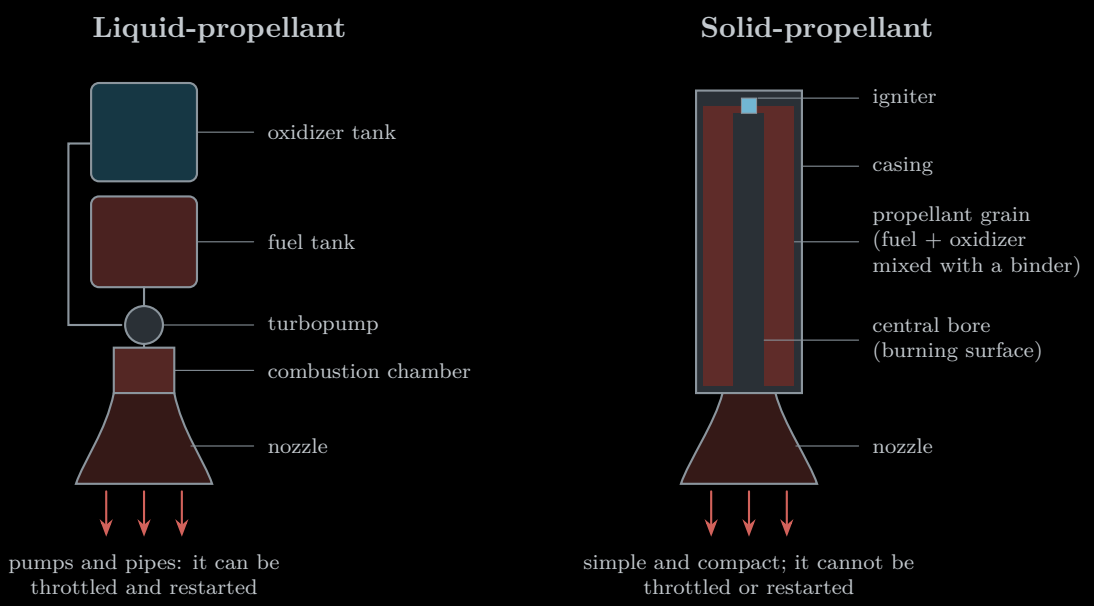


Figure 6.12. Liquid-propellant (left) and solid-propellant (right) rockets, schematic.

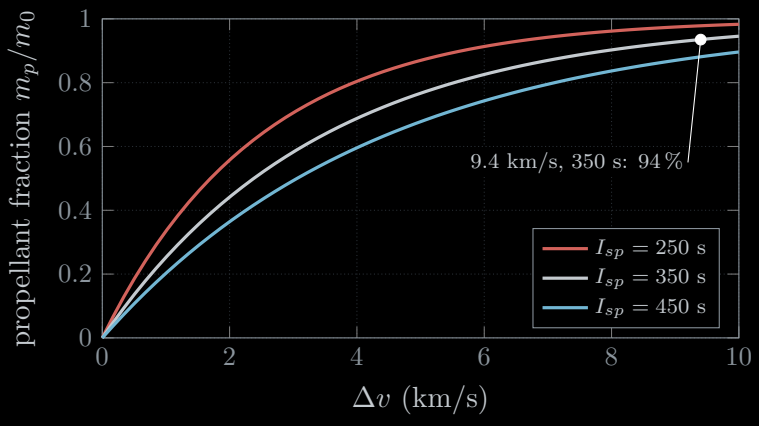


Figure 6.13. Propellant fraction m_p/m_0 required versus Δv for three values of I_{sp} (the dot is the example in the text).

System	Operating range	Characteristics
Propeller (electric, reciprocating or turboprop)	$M < 0.6$–0.7 (up to ~ 0.8)	Large mass flow, small ΔV: highest η_p. Limit: transonic blade tips.
Turbofan	Transport at $M \approx 0.8$; on the slide 23 chart it extends up to $M \approx 1.6$	Higher I_{sp} than the turbojet in that range; **high bypass**.
Turbojet (and with afterburner)	Up to $M \approx 3$ (slide 23 chart)	No fan; less efficient; the afterburner gives extra thrust.
Ramjet	Supersonic, from $M \approx 1$ to ≈ 6 (the chart's I_{sp} is the highest at $M \approx 3$–6)	No moving parts; does not work at rest.
Scramjet	$M \gtrsim 6$	Supersonic combustion.
Rocket	All M	Non-air-breathing; very low I_{sp}.

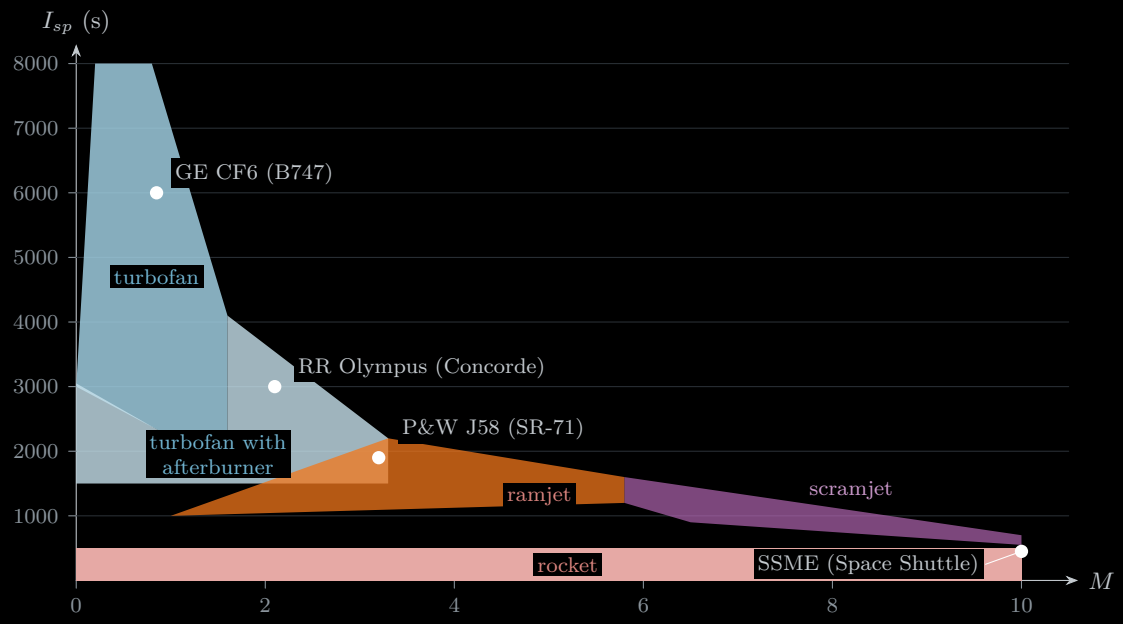


Figure 6.14. Specific impulse versus Mach for each engine type, redrawn and simplified from slide 23 (the regions are approximate; the dots are real engines: GE CF6 of the B747, RR Olympus of the Concorde, P&W J58 of the SR-71 and SSME of the Space Shuttle). The slide also shows the theoretical maxima of hydrocarbons (42 MJ/kg) and hydrogen (143 MJ/kg) in air, which have not been drawn.

5 Solved exam-style problems

1. PROPULSIVE EFFICIENCY OF A PROPELLER AND OF A JET
Same thrust and same flight speed $V_{in} = 200\,\text{m/s}$.

- **Propeller** accelerating the air to $V_{out} = 240\,\text{m/s}$: $\eta_p = \dfrac{2}{1+240/200} = \dfrac{2}{2.2} = 0.91$.
- **Jet** accelerating it to $V_{out} = 600\,\text{m/s}$: $\eta_p = \dfrac{2}{1+3} = 0.50$.

To give the same thrust with $\dot{m}\,\Delta V = T$, the propeller needs **much more mass flow**: $\dot{m}_{propeller}/\dot{m}_{jet} = 400/40 = 10$ (with ΔV of 40 and 400 m/s). The propeller moves 10 times more air, accelerating it 10 times less, and wastes much less kinetic energy.

2. THRUST OF A TURBOJET
An engine takes in $\dot{m} = 20\,\text{kg/s}$ at $V_{in} = 250\,\text{m/s}$ and expels it at $V_{out} = 650\,\text{m/s}$ (pressure terms and the added fuel are neglected):

$$T = \dot{m}\,(V_{out} - V_{in}) = 20 \cdot 400 = 8\,000\ \text{N}$$

Useful power $T \cdot V_\infty = 8\,000 \cdot 250 = 2.0$ MW; power given to the air $= \frac{1}{2}\dot{m}(V_{out}^2 - V_{in}^2) = \frac{1}{2}\cdot 20\cdot(650^2 - 250^2) = 3.6$ MW; $\eta_p = 2.0/3.6 = 0.56$, which agrees with $2/(1+650/250) = 0.556$.

3. BATTERIES VERSUS FUEL (SLIDE 11)
The specific energy of fuel is $\approx 42\,\text{MJ/kg} = 11.7\,\text{kWh/kg}$. If 1 kg of fuel contains the energy of 70–80 kg of LiPo batteries, each kilogram of battery stores $\approx 11.7/75 \approx 0.16\,\text{kWh} = 160\,\text{Wh/kg}$. But this compares *stored energy*; in practice the electric motor converts better (0.72 versus 0.30–0.35 in the UAS Propulsion notes), so the real mass ratio for the same useful energy is somewhat smaller (about 30–35 times instead of 75, indicative).

4. PROPULSION STATEMENTS FROM PAST TESTS (2023/24 AND 2024/25)

- *"The efficiency of turbofans is higher than that of turbojets at M 0.8."* **True**: more mass flow, smaller ΔV, higher η_p.
- *"Ramjets require a starter motor to accelerate their turbines."* **False**: they have no turbines.
- *"Rockets outperform turbojets in specific consumption at supersonic speeds."* **False**: the rocket has a low I_{sp} (much higher consumption).
- *"Mach 3 at 10 km: highest I_{sp}?"* **Ramjet**, not turboprop, rocket or scramjet.
- *"The thrust of commercial turboprops is due to the inlet-outlet pressure difference."* **False**: it is given by the propeller.

WHAT USUALLY APPEARS ON THE EXAM (T6)

Propulsion appeared in none of the six first midterms analysed (2018/19 to 2024/25 except 2021/22), so it is the lowest-priority unit for the first test. It is included because it belongs to the first-midterm syllabus; the questions to expect are **conceptual multiple-choice, no calculations**.

- **Choice of system according to vehicle and speed**: piston engine for ultralights, turboprop for regional aircraft, electric for small UAS, turbofan at M 0.8, ramjet at M 3.
- $\eta_p \simeq 2/(1 + V_{out}/V_{in})$: more mass flow and less ΔV = more efficient (turbofan > turbojet; propeller > jet).
- **Turbojet elements** and nozzle (convergent subsonic, convergent-divergent supersonic); **the compressor rotor** is what gives kinetic energy to the air.
- **Ramjet/scramjet**: no moving parts, subsonic/supersonic combustion; they do not work at rest (the Meteor carries a booster rocket).
- **Rocket**: non-air-breathing, low I_{sp}, whole Mach range.
- **Fixed-pitch propeller at fixed rpm**: as speed rises, *the thrust (and C_T) always drops*; the efficiency does not always.
- **Figures of merit**: BSFC (propeller, per power), TSFC (jet, per thrust), $I_{sp} = T/(\dot{m}_{fuel} g)$ in seconds; high specific consumption = poor efficiency; high I_{sp} = good.
- **Fuel versus battery**: 1 kg of fuel ≈ 70–80 kg of LiPo.

The essentials: $T \approx \dot{m}\,\Delta V$; $\eta_p \simeq 2/(1 + V_{out}/V_{in})$; BSFC $= \dot{m}_f/P$; TSFC $= \dot{m}_f/T$; $I_{sp} = T/(\dot{m}_f g)$; propeller up to M 0.6–0.7; jet at high speeds.

OPEN QUESTIONS AND LOOSE ENDS (T6)

- Slide 23 (the I_{sp}-Mach chart) is an image: the regions and the position of the points have been estimated from the image and the own drawing is approximate.
- In the slides, the propeller Mach limit appears as "less than 0.8, normally less than 0.6" (slide 7) and as "Mach 0.7" (slide 10); both are given.
- The BSFC and TSFC examples are illustrative: the slides give no numerical values.

END OF THE FIRST-MIDTERM MATERIAL

Units 1 to 6 are the course slides, up to propulsive systems.

What follows (Unit 7) is **additional material**: the documents of the *English_recursos* folder and a list of review questions.

It is not confirmed that it belongs to the first midterm. About a third of the first-midterm questions of past years touch its topics (structure, navigation aids, Breguet, ice, kinetic heating, drag, centre of pressure), but those are also covered in Units 2 to 5. Rocket, rotorcraft and surveillance topics did not appear in any first midterm.

UNIT 7

Additional material and review questions

1 Additional course documents

Besides the slides, the course provides eight **additional documents** (short papers by the teaching staff plus a presentation and an assignment). The students were told that this material **also appears in the exam**, so each one is summarised below as exam-oriented notes. Where a topic was already explained in the units, the text refers to it instead of repeating it, but each subsection can be read on its own.

Document	Topic
`architecture`	Primary structure: the five classic structural layouts
`wing_high_low`	Vertical position of the wing; monoplanes and multiplanes
`CA_english`	Centre of pressure, aerodynamic centre, mean aerodynamic chord
`D_English`	Effect of the incident velocity on the aerodynamic drag
`storm`	Ice, wind gusts and storm clouds
`breguet_eng`	Quick estimation of the range (simplified Breguet equation)
`Spanish_industry`	Presentation on the Spanish aerospace industry (2016)
`ProblemAeroTec1`	"Aircraft characterisation problem" (assignment)

1.1 Aircraft general architecture: the five kinds of primary structure

Document: "Aircraft general architecture". Background: Unit 4.

Basic ideas. The **primary structure** is the part that, if it fails, endangers the aircraft. It is made of elements that look like beams or plates (one dimension much smaller than the other two) and that carry tension, compression, shear, torsion and bending. The load per unit area is the **stress** σ and the relative deformation is the **strain** $\varepsilon = \Delta L/L$. In the elastic zone they are proportional (Hooke's law, with E the Young modulus; the document writes the relation upside down, the correct form is the one below):

$$\sigma = \frac{F}{A} = E\,\varepsilon = E\,\frac{\Delta L}{L} \quad \Longleftrightarrow \quad F = \frac{E\,A}{L}\,\Delta L$$

so a bar behaves like a **spring of stiffness** EA/L. Beyond the **elastic limit** the strain is plastic (it stays after unloading); aircraft are designed not to exceed it.

- **Buckling**: a compressed element can collapse *before* reaching the elastic limit. The buckling stress is higher with a larger E, a larger cross-section and a **shorter length**. The cheapest remedy is to **shorten the length by adding supports** (the element then behaves as several short ones).
- In an optimised primary structure the element is designed so that the stress that makes it yield and the stress that makes it buckle are **similar**: nothing is over-sized.

- **Fatigue**: under repeated (cyclic) loads, cracks appear and grow; hence **periodic inspections**.

Materials. Early aircraft used wood, cloth and steel wire: humidity condenses on cloth when climbing and rots it on the ground (and cloth carries little stress); humidity also rots and deforms wood; steel is dense. Today:

- **Aluminium light alloys**: the most important (high specific strength, low density).
- **Titanium alloys**: less common; used where there are thermal loads, since aluminium loses strength at much lower temperatures.
- **Stainless steel**: very strong but dense; only for robust, compact elements with buckling problems, such as the **landing gear**.
- **Composites** (carbon-fibre fabric in an epoxy resin matrix): very light, but poor **impact resistance**.
- **Paint**: needed on aluminium (corrosion) and on composites (water replacing the resin).

The five structures. The document singles out five layouts (Figure 7.1); the first three are **truss** (reticulated) designs and the last two are **shell** designs.

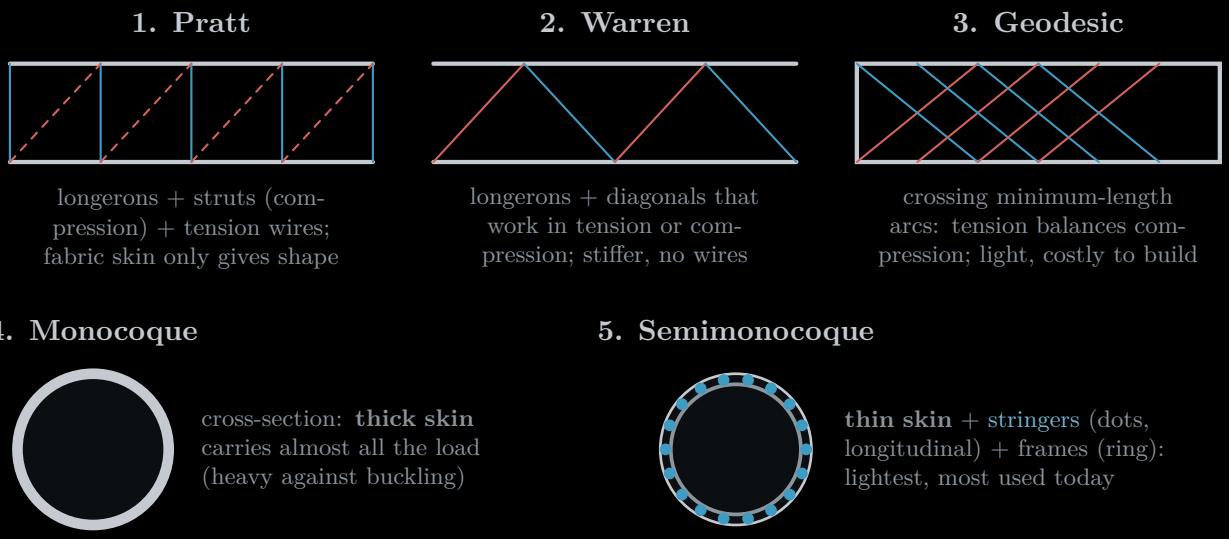


Figure 7.1. The five kinds of primary structure of the document: Pratt, Warren and geodesic (side views of a fuselage section) and monocoque and semimonocoque (cross-sections). Cyan: elements in compression; red: elements in tension (dashed: wires that only work in tension). Own sketch from the document.

- **Pratt**: longerons joined by vertical and horizontal **struts**; shear is carried by **tension wires** (they only work in tension). The cloth skin only gives the shape. Early aviation; without cover it had high drag.
- **Warren**: the wires are replaced by **diagonal beams** that carry tension *and* compression, so the structure is stiffer. Wood or steel, sometimes aluminium; also high drag if uncovered.
- **Geodesic**: beams along the **arcs of minimum length** over the fuselage shape; compression is balanced by tension and the weight is minimal. Much more efficient than Pratt and Warren, but costly to build and maintain. Loads redistribute well after damage (in World War II aircraft flew with large damaged areas).
- **Monocoque**: the **thick skin** carries most of the load; it may be stiffened by **frames** (reinforced monocoque). Wood originally, light alloys and composites now. The skin is poor in compression, so it must be very thick to avoid buckling.

- **Semimonocoque**: **thin skin** that buckles easily, held by **stringers** (longitudinal) and **frames** (transversal). **The most common today**: it weighs less than a monocoque for the same loads, because adding supports is more efficient than thickening the skin.

Unit 4 has the same structures: Figure 4.6, 4.7 and 4.8.

Semimonocoque fuselage: the elements.

- **Stringers**: longitudinal small beams; support the skin against buckling (more stringers, closer together ⇒ higher buckling load) and carry tension/compression, including fuselage bending.
- **Frames**: transversal; give the shape and support skin and stringers (placed normal to them).
- **Longerons**: longitudinal beams carrying bending and axial loads.
- **Pressure bulkheads** (skullcap shape, they close the pressurised zone) and **flat bulkheads** with frames (where loads concentrate: engines, landing gear, wing).
- The **skin** carries tension, shear and torsion; it is joined by bolts or rivets (flush-head rivets to reduce drag) or, in composites, by resin.
- **Windows** go *between* frames, which are about **half a metre** apart (to avoid buckling). They are discontinuities with stress concentration (crack growth, fatigue, corrosion), so their frames are reinforced and the windows are kept small.

Pressurised fuselage. At altitude the oxygen partial pressure is too low for people, so the cabin is pressurised; the inside-outside pressure difference is **the most important stress** of the fuselage. Sharp corners would concentrate stress, hence the **round or oval cross-section**; **floating frames** may move relative to the skin under shear. Doors and windows need very robust frames. They are semimonocoque, because the skin carries the tension caused by the pressure.

Wing and stabilisers. Normally semimonocoque too, with the same elements and three differences: **no pressure bulkheads**; transversal elements are **ribs** (instead of frames); longitudinal closing elements are **spars** (instead of longerons); **fittings** join wing to fuselage or control surfaces to wing.

- **Rib orientation**: parallel to the aircraft axis near the root and normal to the spar near the tip is the usual compromise in swept wings.
- **Types of rib**: *compression ribs* (link the spars and spread local loads); *master ribs* (keep the distance between spars, stiffen skin and stringers against buckling); *common ribs* (lighter; keep the airfoil shape and pass inner forces to the spars); *false ribs* (only keep the skin shape between the spar and the leading or trailing edge).
- Ribs and spars have a **web** and upper and lower flanges ("nerves"); the webs have holes so that a technician's hand fits in.
- **Bending**: in flight the **upper** part of the wing is in **compression** and the lower in **tension**; **on the ground it is the opposite**. Between them there is a **neutral line** with zero stress. Flanges, skin and stringers carry these stresses.
- **Torsion box**: the region formed by skin, stringers and spars; it usually holds the **fuel**, and ribs can separate tanks.

> WHAT TO REMEMBER: STRUCTURES
> Five structures: **Pratt** (wires in tension), **Warren** (diagonals), **geodesic** (minimum-length arcs) are trusses; **monocoque** (thick skin) and **semimonocoque** (skin + stringers + frames) are shells. The semimonocoque is the standard because **supports are a lighter fix for**

buckling than thickness. Wing: ribs instead of frames, spars instead of longerons, no pressure bulkheads; upper skin in compression in flight. Aluminium dominates, titanium for heat, steel for the landing gear, composites for lightness (poor impact resistance).

1.2 Vertical position of the wing; monoplanes and multiplanes

Document: "Vertical position of the wing. Monoplanes and multiplanes". Background: Figure 3.3 (low, mid and high wing and dihedral) in Unit 3.

The vertical position of the wing affects aerodynamics, structural loads, stability under perturbations and even evacuation after a splashdown. Figure 3.3 shows the three positions; the table gathers the effects given in the document.

	Low wing	**Mid wing**	**High wing**
Landing gear	Longer if the engines hang below the wing	Shorter than low wing	Short; easy cargo loading (most of all)
Ground effect	Strong: less induced drag, more lift at take-off and landing	Lower than low wing	Lowest: good for short or unprepared runways
Roll stability	Unstable: CG above the aerodynamic centre ⇒ **positive dihedral** (weight penalty); negative in some military/acrobatic aircraft	Intermediate	Stable: CG below the aerodynamic centre ⇒ often **negative dihedral** (anhedral) to keep manoeuvrability; positive in gliders
Wing–fuselage interference drag	Penalty (upper root surface inside the fuselage)	**Lowest**: easy to fair	Higher than mid wing (like low wing)
Cabin (airliner)	Wing passes through the cargo hold: **cabin clear**	Wing divides the cabin into two parts	Less room at the wing fairing (seats, overhead bins)
Main gear fixings	Gear loads go to the root region, which is strongly reinforced	Not at the torsion box: extra reinforcement	Not at the torsion box: extra reinforcement
Water	Wing acts as a **float** after splashdown	–	Engines above the wing avoid water ingestion and splash damage
Ground handling	Fuel loading and inspections easier (closer to the ground)	–	Cargo loading easier

Ground effect in detail (low wing). Near the ground the induced drag drops and the lift grows. *Take-off:* the lower induced drag shortens the run, but the extra lift hardly matters, because the aircraft must reach the speed that lifts it *without* ground effect before leaving the runway. *Landing:* the extra lift makes it **harder to touch down** (the aircraft tends to float), the lower drag makes it **harder to stop**, and the lift that remains after touchdown **reduces the load on the wheels**, so braking is less effective.

Monoplanes, biplanes and multiplanes. To manoeuvre, the maximum lift must exceed the weight:

$$W < L_{max} = \tfrac{1}{2}\,\rho_\infty U_\infty^2\, S\, C_{L,max}$$

If $C_{L,max}$ is not enough, one can fly faster, fly lower (denser air) or **increase the wing area** S. Speed and altitude are often not options, so S grows. In the early days the required area was too large for a single plane, hence **biplanes, triplanes and multiplanes** (Figure 7.2).

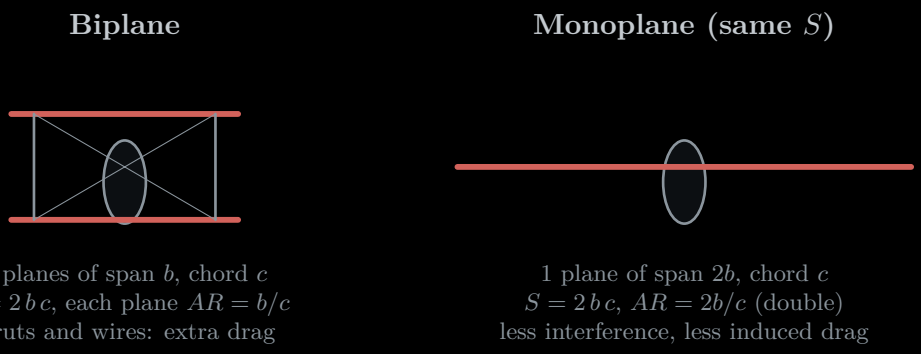


Figure 7.2. Same wing area $S = 2bc$ obtained with two planes (biplane) or with one plane of double span (monoplane): the monoplane has twice the aspect ratio of each biplane plane. Front view. Own sketch.

Multiplane	Monoplane (same area, larger aspect ratio)
High **structural efficiency** (low aspect ratio, struts and wires stiffen the wing)	**Higher structural loads**: the bending moment grows with aspect ratio
High **drag** from struts and wires	Lower **interference drag** (one plane)
Extra drag from **interference between planes**	**Lower induced drag** (high aspect ratio)
Lower $C_{L,max}$ **and lower slope** of the C_L–α curve of each plane because of the interference	**Higher** $C_{L,max}$ **and higher lift slope**

The effects get worse as the number of planes increases. Modern technology builds wings that carry the extra loads of the monoplane without a large weight penalty, and the aerodynamic advantages exceed the structural drawbacks *by far*: **the monoplane is the standard solution**. The document mentions **new biplane concepts without struts or wires** (for instance a ring wing, the PrandtlPlane type), where the interference is minimal and the induced drag can be kept low by limiting the wingtip vortices; they may be interesting when the **span is limited**, such as by the airport parking space (maximum today: **80 m** of wingspan).

WING AREA, MONOPLANE OR BIPLANE (OWN ILLUSTRATIVE NUMBERS)
Aircraft of 5 000 kg that must be able to fly at 40 m/s at sea level with $C_{L,max} = 1.5$:

$$S \geq \frac{2W}{\rho U^2 C_{L,max}} = \frac{2 \cdot 5000 \cdot 9.81}{1.225 \cdot 40^2 \cdot 1.5} = \frac{98\,100}{2\,940} = 33.4 \text{ m}^2$$

With a chord $c = 2$ m: **monoplane** $b = S/c = 16.7$ m, $AR = b/c = 8.3$. **Biplane**: two planes of 16.7 m^2, each with span 8.35 m and $AR = 4.2$: half the aspect ratio, so **more induced drag per plane**, plus struts and interference.

WHAT TO REMEMBER: WING POSITION
Low wing: ground effect (shorter take-off, long landing float), roll-unstable (dihedral), cabin clear, flotation, easy servicing. **High wing**: roll-stable (anhedral possible), lowest ground effect, short gear, easy loading, engines away from water, but less cabin space at the wing. **Mid wing**: lowest interference drag, intermediate in everything else, but the wing splits the cabin. **Monoplane**: higher aspect ratio, less drag, more $C_{L,max}$, heavier structure (accepted).

1.3 Centre of pressure, aerodynamic centre and mean aerodynamic chord

Document: "Center of pressure, aerodynamic center and mean aerodynamic chord". Background: pitching moment in Unit 5 (Figure 5.13) and wing geometry (Figure 5.18).

Why these points exist. The forces (lift, drag, thrust, weight) create **moments** about the centre of mass: roll (about the longitudinal axis), **pitch** (about the axis along the wingspan) and yaw (about the vertical axis). Computing them exactly is complicated, but **for small angles of attack** (lift linear in α) three concepts make it easy. The moment is always "force $\times$ arm": what changes is the point used.

	Definition	**Position and use**
Centre of pressure (CP)	Point about which the aerodynamic moment is **zero**: the resultant force can be applied there.	**Moves with** α (from infinity upstream to infinity downstream). Only for a *symmetric* airfoil at low α it stays fixed, at about $c/4$. To use it you need $x_{cp}(\alpha)$ and $L(\alpha)$.
Aerodynamic centre (AC)	Point about which the moment **does not change with** α.	At about $c/4$ and practically fixed. Symmetric airfoil: AC $\equiv$ CP and $c_{m,ac} \approx 0$. Cambered: $c_{m,ac} \neq 0$ even with zero lift (positive camber: nose-down moment; negative camber: nose-up). You only need $L(\alpha)$, the constant $c_{m,ac}$ and the AC position.
Mean aerodynamic chord (MAC)	Chord of an **equivalent rectangular wing** of the same area that gives the same pitching moment (about the AC) when $c_{m,ac}$ is uniform along the span.	Has a value *and* a position: if the lift of each half-wing acts at the AC of the MAC, the moment about it is independent of α.

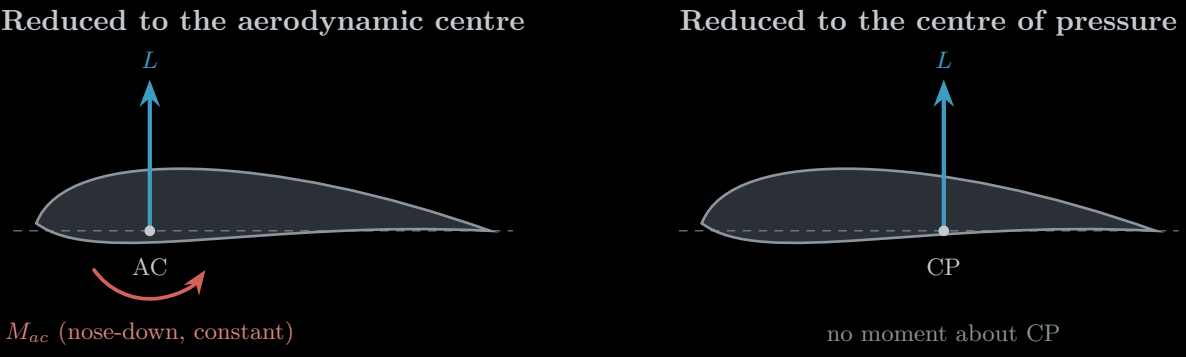


Figure 7.3. The same aerodynamic action seen from two points of a cambered airfoil: lift L at the aerodynamic centre plus a constant nose-down moment M_{ac}, or the lift alone at the centre of pressure, which lies behind the AC. Own sketch.

The formulas of the document. With $L = \frac{1}{2}\rho_\infty U_\infty^2 S\, C_L$ and, for the airfoil at position y along the span, $\dfrac{dL}{dy} = \frac{1}{2}\rho_\infty U_\infty^2\, c_l(y)\, c(y)$ and $c_l = \dfrac{\partial c_l}{\partial \alpha}\alpha + c_{l0}$. The moment of the wing about the **centre of gravity** using the CP is $M = \displaystyle\int \frac{1}{2}\rho_\infty U_\infty^2\, (x_{cg} - x_{cp}(y))\, c\, c_l\, dy$, with the AC it is:

$$M_{cg} = \int_{-b/2}^{b/2} \frac{1}{2}\rho_\infty U_\infty^2 \Big\{ [x_{cg} - x_{ac}(y)]\, c(y)\, c_l(y) + c^2(y)\, c_{m,ac}(y) \Big\} dy \qquad (x_{ac} \approx x_{c/4})$$

that is, *lift × arm to the AC* plus the *moment about the AC*. If $c_{m,ac}$ is uniform along the span, the moment about the AC is

$$M_{ac} = \tfrac{1}{2}\rho_\infty U_\infty^2 \, S \, c_{m,ac} \, \mathrm{MAC}, \qquad \mathrm{MAC} = \frac{1}{S}\int_{-b/2}^{b/2} c^2(y)\, dy$$

Position of the MAC (half-wing, $x_{c/4}(y)$ = quarter-chord line):

$$y_{MAC} = \frac{2}{S}\int_0^{b/2} y\, c(y)\, dy, \qquad x_{MAC/4} = \frac{2}{S}\int_0^{b/2} x_{c/4}(y)\, c(y)\, dy$$

For a **simply tapered (trapezoidal) wing** with root chord c_r, tip chord c_t, taper $\lambda = c_t/c_r$ and half-span $b/2$ (standard results, the document gives the graphical method instead):

$$\mathrm{MAC} = \tfrac{2}{3}\, c_r \, \frac{1+\lambda+\lambda^2}{1+\lambda}, \qquad y_{MAC} = \frac{b}{6}\,\frac{1+2\lambda}{1+\lambda}$$

Graphical method (Figure 7.4): extend the root chord backwards by c_t and the tip chord forwards by c_r; join the two ends; join the **mid-points** of the root and tip chords. The two lines cross at the middle of the MAC, which is the chord at that spanwise station.

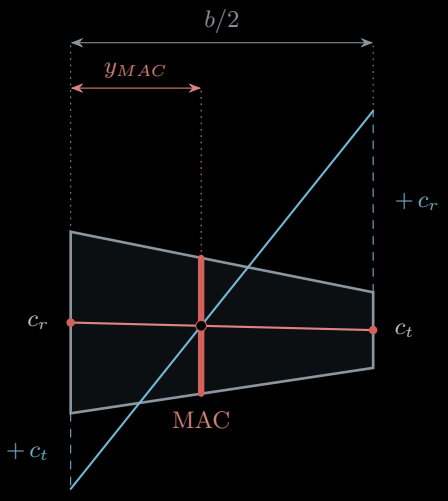


Figure 7.4. Graphical method for the mean aerodynamic chord of a tapered half-wing (here $c_r = 2.4$, $c_t = 1.0$, $b/2 = 4$ in arbitrary length units). Own sketch following the document's figure.

MAC AND MOMENT OF A TAPERED WING (OWN ILLUSTRATIVE NUMBERS)
Wing with $c_r = 2.4$ m, $c_t = 1.0$ m, $b = 8$ m. Area $S = \frac{c_r+c_t}{2}\, b = 13.6$ m^2; $\lambda = 1/2.4 = 0.417$.

$$\mathrm{MAC} = \tfrac{2}{3}\cdot 2.4 \cdot \frac{1+0.417+0.174}{1.417} = 1.80 \text{ m} \qquad y_{MAC} = \frac{8}{6}\cdot\frac{1.833}{1.417} = 1.73 \text{ m from the root}$$

(check with the definition: $c(y) = 2.4 - 0.35y$, $\int_0^4 c^2 dy = 12.21$, MAC $= 2 \cdot 12.21/13.6 = 1.80$ m).
At $U_\infty = 60$ m/s, $\rho = 1.225$, $c_{m,ac} = -0.05$: $M_{ac} = \frac{1}{2}\cdot 1.225 \cdot 60^2 \cdot 13.6 \cdot (-0.05) \cdot 1.80 = -2\,690$ N m (nose-down). If the lift is $L = 20$ kN and the CG is $0.05\,\mathrm{MAC} = 0.09$ m behind the AC, the lift (ahead of the CG) gives $+20\,000 \cdot 0.09 = +1\,800$ N m nose-up, so the wing alone gives a net ≈ -890 N m about the CG. (Sign: nose-up positive.)

WHAT TO REMEMBER: CP, AC, MAC
CP: moment zero, but it **moves** with α (fixed at $c/4$ only for symmetric airfoils). **AC**: moment **independent of** α, at $\approx c/4$; $c_{m,ac} = 0$ for symmetric airfoils and < 0 for positive

camber. **MAC**: $\frac{1}{S}\int c^2 dy$, not the geometric mean; with the AC and $c_{m,ac}$ the moment about any point is "$L\times$ arm + M_{ac}".

1.4 Effect of the incident velocity on the aerodynamic drag

Document: "On the effect of the incident velocity on the aerodynamic drag". Background: Figure 5.22 and Section 3.3.2 of Unit 5.

Set-up. (Notation: the document writes U_∞ and U_{md}; the review questions below write V and V_{md} for the same speeds, and $k = 1/(\pi\, AR\, e)$.) Subsonic, trimmed level flight: $W = L = \frac{1}{2}\rho_\infty U_\infty^2 S C_L$, so $C_L = \dfrac{2W}{\rho_\infty U_\infty^2 S}$, with the parabolic polar $C_D = C_{D0} + \dfrac{C_L^2}{\pi\, AR\, e}$. Multiplying by $\frac{1}{2}\rho U^2 S$ gives the drag split into parasitic and induced parts:

$$D = \underbrace{\tfrac{1}{2}\rho_\infty S\, U_\infty^2\, C_{D0}}_{D_0\ \text{(parasitic)}} + \underbrace{\frac{W^2}{\tfrac{1}{2}\rho_\infty U_\infty^2\, S\, \pi\, AR\, e}}_{D_i\ \text{(induced)}}$$

- **Parasitic drag**: with the flat-plate analogy (laminar, Blasius $C_f = 1.328/\sqrt{Re}$, wetted area $\approx 2S$) $C_{D0} \simeq 2 \cdot 1.328\sqrt{\mu_\infty}/\sqrt{\rho_\infty U_\infty \mathrm{MAC}}$: the **coefficient falls slowly** as U_∞ grows (it stops falling at some point if the boundary layer is turbulent), but the **force increases**: $D_0 \simeq 1.328\, S\sqrt{\rho_\infty \mu_\infty / \mathrm{MAC}}\, U_\infty^{3/2}$ (with constant C_{D0} it would grow as U_∞^2).
- **Induced drag**: it decreases with speed ($D_i \propto 1/U_\infty^2$, and $C_{Di} \propto 1/U_\infty^4$), because less lift coefficient is needed. At low speed it dominates.

Speed for minimum drag. Setting $dD/dU_\infty = 0$ with C_{D0} about constant (only one extreme exists and it is a minimum):

$$U_{md} = \Big(\frac{1}{\pi\, AR\, e\, C_{D0}}\Big)^{1/4}\sqrt{\frac{2W}{\rho_\infty S}} \qquad C_L^* = \sqrt{C_{D0}\,\pi\, AR\, e} \qquad \Big(\frac{L}{D}\Big)_{max} = \frac{1}{2}\sqrt{\frac{\pi AR\, e}{C_{D0}}}$$

The **maximum aerodynamic efficiency** L/D requires the same C_L^* and the same speed (for a given lift, maximum efficiency means minimum drag). At that point $C_{Di} = C_{D0}$ (the two drag components are equal, as in Figure 5.22). Note that it refers to the **whole aircraft**, not to the airfoil.

- U_{md} is **higher at higher altitude** ($\propto 1/\sqrt{\rho}$) and with a **higher wing loading** W/S ($\propto \sqrt{W/S}$).
- Increasing AR, e or C_{D0} **lowers** U_{md} but weakly, because they appear to the power 1/4.
- Valid only before the transonic regime (where C_{D0} stops being constant).

MINIMUM-DRAG SPEED OF THE DOCUMENT'S AIRCRAFT
$W = 10\,000 \cdot 9.81 = 98\,100$ N. At sea level ($\rho = 1.225$): $\sqrt{2W/(\rho S)} = \sqrt{196\,200/36.75} = 73.1$ m/s and $(\pi \cdot 8 \cdot 0.85 \cdot 0.02)^{1/4} = 0.427^{1/4} = 0.808$, so $U_{md} = 73.1/0.808 = 90.4$ m/s. Efficiency $\frac{1}{2}\sqrt{\pi \cdot 8 \cdot 0.85/0.02} = 16.3$, so $D_{min} = W/16.3 = 6.0$ kN. At 11 000 m ($\rho = 0.3639$): $U_{md} = 90.4\sqrt{1.225/0.3639} = 166$ m/s, with the *same* 6.0 kN: the curves of Figure 7.5 shift to the right but their minimum does not drop (constant C_{D0}).

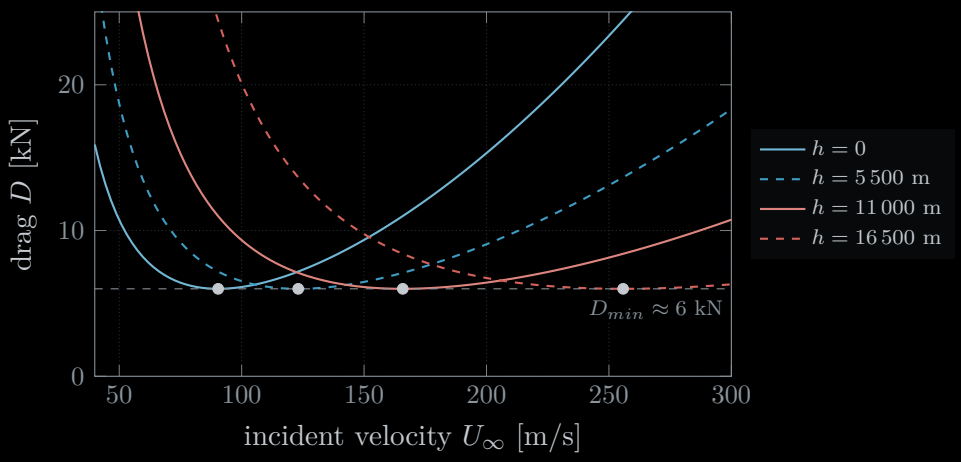


Figure 7.5. Drag of the aircraft of the document ($m = 10\,000$ kg, $AR = 8$, $S = 30$ m^2, $e = 0.85$, $C_{D0} = 0.02$) at four altitudes in ISA. The dots mark U_{md}: the minimum drag is the same, but it moves to higher speed with altitude. At 16 500 m the speed (≈ 256 m/s, $M \approx 0.87$) is already outside the subsonic validity of the model. Recomputed from the document's equation.

Wind. The incident velocity is relative to the air: $\vec{U}_\infty = \vec{U}_{wind} - \vec{U}_{aircraft}$ (vectors relative to the ground), so $\vec{U}_{aircraft} = \vec{U}_{wind} - \vec{U}_\infty$. With the same drag and the same airspeed, a **tailwind increases the ground speed** and a headwind reduces it. Jet streams are used this way: in 1952 a Boeing 377 Stratocruiser flew Tokyo–Honolulu in 11 h 30 min using the jet stream at 7 600 m (6.5 hours less, no stop at Wake Island); a few months later another flight took 9 h 18 min.

Cruise speed in practice. Aircraft do not necessarily fly at the minimum-drag speed (the mission, not the drag, is optimised), but **the cruise speed is usually slightly above it**, with drag only slightly above the minimum. As fuel is burnt, the weight falls and so does $U_{md} \propto \sqrt{W/\rho}$: to keep flying at the minimum-drag speed the **altitude has to change** (climb, so that ρ decreases). For instance, a 10 % weight loss lowers U_{md} by $1 - \sqrt{0.9} = 5\,\%$.

> **WHAT TO REMEMBER: DRAG AND VELOCITY**
> $D = D_0 + D_i$ with $D_0 \uparrow$ and $D_i \downarrow$ as speed grows; minimum at $D_0 = D_i$. $U_{md} \propto \sqrt{W/(\rho S)}$ (higher with altitude and wing loading) and $(L/D)_{max} = \frac{1}{2}\sqrt{\pi AR\, e/C_{D0}}$ does *not* change with altitude. Same speed gives maximum L/D and minimum drag. Tailwind: higher ground speed for the same drag.

1.5 Ice, wind gusts and storm clouds

Document: "Ice formation, wind gusts and storm clouds". Background: Unit 2 (clouds, winds, icing; Figure 2.9).

In the troposphere the temperature falls 6.5 K per kilometre up to 11 km and then stays constant (tropopause, up to 20 km). Flying in the troposphere, sometimes inside clouds, brings three problems (Figure 7.6):

- **Ice formation** (low temperature and some humidity): the water freezes on the leading edges (wing, fuselage, engines). The ice **changes the shape** of the surfaces, which causes flow **separation**, **loss of lift**, **more drag** and **loss of engine performance**. Remedies: avoid medium altitudes where it is frequent, and use **anti-ice/de-ice systems** (Figure 2.9). Near the tropopause the humidity is low and ice is reduced.
- **Gusts and turbulence**: they change the **lift, drag and moments quickly**; they disturb passengers, can cause **loss of control** and **high structural loads**. Important at intermediate

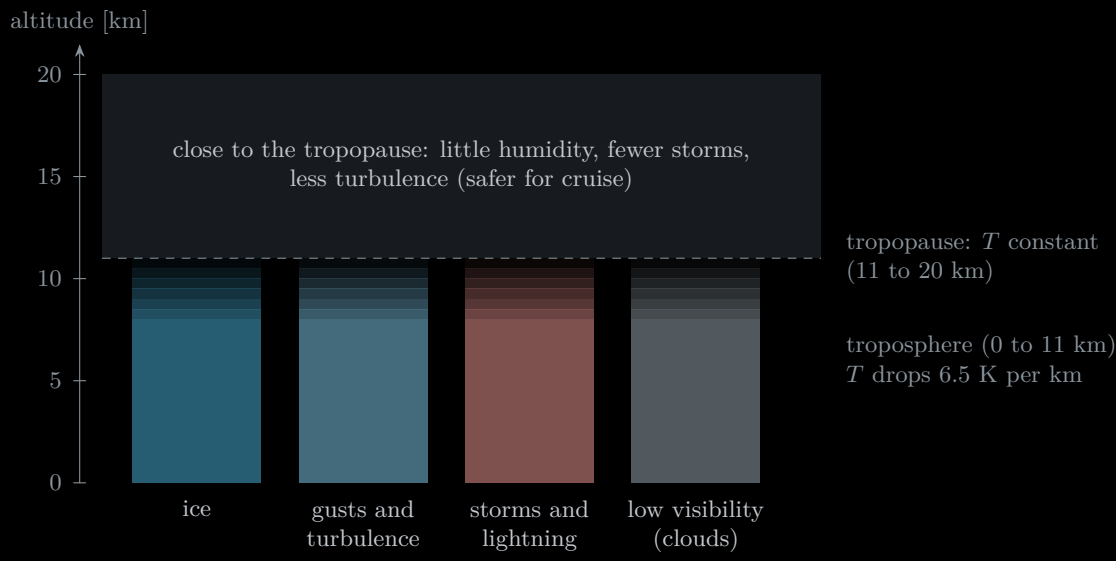


Figure 7.6. Qualitative sketch of the document: ice, gusts and turbulence, storms and low visibility are worst at medium altitudes (strong colour) and fade towards the tropopause. The heights are indicative, not data.

altitudes, smaller near the tropopause.

- **Clouds and storms**: at high altitude low temperature and lots of water give **large amounts of ice**; storms bring strong gusts and **lightning**. An aircraft tolerates an isolated lightning strike but *many* can damage it. Clouds limit visibility, so the flight must be carried out in **instrument mode**. **Storms must be avoided**; they are unusual at high altitude.

> **WHAT TO REMEMBER: WEATHER**
> Ice ⇒ separation, lift loss, drag, engine loss. Gusts ⇒ fast changes of L, D, M and structural loads. Storms ⇒ ice, gusts, lightning, no visibility. The common answer to "why cruise near the tropopause?" is that **ice, turbulence and storms are much less frequent there**.

1.6 Quick estimation of the range: the simplified Breguet equation

Document: "Quick estimation of the range of an aircraft". Background: Section 6.2 of Unit 3. The review questions of this part treat the Breguet equation again; here only the document is summarised.

The range depends on weather, payload, fuel, flight programme (for instance climbing as fuel is burnt) and configuration. A **quick estimate** uses the simplified Breguet equation at the average conditions of cruise. Note that the document writes the specific consumption **per unit of thrust and time (mass of fuel)**, c_e, whereas Unit 3 writes $c_{sp} = g\,c_e$ (per unit of weight, in 1/s); the equations are the same.

Jet aircraft.

- Definition: $c_e = -\dfrac{1}{T}\dfrac{dm_f}{dt}$ (kg of fuel per newton and second).
- Level flight at constant speed: $T = D$ and $dW/dt = g\,dm_f/dt = -g\,c_e\,D$.
- Hence $dt = -\dfrac{1}{g\,c_e}\dfrac{L}{D}\dfrac{dW}{W}$ (using $D = W\,D/L$); multiplying by V_∞ and integrating from the initial weight W_0 to the final weight W_f gives $R = \displaystyle\int \frac{V_\infty}{g\,c_e}\frac{C_L}{C_D}\frac{dW}{W}$.

- Taking the speed, consumption and efficiency at the **middle of the cruise** (half the fuel burnt) as constants, and the limits MTOW and MTOW−FW (all the fuel FW is burnt):

$$R_{jet} = \frac{V_\infty}{g\, c_e} \left(\frac{C_L}{C_D}\right)_{avg} \ln \frac{\text{MTOW}}{\text{MTOW} - \text{FW}}$$

Several engines give the same c_e as one (consumption and thrust both grow). Typical values: $c_e \approx 5$ to $20\ \text{g}\,\text{kN}^{-1}\text{s}^{-1}$ ($= 5 \cdot 10^{-6}$ to $2 \cdot 10^{-5}$ kg/(N s), about 0.18 to 0.71 lb/(lbf h)).

Propeller aircraft (piston or turboprop). Now $c_e = -\dfrac{1}{P}\dfrac{dm_f}{dt}$ per unit **power**, and the power needed is $P = D\,V_\infty/\eta_p$, where η_p is the propeller efficiency (the speed cancels):

$$R_{prop} = \frac{\eta_p}{g\, c_e} \left(\frac{C_L}{C_D}\right)_{avg} \ln \frac{\text{MTOW}}{\text{MTOW} - \text{FW}}$$

Typical: $c_e \approx 250$ to $300\ \text{g}\,\text{kW}^{-1}\text{h}^{-1}$ ($7 \cdot 10^{-8}$ to $8 \cdot 10^{-8}$ kg/J) and $\eta_p \approx 0.8$.

Limits and cautions. It uses *average* values (efficiency, consumption, speed, and η_p do change as the weight falls), assumes the whole fuel is burnt (no reserves), no wind, and **units must be consistent** (SI: V in m/s, c_e in kg/(N s) or kg/J, R in m). Departing from the best-range conditions penalises the range in ways that are hard to estimate.

RANGE OF A JET AND OF A PROPELLER AIRCRAFT (OWN ILLUSTRATIVE NUMBERS)

Jet: $V_\infty = 230$ m/s, $L/D = 17$, $c_e = 15\ \text{g}\,\text{kN}^{-1}\text{s}^{-1} = 1.5 \cdot 10^{-5}$ kg/(N s), MTOW $= 78\,000$ kg, FW $= 18\,000$ kg.

$$R = \frac{230}{9.81 \cdot 1.5 \cdot 10^{-5}} \cdot 17 \cdot \ln \frac{78}{60} = 1.563 \cdot 10^6 \cdot 17 \cdot 0.2624 = 6.97 \cdot 10^6\ \text{m} \approx 6\,970\ \text{km}$$

Propeller: $\eta_p = 0.8$, $L/D = 12$, $c_e = 270$ g/(kW h)$= \dfrac{0.270}{3.6 \cdot 10^6} = 7.5 \cdot 10^{-8}$ kg/J, MTOW $= 3\,200$ kg, FW $= 600$ kg.

$$R = \frac{0.8}{9.81 \cdot 7.5 \cdot 10^{-8}} \cdot 12 \cdot \ln \frac{3200}{2600} = 1.087 \cdot 10^6 \cdot 12 \cdot 0.2076 = 2.71 \cdot 10^6\ \text{m} \approx 2\,700\ \text{km}$$

Because of the logarithm, doubling the fuel does *not* double the range.

WHAT TO REMEMBER: SIMPLIFIED BREGUET

Range = (speed or η_p)$/(g\, c_e) \times L/D \times \ln(\text{MTOW}/(\text{MTOW} - \text{FW}))$, with average mid-cruise values. Jet: proportional to speed; propeller: speed does not appear but η_p does and c_e is per unit power. Several engines do not change c_e.

1.7 The Spanish aerospace industry

Document: "Spain: Business opportunities in aerospace" (ICEX–Invest in Spain, Madrid, 2016, 26 slides; data of 2015, source TEDAE). The numbers below are those of the slides; keep in mind that they are from 2015.

Market figures (slide 4).

- **Fifth in Europe** by turnover (**9 700 million euros** in 2015) and by employment (**more than 54 400** employees).
- **Exports: 84 %** of the turnover. It represents **0.7 %** of the Spanish GDP and **5.8 %** of the industrial GDP.
- Growth of **3.2 %** in 2015 and **almost 100 %** since 2006 (4 607 M euros in 2006; Figure 7.7, left).



Figure 7.7. Left: turnover of the Spanish aerospace and defence sector, 2006–2015 (millions of euros). Right: share of turnover and employment by region. Data from the slides (source: TEDAE).

Clusters (slide 5). **Madrid** 49.4 % of turnover and 42.7 % of employees; **Andalusia** 21.7 % and 28.6 %; **Castilla-La Mancha** 12 % and 9.9 %; **Basque Country** 9.9 % and 10.6 %; **Catalonia** 1.4 % and 2.2 % (Figure 7.7, right).

Industrial infrastructures (slides 6 and 7).

- **Aerópolis** (Seville): Andalusian aerospace technology park, 47 companies and the advanced centre CATEC.
- **Albacete Aeronautics and Logistics Park**: 250 000 m^2, specialised in engines and **helicopters** (Eurocopter plant, ITH maintenance of the ITP group).
- **PLATA** (Teruel, Aragon): airport platform for long-term parking, recycling and maintenance of aircraft; 340 ha, able to operate 747-type aircraft.
- **Delta BCN** (Catalonia): 55 ha, 245 000 m^2 built; wind tunnel, excellence centres on aerodynamics and pilotless aircraft, CTAE laboratories.

Main players (slides 8 and 9). Companies in the whole chain (prime contractors, OEMs, Tier 1, auxiliary industry, maintenance): **335** companies below 250 employees (SMEs), **8** between 250 and 1 000 and **7** between 1 000 and 10 000. Ranking by turnover: **Airbus Operations** (1 299 M euros), **ITP**, Industria de Turbo Propulsores (594), **Aernnova** (374), **Airbus Helicopters España** (266), **Aciturri Composites** (184), then CESA, Internacional de Composites, Aciturri Metallic Parts, Aernnova Composites, Aerometallic Components and CRIsA.

Strengths and drivers (slides 11 to 15).

- Spain leads in **composite aerostructures**, **low-pressure turbines**, **air traffic management** (“3 out of 5 flights in the world use Spanish software for landing”), **military trans-**

port aircraft and medium-capability **space** systems (electronics, telecommunications, mission analysis, instrumentation, satellite navigation).

- **R&D**: 970 M euros in 2015 (11 % of the turnover in aeronautics, 12 % in space), public research centres and a foundation for composites.
- **Airbus** (more than 3 250 employees): **Illescas** (Toledo; composites excellence centre, fuselage sections, landing gear parts, section 19 of the A350 XWB), **Getafe** (A380 horizontal tail plane designed here; assembly and testing of the horizontal tail planes of the A350 XWB, A330 and A320), **Puerto Real** (Cádiz; A380 tail plane final assembly, parts for A350 XWB and A330).
- **Airbus Defence and Space** (7 700 employees): **Seville** (final assembly line of the **A400M** and of the CN235 and C295), **Getafe** (Eurofighter final assembly for Spain and starboard wing for all Eurofighters; conversion of the A330 into the MRTT tanker).
- **Airbus Helicopters** (more than 500): Albacete and Getafe (Tiger, NH90, EC135 assembly).
- **Boeing Research & Technology Europe** (Madrid, 2002): Boeing's first R&D centre outside the US (air traffic management, unmanned systems, safety, environment).

Governmental support (slides 16 to 22). European incentives, the **Aerospace Sector Competitiveness Programme** (minimum budget 75 000 euros, up to 75 % financed, 10 years with a 3-year grace period), **CDTI R&D projects** (12–36 months, minimum 175 000 euros; national cooperation: 2 to 6 companies, 500 000 euros), **FEDER-Innterconecta** (consortia of at least 3 independent companies, two years), **ICEX–Invest in Spain Technological Fund** (up to 80 % of eligible investment for small companies in industrial research, 60 % in experimental development) and **ICO** loans (up to 12.5 M euros per company).

People (slide 23). Qualified employees: **47 %** highly qualified technicians, **40 %** engineers and university graduates, 13 % other technical profiles. Productivity (turnover per employee) rose from 152 to 188 thousand euros between 2009 and 2014 (+38 % since 2005).

Business opportunities (slide 25). Navigation and avionic systems; helicopters (engineering, electronics, maintenance); engines and propulsion engineering; pilot and maintenance **training centres**; **MRO** centres; low-cost flight simulators; aviation safety and human factors; **UAVs**; airport services (aircraft coating, end-of-life dismantling); **ATM**; aerostructures and composites; airport and tower ATC liberalisation.

> WHAT TO REMEMBER: SPANISH INDUSTRY
> 5th in Europe; **9 700 M euros**, **54 400 jobs**, **84 % exports**, 5.8 % of industrial GDP; clusters Madrid > Andalusia > Castilla-La Mancha > Basque Country > Catalonia; Airbus Operations and ITP lead the ranking; strengths are **composites, low-pressure turbines, ATM, military transport**; Airbus sites: Illescas, Getafe, Puerto Real, Seville (A400M).

1.8 Aircraft characterisation problem

Document: "Aircraft characterisation problem" (Pau Varela Martínez, 2026–2027). It is an assignment: each group takes one aircraft and answers six questions. Background: Units 2, 3 and 5 and the subsection on the simplified Breguet equation above.

Where to get the aircraft data. The document does not give the data of every aircraft; it only lists, as an example, the data of the **Cessna Caravan 208** (Section 2 of the document: MTOM 3 629 kg, OEM 2 100 kg, MPL 1 360 kg, MFM 1 009 kg, MLM 3 538 kg, cruise 175 kt TAS, ceiling

25 000 ft, propeller diameter 2.69 m, 675 SHP). For your own aircraft look for the **manufacturer's data sheet or airport-planning manual**, the **type-certificate data sheet** (EASA/FAA), the flight manual, and the **3-view drawings**; the groups must state the source of every number. Do not invent values: where something is not published (for instance C_{D0} or e), **estimate it with the method of the units and state the assumption**.

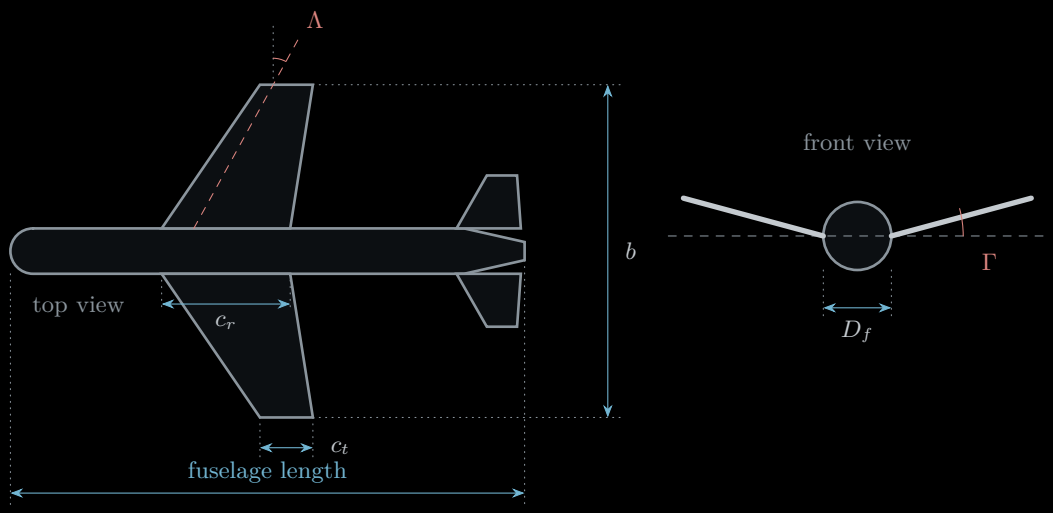


Figure 7.8. What to measure on the 3-view drawings of the assignment (generic aircraft, not the Caravan): top view with fuselage length, span b, root and tip chords and the sweep Λ of the quarter-chord line; front view with fuselage diameter D_f and dihedral Γ. The side view gives the heights. Own sketch.

The six questions and how to answer them.

- **Basic specifications**: (a) weights MTOM, OEM, MPL, MFM, MLM and, for each mission, payload PL and fuel mass FM; (b) cruise speed; (c) cruise altitude; (d) powerplant (kind, manufacturer, take-off thrust or power, specific fuel consumption). *Guide*: the definitions are in Unit 3. The consistency rules are MTOM $\geq$ OEM+PL+FM, PL $\leq$ MPL, FM $\leq$ MFM, MLM $\leq$ MTOM. Analyse **three missions**: maximum payload (fuel limited to MTOM $-$ OEM $-$ MPL), maximum fuel (payload limited to MTOM $-$ OEM $-$ MFM) and, if wished, a ferry flight (no payload). Convert units: 1 kt = 0.5144 m/s, 1 ft = 0.3048 m, 1 SHP $\approx$ 745.7 W.
- **Atmospheric conditions** (temperature, pressure, density) in four cases, **stating the hypotheses**: (a) typical ISA cruise; (b) VLC–MAD cruise on 18 October (use a representative altitude if cruise altitude is not reached); (c) take-off from VLC on 3 August; (d) landing in ANC on 29 November. *Guide*: ISA formulas of Unit 2: $T = 288.15 - 0.0065\,h$ (K, h in m, up to 11 km), $p = p_0\,(T/288.15)^{5.2559}$ with $p_0 = 101\,325$ Pa, $\rho = p/(RT)$, $R = 287.05$ J/(kg K). For (b)–(d): airport elevation and the **actual temperature** of that day and place from a meteorological source (METAR, climate table: the typical value of that month), then $T = T_{ISA} + \Delta T$ and $\rho = p/(RT)$. Hypotheses to state: standard pressure setting (pressure altitude), dry air, ideal gas, typical temperature instead of the real one.
- **Geometry from the 3 views** (at least two dimensions per view, Figure 7.8): fuselage length and diameter, S, b, geometric chord, MAC, AR, taper, sweep at $c/4$ and $c/2$, dihedral, and wetted surfaces of wing, fuselage, engines and stabilisers, each simplified. *Guide*: $\bar{c} = S/b$, $AR = b^2/S = b/\bar{c}$, $\lambda = c_t/c_r$, $S = \frac{1}{2}(c_r + c_t)b$ (trapezoid), MAC as in the subsection on the aerodynamic centre above, sweep from $\tan\Lambda = \Delta x/(b/2)$ measured between the root and tip points at $c/4$ (or $c/2$), dihedral from the front view. Wetted area: wing $S_{wet} \approx 2S(1 + 0.2\,t/c)$ (about $2.04S$ for $t/c = 0.12$, as in the Unit 5 example), fuselage as a cylinder $\pi D_f L_f$ (a bit smaller in practice because of the tapered nose and tail), engine nacelles as cylinders, stabilisers as thin plates $\approx 2S_{surface}$.

- **Cruise aerodynamics**: estimate the parameters, plot the drag polar $C_D(C_L)$ and the efficiency C_L/C_D versus flight speed and altitude, and **locate on the figures**: (a) the nominal cruise altitude and speed; (b) altitude ±20 % at nominal speed; (c) ISA−10 and ISA+10 at nominal speed; (d) speed ±20 % at nominal altitude. If one of these conditions is impossible for the aircraft, replace it **with a justification**. *Guide*: $C_L = \dfrac{2W}{\rho V^2 S}$ (with $W = mg$ at the weight of the cruise), $C_D = C_{D0} + \dfrac{C_L^2}{\pi A R\, e}$, $L/D = C_L/C_D$. C_{D0} by the component build-up of Unit 5 (friction coefficient × form factor × S_{wet}/S, adding all elements) and e as an assumed value (0.7 to 0.85 for conventional aircraft; **state it**). The altitude changes ρ (ISA), ISA±10 changes $\rho = p/(RT)$ at the same pressure altitude, and speeds ±20 % change C_L by $1/V^2$. Check that each point is possible: $C_L \le C_{L,max}$ (stall), altitude below the ceiling and Mach number below the limit. The formulas for $(L/D)_{max}$ and V_{md} of the drag subsection above help to check the curves.
- **Thrust or power required in cruise** for the conditions of Question 4, compared with each other and with the take-off thrust or power. *Guide*: in level cruise $T = D = W/(L/D)$ (jet). For a propeller engine the power required is $P = D\,V/\eta_p = \dfrac{W\,V}{(L/D)\,\eta_p}$ (propeller efficiency typically about 0.8). Present the table of required values and divide by the take-off value: it gives the fraction of power in cruise. Remark: the available thrust or power falls with altitude, so the point must also be possible from the engine side.
- **Breguet range**: identify the assumptions, compute the typical range and compare it with the specifications. *Guide*: use the simplified equation of Section 1.6, propeller version for the Caravan: $R = \dfrac{\eta_p}{g\,c_e}\dfrac{L}{D}\ln\dfrac{W_i}{W_f}$, with W_i = MTOM and W_f = MTOM − FM (use the maximum fuel mission). Assumptions to list: constant average V, L/D, c_e and η_p; the whole fuel burnt; no wind; climb, descent, take-off, reserves and the weight change of the aircraft neglected. The published range is usually shorter than the estimate because of reserves and non-cruise phases; explain the difference.

TEMPLATE WITH THE DATA GIVEN IN THE DOCUMENT (CESSNA CARAVAN 208)

Question 1. MTOM − OEM = 3 629 − 2 100 = 1 529 kg is shared between payload and fuel. *Maximum payload*: PL = 1 360 kg, fuel limited to 1 529 − 1 360 = 169 kg. *Maximum fuel*: FM = 1 009 kg, payload limited to 1 529 − 1 009 = 520 kg. Landing mass limit 3 538 kg (91 kg below MTOM). Cruise 175 kt = 90.0 m/s (324 km/h); ceiling 25 000 ft = 7 620 m; engine 675 SHP = 503 kW.

Question 2 (example at 10 000 ft, an assumed illustrative cruise altitude: the document does not give the cruise altitude). $h = 3\,048$ m: $T = 288.15 - 0.0065 \cdot 3048 = 268.3$ K, $p = 101\,325\,(268.3/288.15)^{5.2559} = 69\,680$ Pa, $\rho = \dfrac{69\,680}{287.05 \cdot 268.3} = 0.905$ kg/m^3. On an ISA+15 day: $T = 283.3$ K, $\rho = \dfrac{69\,680}{287.05 \cdot 283.3} = 0.857$ kg/m^3 (5 % less).

Questions 4 and 5 (chain of formulas). $W = 3\,629 \cdot 9.81 = 35\,600$ N at MTOW. Dynamic pressure $q = \frac{1}{2}\rho V^2 = \frac{1}{2} \cdot 0.905 \cdot 90.0^2 = 3\,670$ Pa. Then $C_L = W/(qS)$ (with the S from your 3 views), $C_D = C_{D0} + C_L^2/(\pi A R\, e)$, $D = C_D\, q\, S$ and $P_{req} = D\,V/\eta_p$ to be compared with 503 kW. For a constant altitude, changing the speed ±20 % multiplies q by 0.64 and 1.44.

Question 6. $\ln\dfrac{3629}{3629 - 1009} = \ln 1.385 = 0.326$; $R = \dfrac{\eta_p}{g\,c_e}\dfrac{L}{D} \cdot 0.326$ with c_e and η_p from the engine and propeller data and L/D from Question 4.

WHAT TO REMEMBER FROM THE ADDITIONAL DOCUMENTS

- **Structures**: Pratt, Warren, geodesic (trusses), monocoque, semimonocoque (shells); the semimonocoque dominates because stringers and frames are a lighter cure for buckling than a thick skin. Wing: ribs, spars, torsion box with the fuel.
- **Wing position**: low wing = ground effect, roll instability (dihedral), cabin clear; high wing = stable (anhedral), short gear, little ground effect; mid wing = lowest interference drag. Monoplanes win through higher aspect ratio, less interference drag and more $C_{L,max}$.
- **CP, AC, MAC**: CP moves, AC is fixed at about $c/4$ with constant $c_{m,ac}$, MAC $= \frac{1}{S}\int c^2 dy$ with a graphical construction.
- **Drag**: $D = D_0 + D_i$, minimum at $D_0 = D_i$, $U_{md} \propto \sqrt{W/(\rho S)}$, $(L/D)_{max} = \frac{1}{2}\sqrt{\pi AR\, e/C_{D0}}$ independent of altitude; tailwind raises ground speed.
- **Weather**: ice, gusts and storms are worst at medium altitude; the tropopause is calmer.
- **Breguet**: $R = \frac{V}{g\, c_e}\frac{L}{D}\ln\frac{\text{MTOW}}{\text{MTOW}-\text{FW}}$ (jet) or $\frac{\eta_p}{g\, c_e}\frac{L}{D}\ln(\cdot)$ (propeller), with mid-cruise averages and consistent units.
- **Spain**: 5th in Europe, 9 700 M euros, 54 400 jobs, 84 % exports; composites, low-pressure turbines, ATM, military transport.
- **Problem**: weights, ISA data, 3-view geometry, polar and L/D, T or P required, Breguet range: each answer uses formulas of the units and states its assumptions and the source of its data.

2 Review questions: environment, aerodynamics and performance

Each subsection answers one review question. The answers reuse the earlier units (they are cross-referenced instead of repeated), but each one is written to stand on its own. Where an answer goes beyond the slides, a box says so.

2.1 ICAO, IATA, EASA, AESA and DGAC

What are the differences between ICAO, IATA, EASA, AESA and DGAC?

The five names are easy to mix up because they all work in civil aviation, but they sit at different levels and do different jobs: one is a global regulator-coordinator, one is an airline trade association, one is the European safety agency and two are Spanish national bodies (see also Unit 1 and Figure 1.2).

Body	What it is	Scope	Main role
ICAO	*International Civil Aviation Organization*, specialized agency of the UN (Chicago Convention, 1944; seat in Montreal)	world, member states	agrees the international standards and recommended practices (SARPs, published as Annexes) that states then transpose into their own law; it does not inspect or fine airlines
IATA	*International Air Transport Association*, trade association of airlines	world, member airlines	industry cooperation: commercial standards (tickets, baggage, cargo), airline and airport codes, safety audits for members; **not** a legislator
EASA	*European Union Aviation Safety Agency*	European Union (plus associated states)	common European safety rules; certification of aircraft types and of organizations; the European counterpart of the US FAA
AESA	*Agencia Estatal de Seguridad Aérea*	Spain	Spanish national aviation authority: applies the EASA and national rules; licences, aircraft registration and airworthiness, operator and drone oversight, inspections and sanctions
DGAC	*Dirección General de Aviación Civil*	Spain (ministry)	policy level of the ministry responsible for transport: proposes national regulation and aviation policy (the same acronym is used by other countries' civil aviation directorates)

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES): THE KEY DIFFERENCES

- **Who makes binding rules?** ICAO sets international standards (states are bound to follow them or notify differences), EASA writes the rules that apply in the EU, and AESA enforces them in Spain. IATA has no legal power: its rules apply because airlines agree to them.
- **Level**: global (ICAO, IATA) → European (EASA) → national (AESA, DGAC).
- **Who is who in Spain?** DGAC is the policy-making directorate; AESA is the agency that supervises, certifies and sanctions.

2.2 Navigation aids: principles and uses

What are the working principles of the barometric altimeter, the radio altimeter, the NDB, the VOR, the ILS, the GNSS and the GNSS with GBAS? What are their differences and when is each one used?

Section 5 of Unit 2 classifies the aids into *in cruise* and *in approach and landing*. Here each principle is explained a little further, and Figure 7.9 contrasts the two altimeters, which are the ones most often confused.

- **Barometric altimeter.** An aneroid capsule senses the static pressure p and a mechanism converts it to altitude with the ISA relation $p(h)$ (Unit 2). It is passive and self-contained. If the day is not standard, or the sea-level pressure setting is wrong, the reading is off ("miscalibrated"). Used in all phases; in cruise all aircraft set the same standard reference, so altitudes become *flight levels* and separation is consistent.
- **Radio altimeter.** A downward-pointing radar sends a radio wave and measures the echo delay Δt: $h = c\,\Delta t/2$. It gives the height *above the ground*, with metre-level accuracy, but

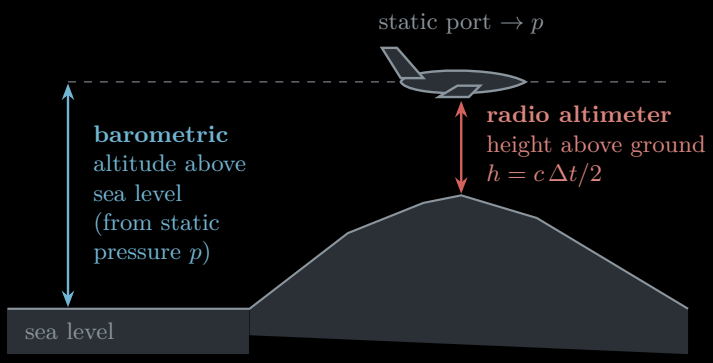


Figure 7.9. The two altimeters. The barometric one reports the altitude above a pressure reference (sea level), so over a hill it does not change; the radio altimeter measures the height above whatever is directly below. Schematic.

only up to a few thousand feet. Used in the final approach, the flare and automatic landing, and by the terrain-warning systems.

- **NDB.** A ground transmitter in the low/medium-frequency band radiates in all directions; the onboard receiver (ADF) has a directional antenna that points towards the station, so it indicates the *direction* to it. Simple but of low accuracy (it is affected by weather, night effects and terrain); used as an older en-route aid or for non-precision approaches.
- **VOR.** A VHF ground station emits a signal that carries a reference phase and a second signal whose phase depends on the direction in which it is emitted (as if a rotating beam). The receiver compares both phases and obtains the *radial* (bearing from the station) on which the aircraft is. Accuracy of about a degree. Combined with a DME (distance) it fixes the position (Figure 2.11). Used on the airways network and in approaches.
- **ILS.** Two VHF/UHF radio beams per runway, each formed by two lobes modulated at different tones: the receiver compares the strength of the tones and knows on which side of the centreline (localizer) or of the glide path (glide slope) it is. It gives lateral and vertical *guidance* (Figure 2.12), so it supports precision approaches in very low visibility, but it only serves the runway where it is installed.
- **GNSS.** Each satellite broadcasts its position and the transmission time. The receiver measures the signal delay to several satellites, obtains the *pseudoranges*

$$\rho_i = \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} + c\,\delta t, \qquad i = 1, 2, \ldots$$

and solves for the position (x, y, z) and its own clock error δt: four unknowns, hence at least four satellites. Worldwide coverage with no ground equipment at the airport; accuracy of a few metres, but with no guarantee of integrity and with sensitivity to interference.

- **GNSS with GBAS** (the ground-based augmentation, also called LAAS in the slides). A ground station at the airport has receivers at *surveyed* positions, so it can compute the error of each pseudorange, and broadcasts the corrections by a VHF data link to the aircraft nearby, which applies them (Figure 7.10). Errors common to both (atmosphere, satellite clock and orbit) cancel, and the system also monitors integrity. Result: sub-metre accuracy, enough for precision approaches, with one station serving several runways.

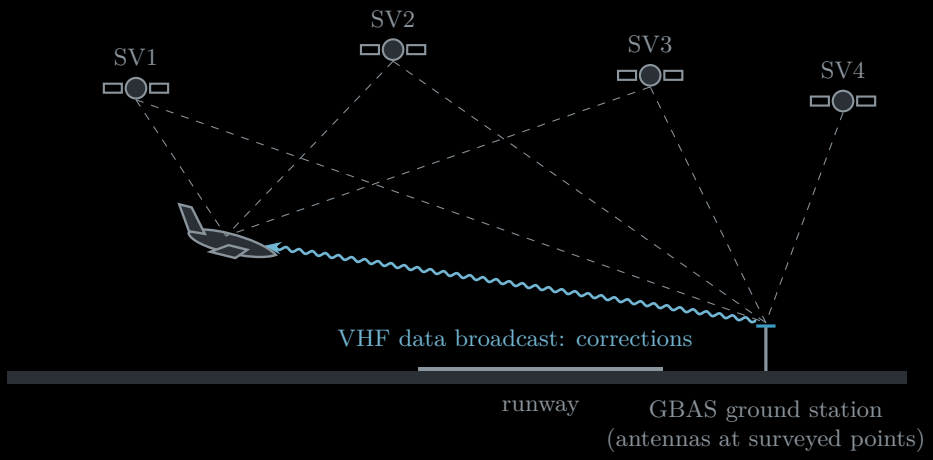


Figure 7.10. GNSS with GBAS: the satellites' signals reach both the aircraft and the ground station, which knows its own exact position, derives corrections and broadcasts them by VHF. Schematic.

Aid	Gives	Typical accuracy	When it is used
Barometric altimeter	altitude above a pressure reference	tens of metres, depends on the setting	all phases; vertical separation (flight levels)
Radio altimeter	height above ground	about 1 m, up to ∼2 500 ft	final approach, flare, autoland, terrain warning
NDB	direction to a station	a few degrees	older en-route and non-precision approaches
VOR (+DME)	radial (and distance with DME)	about 1°	airways, non-precision approaches
ILS	lateral and vertical guidance	angular, very precise near the runway	precision approach, even with very low visibility
GNSS	3D position and time	few metres, no integrity guarantee	en-route and approaches with augmentation
GNSS + GBAS	corrected position with integrity	sub-metre	precision approaches without an ILS per runway

The accuracy figures are typical orders of magnitude, not slide data.

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES): HOW TO CHOOSE

- **Ground-referenced vs pressure-referenced**: only the radio altimeter knows the height above the terrain; the barometric one knows the altitude above a pressure level.
- **Direction/position aids** (NDB, VOR) need the aircraft to be within radio line of sight of ground stations; GNSS does not.
- **Precision approach**: ILS (the established solution) or GNSS with GBAS; plain GNSS lacks the integrity required. In very low visibility the PAPI (visual) is useless but these aids still work.

2.3 Kinetic heating of A350, Concorde and SR-71

What temperature is reached due to kinetic heating (the main cause of which is not friction) on the stagnant air in contact with the surface of an A350 in cruise flight? And a Concorde? And an SR-71?

Why it is *not* mainly friction: the air next to the skin is brought to rest relative to the aircraft, and the kinetic energy per unit mass it had, $V^2/2$, is converted into internal energy (it appears as a temperature rise $V^2/2c_p$). This is the adiabatic compression of Section 6 of Unit 2. Friction (viscous dissipation) is only the mechanism by which part of the boundary-layer air comes to rest; the energy comes from the kinetic energy of the flow, not from rubbing. The stagnation temperature is therefore

$$T_0 = T\left(1 + \frac{\gamma - 1}{2} M^2\right) = T\,(1 + 0.2M^2), \qquad \gamma = 1.4$$

with T the static temperature of the ISA at the flight altitude: 216.65 K from 11 to 20 km, and $216.65 + 1 \cdot (h - 20)$ K above 20 km (so 220.65 K at 24 km). The cruise conditions are approximate.

STAGNATION TEMPERATURE OF THREE AIRCRAFT

A350 ($M \approx 0.85$, 11 to 12 km): $T_0 = 216.65\,(1 + 0.2 \cdot 0.85^2) = 216.65 \cdot 1.1445 = 248.0$ K.
Concorde ($M \approx 2.02$, about 17 km): $T_0 = 216.65\,(1 + 0.2 \cdot 2.02^2) = 216.65 \cdot 1.816 = 393.5$ K.
SR-71 ($M \approx 3.2$, about 24 km): $T_0 = 220.65\,(1 + 0.2 \cdot 3.2^2) = 220.65 \cdot 3.048 = 672.5$ K.

	M	Altitude (km)	T (ISA) (K)	T_0 (K)	T_0 (°C)	T_{aw}, $r = 0.89$ (°C)
A350	0.85	11–12	216.65	248.0	−25	−29
Concorde	2.02	≈17	216.65	393.5	120	101
SR-71	3.2	≈24	220.65	672.5	399	350

The ambient temperature at those altitudes is −56.5 °C for the first two (216.65 K) and −52.5 °C for the SR-71 (220.65 K), so the rise is about 31 K, 177 K and 452 K respectively.

REAL SURFACE TEMPERATURE

T_0 is the *ideal* value for air brought to rest without losses. The temperature the skin actually reaches (adiabatic wall temperature) is somewhat lower: $T_{aw} = T\,(1 + r\,0.2M^2)$ with a *recovery factor* $r \approx \sqrt{Pr} \approx 0.85$ for a laminar and $\approx Pr^{1/3} \approx 0.89$ for a turbulent boundary layer (typical values; the last column of the table uses 0.89). The skin also radiates and conducts heat into the structure, so it is lower still in practice. The trend is what matters: negligible for the airliner (that is why aluminium is enough), serious for Concorde (aluminium alloys limit it to Mach 2) and extreme for the SR-71 (titanium structure). Compare Figure 2.14.

2.4 Breguet range equation for jets and propellers

Develop Breguet's range equation for jet propulsion and for propeller propulsion. In which cases do these expressions depend directly on speed and in which they do not?

The idea (Section 6.2 of Unit 3 gives the result; here it is derived, following the course note on range estimation) is that the aircraft gets lighter as it burns fuel, so the range is obtained by integrating over the weight.

Assumptions: steady, level flight, no wind, so $L = W$ and $T = D$; the only change in weight is the fuel burned, $\mathrm{d}W = -g\,\mathrm{d}m_f$.

Jet propulsion

The thrust-specific fuel consumption is $c_T = \dot{m}_f/T$ [kg/(N s)] (the quantity called c_e in Section 1.6, with $E = L/D = C_L/C_D$ here). Then

$$\frac{\mathrm{d}W}{\mathrm{d}t} = -g\,\dot{m}_f = -g\,c_T\,T = -g\,c_T\,D = -g\,c_T\,\frac{W}{E}, \qquad E = \frac{L}{D} = \frac{C_L}{C_D}.$$

The distance flown is $\mathrm{d}x = V\,\mathrm{d}t$, so

$$\mathrm{d}x = V\,\mathrm{d}t = -\frac{V\,E}{g\,c_T}\,\frac{\mathrm{d}W}{W} \quad\Longrightarrow\quad R = \int_{W_f}^{W_0} \frac{V\,E}{g\,c_T}\,\frac{\mathrm{d}W}{W}.$$

If V, E and c_T are constant (or replaced by average values):

$$R_{jet} = \frac{V}{g\,c_T}\,E\,\ln\frac{W_0}{W_f} \qquad \left(\text{with } c_{sp} = g\,c_T:\ R = \frac{V}{c_{sp}}\,\frac{L}{D}\,\ln\frac{W_0}{W_f}\right)$$

Propeller propulsion

The consumption is now referred to the *shaft* (brake) power: $c_P = \dot{m}_f/P$ [kg/J]. The useful power is the thrust power $TV = DV = \eta_p P$ with η_p the propeller efficiency, so $P = DV/\eta_p$:

$$\frac{\mathrm{d}W}{\mathrm{d}t} = -g\,c_P\,P = -g\,c_P\,\frac{D\,V}{\eta_p} = -g\,c_P\,\frac{W\,V}{\eta_p\,E} \;\Longrightarrow\; \mathrm{d}x = V\,\mathrm{d}t = -\frac{\eta_p\,E}{g\,c_P}\,\frac{\mathrm{d}W}{W}.$$

The speed has cancelled. With constant values:

$$R_{prop} = \frac{\eta_p}{g\,c_P}\,E\,\ln\frac{W_0}{W_f}$$

When does it depend directly on speed?

- **Jet, with c_T constant**: yes, $R \propto V$. The fuel burned per second depends on the thrust (that is, on D), not on the speed, so flying faster covers more distance for the same fuel.
- **Propeller, with c_P and η_p constant**: no, V does not appear. The fuel flow is proportional to the power DV, which grows with speed exactly as much as the distance covered per second, so the distance per kilogram of fuel depends only on D (that is, on E).
- **Both are indirect in any case**: E depends on speed through C_L (with $L = W$, $C_L = 2W/\rho V^2 S$, so each speed fixes an E; the maximum of E is derived in Section 2.8), c_T changes with Mach number and altitude, and η_p depends on the advance ratio $V/(nD_p)$. These variations are what the average values in the simplified equation hide.

A unified way to see it: with the overall efficiency $\eta_0 = TV/(\dot{m}_f Q)$ (Q the fuel's heating value) one has $c_T = V/(\eta_0 Q)$, and then $R = \eta_0\,(Q/g)\,E\ln(W_0/W_f)$ for *any* engine. A jet engine is described with c_T (which grows with V, so the V in front is compensated), and a propeller engine with c_P (almost independent of V).

> **WORKED EXAMPLE (ILLUSTRATIVE DATA)**
> **Jet**: $W_0 = 250\,000$ kg, fuel $80\,000$ kg ($W_f = 170\,000$ kg), cruise at $M = 0.85$ at 11 km

($a = 295.1$ m/s, $V = 250.8$ m/s), $E = 18$ and TSFC $= 0.55$ lb/(lbf h)$= 1.558 \cdot 10^{-5}$ kg/(N s):

$$R = \frac{250.8 \cdot 18}{9.81 \cdot 1.558 \cdot 10^{-5}} \ln \frac{250}{170} = 2.954 \cdot 10^{7}\ \text{m} \cdot 0.3857 \approx 1.14 \cdot 10^{4}\ \text{km}.$$

A real flight is shorter because of reserves, climb and descent, and the changing E and c_T.
Propeller: $W_0 = 5\,000$ kg, fuel 500 kg, $E = 14$, $\eta_p = 0.8$ and $c_P = 0.45$ lb/(hp h)$= 7.60 \cdot 10^{-8}$ kg/J (about 274 g/kWh):

$$R = \frac{0.8 \cdot 14}{9.81 \cdot 7.60 \cdot 10^{-8}} \ln \frac{5000}{4500} = 1.502 \cdot 10^{7}\ \text{m} \cdot 0.1054 \approx 1.58 \cdot 10^{3}\ \text{km}.$$

The result is the same whether it flies at 60 or at 80 m/s as long as E, c_P and η_p do not change; the jet range, in contrast, would grow in proportion to V for constant c_T and E.

2.5 Specific fuel consumption data in SI units

Take a look at thrust and brake-power specific fuel consumption data for aero engines. Convert them to basic SI units to familiarise yourself with typical values.

The catalogues and textbooks use mixed units: TSFC in lb/(lbf h) and BSFC in lb/(hp h) (definitions in Unit 6). The conversion factors follow from the definitions of the units:

$$1\ \frac{\text{lb}}{\text{lbf} \cdot \text{h}} = \frac{0.45359\ \text{kg}}{4.44822\ \text{N} \cdot 3600\ \text{s}} = 2.833 \cdot 10^{-5}\ \frac{\text{kg}}{\text{N} \cdot \text{s}}$$

$$1\ \frac{\text{lb}}{\text{hp} \cdot \text{h}} = \frac{0.45359\ \text{kg}}{745.70\ \text{W} \cdot 3600\ \text{s}} = 1.690 \cdot 10^{-7}\ \frac{\text{kg}}{\text{W} \cdot \text{s}} = \frac{\text{kg}}{\text{J}}$$

(the first agrees with the $2.83 \cdot 10^{-5}$ used in Unit 6; for the second, 1 lb/(hp h) $\approx$ 608 g/kWh, and g/kWh $\to$ kg/J divides by $3.6 \cdot 10^{9}$.)

Engine (cruise)	Typical value	In SI	Other unit
High-bypass turbofan	0.5–0.6 lb/(lbf h)	1.4–1.7·10^{-5} kg/(N s)	14–17 g/(kN s)
Low-bypass turbofan	0.7–0.9 lb/(lbf h)	2.0–2.5·10^{-5} kg/(N s)	20–25 g/(kN s)
Turbojet (no afterburner)	0.9–1.1 lb/(lbf h)	2.5–3.1·10^{-5} kg/(N s)	25–31 g/(kN s)
Turboprop (per shaft power)	0.5–0.6 lb/(hp h)	0.85–1.0·10^{-7} kg/J	300–365 g/kWh
Piston engine	0.4–0.5 lb/(hp h)	0.68–0.85·10^{-7} kg/J	250–300 g/kWh

Typical orders of magnitude from general knowledge; real values depend on the engine, the flight condition and the rating. The course note on range quotes 5–20 g/(kN s) for jets and 250–300 g/(kW h) for propeller engines, which agrees with this table.

CHECKING THE CONVERSIONS
0.6 lb/(lbf h)·$2.833 \cdot 10^{-5} = 1.70 \cdot 10^{-5}$ kg/(N s), i.e. 17 g/(kN s). The corresponding $I_{sp} = 1/(g\,\text{TSFC}) = 1/(9.81 \cdot 1.70 \cdot 10^{-5}) \approx 6\,000$ s. For a piston engine, 0.45 lb/(hp h)·$1.690 \cdot 10^{-7} = 7.6 \cdot 10^{-8}$ kg/J, which is 274 g/kWh and a thermal efficiency of $1/(7.6 \cdot 10^{-8} \cdot 42 \cdot 10^{6}) = 0.31$ for a fuel of 42 MJ/kg.

Note that the two numbers cannot be compared directly: TSFC is per unit of thrust and BSFC per unit of power; they are related by the flight speed through the propulsive efficiency.

2.6 Centre of pressure of a cambered airfoil

How does the position of the centre of pressure behave in a positively cambered airfoil as the angle of attack varies from the zero-lift angle to the critical angle?

For a given α, the aerodynamic force can be replaced by the lift L at the aerodynamic centre (AC, near $c/4$) plus the moment about the AC, $M_{ac} = c_{m,ac}\,\frac{1}{2}\rho V^2 S c$. For a positively cambered airfoil $c_{m,ac} < 0$ and almost constant (far from stall). The CP is the point where the moment about it vanishes, which gives (Unit 5)

$$\frac{x_{cp}}{c} = \frac{1}{4} - \frac{c_{m,ac}}{c_l}, \qquad c_l = c_{l,\alpha}(\alpha - \alpha_0)$$

- **At $\alpha = \alpha_0$** (zero lift): $c_l = 0$ but the moment is not, so the same moment has to be produced by a zero force and $x_{cp} \to \infty$ (downstream, behind the trailing edge).
- **As α grows**: the CP **comes from infinity, crosses the trailing edge** (for example, with $c_{m,ac} = -0.05$, at $c_l = 0.067$) **and moves forward**, approaching the quarter chord from behind. It never goes ahead of it, because $-c_{m,ac}/c_l > 0$.
- **Near the critical angle**: it ends close to the AC; for instance with $c_{l,\alpha} = 2\pi$, $\alpha_0 = -2°$ and $\alpha_{crit} = 14°$ ($c_l = 1.75$), $x_{cp}/c = 0.25 + 0.05/1.75 = 0.28$.

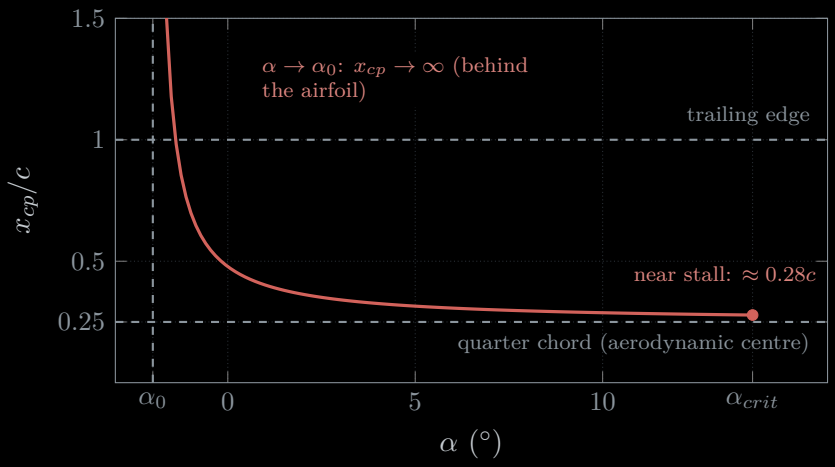


Figure 7.11. Centre-of-pressure position against α for a cambered airfoil ($c_{m,ac} = -0.05$, $c_{l,\alpha} = 2\pi$, $\alpha_0 = -2°$ and $\alpha_{crit} = 14°$; illustrative values). The dashed vertical line is the zero-lift angle, where $x_{cp} \to \infty$.

Practical consequence. The CP moves a lot, which is why aerodynamics uses the AC (a fixed point) instead. For a symmetric airfoil $c_{m,ac} = 0$ and the CP stays at $c/4$ at every angle. In real airfoils the CP moves back again once the flow separates close to stall (a remark from general knowledge; the model above assumes attached flow).

2.7 Skin-friction coefficient and roughness

Examine the skin-friction coefficient curves for both laminar and turbulent flow regimes. How does the turbulent case change when surface roughness increases?

The flat-plate friction coefficient is introduced in the flat-plate analogy of Unit 5 (C_f against Re in Figure 5.21). In summary:

- **Laminar**: $C_f = 1.328/\sqrt{Re_x}$ (Blasius). It falls with Re, but the laminar layer on a plate becomes unstable at $Re_x \approx 5 \cdot 10^5$ and turns turbulent.

- **Turbulent, smooth**: $C_f \approx 0.074/Re_x^{0.2}$ (valid up to about 10^7) or the Prandtl–Schlichting fit $C_f = 0.455/(\log_{10} Re_x)^{2.58}$, which the slides give with 0.445 and a compressibility factor. It is higher than the laminar value at equal Re (the velocity profile is fuller, so the shear at the wall is larger) and decreases more slowly.

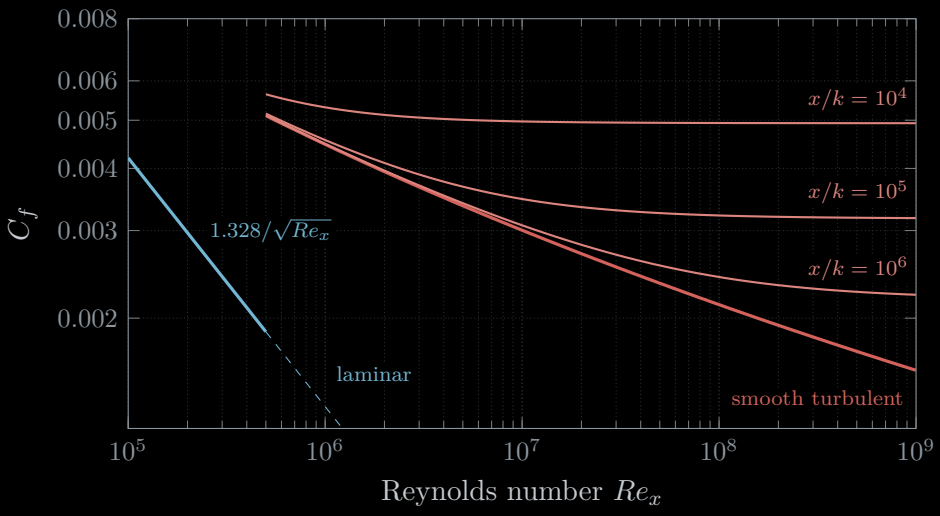


Figure 7.12. C_f of a flat plate against Re_x in log-log scale: laminar (blue), smooth turbulent (red) and turbulent over rough surfaces with different relative roughness x/k (light red, sketched with the fully rough values). Each rough curve separates from the smooth one and then stops decreasing.

Effect of roughness. Roughness is measured by the equivalent height k of the bumps, usually relative to the length, x/k.

- At low Re the viscous sublayer is thicker than the bumps and the surface behaves as smooth: the curve coincides with the smooth one.
- As Re grows the sublayer becomes thinner, the bumps poke out of it, and C_f is *higher* than the smooth value.
- At high Re the drag is dominated by the pressure on the bumps, proportional to $\frac{1}{2}\rho V^2$, so C_f stops depending on Re (a **plateau**). This is the "fully rough" regime, exactly like the horizontal part of the Moody diagram for pipes, and the slides call the start of it the cutoff Reynolds number Re_{cutoff} (Unit 5). For the fully rough regime $C_f \approx [1.89 + 1.62\log_{10}(x/k)]^{-2.5}$ (Schlichting; general knowledge).
- A **rougher** surface (smaller x/k) has a higher plateau and reaches it sooner (at lower Re); a smoother one reaches it later or never, within the range of interest.

The laminar regime is hardly affected, although large roughness can make the flow turbulent earlier.

> **ORDER OF MAGNITUDE (ILLUSTRATIVE DATA)**
> A plate $x = 10$ m long at 100 m/s at sea level: $Re = 1.225 \cdot 100 \cdot 10/1.789 \cdot 10^{-5} = 6.8 \cdot 10^7$. Smooth: $C_f = 0.455/(\log_{10} 6.8 \cdot 10^7)^{2.58} = 0.455/7.83^{2.58} = 2.25 \cdot 10^{-3}$. With rough paint of $k = 0.1$ mm ($x/k = 10^5$): $C_f = (1.89 + 1.62 \cdot 5)^{-2.5} = 3.17 \cdot 10^{-3}$, about 41 % higher.

2.8 Aerodynamic efficiency: derivation and trends

Derive the expression for aerodynamic efficiency and sketch a qualitative graph; examine the trend when the induced drag coefficient is much greater than the parasitic one and when it is much smaller.

With the parabolic polar of Unit 5, $C_D = C_{D,0} + kC_L^2$ and $k = 1/(\pi\, AR\, e)$:

$$E = \frac{L}{D} = \frac{C_L}{C_D} = \frac{C_L}{C_{D,0} + k\,C_L^2}$$

Maximum. Setting $\mathrm{d}E/\mathrm{d}C_L = 0$:

$$\frac{\mathrm{d}E}{\mathrm{d}C_L} = \frac{(C_{D,0} + kC_L^2) - 2kC_L^2}{(C_{D,0} + kC_L^2)^2} = 0 \implies C_{D,0} = kC_L^2 = C_{D,i}.$$

$$C_L^* = \sqrt{\frac{C_{D,0}}{k}}, \qquad C_D^* = 2C_{D,0}, \qquad E_{max} = \frac{C_L^*}{2C_{D,0}} = \frac{1}{2\sqrt{k\,C_{D,0}}}$$

So the maximum efficiency occurs when the induced and parasitic drags are *equal*, and it grows with the aspect ratio (k smaller) and with a clean aircraft ($C_{D,0}$ smaller). Writing $x = C_L/C_L^*$ gives a curve without parameters, $E/E_{max} = 2x/(1+x^2)$, plotted in Figure 7.13.

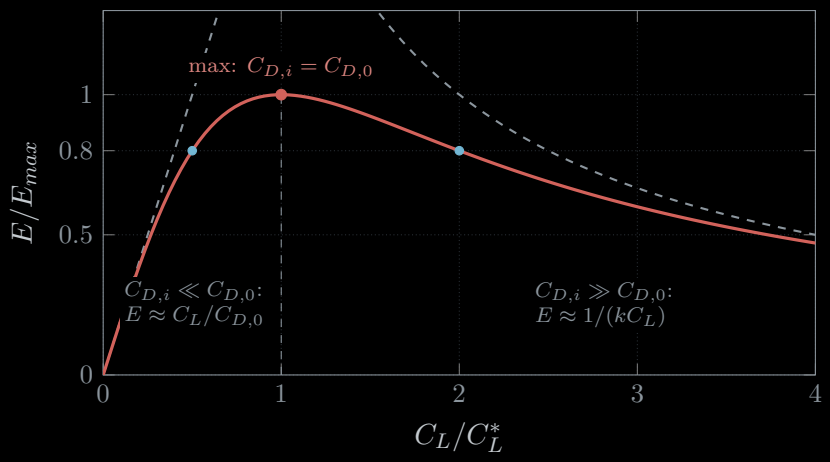


Figure 7.13. Qualitative shape of the efficiency against C_L, normalized with the optimum values. Dashed: the two asymptotes. The two blue dots show that at half or twice the optimum C_L the efficiency is still 80 % of the maximum.

Asymptotic trends

- $C_{D,i} \ll C_{D,0}$ (small C_L: high speed, or a cruise at low C_L): $C_D \approx C_{D,0}$ and $E \approx C_L/C_{D,0}$. The efficiency **grows linearly** with C_L (a straight line through the origin). Parasitic drag dominates.
- $C_{D,i} \gg C_{D,0}$ (large C_L: low speed, near stall): $C_D \approx kC_L^2$ and $E \approx 1/(kC_L)$. The efficiency **falls as a hyperbola**, as $1/C_L$. The induced drag dominates, and a larger aspect ratio helps.

NUMBERS OF THE WING IN UNIT 5
With $C_{D,0} = 0.02$, $k = 0.0468$: $C_L^* = 0.654$ and $E_{max} = 16.3$. At $C_L = 0.327$ ($x = 0.5$): $E = 0.327/(0.02 + 0.0468 \cdot 0.1069) = 0.327/0.0250 = 13.1 = 0.80\,E_{max}$. At $C_L = 1.31$ ($x = 2$): $E = 1.31/(0.02 + 0.0468 \cdot 1.716) = 1.31/0.1003 = 13.1$, also $0.80\,E_{max}$.

Link with range. For the propeller, the range is maximum when E is maximum. For the jet at constant altitude and constant c_T, $R \propto VE \propto \sqrt{C_L}/C_D$, whose maximum is at $C_{D,0} = 3kC_L^2$ ($C_L = C_L^*/\sqrt{3}$), where $E = (\sqrt{3}/2)E_{max} = 0.87E_{max}$ at a higher speed.

3 Review questions: propulsion, structures, surveillance and rotorcraft

This section answers review questions 9 to 20. Each subsection starts with the question in italics and a short answer in a box. Where the topic is explained in a unit, the unit is cited and not repeated, but each answer can be read on its own. Boxes titled *Answer from general knowledge (not in the slides)* contain standard textbook material; numerical values given there are **typical orders of magnitude**, not data from the course.

3.1 Total drag in level flight and the minimum-drag speed

Question 9. Derive the expressions for total drag in steady, level flight and for the speed corresponding to minimum drag. How are these affected by changes in flight altitude, aspect ratio, zero-lift drag coefficient and wing loading?

SHORT ANSWER
In steady level flight $L = W$. Drag is the sum of a parasitic term that grows as V^2 and an induced term that falls as $1/V^2$:

$$D = \tfrac{1}{2}\rho V^2 S\, C_{D0} + \frac{2kW^2}{\rho V^2 S}, \qquad V_{md} = \sqrt{\frac{2W}{\rho S}}\left(\frac{k}{C_{D0}}\right)^{1/4}, \qquad D_{min} = 2W\sqrt{kC_{D0}}$$

Derivation. With the parabolic polar $C_D = C_{D0} + kC_L^2$ and $k = 1/(\pi\, AR\, e)$ (Unit 5, Section 3.3.2), drag is $D = \tfrac{1}{2}\rho V^2 S\, C_D$. In steady level flight the lift equals the weight, so the lift coefficient is fixed by the speed:

$$L = W = \tfrac{1}{2}\rho V^2 S\, C_L \;\Rightarrow\; C_L = \frac{2W}{\rho V^2 S}$$

Substituting in the polar, the induced part is $\tfrac{1}{2}\rho V^2 S\, k\, C_L^2 = \tfrac{1}{2}\rho V^2 S\, k\, \dfrac{4W^2}{\rho^2 V^4 S^2}$, that is

$$D = \underbrace{\tfrac{1}{2}\rho V^2 S\, C_{D0}}_{D_0 \text{ (parasitic) } \propto V^2} + \underbrace{\frac{2kW^2}{\rho V^2 S}}_{D_i \text{ (induced) } \propto 1/V^2}$$

The minimum is found by setting $\mathrm{d}D/\mathrm{d}V = 0$:

$$\rho V S\, C_{D0} - \frac{4kW^2}{\rho V^3 S} = 0 \;\Rightarrow\; V^4 = \frac{4kW^2}{\rho^2 S^2 C_{D0}} \;\Rightarrow\; V_{md} = \left(\frac{2W}{\rho S}\right)^{1/2}\left(\frac{k}{C_{D0}}\right)^{1/4}$$

At this speed the two terms are equal, $D_0 = D_i$ (which is also the condition $C_{D0} = kC_L^2$ of maximum efficiency, so $C_L = \sqrt{C_{D0}/k}$ there). Hence

$$D_{min} = 2D_0 = \rho V_{md}^2 S\, C_{D0} = 2W\sqrt{k\, C_{D0}} = \frac{W}{(L/D)_{max}}, \qquad \left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{kC_{D0}}}$$

Below V_{md} the induced drag dominates; above it, the parasitic drag. Figure 5.22 shows the two curves and their sum for a reference case; Figure 7.14 shows how the total curve moves when one parameter changes.

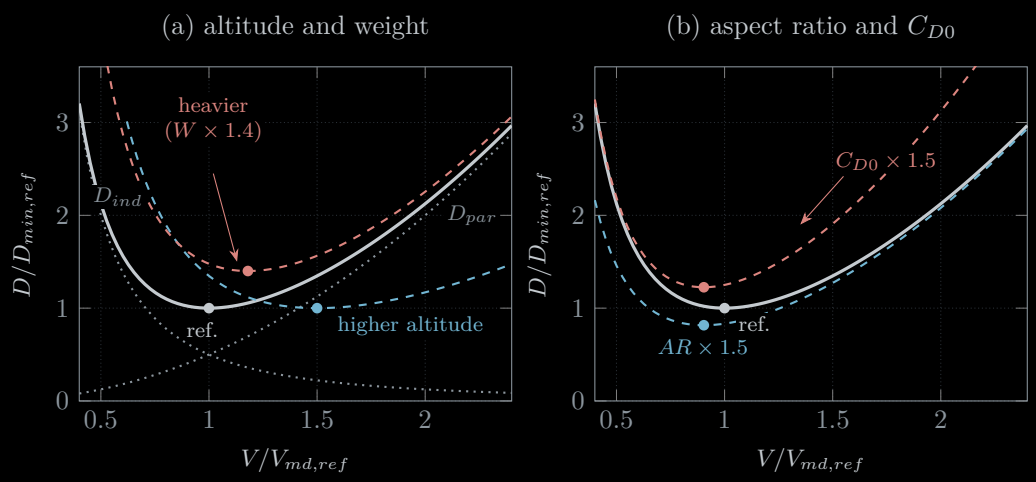


Figure 7.14. How the total drag curve moves (own sketch from the formulas; the dots are the minima). Reference solid; dashed curves change one parameter each, by the factors given in the table below. (a) A higher altitude shifts the curve to the right at the same minimum value; a heavier aircraft shifts it right and up. (b) A higher aspect ratio lowers the minimum and moves it to the left; a higher C_{D0} raises it and also moves it to the left.

Change	V_{md} (true airspeed)	D_{min}	Comment
Higher altitude (lower ρ)	$\propto \rho^{-1/2}$: **increases** (the equivalent airspeed is unchanged)	**unchanged**	curve shifts right; $(L/D)_{max}$ does not depend on ρ
Higher aspect ratio ($k \propto 1/AR$)	$\propto AR^{-1/4}$: decreases slightly	$\propto AR^{-1/2}$: **decreases**	induced drag is smaller at every speed
Higher C_{D0}	$\propto C_{D0}^{-1/4}$: decreases slightly	$\propto C_{D0}^{1/2}$: **increases**	parasitic branch is higher
Higher wing loading W/S	$\propto (W/S)^{1/2}$: **increases**	$\propto W$ if S is fixed (heavier aircraft); **unchanged** if W is fixed and S is reduced (with C_{D0} and k unchanged)	higher V_{md} means higher stall speed too

NUMBERS FOR THE EXAMPLE OF UNIT 5 (OWN CALCULATION)
With $C_{D0} = 0.02$, $AR = 8$, $e = 0.85$ ($k = 0.0468$), $S = 16\,\mathrm{m}^2$, $W = 9810\,\mathrm{N}$ at sea level:

$$V_{md} = \sqrt{\frac{2 \cdot 9810}{1.225 \cdot 16}}\left(\frac{0.0468}{0.02}\right)^{1/4} = 31.6 \cdot 1.237 = 39.1\ \mathrm{m/s}$$

$$D_{min} = 2 \cdot 9810\sqrt{0.0468 \cdot 0.02} = 600\ \mathrm{N}$$

(equal to $W/(L/D)_{max} = 9810/16.3$). At 5 000 m ($\rho = 0.736$) $V_{md} = 50.5$ m/s and D_{min} is still 600 N. If the aspect ratio rises from 8 to 12 (k falls by 8/12): $V_{md} \to 39.1 \cdot (2/3)^{1/4} = 35.3$ m/s and $D_{min} \to 600 \cdot (2/3)^{1/2} = 490$ N (the 18 % reduction in drag is paid for in a longer, heavier wing; C_{D0} is assumed unchanged).

Supplement (not in the slides). For a propeller aircraft the quantity to minimise is power, $P = DV$, whose minimum is at $V_{mp} = V_{md}/3^{1/4} \approx 0.76\,V_{md}$ (it gives the best endurance); for a jet the best range is obtained near $3^{1/4}V_{md} = 1.32\,V_{md}$ (see the specific range subsection below). Everything above neglects compressibility: close to the critical Mach number the wave drag adds a third term.

3.2 Propulsive efficiency of air-breathing engines

Question 10. Develop the expression for the propulsive efficiency of non-autonomous (air-breathing) propulsion. What does it depend on? How does it vary when the airspeed at the engine inlet changes? And at the outlet?

SHORT ANSWER

Using only the air mass flow $\dot{m}$ (fuel mass neglected), with V_0 the flight speed (inlet) and V_e the exit speed:

$$\eta_p = \frac{T\,V_0}{\frac{1}{2}\dot{m}\,(V_e^2 - V_0^2)} = \frac{2}{1 + V_e/V_0}$$

It depends **only on the ratio** V_e/V_0. For a given V_e, a higher flight speed improves η_p; for a given V_0, a higher exit speed worsens it.

Development (it matches the formula of Unit 6, where $V_{in} = V_0$ and $V_{out} = V_e$). The engine takes $\dot{m}$ of air at the flight speed V_0 and expels it at V_e. Neglecting the pressure terms and the added fuel mass (about 2 % of the air mass flow):

- thrust (momentum balance): $T = \dot{m}\,(V_e - V_0)$;
- useful power, the power that moves the aircraft: $P_u = T\,V_0$;
- power given to the flow, the increase of its kinetic energy per second: $P_k = \frac{1}{2}\dot{m}\,(V_e^2 - V_0^2)$.

$$\eta_p = \frac{P_u}{P_k} = \frac{\dot{m}\,(V_e - V_0)\,V_0}{\frac{1}{2}\dot{m}\,(V_e - V_0)(V_e + V_0)} = \frac{2V_0}{V_e + V_0} = \frac{2}{1 + V_e/V_0}$$

The difference $P_k - P_u = \frac{1}{2}\dot{m}\,(V_e - V_0)^2$ is the kinetic energy left in the wake, which is lost. *If the fuel mass flow $f\dot{m}$ is kept*, then $T = \dot{m}[(1+f)V_e - V_0]$ and $\eta_p = 2V_0[(1+f)V_e - V_0]/[(1+f)V_e^2 - V_0^2]$, which differs from the simple formula by a few per cent at most.

What it depends on and how it varies.

- **Inlet speed V_0 (flight speed), V_e fixed:** η_p *increases* with V_0 and tends to 1 when $V_0 \to V_e$. But the thrust $T = \dot{m}(V_e - V_0)$ falls and vanishes at $V_0 = V_e$, so there is no point in flying faster than the jet. At $V_0 = 0$ (static) $\eta_p = 0$: the engine gives thrust but no useful power. This is why jets are efficient at high speed and poor at low speed.
- **Exit speed V_e, V_0 fixed:** η_p *decreases* as V_e grows, but the thrust *increases*. For a fixed thrust, a low V_e means a larger mass flow $\dot{m} = T/(V_e - V_0)$: this is the reason for propellers and high-bypass fans (Figure 6.3).

Figure 6.3 of Unit 6 plots exactly this curve; the typical orders of magnitude of V_e/V_0 to read on it are: propeller about 1.1–1.3 ($\eta_p \approx 0.91$ at 1.2), high-bypass fan about 1.4–1.8 (≈ 0.77 at 1.6), low-bypass turbofan or turbojet about 2–3.5 (0.5 at 3), and still higher for afterburner or ramjet (not data from the slides).

SAME THRUST, TWO ENGINES (OWN CALCULATION)

Flight at $V_0 = 250$ m/s with a thrust of 20 kN. A turbojet with $V_e = 600$ m/s needs $\dot{m} = 20\,000/350 = 57$ kg/s and $\eta_p = 2/(1 + 2.4) = 0.59$. A turbofan whose total exhaust is at $V_e = 330$ m/s needs $\dot{m} = 20\,000/80 = 250$ kg/s and $\eta_p = 2/(1 + 1.32) = 0.86$. Check with the powers: useful $20\,000 \cdot 250 = 5.0$ MW; given to the air $\frac{1}{2} \cdot 57 \cdot (600^2 - 250^2) = 8.5$ MW (turbojet, $5.0/8.5 = 0.59$) or $\frac{1}{2} \cdot 250 \cdot (330^2 - 250^2) = 5.8$ MW (turbofan, $5.0/5.8 = 0.86$).

Supplement. The overall efficiency of an engine is the product $\eta_o = \eta_{th}\,\eta_p$ of the thermal efficiency (fuel energy to kinetic energy of the flow) and the propulsive efficiency; improving the cycle raises η_{th}, lowering the exhaust speed raises η_p.

3.3 Propeller selection (Lab session 2)

Question 11. Complete the propeller selection exercise from Lab session 2.

> **ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)**
> **The exact data and charts of Lab session 2 are not in these notes**, so the exercise cannot be solved here with its own numbers. What follows is the general procedure with the coefficients of Unit 6 (J, C_T, C_P, η) and a worked example with invented data. To do the lab, repeat the same steps with the lab's power, speed, density and propeller chart.

Definitions (Section "Propeller coefficients" of Unit 6): $J = V/(nD)$, $C_T = T/(\rho n^2 D^4)$, $C_P = P/(\rho n^3 D^5)$, $\eta = J\,C_T/C_P$, with n in rev/s, D in m and P the shaft power. A propeller family is characterised by charts of C_T, C_P and η against J for several blade pitch angles (Figure 7.15; compare Figure 6.6).

Procedure.

1. Fix the requirement: shaft power P at the design point (usually cruise), flight speed V, density ρ (altitude).
2. Choose a diameter D (limited by ground clearance and tip speed) and a rotational speed n (limited by the engine, the gearbox and the tip Mach number; helical tip speed $\sqrt{(\pi n D)^2 + V^2}$ should stay below about Mach 0.8 to 0.85).
3. Compute $J = V/(nD)$ and the power coefficient $C_P = P/(\rho n^3 D^5)$.
4. In the propeller charts, enter with J and C_P, and read the efficiency η (and the blade pitch needed). Repeat for each candidate propeller; the best one is that whose peak efficiency lies at the design (J, C_P).
5. Compute the thrust at cruise $T = \eta P/V$ and C_T. Check the consistency $\eta = J\,C_T/C_P$.
6. Check the other conditions: static thrust at take-off ($J = 0$), where the fixed-pitch propeller that is best in cruise gives less thrust than a fine-pitch one; this is why constant-speed (variable-pitch) propellers are used.

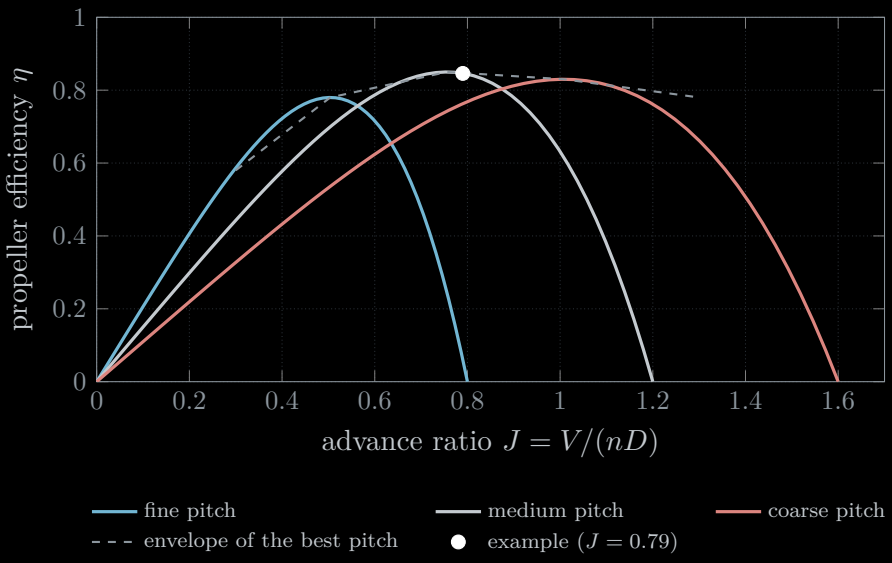


Figure 7.15. Efficiency curves of a propeller family for three blade pitch settings (schematic, no real values). Each pitch has a peak at a different advance ratio; the dashed line is the envelope of the best pitch. The white dot is the example in the text.

ILLUSTRATIVE WORKED EXAMPLE (INVENTED DATA, NOT FROM THE LAB)
Light aircraft: $P = 75$ kW at cruise, $V = 60$ m/s, sea level ($\rho = 1.225$ kg/m^3). Candidate propeller: $D = 1.9$ m at 2 400 rpm ($n = 40$ rev/s).

$$J = \frac{60}{40 \cdot 1.9} = 0.79, \qquad C_P = \frac{75\,000}{1.225 \cdot 40^3 \cdot 1.9^5} = 0.0386$$

Tip speed: $\pi n D = 239$ m/s; helical $\sqrt{239^2 + 60^2} = 246$ m/s, that is Mach $246/340 = 0.72$ (acceptable). In the chart of Figure 7.15 (medium pitch) the point ($J = 0.79$, $C_P = 0.0386$) gives $\eta \approx 0.85$. Then

$$T = \frac{\eta P}{V} = \frac{0.85 \cdot 75\,000}{60} = 1\,060 \text{ N}, \qquad C_T = \frac{1\,062}{1.225 \cdot 40^2 \cdot 1.9^4} = 0.0416$$

$$\eta = \frac{J\,C_T}{C_P} = \frac{0.79 \cdot 0.0416}{0.0386} = 0.85 \ \checkmark$$

If the tip Mach were too high, one would lower n (which raises J and C_P) or use a smaller D, and read the chart again.

3.4 Specific impulse of chemical rockets

Question 12. Find the typical value of the specific impulse of chemical rocket engines.

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)
Chemical rockets have $I_{sp} \approx 250$ to 450 s, that is, effective exhaust speeds of 2.5 to 4.5 km/s ($v_e = I_{sp}\, g_0$). About **300 to 350 s** is a good typical value for the common kerosene/oxygen stages, and about 450 s is the practical maximum of hydrogen/oxygen in vacuum. Compare with the thousands of seconds of air-breathing engines in Figure 6.14: the rocket carries its own oxidizer.

Unit 6 gives the definition $I_{sp} = T/(\dot{m}_{fuel}\, g)$ and the families of propellants. The usual orders of magnitude are (all values are typical, and each engine has its own exact figure):

Propellant type	I_{sp} **(s)**	**Remarks**
Solid (APCP, e.g. boosters)	≈ 250–280	simple, storable, cannot be throttled; about 240 s at sea level
Liquid LOX/kerosene (RP-1)	≈ 300–350	about 280–310 s at sea level, 310–350 s in vacuum; dense
Liquid LOX/LH_2	≈ 420–450	in vacuum (about 360–390 s at sea level); highest of the chemical rockets, but low density
Hypergolic liquid (e.g. N_2O_4 with hydrazine derivatives)	≈ 300–340	vacuum value; storable and reliable, toxic
For comparison, outside the chemical family: monopropellant hydrazine thrusters about 220 s; cold gas 50 to 70 s; electric (ion) thrusters, thousands of seconds.		

Why those values. The exhaust speed grows with $\sqrt{T_c/\bar{M}}$ (combustion temperature over mean molecular mass of the exhaust). Hydrogen gives a light exhaust (water, with unburnt H_2), hence its high I_{sp}. The value in vacuum is higher than at sea level because the nozzle can expand the gas more and there is no back-pressure; so always check which of the two a quoted figure refers to. Link with the Tsiolkovsky equation: $\Delta v = I_{sp} g_0 \ln(m_0/m_f)$; the worked example of the next subsection uses $I_{sp} = 350$ s, in the middle of the range.

3.5 Mass estimate for single-stage and two-stage launch vehicles

Question 13. Carry out a mass estimation exercise for a single-stage-to-orbit and a two-stage-to-orbit launch vehicle.

SHORT ANSWER
With $\Delta v \approx 9.4$ km/s and $I_{sp} = 350$ s, a single stage needs a mass ratio $m_0/m_f = 15.5$ and its structure must be lighter than 6.5 % of the propellant plus structure: it is marginal even with a very optimistic structure ($\varepsilon = 0.05$ leaves a payload of 1.5 % of the launch mass). Two stages need only a ratio of 3.9 each and, with a realistic $\varepsilon = 0.10$, give a payload of 2.9 %. Staging makes orbit feasible.

Model and assumptions (Tsiolkovsky equation of Unit 6; own calculation):

- Required $\Delta v \approx 9.4$ km/s: 7.8 km/s of orbital speed plus about 1.5 to 1.6 km/s of losses (gravity, drag, steering, partly offset by the Earth's rotation). This is a typical order of magnitude, the value of Unit 6.
- Same engine in all the stages with $I_{sp} = 350$ s ($v_e = 3\,434$ m/s); in reality the first stage has lower I_{sp} at sea level and the upper stage a higher one in vacuum.
- **Structural coefficient** $\varepsilon = m_s/(m_s + m_p)$: structure (tanks, engines) over structure plus propellant of the stage. Typical order of magnitude for liquid stages: 0.06 to 0.12. The same ε for all the stages, and the same Δv for each (the optimum split for identical stages).

Equations. For a stage with payload m_{pl} (on top of it): $m_0 = m_s + m_p + m_{pl}$, $m_f = m_s + m_{pl}$, $m_s = \varepsilon(m_0 - m_{pl})$. With the mass ratio $R = m_0/m_f = e^{\Delta v/(I_{sp} g_0)}$ the payload fraction is

$$\lambda = \frac{m_{pl}}{m_0} = \frac{1/R - \varepsilon}{1 - \varepsilon}, \qquad \lambda_{total} = \lambda_1 \, \lambda_2 \cdots$$

and the stage is possible only if $1/R > \varepsilon$.

SINGLE STAGE VERSUS TWO STAGES, PAYLOAD 10 T (OWN CALCULATION)
Single stage. $R = e^{9400/3434} = 15.45$, $1/R = 0.0647$. With $\varepsilon = 0.10$ (typical): $1/R < \varepsilon$, so it is **impossible** (even with no payload the structure alone is heavier than the allowed final mass). With the very optimistic $\varepsilon = 0.05$: $\lambda = (0.0647 - 0.05)/0.95 = 1.5\,\%$, so $m_0 = 10/0.0155 = 645$ t, 94 % of it propellant. With $\varepsilon = 0.06$: $\lambda = 0.5\,\%$, $m_0 = 2\,000$ t.
Two stages, $\Delta v = 4.7$ km/s each. $R = e^{4700/3434} = 3.93$, $1/R = 0.254$. With $\varepsilon = 0.10$:

$$\lambda_1 = \lambda_2 = \frac{0.254 - 0.10}{0.90} = 0.1716, \qquad \lambda_{total} = 0.1716^2 = 2.9\,\%$$

Masses from the top: $m_{0,2} = 10/0.1716 = 58.3$ t (propellant $0.9\,(58.3 - 10) = 43.5$ t, structure 4.8 t); $m_{0,1} = 58.3/0.1716 = 340$ t (propellant $0.9\,(340 - 58.3) = 253$ t, structure 28 t). Total launch mass **340 t** for 10 t of payload, against 645 t for the (almost impossible) single stage. For comparison, hydrogen/oxygen with $I_{sp} = 450$ s and $\varepsilon = 0.08$ gives a single-stage payload of $(0.1189 - 0.08)/0.92 = 4.2\,\%$: this is why single-stage-to-orbit designs aim at hydrogen, and still have been very hard to build.

Reading the result. At $\varepsilon = 0$ all the options are equal (a 6.5 % payload fraction, the $1/R_{total}$ of the whole mission), because staging only helps by discarding dead mass. The higher ε is, the more staging pays; beyond three stages the gain is small compared with the added complexity (Figure 6.13 shows the propellant fraction of a single stage).

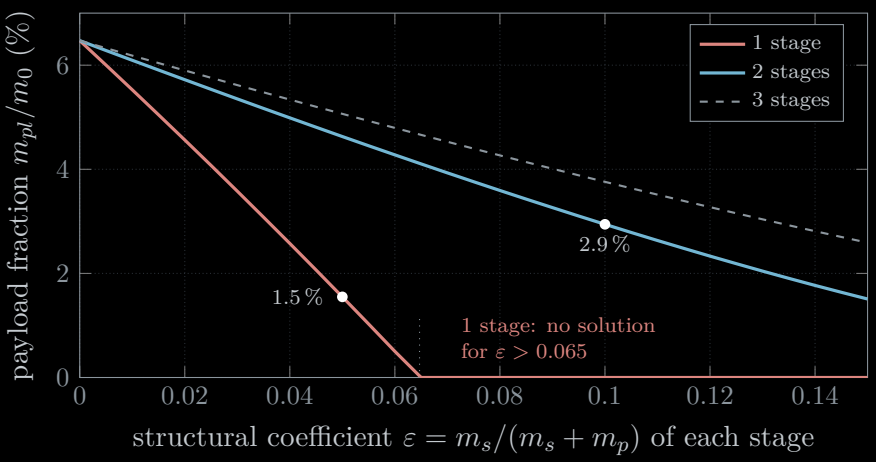


Figure 7.16. Payload fraction m_{pl}/m_0 against the structural coefficient ε of each stage, for $\Delta v = 9.4$ km/s and $I_{sp} = 350$ s (same ε and Δv per stage). A single stage disappears at $\varepsilon = 0.065$; with several stages the same technology can reach orbit. The dots are the two examples of the text. Own calculation.

3.6 Operating Mach ranges and specific impulse of air-breathing engines

Question 14. What are the typical operating Mach number ranges of turbofan, turbofan with afterburner, ramjet and scramjet engines, and their typical specific impulses?

SHORT ANSWER
Turbofan: $M \approx 0$–0.9 (transport aircraft cruise at 0.78 to 0.85), I_{sp} of thousands of seconds, about 5 000 to 10 000 s for modern high-bypass engines. Turbofan with afterburner (military): up to $M \approx 2$–2.5, I_{sp} about 1 500 to 3 000 s. Ramjet: $M \approx 2$–5 (up to about 6), I_{sp} about 1 000 to 3 000 s. Scramjet: $M \approx 5$–15, I_{sp} of the order of 500 to 1 500 s, falling with Mach. These are **typical orders of magnitude**.

The chart of Figure 6.14 (Unit 6, slide 23 redrawn) already shows the regions; the table gives the same information with numbers. The turbofan region of the slide chart extends to about $M \approx 1.6$ for low-bypass military engines, and the I_{sp} of jets is obtained from the thrust specific fuel consumption by $I_{sp} = 1/(\text{TSFC} \cdot g)$, i.e. $I_{sp} \approx 3\,600/\text{TSFC}$ with TSFC in lb/(lbf·h) (for example 0.35–0.7 gives about 10 000 to 5 000 s).

Engine	Mach range	I_{sp} (s), typical	Remarks
Turbofan (high bypass)	0 to 0.9	5 000–10 000	airliners; the higher the bypass, the higher I_{sp} but the lower the top speed
Turbojet / low-bypass turbofan, dry	0 to ≈2	3 000–4 500	fighters and old supersonic aircraft
Turbofan with afterburner (wet)	up to 2–2.5 (3 for a turbojet)	1 500–3 000	extra thrust for short times, at the cost of a much higher consumption
Ramjet	≈2 to 5–6	1 000–3 000 (chart of slide 23: highest at $M \approx 3$–5)	does not work at rest
Scramjet	≈5 to 15	≈500–1 500	supersonic combustion; very few flight tests, values are estimates
Rocket (reference)	all	250–450	independent of Mach

Remember the ordering that follows from the table: below $M \approx 2$ the turbofan is the most economical; from about $M \approx 3$ the ramjet; above $M \approx 6$ only the scramjet or the rocket remain. The trend of the I_{sp} is that it falls for all the air-breathing engines at high Mach, when used at the top of its range, because intake losses and the dissociation of the hot gas grow.

3.7 Primary structure of a fuselage and of a wing

Question 15. Draw the elements of the semi-monocoque primary structure of a fuselage and of a wing. What are the main elements and what are their main functions?

SHORT ANSWER

A semi-monocoque structure is a thin **skin** stiffened by lengthwise members (stringers, longerons in the fuselage; stringers and spars in the wing) and by transverse members (frames and bulkheads in the fuselage; ribs in the wing). The skin carries shear and torsion and, in the fuselage, the pressure; the longitudinal members carry the axial loads of bending; the transverse members keep the shape and spread the loads. Descriptions are in Unit 4; Figure 4.10, Figure 4.11 and Figure 4.12 are the existing figures, and Figure 7.17 and Figure 7.18 are drawn here.

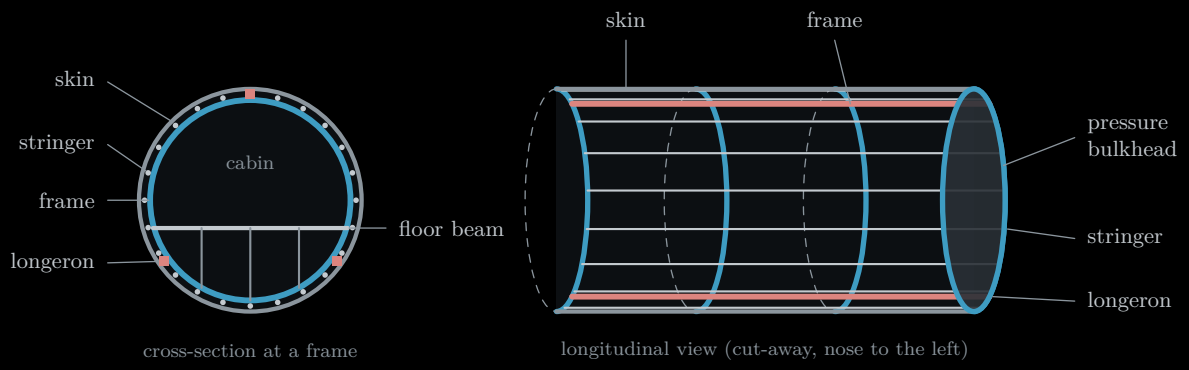


Figure 7.17. Semi-monocoque fuselage: cross-section at a frame (left) and cut-away of a stretch of barrel (right). The skin is the outer shell; the frames are the transverse rings; stringers (small) and longerons (heavy) run lengthwise. Schematic, with only a few stringers drawn.

Fuselage.

Element	Main function
Skin	Thin sheet that gives the aerodynamic shape; carries the **shear** and **torsion** of the fuselage and the **cabin pressure** (hoop tension); it also takes part of the bending loads
Stringers	Light longitudinal members riveted to the skin: carry the **axial loads** of fuselage bending (tension in the lower part, compression in the upper part) together with an effective width of skin, and **stiffen the skin panels**
Longerons	Heavier longitudinal members (floor level, around doors and cut-outs): carry larger axial loads and concentrated loads
Frames (rings, formers)	Transverse rings: keep the **cross-section shape**, support the skin and the stringers at regular intervals, and **transfer concentrated loads** (wing, landing gear, tail, floor) into the skin as shear
Bulkheads	Pressure bulkheads close the pressurised cabin at the ends; flat bulkheads carry large concentrated loads (wing, tail, engine mounts)

SKIN STRESS DUE TO CABIN PRESSURE (ILLUSTRATIVE VALUES)

A pressurised fuselage of radius $r = 1.97$ m, skin thickness $t = 1.6$ mm and a pressure difference $\Delta p = 0.6$ bar: the hoop stress is $\sigma_h = \Delta p\, r/t = 60\,000 \cdot 1.97/0.0016 = 74$ MPa, and the longitudinal stress is half, $\sigma_l = \Delta p\, r/(2t) = 37$ MPa. This is why the skin is thin but must be continuous, and why windows with sharp corners are dangerous (Comet case of Unit 4).

Wing.

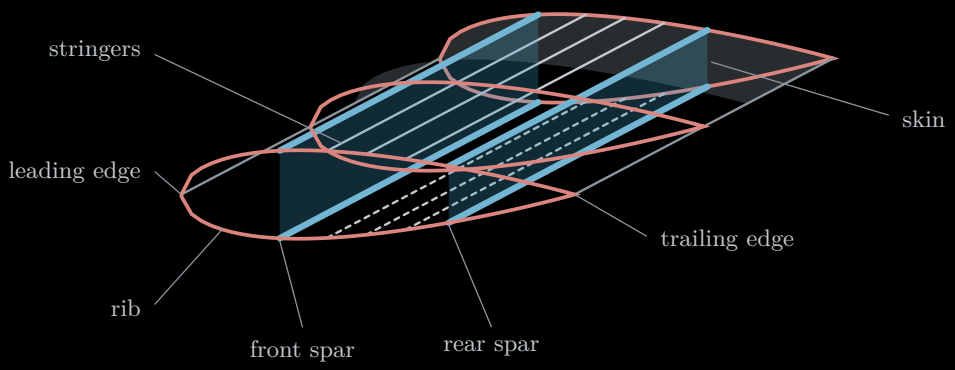


Figure 7.18. Semi-monocoque wing box. The two spars (blue; heavy flanges and a web), the ribs (red outlines) and the stringers (light lines, continuous on the upper skin and dashed on the lower skin) form the skeleton; the skin (grey) is drawn only on the outer bay, so that the inside can be seen. Schematic, not to scale.

Element	Main function
Skin	Aerodynamic surface; together with the spar webs forms the closed **torsion box** that resists **torsion** and carries **shear**; with the stringers, its upper panel works in compression and its lower panel in tension when the wing bends
Stringers	Carry the **axial loads** of wing bending in the skin panels and stiffen them
Spars	Main beams along the span: the **web** carries the **shear** (lift) and the **flanges (caps)** carry the **bending** loads
Ribs	Keep the **airfoil shape**, transfer the load of the skin to the spars, spread concentrated loads (landing gear, engine, flaps), and support the skin and the stringers (a rib is a support point for them)

NOMENCLATURE: WHAT STRINGERS DO ABOUT BUCKLING

Stringers do **not** "reduce buckling": a thin skin loaded in compression or shear tends to buckle, and the stringers (and ribs and frames) **increase the buckling resistance of the panel** by cutting it into narrower panels. The elastic buckling stress of a flat panel of thickness t and unsupported width b is

$$\sigma_{cr} = k_c \frac{\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b}\right)^2$$

so reducing b (stringers close together) raises σ_{cr} strongly, and the stringer also adds area to carry the axial load. In turn, a stringer can buckle as a column between frames (or ribs), so the **frame/rib spacing limits the stringer column length**. Unit 4 uses the shorthand "prevent buckling"; the precise statement is the one given here.

3.8 Secondary radar, TCAS and ADS-B

Question 16. What are the principles of operation of secondary surveillance radar (SSR), TCAS and ADS-B? What is each used for, and how do they differ?

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)

All three rely on the aircraft **transponder**, a radio on board. **SSR** (ground radar): the ground station interrogates (1030 MHz) and the transponder replies (1090 MHz) with an identity code, altitude or address: it is used by air traffic control to know who and where each aircraft is. **TCAS**: an aircraft interrogates the transponders of nearby aircraft with the

same frequencies, to avoid collisions without ground help (traffic and resolution advisories). **ADS-B**: the aircraft itself broadcasts its GNSS position and velocity periodically, without being interrogated; it is received by ground stations and by other aircraft. Figure 1.10 (Unit 1) shows who controls the aircraft in each phase; Figure 1.11 the ground navigation aids.

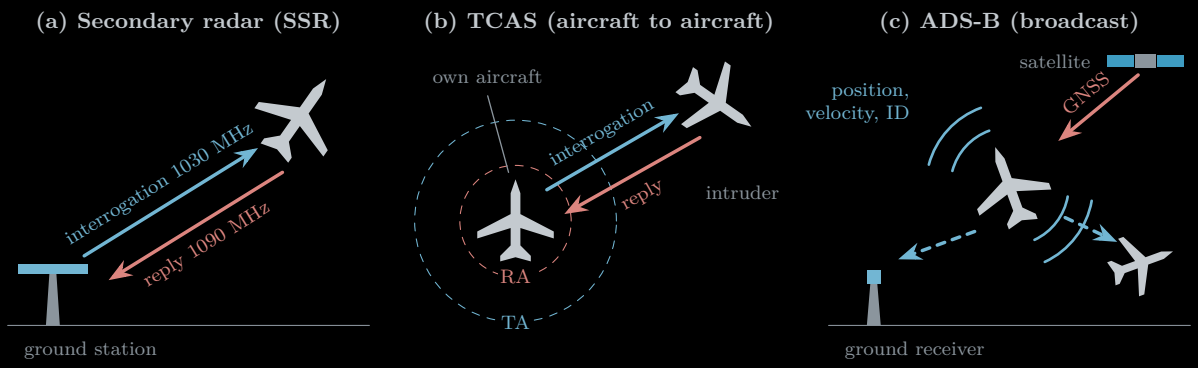


Figure 7.19. Three surveillance systems that use the transponder (schematic). (a) The ground SSR interrogates and the aircraft replies. (b) The own aircraft's TCAS interrogates the intruder's transponder; the dashed circles represent the alert zones (TA outside, RA inside). (c) ADS-B: the aircraft broadcasts what its GNSS receiver computes and nobody needs to ask.

Primary versus secondary radar. A *primary* radar sends a pulse and listens for the echo reflected by the aircraft: it needs no cooperation, but it only gives range and bearing and the echo is weak. A *secondary* radar (SSR) adds a cooperative element: the ground antenna, usually mounted on top of the primary one, transmits an **interrogation** pulse pair at **1030 MHz**; the aircraft transponder answers on **1090 MHz** with a coded message. Range comes from the reply delay, bearing from the antenna direction. The **modes** define what is asked: **Mode A** (an identity code of four octal digits, 4 096 codes, the *squawk*), **Mode C** (pressure altitude, in steps of 100 ft), and **Mode S** (selective: each aircraft has a unique 24-bit address, so it can be interrogated individually, and data such as flight identification can be exchanged).

TCAS (traffic alert and collision avoidance system; ACAS in ICAO terms). Its logic is on board. It interrogates on 1030 MHz and listens to the replies on 1090 MHz of every transponder nearby (modes A/C and S), and computes range, closing rate and relative altitude of each *intruder*. If the estimated time to the closest point of approach (the *tau*) falls below a threshold (typically about 40 s for a **traffic advisory, TA**, and 25 s for a **resolution advisory, RA**; thresholds depend on altitude), it warns the pilot (TA) and then orders a vertical manoeuvre (RA: "climb", "descend", "adjust vertical speed"). If the intruder also has TCAS, both aircraft coordinate their advisories through Mode S, so that one climbs and the other descends. It is mandatory on large transport aircraft. It only sees aircraft that have a working transponder.

ADS-B (automatic dependent surveillance–broadcast). *Automatic*: no interrogation is needed; *dependent*: the data come from the aircraft's own navigation system (GNSS). The aircraft broadcasts its identification, position, altitude, velocity and status, about twice per second, on 1090 MHz (extended squitter of Mode S) or on 978 MHz (UAT, used in some regions for general aviation). Any receiver in range uses it: air traffic control (cheap ground stations, even satellites for oceanic coverage) and other aircraft (ADS-B *In*). Precision is that of GNSS, better than that of a radar for position and velocity, and it needs no rotating antenna.

	SSR	TCAS	ADS-B
Who initiates	ground station interrogates	own aircraft interrogates others	aircraft transmits by itself
Frequencies	1030 MHz up, 1090 MHz down	1030 / 1090 MHz	1090 MHz (or 978 MHz UAT)
Source of position	radar measurement (range, bearing)	relative measurement (range, bearing, altitude)	GNSS on board
Main use	air traffic control surveillance	airborne collision avoidance (last safety net)	surveillance by ATC and awareness of traffic
Who uses it	controllers	pilots (TA/RA)	controllers and pilots
Needs	ground radar network	transponder in the other aircraft	GNSS and a transmitter on board; receivers
Limitations	cost of radars, limited coverage, only cooperative targets	only transponder-equipped targets; vertical advisories	depends on GNSS; data are not authenticated

In short: SSR and TCAS *ask*, ADS-B *tells*; SSR serves the controller on the ground, TCAS serves the pilot with no controller involved, and ADS-B serves both. None of them detects an aircraft with its transponder off; only the primary radar does.

3.9 Specific range of a jet aircraft

Question 17. Draw typical specific range curves as a function of speed and altitude for passenger jet aircraft.

> **ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)**
> The specific range is the distance flown per unit of fuel, $SR = V/(c\,D)$ with c the weight-specific fuel consumption (Unit 3, Breguet). At a given altitude it has a maximum at a **best Mach number**, which is a little above the speed of minimum drag and is limited by the wave-drag rise (near $M \approx 0.8$). The best Mach is higher at higher altitude, and SR increases with altitude up to an **optimum altitude** that **rises as the aircraft burns fuel and gets lighter**, so airliners climb in steps. The curves are in Figure 7.20.

Equation. The fuel weight flow of a jet is $c\,T$, and in steady level flight $T = D = W/(L/D)$, so

$$SR = \frac{\mathrm{d}x}{\mathrm{d}W_{fuel}} = \frac{V}{c\,D} = \frac{V\,(L/D)}{c\,W}$$

(this is the Breguet range equation of Unit 3 before integrating, and the range is $R = \int SR\,\mathrm{d}W_{fuel}$). The speed appears twice: the numerator grows with V (and with Mach, $V = Ma$), while D has a minimum at V_{md} (Question 9). With the parabolic polar, V/D is maximum at $V = 3^{1/4}V_{md} = 1.32\,V_{md}$, where L/D is 87 % of its maximum; with compressibility, the wave drag rises steeply above the drag-divergence Mach number and the maximum is reached below it.

- **Versus Mach, fixed altitude:** a flat maximum; flying faster or slower than the best Mach loses range. A faster cruise costs a few per cent of SR but saves time.
- **Versus altitude:** higher altitude means lower density, so the same C_L is flown at higher true airspeed: the best Mach rises and SR improves, up to the altitude where the C_L at the best Mach equals the value for best L/D. Above it C_L is too high (buffet, higher induced drag). In the troposphere c also falls with temperature; above 11 km it is roughly constant.
- **Weight:** $C_L \propto W/(\rho V^2)$, so as fuel is burned the optimum altitude rises (about 3 000 to 4 000 ft for every 15 t lost in a mid-size airliner, as in Figure 7.20b); the aircraft climbs gradually (*step cruise*).

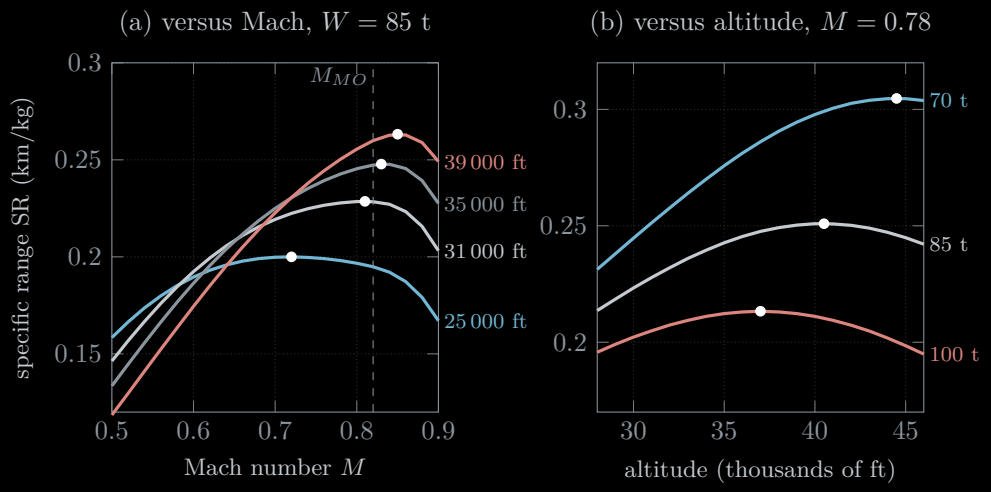


Figure 7.20. Specific range of a twin-jet airliner (own illustrative model, not data of an actual aircraft). (a) Versus Mach for four altitudes at $W = 85$ t; the white dots mark the best Mach of each altitude. (b) Versus altitude at $M = 0.78$ for three weights; the dots mark the optimum altitude. Model: ISA, $S = 122$ m^2, $AR = 9.5$, $e = 0.85$, $C_{D0} = 0.030$ plus a wave-drag term $20(M - 0.78)^4$ above $M = 0.78$, TSFC $= 1.6 \cdot 10^{-5}$ kg/(N s) at the tropopause temperature, varying as $\sqrt{T}$.

A POINT OF THE MODEL (OWN CALCULATION)
$W = 85$ t, $M = 0.78$, 35 000 ft ($T = 218.8$ K, $p = 23.8$ kPa): $V = 231$ m/s, $C_L = 0.67$, $C_D = 0.0478$, $L/D = 14.1$, $D = W/(L/D) = 59.3$ kN. Fuel flow $=$ TSFC $\cdot\, D = 1.61 \cdot 10^{-5} \cdot 59\,270 = 0.95$ kg/s, so $SR = 231/0.95 = 243$ m/kg $= 0.24$ km/kg, i.e. about 4.1 kg of fuel per km.

3.10 The rotor in a few lines (introduction for Questions 18 to 20)

The questions on helicopters use some ideas that are not in the units; this short summary (general knowledge) is enough to follow them.

- The **main rotor** is a set of rotating wings (blades) that produces both lift and propulsion. The thrust is changed with the **collective pitch**, which varies the pitch of all the blades together (the lever in the pilot's left hand).
- With **cyclic pitch** the pitch of each blade varies with its azimuth in each turn, which tilts the rotor disc and so the thrust vector: forward, backward or sideways (the stick in the right hand). A helicopter does not tilt its fuselage to go forward; it tilts the rotor (Figure 7.21b).
- **Torque reaction**: the engine drives the rotor and the fuselage reacts with an opposite torque, which a **tail rotor** (or another device) must cancel; the pedals control it and with it the yaw.
- **Dissymmetry of lift**: in forward flight the blade that advances sees a speed $\Omega R + V$ and the retreating one $\Omega R - V$ (Figure 7.21a), so with equal pitch the lift would be much larger on the advancing side and the helicopter would roll. It is cured by **blade flapping** (the blades are hinged or flexible and rise on the advancing side and descend on the retreating side, which reduces and increases their angle of attack) and by the cyclic pitch. Near the hub, the retreating blade sees reverse flow.

SPEEDS ON THE DISC (ILLUSTRATIVE NUMBERS)
Rotor radius $R = 5.5$ m at 400 rpm: $\Omega = 41.9$ rad/s and $\Omega R \approx 230$ m/s. In forward flight at $V = 60$ m/s (117 kt): advancing tip $230 + 60 = 290$ m/s (Mach 0.85 at sea level, $a = 340$ m/s), retreating tip $230 - 60 = 170$ m/s. The reverse-flow region is a circle of diameter

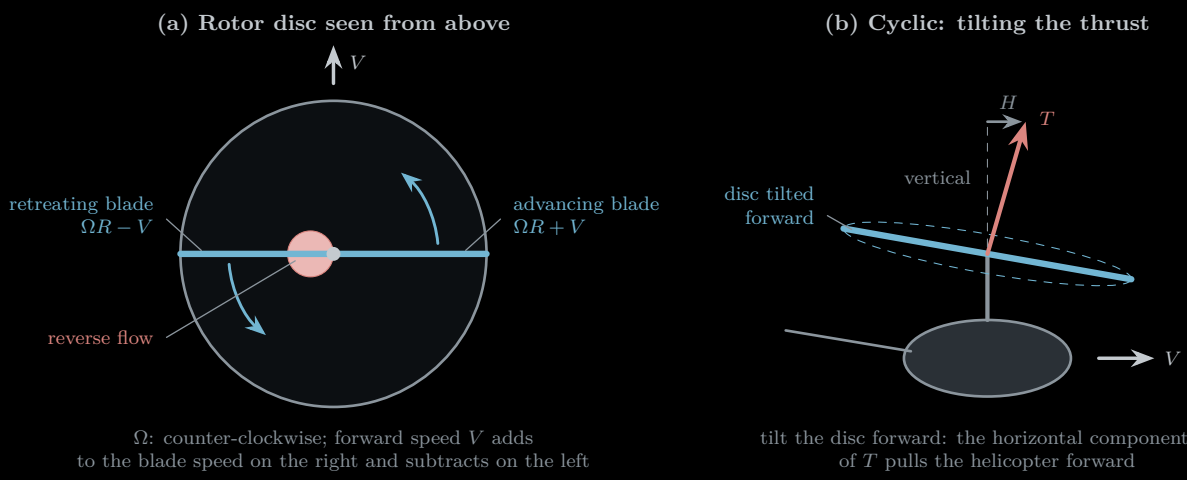


Figure 7.21. (a) Rotor disc seen from above in forward flight at speed V: the advancing blade (right) moves faster relative to the air than the retreating blade (left); a small circle near the hub is the reverse-flow region. (b) Side view: the cyclic tilts the disc and the thrust T forward, giving a horizontal component H. Schematic.

$V/\Omega = 1.4$ m on the retreating side. This is why the top speed of conventional helicopters is limited (Question 19).

3.11 Vertical flight of a helicopter: climb and descent

Question 18. Have a look at the document on vertical flight of helicopters: what happens when they climb or descend at different speeds?

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)
The "vertical flight" document is not available in the files used for these notes, so the answer comes from the standard **momentum theory** of the rotor. In hover the rotor induces a velocity $v_h = \sqrt{T/(2\rho A)}$ through the disc. **In climb** the induced velocity *decreases* (and power is needed to climb); **in slow descent** (about 0 to $2v_h$) the rotor sinks into its own wake and enters the **vortex ring state**, with loss of thrust and control; **in fast descent** (above about $2v_h$) the air passes upward through the disc (**windmill brake state**) and the rotor can turn by itself: **autorotation**.

Momentum theory. Model the rotor as an actuator disc of area $A = \pi R^2$ that accelerates the air; the induced velocity at the disc is v_i, and far downstream the wake velocity is $2v_i$ (in hover). The vertical speed of the helicopter is V_c (positive upward). The mass flow through the disc is $\dot{m} = \rho A\,(V_c + v_i)$ and the thrust is the momentum change, $T = \dot{m} \cdot w$ with w the velocity increment $2v_i$:

$$T = 2\rho A\,(V_c + v_i)\,v_i \;\Rightarrow\; v_h = \sqrt{\frac{T}{2\rho A}}\;(V_c = 0), \qquad \frac{v_i}{v_h} = -\frac{V_c}{2v_h} + \sqrt{\left(\frac{V_c}{2v_h}\right)^2 + 1}$$

The ideal power is $P = T\,(V_c + v_i)$: the sum of the climb power TV_c and the induced power Tv_i. For descent ($V_c < 0$) the same formula holds only in the windmill brake state, written for $V_c \le -2v_h$ as $v_i/v_h = \dfrac{|V_c|}{2v_h} - \sqrt{\left(\dfrac{V_c}{2v_h}\right)^2 - 1}$.

Flight state	V_c/v_h	What happens
Climb	> 0	the air arrives from below with speed V_c, so for the same thrust v_i is smaller than in hover; the power needed is hover power plus climb power
Hover	0	$v_i = v_h$; the most basic reference ($P_{ideal} = Tv_h$)
Vortex ring state (and turbulent wake)	about -2 to 0 (slow descent)	the rotor descends at a speed comparable to its own downwash: the flow recirculates around the blade tips in a doughnut-shaped vortex, the thrust becomes unsteady, the helicopter vibrates and sinks faster; **more collective makes it worse**. Momentum theory is not valid; the curve is empirical
Windmill brake state	≤ -2	the flow goes upward through the disc; the rotor extracts power from the air; this is the regime of **autorotation**, used after an engine failure

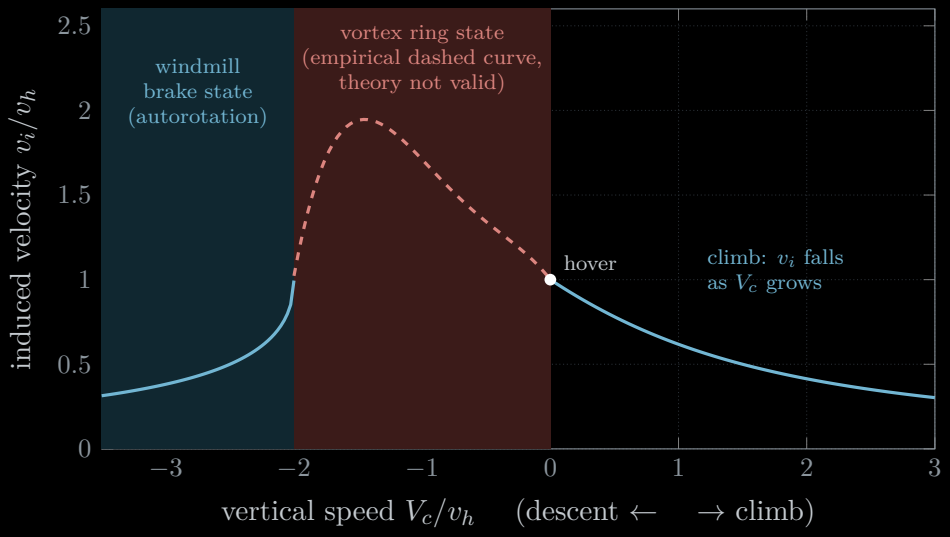


Figure 7.22. Induced velocity over hover value against the vertical speed over the hover velocity (momentum theory in solid lines; the dashed curve is the typical empirical shape in the vortex ring region, drawn from a standard polynomial fit and scaled to equal 1 at hover; only the qualitative shape matters). Climb is on the right, descent on the left. The coloured bands mark the regions.

A LIGHT HELICOPTER (ILLUSTRATIVE NUMBERS)

Mass 1 500 kg ($T = 14\,715$ N), rotor radius 5.5 m ($A = 95.0$ m^2), sea level.

- Hover: $v_h = \sqrt{14\,715/(2 \cdot 1.225 \cdot 95.0)} = 7.95$ m/s; ideal power $Tv_h = 117$ kW (the real one is higher: blade profile drag and other losses).
- Climb at $V_c = 5$ m/s: $v_i = -2.5 + \sqrt{2.5^2 + 7.95^2} = 5.83$ m/s, power $T(V_c + v_i) = 14\,715 \cdot 10.83 = 159$ kW.
- Descent at 4 m/s ($V_c/v_h = -0.5$) or 10 m/s (-1.3): inside the vortex ring band (0 to $-2v_h = 15.9$ m/s). Descent at 16 m/s or more: windmill brake state; at 16 m/s the formula gives $v_i = 7.1$ m/s.

Practical consequences. The vortex ring state arises with a high power setting, a descent rate comparable to v_h and a low forward speed (rule of thumb, order of magnitude: forward speed under about 20 kt); with forward speed the rotor leaves its own wake and the problem disappears. The recovery is to gain forward speed with the cyclic and lower the collective (never raise it); in autorotation the pilot lowers the collective at once to keep the rotor speed and flares near the ground to use the stored rotor energy.

3.12 Flight envelopes of a helicopter and of a jet airliner

Question 19. Draw the speed–altitude flight envelope diagrams for a helicopter and a passenger jet aircraft. What are the limits in each case and what are they due to?

> **ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)**
> **Jet:** on the left the **stall** (and, high up, the **buffet** boundary); on the right the V_{MO}/M_{MO} limits (structure, then compressibility); on top the **thrust (service) ceiling**. Where the two buffet limits converge is the "**coffin corner**". **Helicopter:** no low-speed limit (it hovers), but a **power ceiling** on top (hover ceiling at zero speed) and a **never-exceed speed** V_{NE} on the right due to **retreating-blade stall** and **advancing-tip compressibility**; at low height and speed lies the **height–velocity "dead man's curve"** where an engine failure cannot be survived.

Passenger jet (Figure 7.23). The diagram is plotted against Mach number, so that each limit is a curve.

- **Low-speed limit (stall):** lift $L = W$ needs $C_L = 2W/(\rho V^2 S) \le C_{L,max}$, so the stall speed is $V_s \propto \sqrt{W/(\rho\, S\, C_{L,max})}$. At altitude the true speed (and the Mach number) at stall is higher because ρ is lower. At high Mach the buffet boundary replaces the stall: shock waves on the wing separate the flow at a C_L well below $C_{L,max}$.
- **High-speed limit:** up to a crossover altitude it is V_{MO} (about 340 kt of calibrated airspeed in many airliners), set by dynamic pressure, structural loads and aeroelastic flutter; above it, M_{MO} (typically 0.82 to 0.86), set by the rise of wave drag, the buffet and the loss of control (*Mach tuck*).
- **Ceiling:** the thrust decreases with altitude roughly in proportion to the density, while the drag at the cruise C_L stays similar: when the excess thrust is zero (in practice, below a minimum rate of climb) the **service ceiling** is reached. A certified maximum altitude (cabin pressurization) also applies. Cruise is at about FL350 to FL400.
- **Coffin corner:** with increasing altitude, the low-speed buffet Mach increases and the maximum Mach decreases, until they meet; in the corner the margin between stall and Mach buffet is nearly zero, and the corner is lower for heavier aircraft. In the figure it lies above the thrust ceiling (dotted), which is binding first for this weight.

Helicopter (Figure 7.24).

- **No low-speed limit.** A helicopter can hover, so the left boundary is $V = 0$. The envelope is plotted against altitude (density altitude) to include the effect of temperature.
- **Power ceiling.** Power required exceeds power available above a certain altitude: at $V = 0$ it is the **hover ceiling** (lower out of ground effect than in ground effect). The ceiling is higher at moderate speed, because the induced power falls with forward speed (translational lift) up to the speed of minimum power (about 60 to 80 kt), and then falls again as the parasite power grows with V^3.
- V_{NE} **(never-exceed speed).** The advancing blade tip approaches the speed of sound ($\Omega R + V$, Mach 0.85 to 0.9) and the drag rises (compressibility); the retreating blade has little relative speed ($\Omega R - V$) and needs a large angle of attack to produce its lift, so it **stalls** (vibration, pitch-up, roll). Both worsen with altitude and weight, so V_{NE} is reduced there. Typical maximum speeds of conventional helicopters are 150 to 170 kt, an order of magnitude.
- **Height–velocity ("dead man's") diagram.** Panel (b) shows in red the combinations of height above ground and speed to avoid: if the engine fails at low speed and high height (above the low hover) there is not enough time or speed to build up rotor speed and flare, and

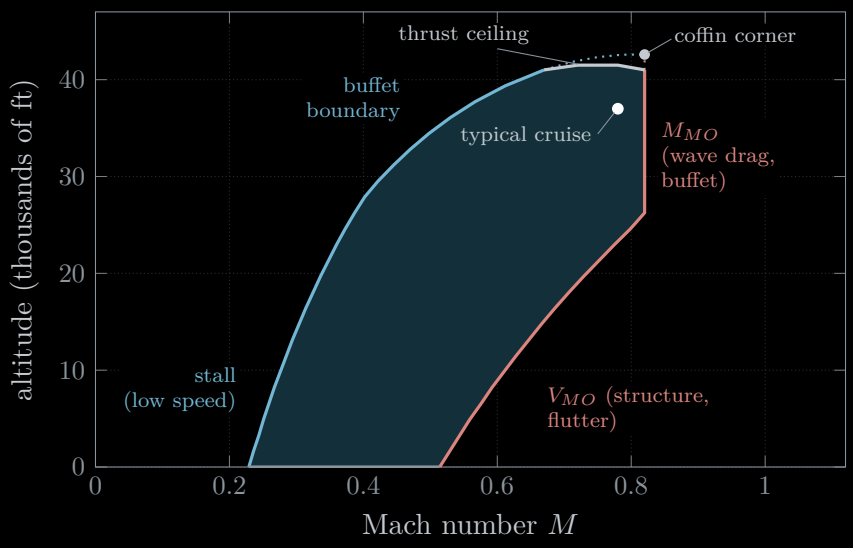


Figure 7.23. Speed–altitude flight envelope of a passenger jet (own illustrative model for $W = 65$ t and $S = 122$ m^2: a C_L limit falling from 1.4 to 0.67 between $M = 0.4$ and 0.82, $V_{MO} = 340$ kt, $M_{MO} = 0.82$, ceiling at 41 000 ft). The shaded area is the permitted region.

if it fails at high speed and very low height there is no room to flare. The exact curve is in the flight manual of each helicopter (typical orders of magnitude: low hover under about 15 ft; avoid area up to 400 to 500 ft near zero speed).

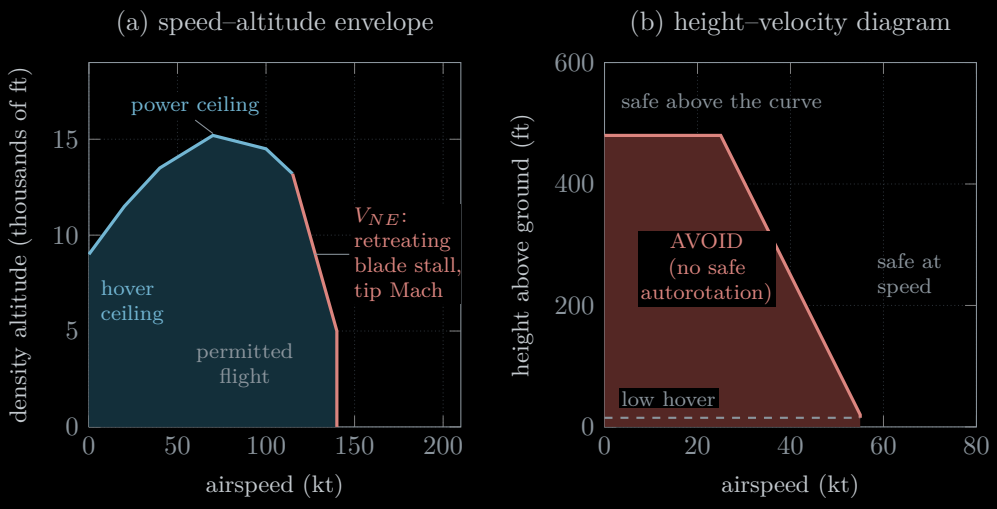


Figure 7.24. (a) Speed–altitude envelope of a conventional helicopter (illustrative). (b) Height–velocity diagram of a single-engine helicopter (illustrative shape): the red region is to be avoided. Both are schematic and the numbers are only typical orders of magnitude.

3.13 Yaw control of a tandem twin-rotor helicopter

Question 20. How do you control yaw in a tandem twin-rotor helicopter?

ANSWER FROM GENERAL KNOWLEDGE (NOT IN THE SLIDES)
By **differential lateral cyclic pitch**: the two rotors are tilted sideways in opposite directions, so the front rotor pushes to one side and the rear rotor to the other. The two lateral forces, separated by the fuselage length, give a **yawing moment**. There is no tail rotor: the pedals command this differential cyclic. In a tandem the rotors turn in opposite directions, so their torques cancel; **pitch** is controlled by **differential collective** (more thrust on one

rotor than on the other), and **roll** by lateral cyclic in the same direction on both rotors.

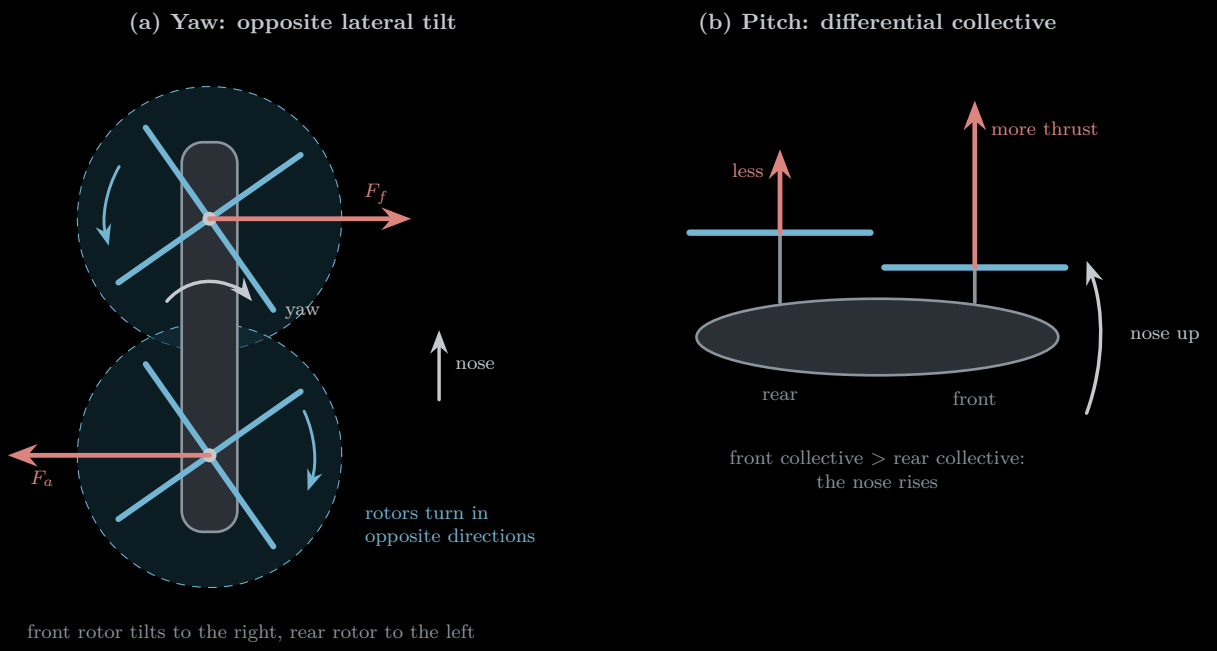


Figure 7.25. Tandem rotor (Chinook type). (a) Top view: the front rotor tilts its thrust to the right and the rear rotor to the left; the couple turns the nose to the right. (b) Side view: more collective on the front rotor than on the rear raises the nose. Schematic.

Explanation. Each rotor has its own swashplate. For the pedals, the lateral cyclic of the front rotor is tilted, for example, to the right and that of the rear rotor to the left by the same amount. The net lateral force is zero (the aircraft does not move sideways) but the two forces F_f and F_a act at a distance L from the centre of gravity, so the yawing moment is $M_z \approx (F_f + F_a)\, L/2$, in magnitude. For the other axes: the **collective lever** raises or lowers the collective on both rotors equally (climb); the **cyclic stick** forward–back tilts both discs the same way (forward flight) and sideways tilts them the same way (roll); the **pedals** give the differential lateral cyclic.

Compared with a single rotor. The tail rotor of the conventional layout takes power that does not contribute to lift, and the tandem uses all the power for lift (**no anti-torque losses**); it also allows a wide centre-of-gravity range and a large cabin. The disadvantages are the complexity of two rotors with their transmission, and the rear rotor working in the wake of the front one (loss of efficiency and vibration). Other configurations that avoid a tail rotor, such as coaxial rotors, also cancel torque with counter-rotating rotors.